



**GREEN
CLIMATE
FUND**

Meeting of the Board
23 – 26 October 2023
Tbilisi, Georgia
Provisional agenda item 12

GCF/B.37/02/Add.06

2 October 2023

Consideration of funding proposals – Addendum VI

Funding proposal package for FP215

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Community Resilience Partnership Program";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Community Resilience Partnership Program
Country(ies):	Multiple Countries: Asia and Pacific Region
Accredited Entity:	Asian Development Bank
Date of first submission:	2020/12/20
Date of current submission	2023/09/25
Version number	V.12



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

FP-ADB-Regional-2022/08/05

PROJECT/PROGRAMME SUMMARY				
A.1. Project or programme	Programme	A.2. Public or private sector	Public	
A.3. Request for Proposals (RFP)	Not applicable			
A.4. Result area(s)			GCF contribution	Co-financers' contribution¹
	Mitigation total		Enter number %	Enter number %
	☐ Energy generation and access		Enter number %	Enter number %
	☐ Low-emission transport		Enter number %	Enter number %
	☐ Buildings, cities, industries and appliances		Enter number %	Enter number %
	☐ Forestry and land use		Enter number %	Enter number %
	Adaptation total		Enter number %	Enter number %
	☒ Most vulnerable people and communities		30 %	30 %
	☒ Health and well-being, and food and water security		30 %	30 %
	☒ Infrastructure and built environment		20 %	20 %
	☒ Ecosystems and ecosystem services		20 %	20 %
A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	Not applicable	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	A minimum of 3,219,400 direct beneficiaries Cambodia – 1,300,000 Indonesia – 1,200,000 Lao PDR – 141,000 Pakistan – 488,000 Papua New Guinea – 10,000 Timor-Leste -60,000 Vanuatu - 20,400	A minimum of 4,465,000 indirect beneficiaries Cambodia - 1,365,000 Indonesia – 450,000 Lao PDR – 1,143,000 Pakistan – 1,200,000 Papua New Guinea – 25,500 Timor-Leste - 241,500 Vanuatu - 40,000
			A minimum of 0.63% of total population Cambodia – 8.13% Indonesia – 0.45% Lao PDR – 2.04% Pakistan – 0.23% Papua New Guinea – 0.11% Timor-Leste – 4.62% Vanuatu – 7.02%	A minimum of 0.87% of total population Cambodia – 8.53% Indonesia – 0.17% Lao PDR – 16.57% Pakistan – 0.57% Papua New Guinea – 0.27% Timor-Leste - 18.58% Vanuatu – 13.76%
A.7. Total financing (GCF + co-finance²)	750 million USD	A.9. Project size	Large (Over USD 250 million)	
A.8. Total GCF funding requested	120 million USD			
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.			

	☒ Grant 100 million USD ☒ Loan 20 million USD ☐ Guarantee <u>Enter number</u>	☐ Equity <u>Enter number</u> ☐ Results-based payment <u>Enter number</u>	
A.11. Implementation period	9 years	A.12. Total lifespan	20 years
A.13. Expected date of AE internal approval	The Community Resilience Financing Partnership Facility, being managed by ADB, will operationalize the CRPP program. The facility was approved by ADB on 12 August 2021. Individual projects to be financed by the CRPP have their own approval dates, with initial projects expected to be approved in 2024.	A.14. ESS category	I-2
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☐ No ☒	A.18. Is this FP included in the country programme?	Yes ☐ No ☒
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	The Accredited Entity is the Asian Development Bank (ADB), 6 ADB Avenue, Mandaluyong City 1550, Metro Manila, Philippines. ADB will be the executing entity. For Projects implemented in Host Countries, the government of the Host Country, acting through its relevant ministry or department, may be the executing agency and/or implementing agency (each as defined by ADB) as further described in this Funding Proposal and, to the extent such Host Country government will use or channel GCF Proceeds for the purposes of the Funded Activity and/or execute, carry out or implement Funded Activities, the relevant Host Country government will be the Executing Entity.”		

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Refer to the Policy of Co-financing of the GCF.

A.21. Executive summary

1. Climate Change Context

Increasing climate shocks and stresses in Asia and the Pacific region require local adaptation solutions that deliver at scale. Evidence indicates that the impacts of heat waves, sea-level rise and storm surges, changes in rainfall patterns, and increases in the intensity and frequency of extreme weather events are already affecting the livelihoods, food security and nutrition, water security and health of millions of people in the region, including the populations of Cambodia, Indonesia, Lao People's Democratic Republic (Lao PDR), Pakistan, Papua New Guinea (PNG), Timor Leste, and Vanuatu.³ Looking ahead, climate change is projected to increase climate risk with changes in the frequency, intensity, extent, duration, and timing of extreme weather events, including unprecedented extremes, and alteration of trends and shifts in the long-term average climate (see Box 1 in section B1). Assessments shows that climate risk is likely to intensify and occur both sooner and at a larger scale than previously assessed. Estimates for the Asia and the Pacific region show that growth will be significantly impacted by climate change. Without climate action, the gross domestic product (GDP) in the region could decrease by as much as 3.3 per cent by 2050 and 10 per cent by 2100, relative to the base case.¹ There is thus a clear and immediate need for adaptation solutions in the region that can be implemented **at scale**. Moreover, the adverse impacts of climate change are manifested largely at the local level due to the local socioeconomic, geographical, ecological, institutional and governance characteristics. Hence, the need for **local adaptation** solutions that are tailored to the specific physical and socioeconomic characteristics of local contexts.

Poor and vulnerable populations face the brunt of climate impacts and require targeted adaptation solutions. Estimates suggest that climate change could push more than 100 million people in developing countries below the poverty line by 2030, with Asia and the Pacific being one of the main hotspots.² This would undermine the progress in poverty reduction in the region and pose a threat to achieving the Sustainable Development Goals (SDGs). The poor and vulnerable are affected disproportionately (in relative terms) because of high levels of **exposure and vulnerability** (due to their sources of livelihoods, incomes, and location) and low **adaptive capacity** (the adaptation and poverty nexus).³ These populations have limited access to climate information, are often left out from formal decision-making processes, and have limited knowledge and financial capacity to undertake adaptation solutions. Thus, their local needs and priorities for adaptation often remain unmet. **Women and men are impacted differently**, with women typically having lower levels of adaptive capacity than men due to pre-existing gender inequalities dictated by cultural norms, and different roles and responsibilities at the household and community levels.⁴ For these reasons **adaptation solutions that are explicitly targeted at poor and vulnerable people** are required, with specific provisions for their empowerment, especially of women.⁵ The pandemic due to coronavirus disease (COVID-19) has reinforced these inequalities, as it is having a disproportionate impact on the poor and especially poor women, who are already burdened with the ill-effects of climate shocks and stresses. Estimates suggest that severe long-term effects of this pandemic could push an additional 78 million people into extreme poverty (under USD1.90 per day) in developing Asia.⁶

Climate finance plays a critical role in facilitating transformational adaptation among the poor and the vulnerable. Climate finance plays a key role in supporting adaptation solutions that improve the well-being of the poorest. However, at present less than USD1 in every USD10 of climate finance committed from dedicated climate funds explicitly seeks to support local climate action.⁷ Global financial institutions can play a key role in addressing this imbalance by (i) making climate finance relevant and unlocking the potential for to the poorest and most vulnerable to **replicate** adaptation measures; (ii) facilitating **systemic change** by strengthening the meaningful participation of the poor and vulnerable, especially women, in adaptation-related decision-making activities; (iii) acting at **scale** across all levels – household, community, local and national governments; and (iv) ensuring **sustainability** by

³ Annex 2.1 of this proposal provides a detailed presentation of the country level climate risk assessments carried out as part of the design and development of the program.

helping local institutions mainstream adaptation and resilience in development processes. It is only by such approach that climate finance can facilitate transformational change at the local level.

2. Proposed Intervention

A regional climate adaptation program targeted at the poor. The **Community Resilience Partnership Program (CRPP)** is a regional partnership program of the Asian Development Bank (ADB) which aims to help countries and communities in Asia and the Pacific region scale up investments in climate adaptation, especially investments at the community level, that explicitly target the **nexus between climate change, poverty, and gender**. The CRPP aims to contribute to transformational change by: (i) mobilizing large-scale public investments that support community level adaptation of poor and vulnerable people; (ii) developing national and local policies, plans, and programs that promote financing for community-led adaptation; and (iii) increasing the meaningful participation of poor women and men in resilience related decision-making. In doing so, the CRPP will address the points of procedural and distributive justice so that the people most vulnerable to the impacts of climate change can engage in a fair process and receive a fair share of the benefits of adaptation efforts.

The CRPP will support countries to meet their commitments to the Paris Agreement, the Sendai Framework for Disaster Risk Reduction, and the Sustainable Development Goals. The program responds to the **COP26 goal** of scaling up adaptation to protect communities and natural habitats, and mobilizing finance for climate adaptation measures that meet the needs of the poor and vulnerable communities and to the decision of **COP 27** which calls for scaling up of approaches in support of transformational adaptation. The CRPP is also identified as flagship program under the **locally led action track of the Global Commission on Adaptation**, which aims to spur financing for adaptation measures at the local level, and to help create structures that support appropriate subsidiarity and give local actors greater influence for adaptation-related decision-making. The CRPP is directly aligned with the core recommendations of the Intergovernmental Panel on Climate Change Working Group 2 Report “Climate Change 2022: Impacts, Adaptation and Vulnerability”, which highlights that largest adaptation gap exists in low-income population group and that for adaptation efforts to be effective, they must be concentrated on promoting climate justice and supporting the most economically and socially marginalized populations and calls for integrated, multi-sectoral solutions that address social inequalities and cut across systems.⁸ Further, with its integrated multi-sector, regional programmatic approach and focus on communities and locally implemented adaptation, the CRPP will directly contribute to a number of **GCFs results as outlined in the 2024-2027 strategic plan**, including those related to Climate Information and Early Warning Systems (CIEWS), Food, Ecosystems, Infrastructure, and elements of locally led adaptation principles.

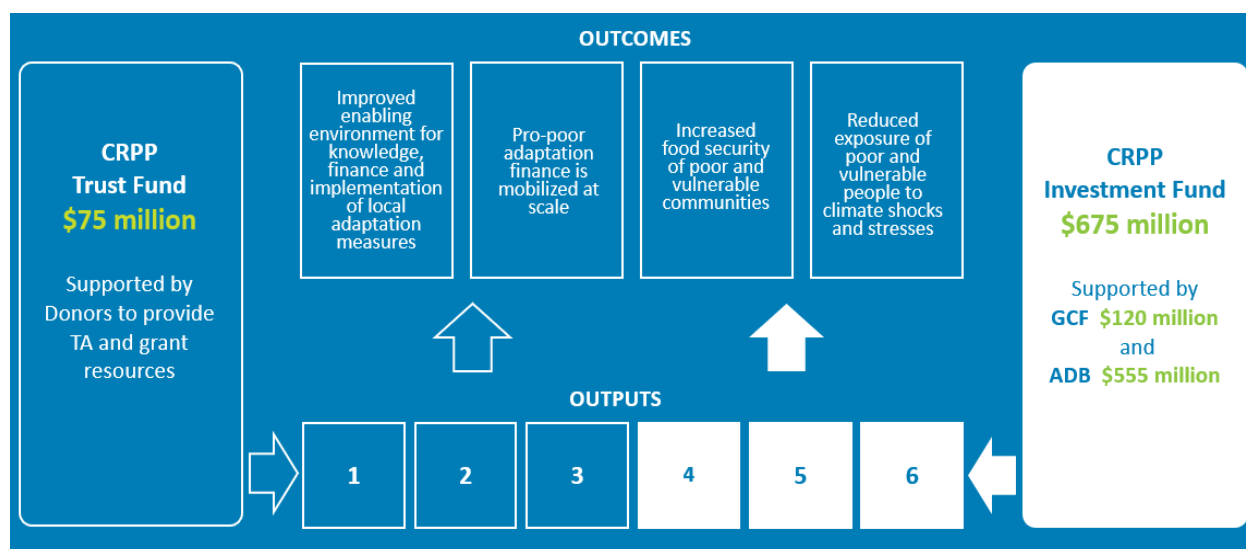
A programmatic approach for delivering pro-poor adaptation solutions at scale. The CRPP is operationalized through the [Community Resilience Financing Partnership Facility \(CRFPF\)](#) which was established by ADB in August 2021 and comprises two separate but interlinked components; the **CRPP Trust Fund (TF)** focusing on upstream support to strengthen the enabling environment required for implementing local adaptation measures at scale; and the **CRPP Investment Fund (IF)** focusing on the efficient roll-out of local adaptation measures through a downstream investment program. CRFPF is not a separate legal entity, but a facility administered within and by ADB. (See Figure 1)

The TF provides technical assistance and grant resources financed by development partners and administered by ADB, to selected developing member countries (DMC) of the ADB to implement the following three **outputs**: (1) Knowledge and action research on climate risk informed pro-poor community-level solutions strengthened; (2) Institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened; and (3) Inclusive and pro-poor adaptation investment projects identified and prepared. The TF includes a dedicated **Gender Window** with funds earmarked specifically for providing technical assistance and grants for identifying, developing, and implementing investments that explicitly strengthen the resilience of women, particularly poor women, and/or adaptation investments led by women that have a specific focus on supporting women to build their resilience. The TF will have a budget of USD75 million with roughly 15% earmarked for the Gender Window. To date, Government of United Kingdom, the Nordic Development Fund, and

the French Development Agency has confirmed financial support for the TF, among to a total of ~ USD68 million. The TF will **not** be funded by GCF.

The IF will be a program to provide grant and loan financing to seven selected DMCs, namely, Cambodia, Indonesia, Lao PDR, Pakistan, Papua New Guinea, Timor-Leste, and Vanuatu, to implement local adaptation measures as part of ADB financed projects that are targeted at the poor and vulnerable population. The IF will deliver three climate change related **outputs** that are the focus of this proposal (4) **information and systems** for delivering applied climate-risk informed local investments at scale improved; (5) climate resilient pro-poor **livelihoods** investments implemented; and (6) pro-poor climate adaptation **infrastructure** investments implemented. All projects financed by the IF will be implemented by the respective governments following ADB's procedures. The GCF is requested to provide USD100 million as grants and USD20 million as loan for the seven countries identified above in support of the IF of this program, while the ADB will provide USD 555 million of grant and loan financing. The program will utilize GCF resources to co-finance 10-15 projects implemented under the IF only, for outputs 4, 5 and 6 as shown in **Table 1** below.

Figure 1: CRPP Program Architecture



Source: ADB

The GCF funds for the CRPP IF will be ring-fenced within the ADB GCF Trust Fund which was created pursuant to the Accreditation Master Agreement. Usage of GCF funds for individual projects will be based on criteria outlined in this proposal which will be included in a dedicated CRPP IF Implementation Guidelines to be developed during inception. The 7 countries identified above as recipients of the IF grant and loan financing have been selected based on a combination of factors including (i) high levels of climate risk and pockets of poverty; (ii) national commitments to scale up climate adaptation investments targeted at the poorest populations; (iii) advanced levels of decentralization that promotes local actions, including local climate actions and ensures sustainability of such actions; (iv) climate adaptation and resilience priorities in ADB's country partnership strategy for the country and indicative pipeline of investments.

Value added of a regional program. The ADB recognises that transformational change often benefits from the kinds of strategic, coordinated response that can be pursued through programs. By adopting a programmatic approach, the CRPP will be able to demonstrate stronger coordination and strategic alignment between its projects in different countries, unified by the overarching vision and common objectives of strengthening community resilience, and thereby allow delivering sustained climate results and impacts in the GCF results areas efficiently, effectively and at scale. The programmatic approach

will bring cost efficiency, higher implementation effectiveness and facilitate shared learning and knowledge generation between participating countries and at a regional level, all of which will be captured in the program's monitoring and reporting arrangements. The proposed inclusive governance structure of the program will facilitate stronger support from participating governments and development partners (that would be financing the CRPP TF) towards the successful implementation of the CRPP.

3. Climate Impacts/Benefits

Climate resilient sustainable development. The anticipated **impact** of the CRPP is that climate resilient sustainable development in Asia and the Pacific is advanced. This represents a paradigm shift from current development pathways that are failing to adequately respond to current and future climate risks. Improved resilience of poor and vulnerable communities is considered a key pre-requisite for bringing about a paradigm shift towards climate resilient sustainable development. As such, the outcome of the program is expected to be that climate resilience of approximately **3.2 million** poor and vulnerable people, in seven countries in the Asia and the Pacific region is strengthened. This outcome marks the pathway towards achieving the programs impact and will be the result of investment in adaptation options that support **systemic change and transitions in socio-ecological systems** that are necessary processes for large-scale transformations and a core enabler of climate resilient development. Table 1 below provides a summary of the outputs that have been identified as necessary for achieving this outcome and the types of activities that would be financed by GCF under the CRPP IF and their anticipated contribution to building climate resilience.

Unique programmatic concept for scaling up local adaptation measures in the region. The CRPP is unique in its ability to develop localized climate adaptation solutions tailored to the needs of the poor and vulnerable populations; and to leverage a range of large ADB supported investments (projects) in social development, gender equality, livelihoods and governance to deliver on adaptation outcomes. The CRPP will directly contribute to the implementation of ADB's ambitious climate targets as set out in its Strategy 2030 and its ambition of providing climate finance of USD100 billion between 2019-2030, of which USD34 billion will support adaptation.⁴ As a multi-country program that comprises a varied but strongly interconnected set of outputs supporting upstream actions to strengthen the enabling environment for adaptation and actions for rolling out of adaptation measures through downstream investments (projects), the CRPP will enable a **coordinated approach** to the deployment of adaptation financing across different geographies and contexts. This programmatic approach allows for the accumulation and **transfer of knowledge** and skills between countries and within countries, **facilitating scalability, replication and sustainability** of adaptation solutions that are demonstrably effective over the course of the program implementation period. It also allows for the reduction of costs for key functions such as monitoring, evaluation and learning and gender mainstreaming.

⁴ ADB has set targets of providing \$80 billion in climate finance from its own resources over the period of 2019 to 2030 and of ensuring that 75% of the number of its committed operations will support climate change mitigation and adaptation by 2030. ADB 2018. *Strategy 2030. Achieving a Prosperous, Inclusive, Resilient, and Sustainable Asia and the Pacific*. Manila.

Table 1: CRPP Outputs, Activities, and their Contribution to Building Climate Resilience

CRPP Outputs		Typical activities to be financed	Contribution to Building Climate Resilience
Trust Fund (Not Financed by GCF)			
1	Knowledge and action research on climate risk informed pro-poor community-level solutions strengthened.	- Knowledge and evidence generation on community-led adaptation solutions. - Evidence sharing and awareness raising on community-led adaptation solutions.	Investing in coordinated and structured knowledge generation and learning, creating a wide range of evidence and knowledge products on climate risk (hazard, exposure and vulnerability) informed pro-poor local solutions, and carrying out policy dialogues at sub-national, national, and regional level will support shifts in attitudes, values, consciousness, and behaviour required for transitions in societal systems towards climate resilient development pathways.
2	Institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened.	- Development of climate risk assessment and decision support tools for local level decision-making. - Training and capacity-building on designing and implementing adaptation investments at the community level.	Creating policy incentives and procedures for the inclusion of local people in adaptation decision-making and building the capacity of governments and communities to use participatory tools and procedures for climate risk (hazard, exposure, and vulnerability) informed development planning and implementation empowers local actors to act on adaptation and supports transitions in societal systems towards climate resilient development pathways.
3	Inclusive and pro-poor adaptation investment projects identified and prepared.	- Preparation of large-scale investment projects with local adaptation solutions - Pilot innovative transformational solutions for building local resilience	Catalyse adaptation finance for implementing local solutions at scale by mobilising large-scale domestic and international resources for pro-poor and gender-responsive adaptation at the local level.
Investment Fund (Financed by GCF)			
4	Information and systems for delivering applied climate-risk informed investments at scale	- Climate information services for key sectors to meet local adaptation needs - Climate risk informed social protection systems - Climate risk information decentralization policies	Suitable, locally relevant, and accessible climate risk information (hazard, exposure, and vulnerability) and understanding of current and future impacts plays a critical role in aiding adaptation decision-making at local levels (household, community and local government), informing local adaptation and development plans, mobilising both domestic and international climate financing, and supporting systems transitions. Climate risk informed social protection systems can protect poor and vulnerable households from the impact of climate change related shocks and build adaptive capacity and reduce vulnerability especially when combined with complementary interventions such as early warning systems, resilient livelihoods, and disaster risk insurance. (Note: GCF resources will only be used to strengthen social protection systems from resilience angle and will not be used to cover costs of actual transfers (such as cash transfers) made under social protection programs or to pay the premium of disaster risk insurance programs.) Climate risk informed decentralization processes (planning, budgeting, implementation, and monitoring) can support coordinated development of effective local responses to climate change and pursue procedural justice in ensuring community engagement, especially engagement of women's group and the delivery of demand-side measures and adaptation implementation thereby increasing adaptive capacity of local communities.
5	Climate resilient pro-poor livelihoods investments implemented	- Climate resilient agroecological systems.	By investing in adaptation options implemented by small holder farmers, especially women farmers such as conservation agriculture and agroforestry, and integrated land and water management this output builds adaptive capacity of poor and vulnerable people, contributes to ensuring access to sufficient, quality food through increasing yields and promotes sustainable use of natural resources, thus facilitating transitions in land and ecosystems towards climate resilient development pathways.

		- Information technology and risk management services for climate resilient livelihoods	Investing in information technology systems to facilitate rapid information flows to small holder farmers, especially women farmers and sharing and spreading financial burdens that result from the impacts of climate change related shocks and stresses through tools such as index based micro-crop and livestock insurance programs, builds adaptive capacity and spreads risk and provides a financial buffer against the impacts of climate hazards. (Note: GCF resources will not be used to pay premium of disaster risk insurance programs).
		- Climate resilient agriculture supply chains.	Investing in transport and logistics infrastructure, protective measures for the siting and storage of seeds and providing technological solutions and capacity building for improved food processing strategies can reduce the risk of decrease in mass and nutritional value of food as a result of climate change, promoting food security and resilient livelihoods.
6	Pro-poor climate adaptation infrastructure investments implemented	☐ Ecosystem-based infrastructure	Investing in ecosystem-based infrastructure can minimize the exposure and physical vulnerabilities of the poor and vulnerable communities to the impacts of climate change and protect their productive assets, while providing additional regulating, provisioning, cultural and support services that promote improvements in adaptive capacity. Ecosystem-based infrastructure can also contribute to local livelihoods, especially for women. Support transitions in land and ecosystems and urban and infrastructure systems.
		☐ Flood and landslide protection infrastructure	Investments in protection infrastructure, especially nature-based solutions in rural and urban communities can limit exposure to climate hazards and thereby reduce the high mortality rate, degradation and irreparable damages and disruption to services associate with floods and landslides promoting transitions in urban and infrastructure systems.
		☐ Multipurpose emergency shelters	Investments in adequately designed emergency shelters that are constructed close to where the poor and vulnerable reside and with inclusive and gender-responsive design features will reduce the vulnerabilities by reducing casualties associated with the impacts of extreme events, safeguarding human and social capital and adaptive capacity. Support transitions in society and urban and infrastructure systems.

PROJECT/PROGRAMME INFORMATION

B.1. Climate context

1. Climate Change Context

Climate risk is a key threat to sustainable development. Countries in Asia and the Pacific region are highly exposed to the damaging impacts of climate change. Prolonged heat waves, sea-level rise, changes in rainfall patterns, and increases in the intensity and frequency of extreme weather events are having a major impact on economies, natural and physical assets, and are compounding development challenges. Between 2000-2019, the region experienced 3,068 disaster events accounting for 42% of global events. The region experienced 41% of all flooding events and with a total of 1.5 billion people affected, accounting for 93% of people affected by floods worldwide; followed by tropical cyclones accounting for 79% of people affected and 90% of deaths globally.⁹ For example, in **Cambodia**, annual flooding along the Mekong River and Tonle Sap basins results in yearly losses of between USD100-170 and in 2015, climate related disasters resulted in losses of USD1.5 billion (equivalent to 10% of GDP).^{10,11} In 2015, Cyclone Pam, a category 5 tropical cyclone, struck **Vanuatu** killing 11 people, damaging over 15,000 buildings and causing economic losses around 64% of GDP.¹² The economic costs associated with disasters across the region are increasing. Estimates suggests that on an annual average basis, the costs of disasters to the region have increased from USD1.8bn during the 1970s, to USD73.8bn between 2004 and 2013, a forty-fold increase, representing 49% of global average annual losses.¹³

Climate risk is likely to intensify and occur both sooner and at a larger scale than previously assessed with vast adverse impacts on development. Looking ahead, ongoing climate change is projected to alter the frequency, intensity, extent, duration, and timing of extreme weather events, and is likely to result in unprecedented extremes.¹⁴ The Intergovernmental Panel on Climate Change (IPCC) Working Group 2 Report “Climate Change 2022: Impacts, Adaptation and Vulnerability” states with high confidence that global warming, reaching 1.5°C in the near-term, would cause unavoidable increases in multiple climate hazards and present multiple risks to ecosystems and humans (Box 1 at the end of this section highlights how the CRPP climate rationale is closely aligned to key headline messages from the IPCC’s Sixth Assessment Report (AR6)). In **PNG**, for example, intensive rainfall events are expected to increase by 20-30% by 2090;¹⁵ and in **Lao PDR** due to increased rainfall in its sub-basins, the Mekong River will have higher discharge leading to increased floods.^{16,17} Climate change is also projected to alter trends and induce shifts in the long-term average climate that will have major impacts in the region. For example, a study estimates that **Pakistan** experienced 126 heat waves between 1997–2015, around 7 per year, and these numbers are expected to increase in the future, with estimates suggesting that the country will experience 12 heat wave events annually by 2030, 20 such events by 2060 and 26 events by 2090.¹⁸ Similarly, it is likely that climate change increased the impact of the 2022 flood disaster in Pakistan. Monsoon rainfall in Balochistan and Sindh is estimated to be 50%–75% more intense than it would be in the absence of climate change.¹⁹ The impacts are projected to be disproportionately high for Asia compared to other regions, with estimates suggesting that the economic impact of climate change on Asia will be double those of the global average by 2060.²⁰ The Pacific Islands are recognised as among the most vulnerable, with studies estimating that climate change impact could be equivalent to 12-13% of the GDP for the region under a high global emission scenario, and 5% of GDP even with average temperature rises of under 2°C.²¹ Recent evidence suggests these may be under-estimates, because of the increased threat of sea-level rise,²² which will pose very large risk for atolls and small islands. Recent assessments highlight that even under the Paris consensus scenario in which global warming is limited to 1.5°C to 2°C above preindustrial levels, much of the land area, ecosystems, and socioeconomic sectors of Asia and the Pacific will be significantly affected by climate change impacts,²³ thus, emphasizing the need for adaptation. While the level of risk significantly varies, the countries selected under this program are all at high risks. According to Notre-Dame GAIN Index 2020, the rank of the selected countries range between 97 and 152 out of 192 countries (Cambodia 140,

Indonesia 97, Lao PDR 142, Pakistan 152, PNG 149, Timor Leste 111 and Vanuatu 142).⁵ The higher the country score, the higher the vulnerability to climate change and other global challenges in combination with its readiness to improve resilience.

Adverse impacts of climate change will be largely manifested at the local level. This is because, the underlying drivers of exposure and vulnerability, including socio-economic, ecological, institutions and governance are inherently context-specific and reflected at local scale.²⁴ Thus, adaptation solutions will be needed at the local level, which captures a granular knowledge on hazards (historic and observed changes in weather patterns) and their interaction with factors contributing to exposure (such as land use changes) and vulnerability (such as existing social norms which decides women's social and productive role in the community) of and how such interaction leads to adverse impacts on people, assets and livelihoods.. It will also require understanding traditional practices and resources deployed by local communities to deal with climate shocks and stresses, and how such practices can be strengthened and scaled up.

Poor and vulnerable populations face the brunt of climate impacts. Estimates suggest that climate change could push more than 100 million people globally in developing countries below the poverty line by 2030, with Asia and the Pacific being the main hotspot (along with sub-Saharan Africa).²⁵ For example, studies show that while **Indonesia's** poverty level has declined significantly in the last decade, about 27% of the country's population have incomes that are marginally higher than the poverty line and could slide back into poverty if food prices surge or if farm incomes suddenly decline due to prolonged drought from El Niño.²⁶ Many households in **Lao PDR** have a high probability of falling into extreme poverty even when exposed to relatively low frequency flood and drought events and in **Cambodia**, an event occurring once in every five years has approximately a 50% chance of pushing a household into extreme poverty.²⁷ The poor and vulnerable are affected disproportionately (in relative terms) because they have high climate sensitivity (due to their activities, incomes and often their location in high-risk areas), they are often close to climate thresholds already, and they have low adaptive capacity.²⁸ They typically rely on fewer and more vulnerable assets and basic services, depend on climate sensitive sectors for livelihoods, have limited access to climate information, have limited technical and financial capacity to implement adaptation measures, and are often left out from formal decision-making processes, which increases their vulnerability to climate impacts. Subsistence-oriented communities around Asia and the Pacific are already observing clear-cut impacts of anthropogenic climate change.²⁹ Women in Asia and the Pacific represent the majority of small-scale and subsistence farmers and are often responsible for the use and management of natural resources. In many cases, the effects of climate impact require the poor to adopt negative coping strategies including reducing quantity and nutritional quality of food consumption, incurring high level of indebtedness, unsustainable use of scarce natural resources, and making trade-offs in investments regarding education, health, and livelihoods that may impact their long-term well-being and further affect their poverty level. Further, climate impacts could disrupt ecosystem services and lead to severe effects on livelihoods which in turn would affect human health, migration dynamics and the potential of conflicts.³⁰

Women and men are impacted differently by climate change, with women typically lacking equal preventative and coping capacity due to pre-existing gender inequalities. Multidimensional poverty significantly limits women's ability to confront climate change. In Asia, women account for 80% of all people living on less than USD2 per day (UN Women 2018)³¹, and their economic activities are often highly concentrated in sectors that are dependent on natural resources such as agriculture. For example, agriculture accounts for 65% of women's total employment in **Pakistan**, 64% in **Lao PDR**, and 60% in **Papua New Guinea**.³² Women also tend to be overly burdened with household work and caring for children, the sick, and the elderly; women do not always have the same rights as men to land; women often have less access to financial services, technology, information, and early warning systems; and women, particularly rural women, are often held back from fully participating in their community and in formal decision-making processes^{33,34,35}. All of these factors contribute to increased vulnerability and lower adaptive capacity, and are closely related to socially created gender functions, systemic discriminations, and the power imbalances between women and men. While it is crucial to recognize and address women's vulnerabilities to climate

⁵ The Notre-Dame GAIN Index summarizes a countries vulnerability to climate change. Higher the country score, higher the vulnerability to climate change and other global challenges in combination with its readiness to improve resilience.

change and understand the structural causes for them, it is fundamental to refrain from categorizing women as innately weak, passive receivers of protection and support. In the past, this approach has prevented women from being recognized as active agents in resilience-building programs. The adoption of programs designed on the assumption that women are powerless further exposes them to climate change shocks and disasters, as well as hinders the realization of women's rights to full and equal participation and decision-making³⁶. Women's unique knowledge, skills, and perspectives, mean that they can contribute positively to the identification of appropriate adaptation measures and ensure that they are implemented effectively and efficiently. Women can also play a leadership role in accessing public and private resources for adaptation, which in turn can give them recognition within the community and promote their position in the family and wider society, creating a positive feedback loop for women's empowerment. These different experiences between men and women in terms of their vulnerability and role of resilience building within communities are not adequately reflected in exiting adaptation policy and practice in CRPP countries, and this require specific attention.

The COVID-19 pandemic has further highlighted the need for building resilience of the poor and vulnerable population. The pandemic is having a disproportionate impact on the poor and the vulnerable, including women, already burdened with the ill-effects of climate shocks and stresses. The poor, who often migrate internally to deal with seasonal lean periods are being forced to make difficult choices to deal with the crisis, which will have implications on their well-being. During this crisis, women and girls can also face higher risks of losing jobs or income, dropping out of school, increased unpaid care work, and domestic violence.³⁷ Estimates suggest that in **PNG**, the COVID-19 pandemic could result in an increase of the poverty rate from 27.2% to 31.7%; and in **Vanuatu**, from 13.9 % to 18.3% in 2020.³⁸

Box 1: CRPP climate change context echoes headline messages from the Working Group 2 contribution to the IPCC Sixth Assessment Report (AR6-WG2)

Headline messages from the IPCC AR6-WG2 contribution includes:

- There is **high confidence** that global warming, reaching 1.5°C in the near-term, would cause unavoidable increases in multiple climate hazards and present multiple risks to ecosystems and humans. For 127 identified key risks, assessed mid and long- term impacts are up to multiple times higher than currently observed.
- In Asia, climate change will exacerbate fundamental challenges around water and food insecurity, poverty and inequality, and increased frequency and severity of extreme events. Even at 1.5°C and more so at 2°C, climate change will significantly constrain the functioning and wellbeing of human and ecological systems in some places and for some people.
- There is **high confidence** that vulnerability is higher in locations with poverty, governance challenges and limited access to basic services and resources, and high levels of climate-sensitive livelihoods (e.g., smallholder farmers).
- There is **high confidence** that vulnerability at different spatial levels is exacerbated by inequity and marginalization linked to gender, ethnicity, low income or combinations thereof.

Source: IPCC 2022^{39,40}

2. Context

The paragraphs below provide brief description of climate risk context of CRPP IF countries, with more details provided in **Annex 2.1**.

Cambodia: Cambodia is ranked 140th out of 181 countries in the ND-GAIN Index⁴¹, due a combination of high vulnerability and low readiness. The country has a tropical monsoon climate, with two distinct seasons, a rainy season from May to October and a dry season from November to April. Average temperatures are relatively uniform across the country and annual average rainfall is high (especially in the highlands and along the coast), most of which falls during the Monsoon. However, there is high annual variability due to the influence of the Monsoon and the El Niño Southern Oscillation (ENSO).

The country has a varied topography, with mountainous and highland regions, the central plains, and a coastal zone, but it is dominated by the Mekong River, which flows directly from the north to Vietnam in the south. The Tonle Sap Lake, an outlet of the Mekong, is located in the northwest region, and covers almost 10% of the country's surface area during the monsoon peak. The central alluvial plains of the Mekong and Tonle Sap represent approximately 80% of the country. These basins are characterized by a flood-pulse hydrology with fluctuations in water levels between seasons, and are critical for agriculture and fisheries. The combination of the monsoon, climate variability from the ENSO, and the geographical terrain makes Cambodia extremely vulnerable to weather-related extremes (floods, storms, and periodically droughts). These occur almost every year and lead to high losses, reported at USD1.4 billion over the last twenty years (EM-DAT, 2021⁴²; DESINVENTAR, 2021⁴³) with annual damages of USD100 to 250 million (UNDRR 2019⁴⁴). Seasonal flooding is essential to agricultural and fisheries production, however, river and flash floods lead to major losses. Cambodia ranks as one of the world's most flood exposed countries in the world⁴⁵ in terms of the proportion of the population affected by floods (at 20-25%). There are also periodic droughts. During the 2015-16 El Nino drought event, 2.5 million people were affected, and three-quarters of the wet season paddy rice production experienced a loss of more than 25% in yield (World Bank, 2017⁴⁶). Tropical storms are rare, though in 2009 Typhoon Ketsana had a large impact (USD132 million in Cambodia, UNDRR, 2019). These weather-related impacts particularly affect rural people because of the reliance on rain-fed agriculture, and the disproportionate impact on the poor. Analysis by ADB (2017) finds that many households in Cambodia have a high probability of falling into extreme poverty even when exposed to relatively low intensity flood and drought events⁴⁷.

Recent observations show that the climate of Cambodia is changing (UNDP, 2012⁴⁸; Cambodia 2nd NC, 2015⁴⁹; World Bank, 2019⁵⁰). There is a clear warming trend, with 0.2°C per decade reported since the 1960s, and slightly above this in the dry season, and an increase in the number of 'hot days'. The patterns of annual rainfall trends are more uncertain, with some areas experiencing increases and others decreases, although there are indications of increasing variability and increasing extreme rainfall. The projected temperature increases for Cambodia indicates a further rise of 1°C to 2°C by mid-century (central multi-model ensemble values, for RCP2.6 and RCP8.5, relative to baseline 1986-2005 period, World Bank, 2020⁵¹; World Bank/ADB, 2021⁵²) and similar increases are found in downscaled modelling analysis (Thoeun, 2015⁵³). It is difficult to model rainfall changes due to the influence of Monsoon dynamics, but most climate models (and also the downscaled analysis) project increases in annual rainfall by the mid-century (2050s). There is more confidence from the climate models that heavy precipitation events will increase, which would increase the flood risks across the country. Modelling analysis for the Mekong indicates higher discharge leading to increased floods^{54,55,56} with greater areas of land and more people at risk of flooding (USAID, 2014⁵⁷). There has also been modeling of the potential climate change impacts on Tonle Sap lake and floodplain (Oeurng et al., 2019⁵⁸), which indicates large possible impacts on the fisheries sectors. For the agriculture sector, Chun et al (2016⁵⁹) report that Cambodia could suffer the largest reductions in rice yield in all Southeast Asia, and Jim et al (2018) also project large yield reductions for the country. These could be amplified by the possible increased annual likelihood of drought⁶⁰ and also by increases in heat and humidity which will reduce labour productivity (Dunne et al., 2013⁶¹). There are also projected increases in health impacts, driven by a projected increase in vector borne disease (WHO, 2015⁶²). Finally, there will be impacts from sea-level rise and storm surge along the 435 kilometres long coastal belt. Economic modelling studies estimate that climate change will reduce Cambodia's total GDP by 1.5 to 2.5% by 2030 and 3.5% to 10% by 2050 (RGC, 2015⁶³; UNDP, 2018⁶⁴).

Indonesia: Indonesia is ranked in the top-third of countries in terms of natural hazard risk (59 out of 191) by the 2020 INFORM Risk Index. In particular, Indonesia has high exposure to flooding, tropical cyclones and tsunamis. Between 2005 and 2015, Indonesia recorded more than 1,800 disaster events.⁶⁵ Studies have estimated the costs of climate change in Indonesia could be more than 6% of GDP annually by 2100, more than double the global average loss.⁶⁶ Despite high exposure to natural hazards, Indonesia ranks moderately in terms of its vulnerability and coping capacity, where it is ranked in the top half (102 and 88 respectively). However, vulnerability and coping capacity among the poorest groups is likely to be quite different, and inequality in the country remains high, with the bottom 40% of the population accounting for 17.7% of consumption⁶⁷. Prior to COVID-19, an estimated 61.8 million people, or 21.5% of the population, were

considered poor or have a high risk of falling into poverty, making them vulnerable to shocks.⁶⁸ This figure is likely to be significantly higher post COVID-19 impacts. Roughly 7% of urban population live below the poverty line and almost equal percentage of population live just above the poverty line. The rural and urban poor will bear the brunt of the climate change burden as poorer households in Indonesia are much less likely to take adaptive measures and plan for longer-term horizons than wealthier households, making them more exposed to environmental shocks and disasters.⁶⁹ The impact on livelihoods and food security is a particular concern. For example, Indonesia is positioned as one of the most vulnerable countries to potentially extreme heatwaves according to climate model projections.^{70,71} Heavy manual labour jobs are commonly among the lowest paid whilst also being most at risk of productivity losses due to heat stress.⁷² The poor in Indonesia are also particularly vulnerable to food price rises, which could become more common and pronounced due to climate's impact on food production and widespread harvest failure resulting from unusual and extreme climate conditions. For the 20-25% of the country's population that have incomes marginally higher than the poverty line, if food prices surge or if farm incomes suddenly decline due to the impact of climate extremes they could slide back into poverty.⁷³ A 100% increase in food prices could increase the number of Indonesians in extreme poverty by just over 25%.⁷⁴

Poor and vulnerable households in rural and urban areas are particularly exposed to flood risk as they frequently reside on low lying lands prone to flooding and suffer from the lack of access to good quality infrastructure and services. As of 2010, assuming protection for up to a 1-in-25 year event, the population annually affected by flooding in Indonesia is estimated at 1.5 million people and expected annual urban damage is estimated at USD1.4 billion.⁷⁵ There are not enough facilities and infrastructure available to abate these existing flooding conditions⁷⁶ and the risks are expected to increase with climate change, with large numbers of coastal cities facing sea level rise and high-density built environment. Climate change could amplify coastal flood risk 19-37% by 2030 (measured by Expected Annual Damage).⁷⁷ Indonesia is particularly exposed to sea-level rise, with the country ranked the fifth highest in terms of population inhabiting the lower elevation coastal zone⁷⁸. It is estimated that by 2030 around 5.5-8 million people could reside in a 100-year floodplain (an area exposed to 1-in-100-year coastal floods resulting from storm surges), growing to 9.5 – 14 million by 2060. These estimates assume a modest sea-level rise of 10cm by 2030 and 21cm by 2060.⁷⁹ Estimates suggest that the number of Indonesians exposed to river flood risk will increase by 75% in four decades between 2015 and 2055.⁸⁰ Climate change is expected to increase the annual urban damage from flooding by USD6.1 billion under the RCP8.5 emissions pathway (AQUEDUCT Scenario B).⁸¹ Even under lower emissions pathways consistent with the Paris Climate Agreement, almost all Asian countries face an increase in the frequency of extreme river flows. What would historically have been a 1-in-100 year flow, could become a 1-in-50 year or 1-in-25 year event in most of South, Southeast, and East Asia.⁸²

In rural areas, the dependence on agriculture for livelihoods is of particular concern for climate change impacts. Nationally, roughly 40 percent of livelihoods are dependent on agriculture, however in some areas, such as East Nusa Tenggara, rural households' have a historically much higher dependence on agriculture (more than 80% as of 2010), and here food security related concerns such as malnutrition is very high.⁸³ Of the 25 million farm households nationally, 17 million are rice farmers with an average land possession of 0.6 ha.⁸⁴ Rice is a major staple in Indonesia and its production is vulnerable to changes in rainfall seasonality and water scarcity. The climate of Indonesia is mainly influenced by the El Niño Southern Oscillation (ENSO), where drier conditions are experienced during El Niño events and wetter conditions during La Niña event.⁸⁵ Droughts are expected to increase in frequency and intensity given the association of accentuated drought with El Niño events, which are expected to increase in frequency and intensity through warmer global temperatures.^{86,87} At present, 14% of Indonesia 453 districts are recorded as having no months of surplus water. By 2025, this is projected as approximately 20% and 31% by 2050.⁸⁸ Crop water demand estimates suggest projected changes will impact the rice growing period between November and March and that there will likely be insufficient water for the second growing period, March to June, when chillies and tobacco are grown.⁸⁹ Delayed rainfall can increase the risk of annual rice deficits, with such events projected to increase from climate change. Naylor et al. (2007) project an increase in the probability of 30-day delay in the wet season from 9-18% today to 30-40% by 2050 (for the main rice-producing areas of Java and Bali).⁹⁰ In South Kalimantan the future water requirement in rice production will increase by around 56% in dry season (July) and 25% during wet season in September and October.⁹¹ Further, USAID assumes the

burden to these rural households as total agricultural loss among the provinces of East, Central, and West Java provinces can account for 19 percent, 15 percent, and 9.5 percent, respectively.⁹² Rice is also particularly sensitive to temperature changes, with some estimates suggesting that an increase of 1oC could reduce national production by 10–25%⁹³ which is well within the range of projected increase over the next century⁹⁴.

Lao People's Democratic Republic (Lao PDR): Lao PDR is a landlocked country in Southeast Asia. A large part of the country is mountainous, but there is a large central area, as well as low-lying alluvial plains along the Mekong River. It is ranked 142th out of 181 countries in the ND-GAIN Index⁹⁵, and as the 36th most vulnerable country globally. Rural areas in Lao PDR account for nearly 90% of the poor population, and rural poverty is almost three times higher than in urban areas. The poverty head count rate in the central provinces is some of the highest in the country and is showing the slowest rate of reduction.⁹⁶

Lao PDR has a tropical monsoon climate. The climate is hot and tropical, with the rainy season between May and October, a cool dry season from November to February, and a short hot, dry season in March and April. The central region experiences high temperatures and high average annual rainfall (which is concentrated in monsoon months), however, there is high variability due to the influence of Monsoon dynamics and the El Niño Southern Oscillation (ENSO). The combination of the monsoon climate, high variability, and the geographical terrain makes Lao extremely vulnerable to weather related extremes, notably from floods, storms, landslide and droughts (UNDP, 2010⁹⁷). The whole country experiences regular flooding and there are also frequent storms: three of the five costliest events on record having taken place since 2009 (EM-DAT, 2021⁹⁸; DESINVENTAR, 2021⁹⁹). Typhoon Ketsana in 2009 and Haima in 2011 caused damages of USD 94 million and USD 66 million respectively, and in 2018 a confluence of storm and flood disasters affected over 600,000 people and led to losses of USD 371 million (UNDRR 2019¹⁰⁰). There are also periodic droughts: in, 2015 a severe drought, driven by a strong El Niño event, damaged tens of thousands of hectares of rice, and fruit crops (Sutton et al., 2019)¹⁰¹ including in the central region. Historical damage data indicates that annual expected losses range between 3.3% and 4.1 percent of GDP (World Bank, 2017¹⁰²).

Disasters disproportionate impact on the poor, and the rural poor are particularly vulnerable to ENSO-related shocks (Sutton et al., 2019¹⁰³). Rural areas experience widespread food insecurity due to the reliance on rain-fed agriculture, and these households have a high probability of falling into extreme poverty even when exposed to relatively low frequency flood and drought events (ADB, 2017).¹⁰⁴

Recent observations show that the climate of Lao PDR is changing. There is a clear warming trend, with 0.1 to 3°C per decade reported (GFDRR, 2011¹⁰⁵, World Bank, 2019¹⁰⁶) across the entire country. The patterns of annual rainfall trends are more uncertain (Lao PDR, 2013¹⁰⁷, World Bank, 2019, Westra et al., 2014¹⁰⁸), although there are trends of increasing variability and increasing extreme rainfall.

Future climate change will exacerbate existing risks. The projected temperature increases for Lao PDR indicates a further rise of broadly 1°C to 2°C by mid-century (central multi-model ensemble values, for RCP2.6 and RCP8.5 respectively, relative to a baseline 1986-2005 period (World Bank, 2020¹⁰⁹; World Bank/ADB, 2021^{110,111}). It is difficult to model rainfall changes due to the influence of Monsoon dynamics, but most climate models project increases in annual rainfall by the mid-century (2050s) and there is more confidence that heavy precipitation events will increase. This will increase the flood risks across the country. Modelling for the country's rivers – from northern watershed, through the central region, and in the Mekong - indicate higher discharge leading to increased floods^{112,113,114} with greater areas of land and more people at risk of flooding, and higher value at risk (USAID, 2014¹¹⁵). These effects will be exacerbated by climate change, notably increases in heat and humidity (including heatwaves) which will reduce labour productivity (Dunne et al., 2013¹¹⁶). There are also projected direct and indirect impacts of climate change on agriculture itself, with reductions in yields in Lao PDR (Behnke et al., 2018)¹¹⁷, noting these could be amplified by the possible increased annual likelihood of drought¹¹⁸.

Pakistan: Ranked 18 out of 191 countries by the 2020 Inform Risk Index¹¹⁹, Pakistan faces some of the highest climate related disaster risk levels in the world. It has high exposure to flooding, including, riverine,

flash, and coastal, as well as exposure to tropical cyclones, droughts and heatwaves. Disaster risk in Pakistan is also driven by its social vulnerability. Pakistan's vulnerability ranking (37th) is driven by its high rates of multidimensional poverty. Geographically, the largest area of vulnerability is the Indus River Delta, which is a hotspot for storm surge, soil erosion and cyclones¹²⁰. This area has very high vulnerability¹²¹, as it is particularly low-lying and flat, with around 4,750 km² below 2m above sea-level¹²². Around 5 million people in Pakistan live in this low-lying coastal area, but because of population growth, it is projected that the population could increase six-fold to reach 30 million people by 2060. The coastal provinces of Sindh and Balochistan, in which Pakistan's 990km coastline is located, have the highest rural poverty rates among Pakistan's provinces and the highest urban-rural poverty gaps.¹²³ Coastal erosion and storm surge waves are already threatening towns and farms situated along the coast and saline intrusion continues to be a major challenge in the coastal zone, degrading land quality and agricultural yields^{124,125}. Cyclones do arise the Arabian sea, and when they hit landfall in Pakistan in this area, they can cause major impacts. For example, a strong cyclone hit Pakistan in 1999, killing some 6200 people, and there have been major storms recorded in 2004, 2007 and 2010, all of which affected the project area. In 2007, storm surges caused by Cyclone Yemyin had significant impacts on lives and property¹²⁶.

Recent observations show that the climate of Pakistan and Sindh region are changing already and many of the issues described above are likely to intensify due to climate change, affecting many marginal and deprived communities. There is a clear warming trend, which has accelerated since the 1960s (with reported 0.5 to 1°C of warming), and higher numbers of heatwaves and hot days.^{127,128,129,130,131} Data from the World Bank downscaled analysis¹³² shows that the projected temperature increases for Pakistan will be above the global average. By mid-century (2050s) a further rise of 1.5°C to 2.5°C (central estimate), relative to a baseline 1986-2005 period is projected. There are also projected increases in hot and very hot days. Multiple sources report that rainfall in the coastal area has declined by 10 – 15% since the 1960s. However, heavy rainfall events have increased, with the heaviest rain intensities recorded over the last decade.^{133,134} Changes in the seasonality, regularity, and extremes of precipitation highlights the risk of increased frequency and intensity of flood and drought events¹³⁵. Coastal areas of Pakistan have experienced changes in mean sea level in the past, with observations at Karachi indicating an increase in the mean sea level of about 1 mm/yr during the 20th century.^{136,137} Under low emission scenarios, it is likely that the coastal areas will experience significant sea level rise over the long-term. However, an upper range, with 0.6 to 1.1 metres has been shown as likely by 2100 under a high-emission scenario.¹³⁸ Recent analysis has also found that climate change has had a major role in increasing cyclone activity in the Arabian sea.¹³⁹ The IPCC SROCC¹⁴⁰ projects that the average intensity of tropical cyclones, the proportion of Category 4 and 5 tropical cyclones and the associated average precipitation rates are projected to increase for a 2°C global temperature rise above any baseline period (medium confidence). This, combined with rising mean sea levels will contribute to higher extreme sea levels associated with tropical cyclones (very high confidence). Coastal hazards will also be exacerbated by an increase in the average intensity, magnitude of storm surge and precipitation rates of tropical cyclones.

These changes in climate are likely to have a significant impact on water resource availability, heat stress, land availability and soil quality, affecting agricultural yields and rural agrarian livelihoods, in a country where Agriculture employs 42.3% of the Pakistan's workforce and contributes 21% to gross domestic product (GDP). Of the 24.3% of population below the national poverty line, over 80% are in rural areas, and comprise a large proportion of small-scale farmers (20%), landless farmers (10%) and agricultural laborers (12%)¹⁴¹. The projections of sea level rise are a particular concern for the communities living in the coastal zone. Work by the UK Met Office (2014) estimated that without adaptation, around one million people will face coastal flooding annually by the period 2070–2100. As sea level rises, there is an increased risk of river banks being overtopped and of flooding of adjacent land further up the estuaries. Without intervention, saltwater will penetrate further upstream and inland. A key issue for livelihoods is thus the potential increase in salt water intrusion affecting water supplies, coupled with broader issues of water availability from the changes due to climate change and demand increases. The Second National Communication from GoP (2018) reports that the Sindh area will suffer the most from water logging and salinity because of poor drainage systems.

Papua New Guinea: The high levels of disaster risk faced in Papua New Guinea (PNG) are reflected in its ranking in the 2019 INFORM Risk index (26th out of 191 countries). PNG's ranking is driven by moderate

levels of exposure to flood, drought and cyclone, but particularly by its lack of coping capacity. On this indicator PNG is ranked as having the 11th lowest coping capacity in the world. PNG was also ranked 158th out of 181 countries in the 2016 ND-GAIN climate change vulnerability ranking. This heightened vulnerability is largely due to widespread poverty and poor infrastructure, among other factors. PNG has consistently ranked among the lowest 30 countries in the Human Development Index. The country ranked 153rd out of 189 countries surveyed in the 2018 UNDP Human Development Report¹⁴² and, although precise figures are difficult to come by, poverty rates of between 30-40% have been reported and an adult literacy rate of around 50%¹⁴³. The monopolization of natural resources by a wealthy minority, is believed to be a significant contributor to income and wealth inequality which are also very high in PNG, and recent global research suggests this may also contribute to reduced resilience to disasters.¹⁴⁴ One assessment suggests the financial burden of the impact of disasters is around 3% of GDP annually.¹⁴⁵ That same study estimates the number of people affected by natural hazards at 7 million between 1990-2012.

The UNDP (2018) estimate that 70% of households in PNG remain dependent on subsistence agriculture¹⁴⁶. These communities are highly vulnerable to climate-related hazards and their impacts on livelihoods and food security. For example, a major drought associated with ENSO in 1997 highlighted the nation's food insecurity, with over a million people (around 20% of the population) facing food shortages. Saline intrusion resulting from sea-level rise has already been documented damaging the production of small-holder farmers.¹⁴⁷ PNG's population is particularly dependent on sweet potato production, with its Second National Communication to the UNFCCC suggesting 63% of calories in rural areas derive from this source. It is significant then that a review of multiple studies conducted by ADB suggested that sweet potato yields in PNG may decline by as much as 10% by 2050 as a result of changing climate conditions. Indeed, all the crops assessed, including cassava, maize, rice, sugarcane and taro, showed negative yield trends of a similar magnitude.¹⁴⁸ The rise in night-time minimum temperatures is believed to be a particularly important driver of these losses, with staple crops such as rice reported to have vulnerability in this regard.¹⁴⁹ Maximum and minimum temperatures are projected to rise notably faster than average temperatures, pointing towards a less stable and more frequently heat stressed climate in the future. Historical data for PNG suggest that temperature rise has been fastest in the minimum temperatures, with daily maximum temperatures rising broadly in line with the average.¹⁵⁰ In addition, under higher emissions pathways (particularly RCP8.5) PNG may transition to a chronically heat-stressed environment, with the CCKP model ensemble projecting a potential 100 day increase in the number of days surpassing 35°C by 2080-2099.¹⁵¹ Even increases in the range of 9-26 days above 35°C (ensemble median estimate) under the three lower emissions pathways represent a potential hazard to livelihoods in the country.

In addition to losses due to changes in the suitability of the climate conditions, production losses are already being experienced in coastal areas due to saltwater contamination of soils and groundwater, and across the country due to flood and drought. At present PNG faces an annual median probability of severe meteorological drought of around 4%, as defined by a standardized precipitation evaporation index (SPEI) of less than -2. Past drought incidents in PNG have led to significant disaster events, with a notable drought in 1997 resulting in widespread famine.¹⁵²

Coastal, river (fluvial), and surface (pluvial) flooding also represent a major risk. The UNDP suggest that 18% of the country's landmass is either permanently or regularly inundated¹⁵³. Lafale et al. (2018) suggest medium confidence should be placed in the projected future trend of increased annual precipitation. In contrast, the projections which suggest the frequency and intensity of extreme rainfall events will increase are associated with high confidence.¹⁵⁴ The intensity of sub-daily extreme rainfall events appears to be increasing with temperature, a finding supported by evidence from different regions of Asia and the Pacific.¹⁵⁵ As of 2010, assuming protection for up to a 1-in-10 year event, the population annually affected by river flooding in PNG is estimated at 33,000 people and the damage to GDP is estimated at USD73 million.¹⁵⁶ By 2030 under the RCP8.5 emissions pathway (AQUEDUCT Scenario B) climate change is expected to increase the annually affected population from river flooding by 20,000 people, and the GDP impact by USD90 million. What would historically have been a 1-in-100 year flow, could become a 1-in-50 year or 1-in-25 year event.¹⁵⁷ Willner et al. (2018) estimate that the population affected by an extreme flood could increase by 35,000-56,000 people by 2035-2044 (assuming present day population distributions) as a result

of climate change. Population growth and poorly planned development have the potential to increase this figure further.¹⁵⁸

PNG's Second National Communication suggests a further 8,000 people are affected by coastal flooding every year, causing over USD10 million in damages. Every decade around 15 tropical cyclones pass through PNG's exclusive economic zone, around a quarter of which are category 3 or stronger (severe events). A 1-in-100 year cyclone landfall was estimated to cost around 8.4% of GDP as of 2011.¹⁵⁹ However even cyclones passing a long way from PNG's land mass can have a significant local impact, with flooding and inundation of low lying islands documented as a result of elevated water levels.¹⁶⁰ Climate change is expected to interact with cyclone hazard in complex ways which are currently poorly understood, however, modelling of climate change impacts on cyclone intensity and frequency conducted across the globe also points to a general trend of reduced cyclone frequency but increased intensity and frequency of the most extreme events.¹⁶¹ Known associated risks include the action of sea-level rise to enhance the damage caused by cyclone-induced storm surges, and the possibility of increased windspeed and precipitation intensity. Sea-level rise can vary spatially, and work by the Australian Bureau of Meteorology and CSIRO has suggested that end-of-century sea-level rise could vary by as much as 10cm across the South Pacific region, meaning countries such as PNG should be prepared for above average sea-level rises¹⁶².

Timor-Leste: Floods, droughts, landslides and cyclones are the major natural hazards in Timor-Leste. According to the 2020 World Risk Report, in terms of exposure and vulnerability to natural hazards, Timor-Leste ranks 20th globally (24th in exposure, 51st in vulnerability (of which 43rd, 42nd and 82nd in susceptibility, lack adaptive capacity and lack coping capacity)¹⁶³. In terms of climate disruptions and adaptation, Timor-Leste is assessed as having high vulnerability and low readiness based on its scores on the ND-GAIN Index. The overall country score of 43.8 ranks Timor-Leste as position number 111 – the 47th most vulnerable, and 89th least ready country on the index. Flooding is the most frequent disaster, with riverine and flash flooding both occurring during periods of extreme rainfall; significant flood events were recorded in 2001, 2003, 2006, 2013, 2019 and 2020¹⁶⁴. Droughts also occur regularly in Timor-Leste, and pose a particular threat to agricultural production in the provinces of Oecusse, Dili and Atauro. The country experienced a major drought from 2015-2017 which caused sharp reductions in agricultural yields, with a decrease of 40% in Maize production, and 57% in Rice production¹⁶⁵. It impacted approximately half the area of maize and rice planted in the country, with 50% of households, reporting loss of livestock, and 50% suffering from food insecurity as a result¹⁶⁶. Such events have significant implications for poverty, food security and well-being in a country where estimates suggest that 30% of the population were living on less than the internationally recognised poverty line of USD1.90/day, 45% were classified as living in multi-dimensional poverty¹⁶⁷, and approximately 80% of the population are reliant on the agriculture sector.¹⁶⁸ Most agriculture is subsistence, and many rural areas suffer from food insecurity, due to a combination of low productivity, high population growth and unpredictable rainfall coupled with extreme climate events¹⁶⁹.

Staple crops in Timor-Leste are sensitive to rising temperatures and the impacts associated with extreme rainfall, changes in the seasonality of rainfall, seawater inundation and salinization of coastal aquifers.¹⁷⁰ The majority of agriculture in the country is rain-fed, and as such is vulnerable to changes in the timing and amount of rainfall, and increasing variability. The supply of water (particularly water scarcity during the dry season) will likely be a significant constraint on agricultural output¹⁷¹. Agricultural systems, including livelihood conditions and food insecurity issues, will be affected both by increases in extreme events, as well as changes in temperature and rainfall resulting from climate change¹⁷². According to climate models¹⁷³, the months of Jun-Sep, largely corresponding to the current dry season, are likely to become even drier, and there is a clear signal for increases in rainfall from February-May. There are indications that the onset of the rainy season may be delayed, with increased rainfall occurring during a shorter rainy season¹⁷⁴. The intensity of rainfall is expected to increase in Timor-Leste, with larger increases in extreme rainfall towards the end of the century. Under RCP4.5 there is a clear signal of increased heavy rainfall, with an average +14.5% increase in the number of days exceeding the 99th percentile by mid-century, and a 38.6% increase in days for the 2066-2095 period.

The El Niño Southern Oscillation (ENSO) exerts a strong influence over rainfall variability in Timor-Leste. El Niño conditions bring drier conditions and shorter, delayed wet seasons, often causing droughts, whilst La

Niña conditions cause higher rainfall even in the dry season and are linked to higher tropical cyclone activity. For the period 1950-1999 Dili recorded its two record low rainfall totals during the El Niño events of 1982-83 and 1997-98, and central provinces can see rainfall totals reduce by up to 50% of the annual average¹⁷⁵. A significant source of agricultural losses during El Niño years is false starts to the rainy season, where the initial onset of rains is followed by an abnormal dry period before the season reliably starts. These false starts can cause crop failure as there is not enough rainfall to support crop growth following planting, for example in Manatuto, farmers reported frequent false starts during previous El Niño events, and having to plant 3-4 times due to the losses suffered. Modelling of maize indicates reductions in yield in the range of 5-20% in the absence of adaptation, noting that much of this loss could be offset by supporting agricultural water supply through activities such as irrigation and water harvesting¹⁷⁶. These losses may be conservative, as they do not attempt to capture any increase in losses from extreme events. The evolution of ENSO under climate change will play an important role in determining drought and inter-annual variability in rainfall in the country. Although, changes in the ENSO cycle are still not well captured by climate models, the latest research indicates that El Niño events may become both more frequent and more severe over the course of the century^{177,178}.

Vanuatu: Disasters resulting from natural hazards are a very significant contributor to issues of social deprivation and impede economic development in Vanuatu, costing on average an estimated 6% of GDP every year.¹⁷⁹ The nation's exposure to natural hazards, extensive low-lying coastal zone, development context, and precarious natural resource base mean it is amongst the most vulnerable nations to climate change in the world, ranked 142nd out of 191 countries in the ND-GAIN Index. Vanuatu has very high exposure and vulnerability to the impact of flood, tropical cyclones, and storm surge which can strike in combination or isolation. Cyclones bring coastal and flash flooding even when only passing in the vicinity of the island. According to its Second National Communication to the UNFCCC around 20-30 cyclones pass over Vanuatu every decade, around 3-5 of which will cause severe damage. In 2015 Cyclone Pam, a category 5 tropical cyclone, struck Vanuatu killing 11 people, destroying or damaging over 17,000 buildings and displacing 65,000 people. The economic damages associated with the incident were estimated to be equivalent to around 64% of GDP.¹⁸⁰

The economy of Vanuatu, like surrounding island states, is based predominantly on the service sector (67% of GDP value added), followed by agriculture (22%) and industry (11%). However, there is significant local dependence on subsistence agriculture¹⁸¹ and issues of poverty and undernourishment are prevalent. Fishing – both artisanal (in which a large proportion of the population participates) and commercial - and offshore financial services are other mainstays of the economy.¹⁸² Vanuatu has a food trade deficit and dependence on imports for food security equivalent to over USD50 per capita in 2011 giving it high vulnerability to price shocks and disaster events.¹⁸³ This heavy reliance on both natural resources for agriculture and on tourism for livelihoods means that impact of extreme climate events in Vanuatu can be catastrophic for, food security, poverty reduction and general well-being of the population.¹⁸⁴ Cyclones can cause devastating damage to agricultural products¹⁸⁵ and known climate risks include the action of sea-level rise to enhance the damage caused by cyclone-induced storm surges, and the possibility of increased wind speed and precipitation intensity. Modelling of climate change impacts on cyclone intensity and frequency conducted across the globe points to a general trend of reduced cyclone frequency but increased intensity and frequency of the most extreme events.^{186, 187} Sea-level rise is not just a threat due to long-term encroachment on coastal areas, but also due to the projected increase in the frequency of extreme sea-level events.¹⁸⁸ The return period of exceptionally high sea-levels, driven by climate circulations, is expected to reduce, and low-lying Pacific island nations are particularly at risk.¹⁸⁹ Vanuatu's Second National Communication reports sea-level rise at a rate of around 6 mm/year between 1990 and 2010 – slightly faster than the global average over the same period.

Slower onset hazards also represent significant challenges. Most Pacific Island states are almost entirely dependent upon rainfall to support agricultural production, and thus are highly vulnerable to rainfall variability.¹⁹⁰ Meteorological drought is expected to increase in frequency, with one study conducted for ADB suggesting a potential doubling in the frequency of severe and extreme droughts by 2080 under RCP8.5.¹⁹¹ This represents a major risk as there is very high dependence on rainwater for subsistence.¹⁹² The future projections of sea level rise are associated with increased risks of flooding, loss

of land, erosion, salt water intrusion (and loss of drinking water supplies), combined with the loss of coastal ecosystems and associated services. These will affect island livelihoods, and are now a much greater concern given the higher projections of sea level rise in the IPCC SROCC¹⁹³ and the risk of extreme sea level events. Given the significance of agriculture to Vanuatu's economy, saltwater intrusion poses a present danger to food production with much of the land currently cultivated situated in coastal areas.¹⁹⁴ Saltwater intrusion, which allows salt from seawater to be absorbed into the soil, has been observed to cause reduced crop efficiency and even lead to crop failure.^{195,196}

Proposed interventions to scale up adaptation solutions targeted at the poor. It is in this context of climate risk that the CRPP IF has been designed to support governments of these seven countries to scale up adaptation solutions at local level, through the implementation of projects that are closely aligned with national climate priorities. These projects (detailed **examples provided in annex 2.3**) will promote adaptation solutions at local level that explicitly address the climate and poverty nexus and can be replicated and scaled up through national policies and programs and thereby bring transformational change. It is expected that CRPP supported solutions will allow overcoming a number of adaptation-related barriers (described in the **section B2**), especially in the context of local level adaptation measures that can be delivered at scale. Importantly, there is a current window of opportunity to implement such solutions hand-in-hand with poverty reduction measures as countries invest in reaching the SDGs, and work to fulfil their commitments to the Paris Agreement and Sendai Framework for Disaster Risk Reduction, and COVID-19 recovery. This can ensure climate finance will reach the people most in need; and local institutions and communities, especially women, will build adequate capacity to manage climate risk without jeopardizing their hard-earned development gains. As described in Box 2 below, this approach proposed by the CRPP directly responds to the IPCC's call for integrated adaptation and poverty reduction solutions for increasing adaptation effectiveness and enabling climate resilient development. In the absence of the CRPP, this opportunity will be lost, and the countries will lock-in development to growing climate risks. This opportunity is particularly important in the post-COVID world, where burdened governments may not prioritize integration of inclusive and resilience considerations in the recovery process.

Box 2: The CRPP's approach directly responds to the IPCC's call for integrated adaptation and poverty reduction solutions for increasing adaptation effectiveness and enabling climate resilient development

The CRPP's proposed approach towards implementing adaptation solutions through large-scale investment projects targeted at the poor directly responds to the IPCC's call for integrated, multi-sectoral solutions that address social inequities, differentiate responses based on climate risk and cut across systems. There is **high confidence** that such approaches increase the feasibility and effectiveness of adaptation in multiple sectors. There is also **high confidence** that climate resilient development is enabled when governments, civil society and the private sector make inclusive development choices that prioritise risk reduction, equity and justice, and when decision-making processes, finance and actions are integrated across governance levels, sectors and timeframes.¹⁹⁷ Something that is fundamental to the CRPP's proposed approach.

Table 2 below presents a summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects which are further detailed in **section B2**. The information demonstrates the relevant climate hazards in CRPP countries and how those hazards in interactions with factors contributing to exposure and vulnerability cause the specific risk/impact that CRPP related activities will address.

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
Indonesia	Climate risk informed social protection systems	Increasing trends in mean and extreme temperature including heatwaves. Mora, C., et.al. <i>Global risk of deadly heat</i>. <i>Nature Climate Change</i>, 7, 501-507. 2017. Matthews, T., R.L. Wilby and C. Murphy. <i>Communicating the deadly consequences of global warming for human heat stress</i>. <i>Proceedings of the National Academy of Sciences</i>, 114, 3861-3866. 2017). Significant decrease in rainfall in the dry season (July–August–September), increase in rainfall in December–January–February (DJF), except in northern Sumatera and Kalimantan. Significant changes especially in the 2080s period with a tendency of rainfall increase in the wet season and decrease in the transition months. <i>National Action Plan for Climate Change Adaptation (RAN-API) Synthesis Report</i>. Jakarta: Ministry of National Development Planning of the Republic of Indonesia, 2014. Increase in frequency and intensity of drought events with occurrences of El Niño events. <i>Climate Change and El Niño Increase Likelihood of Indonesian Heat and Drought</i>. <i>Bulletin of the American Meteorological Society</i>. King, Andrew & Karoly, David & Van Oldenborgh, Geert Jan. (2016). Sea-level rise is projected to reach 35-40 cm in 2050 and can reach up to 175 cm in 2100, relative to year 2000. <i>National Action Plan for Climate Change Adaptation (RAN-API) Synthesis Report</i>. Jakarta: Ministry of National Development Planning of the Republic of Indonesia, 2014.	Productivity losses due to heat stress. <i>Kjellstrom, T., Briggs, D., Freyberg, C., Lemke, B., Otto, M., Hyatt, O. (2016) Heat, human performance, and occupational health: A key issue for the assessment of global climate change impacts</i>. <i>Annual Review of Public Health</i>: 37: 97-112 75% increase of number of Indonesians exposed to river flood in four decades between 2015 and 2055. (Jian Vun, J.Kryspin-watson, & K. S. Alyono. 2019. <i>Urban flood resilience in Indonesia: New approaches through an urban design lens</i>. <i>World Bank Blogs</i>. URL: https://blogs.worldbank.org/eastasiapacific/urban-flood-resilience-indonesia-new-approaches-through-urban-design-lens) Coastal flood for a 1-in-100-year risk to increase from 19-37% (around 5.5-8 million people) by 2030 and to 9.5 – 14 million by 2060 <i>Le Bars, D., Drijhout, S., de Vries, H. (2017) A high-end sea level rise probabilistic projection including rapid Antarctic ice sheet mass loss</i>. <i>Environmental Research Letters</i>: 12:4 Muis, S. & A., and J. & W. Philip. <i>Flood risk and adaptation strategies under climate change and urban expansion: A probabilistic analysis using global data</i>. 2018 1°C increase in temperature changes, could reduce national rice production by 10–25% USAID (2015) <i>Climate Risk Profile: Indonesia Fact Sheet</i>. https://www.climatelinks.org/resources/climate-risk-profile-indonesia Food price surges or sudden decline in incomes due to the impact of climate	Adaptive social protection - climate risk-informed targeting of social protection program beneficiaries, improving shock-responsiveness of social assistance programs through strengthened early warning systems. Gaol, Harapan Lumban; Karyono, Adhy; <i>Adaptive Social Protection- Evidence from Indonesia</i>, Ministry of Social Affairs, Republic of Indonesia https://pubdocs.worldbank.org/en/202301520536557841/SSLF18-Nuts-and-Bolts-2-Indonesia.pdf Bowen, Thomas Vaughan; Del Ninno, Carlo; Andrews, Colin; Coll-Black, Sarah; Gentilini, Ugo; Johnson, Kelly; Kawasoe, Yasuhiro; Kryeziu, Adea; Maher, Barry Patrick ; Williams, Asha M <i>Adaptive Social Protection : Building Resilience to Shocks</i> World Bank Group. http://documents.worldbank.org/curated/en/579641590038388922/Adaptive-Social-Protection-Building-Resilience-to-Shocks Creating opportunities for program beneficiaries, especially women, to engage in climate resilient livelihoods; and building awareness of social workers and program beneficiaries on climate adaptation and disaster risk reduction. IPCC, 2018: <i>Global Warming of 1.5°C</i>. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty [Masson-Delmotte, V., P. Zhai, H.-O. Pörrner, D. Roberts, J. Skea, P.R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J.B.R. Matthews, Y. Chen, X. Zhou, M.I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, and T. Waterfield (eds.)]. In Press.

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
			extremes could increase the number of Indonesians in poverty and extreme poverty. <i>Asian Development Bank. Preparing for El Niño: Policy Options. Manila, Philippines. 2015.</i>	
Lao PDR	Climate resilient agroecological systems.	Increase in temperature across seasons in particular during April and May. Elevated night-time minimum temperatures <i>Lao PDR Nationally Determined Contribution, 2021</i> https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Lao%20People%27s%20Democratic%20Republic%20First/NDC%202020%20of%20Lao%20PDR%20(English).%2009%20April%202021%20(1).pdf <i>Climate Risk Country Profile: Lao PDR (2021): The World Bank Group and the Asian Development Bank.</i> https://www.adb.org/sites/default/files/publication/709846/climate-risk-country-profile-lao-pdr.pdf Increase in precipitation rates in particular sub-daily extreme rainfall. <i>Climate Risk Country Profile: Lao PDR (2021): The World Bank Group and the Asian Development Bank.</i> https://www.adb.org/sites/default/files/publication/709846/climate-risk-country-profile-lao-pdr.pdf Projected increase in deficit of available water leading to increased annual probability of drought https://climateinformation.org. 2022. Indicator: Aridity actual (annual mean), Time period: 2041 2070, Historical period: 1981–2010, RCP 8.5, Model: CORDEX South Asia - WWHYPE Ensemble Mean, Model results for an area covering the location: Paksan, Bolikhamxai (18.38, 103.66) <i>Climate Risk Country Profile: Lao PDR (2021): The World Bank Group and the Asian Development Bank.</i> https://www.adb.org/sites/default/files/publication/709846/climate-risk-country-profile-lao-pdr.pdf Increased incidence of extreme heat.	Changes in precipitation impacting hydrology and water availability. <i>Lao PDR Nationally Determined Contribution, 2021</i> https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Lao%20People%27s%20Democratic%20Republic%20First/NDC%202020%20of%20Lao%20PDR%20(English).%2009%20April%202021%20(1).pdf Reductions in yields of staple crops due to rise in temperatures and water deficits. <i>Drew Behnke, Sam Heft-Neal, and David Roland-Holst (2018). Early Warning Techniques for Local Climate Resilience: Smallholder Rice in Lao PDR. In Climate Smart Agriculture Building Resilience to Climate Change. Natural Resource Management and Policy Volume 52, Editors Lipper, L et al., Published by Springer Nature.</i> <i>Welch, J. R., et.al. Rice yields in tropical/subtropical Asia exhibit large but opposing sensitivities to minimum and maximum temperatures. Proceedings of the National Academy of Sciences, 107(33), 14562–14567.2010.</i> URL: https://www.pnas.org/content/pnas/107/33/14562.full.pdf <i>USAID. (2014). Climate Change in the Lower Mekong Basin: An Analysis of Economic Values at Risk. USAID Mekong Adaptation and Resilience to Climate Change (USAID Mekong ARCC). Published by USAID, 2014.</i> https://www.unep.org/resources/report/climate-change-lower-mekong-basin-analysis-economic-values-risk <i>Li, S., Q. Wang, & J.A. Chun. Impact assessment of climate change on rice productivity in the Indochinese Peninsula using a regional-scale crop model. International Journal of Climatology, 37(April). 2017.</i> URL: https://rmets.onlinelibrary.wiley.com/doi/epdf/10.1002/joc.5072	Promote appropriate technologies including climate-smart and water-saving agricultural practices and nature-based solutions. <i>IPCC, 2018: Global Warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty [Masson-Delmotte, V., P. Zhai, H.-O. Pörrner, D. Roberts, J. Skea, P.R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J.B.R. Matthews, Y. Chen, X. Zhou, M.I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, and T. Waterfield (eds.)]. In Press.</i> Switch from rice to other crops to reduce water requirements and improve flood, drought and temperature tolerance. <i>Lao PDR Nationally Determined Contribution, 2021</i> https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Lao%20People%27s%20Democratic%20Republic%20First/NDC%202020%20of%20Lao%20PDR%20(English).%2009%20April%202021%20(1).pdf <i>Lao PDR National Strategy on Climate Change 2012</i> https://www.la.undp.org/content/laopdr/en/home/librariy/environmentenergy/climate_change_strategy.html <i>Agricultural strategy to 2025 and vision to the year 2030</i> https://www.maf.gov.la/wp-content/uploads/2016/01/MDS-2025-and-Vision-to-2030-Eng.pdf Increasing vegetative cover (natural reforestation and application of bioengineering) in order to prolong water discharge from catchments and regulate

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
		<i>Climate Risk Country Profile: Lao PDR (2021): The World Bank Group and the Asian Development Bank.</i> https://www.adb.org/sites/default/files/publication/709846/climate-risk-country-profile-lao-pdr.pdf	Increased drought risk and productivity loss. Adamson, P. and J. Bird (2010). <i>The Mekong: a drought-prone tropical environment?</i> International Journal of Water Resources Development, 26(4), pp.579-594. 2010. URL: https://www.tandfonline.com/doi/abs/10.1080/07900627.2010.519632 Decline in water resource availability and seasonality, soil organic matter transformation, arable areas due to flooding or desertification and increased soil erosion. <i>Climate Risk Country Profile: Lao PDR (2021): The World Bank Group and the Asian Development Bank.</i> https://www.adb.org/sites/default/files/publication/709846/climate-risk-country-profile-lao-pdr.pdf Labour productivity loss due to heat stress. Kjellstrom, T., et.al. <i>Heat, human performance, and occupational health: A key issue for the assessment of global climate change impacts.</i> Annual Review of Public Health: 37: 97-112. 2016. URL: https://www.annualreviews.org/doi/pdf/10.1146/annurev-publhealth-032315-021740	the impacts of changing precipitation patterns and the likely increase in flooding, resulting from intense periods of rainfall. <i>World Bank, 2021. A Catalogue of Nature-based Solutions for Urban Resilience. Washington, D.C. World Bank Group</i>
Pakistan	Ecosystem-Based Adaptation	Rise in daily maximum and minimum temperature which are above average. Increase in frequency of days with a heat index. Asian Development Bank (ADB). <i>Climate Change Profile of Pakistan.</i> Asian Development Bank. 2017. URL: https://www.adb.org/publications/climate-change-profile-Pakistan Increase in extreme rainfall and changes in the seasonality and regularity of precipitation and water discharge.	Increase in compound / concurrent events impacting on ecosystems, health, crops yield, labour productivity. Decline in living standards across the country in particular Sindh province being the worst affected. Asian Development Bank (ADB). <i>Climate Change Profile of Pakistan.</i> Asian Development Bank. 2017. URL: https://www.adb.org/publications/climate-change-profile-Pakistan Working Group I- Sixth Assessment Report Intergovernmental Panel on Climate Change	Implement integrated coastal zone management and adaptive management approach to flood risk. Klijn, F., Kreibich, H., de Moel, H. et al. <i>Adaptive flood risk management planning based on a comprehensive flood risk conceptualisation. Mitig Adapt Strateg Glob Change</i> 20, 845–864 (2015). https://link.springer.com/article/10.1007/s11027-015-9638-z#citeas Enhance natural coastal defenses through reforestation coastal zones by planting mangrove species along with other fresh and saline tolerant tree

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
		Amin, A., et.al. <i>Regional climate assessment of precipitation and temperature in Southern Punjab (Pakistan) using SimCLIM climate model for different temporal scales. Theoretical and Applied Climatology</i>: 131: 121–131. 2018. Asian Development Bank (ADB). <i>Climate Change Profile of Pakistan</i>. Asian Development Bank. 2017. URL: https://www.adb.org/publications/climate-change-profile-pakistan USAID (2018). <i>Climate Change Risk Profile. Pakistan</i> Increase in mean sea level. GERICS (2015). <i>Climate-Fact-Sheet Pakistan</i>. Climate Service Center Germany M. M. Rabbani et al. 2008. <i>The Impact of Sea Level Rise on Pakistan's Coastal Zones - In a Climate Change Scenario</i>. 2nd International Maritime Conference at Bahria University, Karachi. Oppenheimer, M., B.C. Glavovic, J. Hinkel, R. van de Wal, A.K. Magnan, A. Abd-Elgawad, R. Cai, M. Cifuentes-Jara, R.M. DeConto, T. Ghosh, J. Hay, F. Isla, B. Marzeion, B. Meyssignac, and Z. Sebesvari, 2019: <i>Sea Level Rise and Implications for Low-Lying Islands, Coasts and Communities</i>. In: <i>IPCC Special Report on the Ocean and Cryosphere in a Changing Climate</i> [H.-O. Pörtner, D.C. Roberts, V. Masson-Delmotte, P. Zhai, M. Tignor, E. Poloczanska, K. Mintenbeck, A. Alegría, M. Nicolai, A. Okem, J. Petzold, B. Rama, N.M. Weyer (eds.)]. In press. Increasing cyclone activity in the Arabian sea. (Murakami, H., Vecchi, G.A. & Underwood, S. <i>Increasing frequency of extremely severe cyclonic storms over the Arabian Sea. Nature Clim Change</i> 7, 885–889 (2017). https://doi.org/10.1038/s41558-017-0008-6)	Changes in the seasonality, regularity, and extremes of precipitation results in risk of increased frequency and intensity of flood and drought events with uncertainty associated with long term outlook for water resources. Amin, A., et.al. <i>Regional climate assessment of precipitation and temperature in Southern Punjab (Pakistan) using SimCLIM climate model for different temporal scales. Theoretical and Applied Climatology</i>: 131: 121–131. 2018. Asian Development Bank (ADB). <i>Climate Change Profile of Pakistan</i>. Asian Development Bank. 2017. URL: https://www.adb.org/publications/climate-change-profile-pakistan Greater areas of land and more people at risk of flooding. (Food and Agriculture Organization (FAO). No Date. <i>The Mekong hydrology programme</i>. In: <i>Technical Session III Flood forecasting and river modelling</i>. URL: http://www.fao.org/3/ac146e/AC146E03.htm Try, S., Tanaka, S., Tanaka, K. et al. 2020. <i>Assessing the effects of climate change on flood inundation in the lower Mekong Basin using high-resolution AGCM outputs</i>. <i>Prog Earth Planet Science</i>. 7, 34 (2020). DOI: https://doi.org/10.1186/s40645-020-00353-z) Rattikone Sayasane, Akiyuki Kawasaki, Sangam Shrestha, Masatsugu Takamatsu; <i>Assessment of potential impacts of climate and land use changes on stream flow: a case study of the Nam Xong watershed in Lao PDR</i>. <i>Journal of Water and Climate Change</i> 1 March 2016; 7 (1): 184–197. doi: https://doi.org/10.2166/wcc.2015.050 Storm surge waves. Increase in frequency and severity of coastal flooding in low-lying areas and coastal erosion along sandy coasts. Saline intrusion degrading land quality and agricultural yields on the	species to reduce coastal erosion rate, provide protection from storm surge, and reduce saline intrusion. IUCN. 2013. <i>Valuation Study of Mangrove Plantations Established under Sindh Coastal Community Development Project (SCCDP)</i>. https://www.iucn.org/sites/dev/files/import/downloads/p_k_sccdp_mf_final_report_mangrove_valuations.pdf. Kimball, S. et al., 2015: <i>Cost-effective ecological restoration</i>. <i>Restoration Ecology</i>, 23(6), 800–810, doi:10.1111/rec.12261. Bayraktarov, E. et al., 2016: <i>The cost and feasibility of marine coastal restoration</i>. <i>Ecological Applications</i>, 26(4), 1055–1074, doi:10.1890/15-1077. <i>Coastal Resilience in the Indus Delta; Upscaling Nature-Based Solutions</i>, Coastal Engineering Proceedings

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
			coastal zone affecting marginal and deprived communities. <i>Working Group I- Sixth Assessment Report Intergovernmental Panel on Climate Change Asian Development Bank (ADB). Climate Change Profile of Pakistan. Asian Development Bank. 2017. URL: https://www.adb.org/publications/climate-change-profile-pakistan</i> <i>Zeng, L. & M.C. Shannon. Salinity effects on seedling growth and yield components of rice. Crop Sci. 40, 996–1003. 2000.</i> <i>Letey, J. & A. Dinar. Simulated crop-water production functions for several crops when irrigated with saline waters. Hilgardia. 54, 1–32. 198</i> <i>Redaelli, Silvia. 2019. Pakistan@100 From Poverty to Equity . World Bank Group Policy Note. March 2019. URL: http://documents1.worldbank.org/curated/ar/868741552632296526/pdf/135319-WP-P163618-14-3-2019-20-44-35PakPNFromPovertytoEquityFinal.pdf</i>	
Papua New Guinea	Climate resilient agroecological systems.	Increase in high-time, and daily minimum temperatures. Maximum and minimum temperatures are projected to rise notably faster than average temperatures, pointing towards a less stable and more frequently heat stressed climate in the future. <i>(Australian Bureau of Meteorology and CSIRO (2014) Climate Variability, Extremes and Change in the Western Tropical Pacific: New Science and Updated Country Reports. Pacific-Australia Climate Change Science and Adaptation Planning Program Technical Report, Australian Bureau of Meteorology and CSIRO, Melbourne, Australia)</i> <i>Climate Risk Profile: Papua New Guinea (2021): The World Bank Group.</i> <i>https://climateinformation.org (2022) Indicator: Tropical nights (annual mean), Time period: 2041–2070, Historical period: 1981–2010, RCP 4.5, Model: CORDEX East Asia Ensemble Mean, Model results for an area covering the location: Mount Hagen, Western Highlands (-5.86, 144.23)</i>	70% of households in PNG remain dependent on subsistence agriculture. Sweet potato is particularly important in the highlands, where the crop contributes to the food security and cash incomes of more than 90% of the highland population. Production losses are already being experienced in coastal areas due to saltwater contamination of soils and ground water, and across the country due to flood and drought. <i>UNDP (2018) National Adaptation Plan process in focus: Lessons from Papua New Guinea. United Nations Development Programme.</i> <i>Asian Development Bank. The economics of climate change in the Pacific: Mandaluyong City, Philippines: Asian Development Bank, 2013. [online] available at https://www.adb.org/sites/default/files/publication/31136/economics-climate-change-pacific.pdf</i> <i>Climate Risk Profile: Papua New Guinea (2021): The World Bank Group.</i>	Training and vocational education for vulnerable people in the agriculture sector based on conservation agriculture and agroforestry principles. Mainstream climate change adaptation into the governance, the incentive systems, the management, the curricula, the training modules and the training programs of the TVE system. <i>UNESCO-UNEVOC International Centre for Technical and Vocational Education and Training. Skills development and climate change action plans(2021). https://unesdoc.unesco.org/ark:/48223/pf0000376163</i> <i>Asian Development Bank. Initial poverty and social analysis. Improved Technical and Vocational Education and Training for Employment (2019). https://www.adb.org/sites/default/files/project-documents/53083/53083-001-ipsa-en.pdf</i>

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
		Increase in frequency and intensity of extreme rainfall events. <i>Lafale, P., Diamond, H., Anderson, C. (2018) Effects of climate change on extreme events relevant to the Pacific Islands. Science Review 2018: 50-73</i> <i>Westra, S., Fowler, H. J., Evans, J. P., Alexander, L. V., Berg, P., Johnson, F., Kendon, E. J., Lenderink, G., Roberts, N. (2014). Future changes to the intensity and frequency of short-duration extreme rainfall. Reviews of Geophysics, 52, 522–555.</i> https://climateinformation.org (2022) Indicator: Precipitation (annual mean), Time period: 2041–2070, Historical period: 1981–2010, RCP 4.5, Model: CORDEX East Asia Ensemble Mean, Model results for an area covering the location: Mount Hagen, Western Highlands (-5.86, 144.23) Decrease in frequency of moderate intensity tropical cyclone and increase in wind speeds associated with category 5 cyclones. <i>Climate Risk Profile: Papua New Guinea (2021): The World Bank Group.</i>	Reduced yields and nutritional content of staple crops such as sweet potato and rice due to changes in the suitability of climate conditions. <i>A. McGregor, et al. (2016) Vulnerability of staple food crops to climate change M. Taylor, A. McGregor, A., Taylor, M., B. Dawson, B (Eds.), Vulnerability of Pacific Island Agriculture and Forestry to Climate Change, Pacific Community (SPC), Noumea, New Caledonia (2016), pp. 161-238</i> <i>Lese, V., Holland, E., Wairiu, M., Havea, R., Patolo, S., Nishi, M., Hoponoo, T., Bourke, M., Dean, A., and Wangainbete, I. 2018 Facing food security risks: The rise and rise of the sweet potato in the Pacific Islands. Global Food Security. Vol 18. Pp 48-56</i> <i>Welch, J. R., Vincent, J. R., Auffhammer, M., Moya, P. F., Dobermann, A., & Dawe, D. (2010). Rice yields in tropical/subtropical Asia exhibit large but opposing sensitivities to minimum and maximum temperatures. Proceedings of the National Academy of Sciences, 107(33), 14562–14567.</i> <i>Australian Bureau of Meteorology and CSIRO (2014) Climate Variability, Extremes and Change in the Western Tropical Pacific: New Science and Updated Country Reports. Pacific-Australia Climate Change Science and Adaptation Planning Program Technical Report, Australian Bureau of Meteorology and CSIRO, Melbourne, Australia</i> Impact on national food consumption patterns both through direct impacts on internal agricultural operations, and through impacts on the global supply chain. <i>Climate Risk Profile: Papua New Guinea (2021): The World Bank Group.</i>	<i>Kombra Uke, Papua New Guinea's TVET: at the Crossroads</i> https://www.cpsctech.org/2013/07/papua-new-guineas-tvet-at-crossroads.html
Timor-Leste	Climate resilient agroecological systems.	Heavy rainfall events of greater intensity interspersed with dry periods resulting in greater erosion and more frequent landslides <i>Climate Risk Country Profile: Timor-Leste (2021): The World Bank Group and the Asian Development Bank.</i>	Changes in climate combined with large-scale deforestation contributing to loss and drying up of water sources <i>Oxfam (2012). Weathering Change in Timor-Leste: Participatory Action Research in Timor-Leste identifying climate change and associated impacts experienced at</i>	Sustainable land management promoting fixed / permanent agriculture, reduced burning, reduced erosion and increased soil fertility. <i>Lasco, R.D., R.J.P. Delfino, and M.L.O. Espaldon, 2014: Agroforestry systems: helping smallholders adapt to</i>

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
		https://www.adb.org/sites/default/files/publication/75124/1/climate-risk-country-profile-timor-leste.pdf The current dry season is likely to become even drier and the onset of the rainy season may be delayed, with increased rainfall occurring during a shorter rainy season. (Government of Timor Leste (2014) Initial National Communication to the UNFCCC.) https://climateinformation.org. Indicator: Precipitation (annual mean), Time period: 2041–2070, Historical period: 1981–2010, RCP 4.5, Model: CORDEX East Asia Ensemble Mean, Model results for an area covering the location: Dili (-8.55, 125.57). El Niño events may become both more frequent and more severe over the course of the century. McPhaden, M. J., A. Santoso, and W. Cai (2019), <i>Understanding ENSO in a changing climate</i>, <i>Eos</i>, 100, https://doi.org/10.1029/2019EO124159. Published on 23 May 2019. Wang, B. et al. (2019) <i>Historical change of El Niño properties sheds light on future changes of extreme El Niño</i>. <i>Proceedings of the National Academy of Sciences</i> 116 (45) 22512-22517 Increase in both intensity and duration of heatwaves by 2050. Climate Risk Country Profile: Timor-Leste (2021): The World Bank Group and the Asian Development Bank. https://www.adb.org/sites/default/files/publication/75124/1/climate-risk-country-profile-timor-leste.pdf	the community level. https://www.preventionweb.net/files/24686_24686climatechangereportexecutivesu.pdf Food production most affected by changes in rainfall. Shifts in the suitable growing area for different crops. Impact on the health and productivity of the labor force Climate Risk Country Profile: Timor-Leste (2021): The World Bank Group and the Asian Development Bank. https://www.adb.org/sites/default/files/publication/75124/1/climate-risk-country-profile-timor-leste.pdf The supply of water (particularly water scarcity during the dry season) will likely be a significant constraint on agricultural output. (Scherer, N. & D. Tänzler. <i>The Vulnerable Twenty – From Climate Risks to Adaptation: A compendium of climate fragility risks and adaptation finance needs of the V20 countries</i>. Federal Foreign Office, Berlin Germany. October 1, 2018. Aldephi. 2018.) False starts to the rainy season can cause crop failure as there is not enough rainfall to support crop growth following planting. Potential reductions in maize yield in the range of 5-20% in the absence of adaptation. Government of Timor Leste (2014) Initial National Communication to the UNFCCC. Timor Leste National Communication, 2014	climate risks while mitigating climate change. Wiley Interdisciplinary Reviews: Climate Change, 5(6), 825–833, doi:10.1002/wcc.301. Integrated agroforestry and watershed management. Reforestation of degraded land. Coulibaly, J.Y., B. Chiputwa, T. Nakelse, and G. Kundhlande, 2017: Adoption of agroforestry and the impact on household food security among farmers in Malawi. <i>Agricultural Systems</i>, 155, 52–69, doi:10.1016/j.agsy.2017.03.017. Climate proofed and environmentally sustainable infrastructure to protect water sources. Enhance water harvesting and water distribution. Construct structures (rainwater harvesting and storage) to improve the supply of year-round water for agriculture to offset potential reductions in yield of staple crops. Government of Timor-Leste (2014) Initial National Communication to the UNFCCC. IPCC, 2018: <i>Global Warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty</i> [Masson-Delmotte, V., P. Zhai, H.-O. Pörtner, D. Roberts, J. Skea, P.R. Shukla, A. Pirani, W. Moufouma-Okia, C. Péan, R. Pidcock, S. Connors, J.B.R. Matthews, Y. Chen, X. Zhou, M.I. Gomis, E. Lonnoy, T. Maycock, M. Tignor, and T. Waterfield (eds.)]. In Press.

Table 2: A summary of country-level climate hazard and risk information from **annex 2.1** that is pertinent to the types of activities envisaged for CRPP projects.

Country	Sector	Climate hazard	Risk/Impact	Indicative or example activity and how it reduces risk
Vanuatu	Climate responsive social protection systems and climate resilient livelihood support mechanisms	Increase in intensity of sub-daily rainfall and weather events. <i>Climate Risk Country Profile: Vanuatu (2021): The World Bank Group</i> Meteorological drought is expected to increase in frequency. <i>Y. Li.; P. Ulrich. 2017. Climate Change Impact Assessment for Luganville, Vanuatu: ADB/CLIMsystems Ltd.</i> Potential intensification of the most extreme tropical cyclone events threatens significant damage and loss to Vanuatu's population. <i>Climate Risk Country Profile: Vanuatu (2021): The World Bank Group</i> Sea level rise faster than the global average. Rising sea levels, both on average and during extreme events such as storm surge, king tides, and windwaves. <i>Vanuatu's Second National Communication USAID. Climate Risk Profile – Pacific Islands. 2018</i>	Flooding and landslides damage crops, especially small-scale subsistence farming, affecting food security and reducing income. Marine heatwaves impacting marine ecosystems and the dependent community livelihoods. <i>Frölicher, T. L., Fischer, E. M., & Gruber, N. (2018). Marine heatwaves under global warming. Nature, 560(7718), 360–364. URL: https://www.nature.com/articles/s41586-018-0383-9</i> Increased exposure to riverine flooding and storm surge due to sea-level rise. <i>Climate Risk Country Profile: Vanuatu (2021): The World Bank Group</i>	To increase climate change adaptive capacity and reduce vulnerability to climate hazards so that vulnerable groups - Social protection systems for highly climate vulnerable groups. <i>Bowen, Thomas Vaughan; Del Ninno, Carlo; Andrews, Colin; Coll-Black, Sarah; Gentilini, Ugo; Johnson, Kelly; Kawasoe, Yasuhiro; Kryeziu, Adea; Maher, Barry Patrick ; Williams, Asha M</i> <i>Adaptive Social Protection : Building Resilience to Shocks World Bank Group.</i> http://documents.worldbank.org/curated/en/579641590038388922/Adaptive-Social-Protection-Building-Resilience-to-Shocks

3. Related Interventions

The paragraphs below describe selected ongoing climate change adaptation projects in the target countries, financed by the Adaptation Fund, GCF and the Global Environment Facility (GEF), that complement activities being proposed under the CRPP and/or have the potential to be replicated by projects financed by the CRPP. Any investments under the CRPP directed towards the replication and scaling-up of previous or on-going interventions will be subject to a detailed analysis of evaluation results and feasibility studies that take into account the context specific nature of adaptation interventions.

Cambodia. The trend of adaptation investments in Cambodia from large-scale funds reflects the immediate need for increasing the resilience of the livelihoods of the rural poor and the importance of adaptation of the agricultural sector for protecting livelihoods and improving food security and income levels despite the impact of climate change. The GCF funded *Climate-Friendly Agribusiness Value Chains Sector Project* and *Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood in Northern Tonle Sap Basin Project* along with the GEFs investments in *Promoting Climate-Resilient Livelihoods in Rice-Based Communities in the Tonle Sap Region* and the Adaptation Fund's recently completed project on *Enhancing Climate Resilience of Rural Communities Living in Protected Areas of Cambodia* represent significant investment on increasing food supply and incomes through enhancing the resilience and productivity of target crops, and increasing agricultural competitiveness. Such interventions present an opportunity to use this momentum to invest further in targeted climate resilient rural livelihoods activities in the poorest and highest climate risk populations in the country and capitalize on the related knowledge and enabling environment being created under these projects, while filling in the funding deficits. Investments that have an explicit and targeted focus on women's empowerment would be a particular asset in a country where the transition of employment away from agriculture has been greater for men than for women, but this is not necessarily reflected adaptation interventions where ¹⁹⁸ Cambodia is currently undergoing a process of change in which government structures and administrative procedures are being adapted to develop local self-governance, as financial and human resources and decision-making capacities are gradually being transferred to the local level. This on-going decentralization process is seen as a significant opportunity to expand-on previous and on-going adaptation interventions by ensuring that the processes and mechanisms of local-level government investment decision-making incorporate the tools, information and methods from previous and on-going adaptation projects, and thus scale-up in a sustainable way the progress made so far.

Indonesia. The Adaptation Fund is financing three projects in Indonesia, two of which are highly targeted towards the most vulnerable populations in rural Sulawesi (*Adapting to Climate Change through Sustainable Integrated Watershed Governance in Indigenous People of Ammatoa Kajang Customary Area* and *Community Adaptation for Forest-Food Based Management in Saddang Watershed Ecosystem*) and the third, *Safekeeping-Surviving-Sustaining towards Resilience: 3S Approach to Build Coastal City Resilience to Climate Change Impacts and Natural Disasters in Pekalongan City*, has an urban focus. Investments in Sulawesi reflect the emerging realization that there is a clear and present threat posed by climate change to poor rural populations in remote and isolated parts of the country that depend heavily on subsistence agriculture for their food security. Moreover, these projects show that there is a clear recognition that local and context specific solutions are called for in such an environmentally and socio-culturally diverse nation. However, these relatively small-scale investments by the Adaptation Fund in Indonesia represent only a small fraction of the local-level adaptation financing needs targeted at poor communities, in the world's fourth most populous country where nearly 5% of the population have a power purchasing parity of USD1.9 a day or less. The poorest and most vulnerable populations in Indonesia are being left behind in terms of adaptation financing, and thus there is a need for investment in projects that have the capacity to reach a significant number of people living with high climate risk using existing mechanisms such as social protection and decentralized planning to deliver adaptation financing directly to the end users. More specifically, the Pekalongan City Resilience Project, which seeks to address urban flood risk, presents an opportunity for replication via the transfer of

knowledge and capacity to other cities on the Island of Java, thus strengthening the ability to generate transformational change through delivering complimentary investments.

Lao PDR. Current climate change adaptation financing in Lao PDR from large scale adaptation funds is targeted towards activities which generally focus on urban and peri-urban settings, such as the GCF funded *Building resilience of urban populations with ecosystem-based solutions in Lao PDR*, the GEF's *Integrated Water Resource Management and Ecosystem-based Adaptation in the Xe Bang Hieng River Basin and Luang Prabang City*, and to a large extent the Adaptation Fund's *Enhancing the climate and disaster resilience of the most vulnerable rural and emerging urban human settlements in Lao PDR*. Those interventions that have a specific rural livelihoods targeted approach, such as the GEF funded *Climate Smart Agriculture alternatives for upland production systems in Lao PDR* are focused in the northern highlands. Rural areas in Lao PDR account for nearly 90% of the poor population, and rural poverty is almost three times higher than in urban areas. Furthermore, the poverty head count rate in the central provinces is some of the highest in the country and is showing the slowest rate of reduction.¹⁹⁹ With the current focus of adaptation investment either in urban settings or in the rural northern highlands there is the potential of neglecting large swaths of important agricultural communities in the central provinces that are experiencing devastating impacts from floods and droughts, which are likely to worsen due to climate change. However, there is a clear opportunity to capitalize on the nascent national frameworks and knowledge base regarding the fore mentioned intervention areas, and to transfer, replicate and scale up similar initiatives to other geographies and sectors with high climate risk. Specifically, there is need to bridge the gap related to the conservation and expansion of vegetation in wetlands for flood protection and watershed management in rural areas, by expanding on existing activities that have a more urban focus, and thus addressing the pressing adaptation needs of the rural livelihoods and agriculture sector. This can be complemented by replicating climate smart agriculture activities that are currently focused in the north of the country, by transferring the knowledge and lessons learned to, the lesser funded, central provinces that are highly exposed and vulnerable to the impacts of climate change.

Pakistan. There is an identifiable gap in large scale climate adaptation investments in Pakistan that relates to the coastal zones, and specifically protecting lives and rural livelihoods from extreme climate events. Much of the focus of investment from large-scale adaptation funds in Pakistan has been in the inland districts of the Indus River Basin and the northern provinces, and related to flood and drought risk management, with Glacial lake outburst floods in the northern mountainous areas of the country receiving particular attention (*GCF funded Scaling-up of Glacial Lake Outburst Flood risk reduction in Northern Pakistan*; *Adaptation Fund's now completed Reducing Risks and Vulnerabilities from Glacier Lake Outburst Floods in Northern Pakistan*; and their new project *Enhance community, local and national-level urban climate change resilience to water scarcity, caused by floods and droughts in Rawalpindi and Nowshera, Pakistan*.) By focusing investment on climate adaptive infrastructure and livelihood adaptation in the coastal zones of the Indus River Delta, interventions in Pakistan will not only fill a funding gap in terms of geographical distribution, but also directly complement existing investments by GCF on *Transforming the Indus Basin with Climate Resilient Agriculture and Water Management*, that focus on the provision of climate information services for livelihood decision-making (from federal to farm level). Balancing the geographical and sectoral distribution of adaptation financing in such a way is crucial as Pakistan's 990km coastline is located in the provinces of Sindh and Balochistan, which have the highest rural poverty rates among Pakistan's provinces and the highest urban-rural poverty gaps.²⁰⁰ The recently approved GCF funding for the *Recharge Pakistan: Building Pakistan's resilience to climate change through Ecosystem-based Adaptation and Green Infrastructure for integrated flood risk management* will provide examples of community-based nature-based solutions which can be potentially scaled up in other parts of the country.

Papua New Guinea. Papua New Guinea is yet to receive adaptation funding from GCF, although the recently published National Adaptation Plan process was supported by the GCF Readiness and Preparatory Support Programme. The programmatic, large-scale approach proposed by the CRPP presents an important opportunity to channel GCF's adaptation resources into a country that is considered high climate risk, and where over 85% of the population live in rural areas where access to

markets, services and income-generating opportunities is limited.²⁰¹ There is a great deal of scope to scale-up and transfer knowledge and capabilities from investments from other large-scale adaptation funds, such as the Adaptation Fund's two projects on *Enhancing adaptive capacity of communities to climate change-related floods in the North Coast and Islands Region of Papua New Guinea*, and *Adaptation of Small-Scale Agriculture for Improved Food Security of Resilient Communities in Papua New Guinea (ASSA)* and GEF funded *Enabling sustainable production landscapes in Eastern Highlands and Western Highlands Provinces for Biodiversity, Human Livelihoods and Well-being*, which respectively focus on the installation of flood protection measures, supporting the uptake and commercialisation of climate resilient crop varieties, and improved information and decision-support systems for livelihoods. Channeling new adaptation resources to the country as part of large-scale poverty reduction projects will provide the possibility for delivering resilient livelihood interventions at-scale and capitalize on the smaller, more targeted investments going into the country to date. This would be an important and necessary step forward for building the adaptive capacity of subsistence and small-scale farmers that make up over 70% of the PNG population. Furthermore, it is self-evident that with 37.5% of the population living below the national poverty line (USD1.90 PPP/day)²⁰² gender inequality at extremely high levels, with the country ranking 159 out of 160 countries on the Gender Inequality Index,²⁰³ and one of the most culturally and geographically diverse nations on earth, Papua New Guinea, requires carefully considered and specific investment in adaptation solutions that are explicitly pro-poor and gender empowering and that are driven by a local approach that promotes context specificity.

Timor-Leste. Climate change adaptation financing by large scale funds to date in Timor-Leste has leaned towards the installation of localized protective measures for rural infrastructure (GCF funded *Safeguarding rural communities and their physical and economic assets from climate induced disasters*) and larger-scale protective works (GEF Funded *Strengthening Community Resilience to Climate Induced Natural Disasters in the Dili to Ainaro Road Development Corridor*). Increasing investment in agricultural diversification, and climate resilient value chain development would balance the portfolio of investments in the country and complement the more infrastructure focused interventions to date and address the high risk posed by climate change to the countries important agriculture sector of which more than 80% of the population rely as a source of income and for their food security. This need has been acknowledged with the recently approved GEF funded *Adapting to climate change and enabling sustainable land management through productive rural communities in Timor-Leste* Project. This is a sector that will require significantly more adaptation investment going forward. The benefit of having as yet unallocated resources as part of a pragmatic approach, also leaves open the opportunity to channel further investments to expand on demonstrably successful infrastructure activities and replicate or scale them up in areas of high climate risk where new resilient rural livelihood activities are being undertaken.

Vanuatu. Due to its status as a small island developing state, with high climate risk, Vanuatu is a prime target for climate adaptation financing. Large-scale investments include projects such as *Adaptation to Climate Change in the Coastal Zone in Vanuatu*, which is due to start phase II implementation with GEF co-financing. These investments represent large contributions to financing of climate risk management activities but place a strong emphasis on the installation of protective infrastructure along with strengthening national and local government frameworks for climate risk reduction. To date, GCF has complemented these investments by targeting financing at strengthening climate information services in the country through the Climate Information Services for Resilient Development in Vanuatu project. The current landscape of adaptation investment activities in the country presents a significant opportunity to extend the application of the enhanced CIS mechanisms and products from on-going GCF funded activities and compliment the more infrastructure and strategic framework focused investments of GEF, by developing national level mechanisms that can support the delivery of targeted and timely adaptation finance directly to the hands of the poorest and most vulnerable populations. Such a systemic approach to adaptation will enable those people that rely on fragile livelihoods that are frequently and increasingly disrupted by climate change related shocks and stresses, to take steps both before and after the impact of extreme events to ensure minimal negative impacts to their sources of income and food security and take advantage of new opportunities specific to their individual needs.

B.2 (a). Theory of change narrative and diagram

1. CRPP Vision and Objectives

The CRPP aims to build the resilience of poor and vulnerable people to the impacts of climate change in 7 countries in Asia and the Pacific region. This is considered a critical step towards achieving the desired impact of the CRPP, which is that **climate resilient sustainable development in Asia and the Pacific region is advanced**. The program theory states that in order to build the resilience of poor and vulnerable people to the impacts of climate change, four outcomes must be achieved; **(i) the enabling environment for knowledge, finance and implementation of local adaptation measures is improved; (ii) pro-poor adaptation finance is mobilized at scale; (iii) food security of poor and vulnerable communities is increased; and (iv) exposure of poor and vulnerable people to climate shocks and stresses is reduced**. The CRPP's anticipated impact is closely aligned with the GCF adaptation paradigm shift objective of achieving climate-resilient sustainable development and the CRPP's outcomes encompass all four of the GCF's adaptation result areas.

Goal Statement: **IF** the CRPP Trust Fund and the CRPP Investment Fund are established and provide upstream and downstream efforts to strengthen local adaptation measures in target countries **THEN** development in Asia and the Pacific region will be more climate resilient and sustainable **BECAUSE** the enabling environment for knowledge, finance and implementation of pro-poor local adaptation measures at scale in CRPP countries is strengthened **AND** pro-poor adaptation finance is mobilised at scale **AND** food security of poor and vulnerable communities is increased **AND** exposure of poor and vulnerable people to climate shocks and stresses is reduced.

2. Barriers to Strengthening the Climate Resilience of Poor and Vulnerable People in Asia and the Pacific Region.

Research that has formed the basis of the program design, has highlighted that strengthening the climate resilience of poor and vulnerable people in Asia and the Pacific region requires large-scale adaptation interventions that are explicitly targeted at the poor, promote empowerment, especially of women and girls, and that focus on local level solutions. In the absence of such interventions, future climate impacts (described in **section B1** and **Annex 2.1**) are likely to have catastrophic consequences for poverty reduction, growth and sustainable development in Asia and the Pacific region. Assessments and consultations undertaken with a range of stakeholders, including national governments, local governments, civil society organizations, grassroots women's organizations, and development partners, during the preparation of the CRPP, have identified the following barriers to implementing such interventions. These barriers are common to all seven CRPP countries.

1. **Lack of suitable, locally relevant, and accessible climate information to inform resilience investments targeted at the poor.** Programs targeted at the poorest population, such as social protection, urban informal settlement upgrade, and community-livelihoods, typically do not adequately consider how climate hazards (current and future) will interact with underlying drivers of exposure and vulnerability (such as poverty) and may impact the achievement of the objectives of such programs. This is due to a combination of reasons, including information deficits that stem from limited nation-wide availability of climate information in a format that can be used for decision-making, a lack of government capacity, especially at local level to use climate information, along with its uncertainties, for planning, budgeting and investments, and a lack of resources to regularly update such information. For example, in **Timor-Leste**, barriers for adaptation as identified in national climate plans includes insufficient hydro-meteorological services, limited capacity to forecast potential threats, weak coordination within and across sectors. It also reflects an underutilisation of weather and climate information to support decision-making, including the limited use of impact-based forecasting tools that communicate the potential impact of hazard/s on a community or area to inform planning; and forecast-based financing, which uses forecasts to trigger timely preparedness action before a hazard hits the exposed population. As a result, the longer-term objectives of these programs are often compromised due to climate shocks and stresses. For example, in **Indonesia**, it

is recognized that the current systems for social protection are not yet fully prepared to deal with new trends adequately and calls for adaptive systems that includes early warning systems.²⁰⁴ There is also a major knowledge gap, particularly at local level, regarding climate resilient measures, among planners and extension agents, including how best to support, facilitate and scale-up good practice. For example, the **Vanuatu** Framework for Climate Services developed in 2016 recognize the need to build sufficient knowledge and skills to prepare for and respond to climate threats and the need to improve the capacities of both providers and receivers and development of platforms.²⁰⁵

2. **Limited access to adaptation finance for implementing local solutions at scale.** The government budgets are typically insufficient to address the rising climate impacts, especially when the measures required are not just incremental adaptation (e.g., to make infrastructure resilient), but transformational adaptation (e.g., system-level green infrastructure which is prioritized by the local stakeholders and implemented and monitored by local communities). Moreover, there is a typical preference for the use of limited public resources for projects that address short-term development priorities, yield assured near-term benefits, and generate positive streams of direct or indirect incomes. Investments in climate adaptation are further stymied in some cases by their nature as a public good, where investors do not take into account their positive (and negative) externalities, such as the benefits to downstream communities and neighbouring countries. Thus, there is a need to address the barriers that prevent government and communities taking effective action. Without sustainable and predictable long-term finance, it will not be possible for governments to deliver the scale of local climate adaptation needed to address impacts on the poor and most vulnerable. For example, lack of financing, especially sustained and predictable financing for adaptation is identified as a constraint by **Timor-Leste** due to limited government budget.²⁰⁶ The Framework for Implementation of Climate Change Policy of **Pakistan** notes that financing for climate change adaptation is a challenge and there has been lack of interest from donors to support adaptation actions.²⁰⁷ The **Vanuatu** Climate Change and Disaster Risk Reduction Policy (2016-2030) recognizes that though it is one of the most vulnerable countries in the world to climate change and disaster risks, it received little funding for climate change adaptation.²⁰⁸ The COVID-19 pandemic will increase pressure on domestic finances, thereby making it even more difficult to finance transformational adaptation measures at the local level.
3. **Weak institutional and technical capacity to mainstream climate adaptation in policy and planning and implementing adaptation solutions.** Factors such as inappropriate institutional architecture, lack of capacity for implementation, limited coordination, and limited involvement of local stakeholders (including local private sector and civil society organizations) lead to policy failures that make it difficult to deliver adaptation at scale. For example, in the case of **Indonesia**, coordination has been identified as one of the main challenges at national, local / regional level in planning and implementation of adaptation and the RAN-API has identified increased capacity building as one of its four priority actions.²⁰⁹ While the political will for adaptation exists and a policy and legal framework is often in place, there is typically limited institutional capacity to steer pro-poor development in a climate resilient direction. This stems from a lack of expertise in climate risk informed development among policy makers and planners. While the National Adaptation Planning process is building capacity and provides a key starting point, there is a need to integrate climate risk considerations into all aspects of pro-poor policy, planning and budgeting, including decisions on fiscal transfers, which will in turn lead to scale-up of transformational adaptation measures that helps in alleviating poverty in the face of increasing climate shocks and stresses. There is also limited knowledge on innovative adaptation measures, especially at the local level.
4. **Negative perceptions - risk aversion around community-level actions.** Under-investment in climate adaptation at the local level is reinforced by a typical preference for the use of limited public resources for national level projects that align to the existing political economy (i.e. to support existing ministry budgets and development plans) The direct benefits (avoided costs), as well as the economic, social and environmental co-benefits that local adaptation generates, such as alleviating poverty, ecosystem services, health and other welfare benefits, even though they are greater than less well-targeted national interventions, are rarely identified and captured to encourage greater

investment. There is also lower appetite to finance local level interventions as they are subject to implementation challenges and require longer time and higher overhead. These barriers can be addressed by introducing support to ensure high quality bankable investments and delivering at scale to achieve impact. For example, utilizing tried and tested poverty reduction programs (e.g., community-driven development programs) can be efficient channel to scale up adaptation at the local level. Information on the business case for adaptation (the direct and indirect benefits) can help to encourage uptake.

3. Pathways of Change

The CRPP theory of change described in this section represents the theory that underpins the design of the program and has formed the basis for the identification of the activities and outputs that will help to overcome the barriers identified above and achieve the desired outcome and impact. The theory of change posits that in order to achieve the desired outcome of strengthened climate resilience of poor and vulnerable people in Asia and the Pacific Region, investments under the CRPP need to be made across three different but closely interlinked areas of intervention, hitherto referred to as pathways. The identification of **three key program objectives** has guided the development of the program theory and they are closely integrated into the three pathways of change described below. Meeting these objectives is considered critical to strengthening resilience and redirecting the lives and livelihoods of poor and vulnerable people toward a more climate-resilient and sustainable future. The three objectives are:

- Mobilizing large-scale public investments that support community-level adaptation of poor and vulnerable people.
- Developing national and local policies, plans, and programs that promote sustainable financing for community-led adaptation.
- Increasing the meaningful participation of poor women and men in resilience-related decision-making.

By ensuring that these objectives are met, the CRPP will contribute to transformational change through bringing about changes in culture and decision-making, reaching beneficiaries on a large scale, and ensuring sustainability of adaptation financing at the local level.

Pathway 1: Strengthening the enabling environment for financing and implementing pro-poor local adaptation measures at scale in CRPP countries. The objective of pathway 1 is to overcome the current barriers related to (i) negative perceptions and risk aversion around community-level actions; (ii) weak institutional and technical capacity to mainstream climate adaptation in policy and planning and implementing adaptation solutions; and (iii) lack of suitable, locally relevant, and accessible climate information to inform resilience investments targeted at the poor. The pathway leads to the achievement of **Outcome 1**; the enabling environment for knowledge, finance and implementation of local adaptation measures is improved. The path towards achieving this outcome and overcoming the above stated barriers includes implementing the following activities under two outputs.

Output 1: Knowledge and action research on climate risk informed pro-poor community-level solutions strengthened.

- Bringing together international research organisations, national academic institutions, grassroots community development organisations, national and local governments, financial institutions, and ADB, for coordinated and structured knowledge generation and learning, through action research and the creation of a wide range of evidence and knowledge products on climate risk informed pro-poor local solutions. This will increase awareness and understanding among national and local decision-makers on the benefits of local level adaptation interventions.
- Carrying out policy dialogues at sub-national, national and regional level, bringing key stakeholders together to exchange experiences and ideas and disseminate knowledge products. This will provide an opportunity for local level actors including local governments, communities and civil society

organisations to share their experiences, raise awareness and influence national and local policy related to adaptation decision-making.

Output 2: Institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened.

- Developing participatory climate risk assessment (covering hazard, exposure and vulnerability and how these interact and evolve with changing socioeconomic conditions) and decision support tools and procedures and building the capacity of governments and communities to use them for climate risk informed development planning and implementation. By strengthening decision-making processes, women and men will have the opportunity to participate in a meaningful way in decision-making on local adaptation measures. Capacity building and increased access to climate information will strengthen knowledge and skills which empowers local actors to act on adaptation.
- Building capacity of local governments and communities to access international climate finance for adaptation projects. This will support the channeling of international climate finance to the local level so that adaptation measures prioritized by communities can be implemented.

It is expected that the proposed interventions under the outputs presented above will combine to support changing the culture and decision-making related to adaptation investments at the local level, including increased participation of communities in adaptation decision-making processes. Changes in decision-making is anticipated to result in new and improved climate risk informed policies, budgets and programming, and alter the wider national and local policy environment, regulatory frameworks and governance structure that directly and indirectly influence public and private investment in local level pro-poor adaptation. This represents systemic improvements and in turn links to outcome 2 as it will facilitate the mobilisation of large-scale investment and sustainable financing of local level adaptation measures, and thus contribute to transformational change.

Supporting assumptions are that (i) research and knowledge products address critical gaps in understanding and awareness with regards to inclusive, pro-poor local level adaptation (ii) policy dialogues and evidence sharing workshops are effective in building a culture of collaboration and learning between CRPP stakeholders, and (iii) climate risk assessments and other participatory tools and procedures for climate risk informed development planning and implementation are appropriately designed and become institutionalized within the planning process at the local level. The key potential risk in this pathway is that changing culture and building capacity is slower and more resource intensive than expected.

Pathway 2: Mobilising pro-poor adaptation finance at scale. The objective of this pathway is to overcome the barrier of limited access of governments and communities to adaptation finance for implementing local solutions at scale by mobilising large-scale domestic and international resources for pro-poor adaptation at the local level in CRPP countries. A lack of suitable, locally relevant and accessible climate information to inform pro-poor investments is also considered a significant barrier to mobilizing adaptation finance at scale. Investments in improving CIS and integrating them into pro-poor climate financing decision-making frameworks is therefore a key activity for overcoming this barrier under this pathway. The pathway leads to the achievement of **Outcome 2:** Pro-poor adaptation finance mobilized at scale. The path towards achieving this outcome and overcoming the above stated barriers includes implementing the following activities under two outputs.

Output 3: Investments supporting inclusive pro-poor, community level adaptation prepared.

This output focuses on **mobilizing international climate finance**, and will;

- Provide technical assistance to support DMCs to develop new, large-scale, investment projects that implement adaptation measures at the local level. This will include providing TA resources for climate risk assessments as part of investment project preparation and feasibility studies.
- Provide a platform for coordinating the mobilisation of government and financing partners', including ADB's, resources for investing in adaptation solutions that benefit poor and vulnerable women and men.

Output 4: Information and systems for delivering applied climate-risk informed investments at scale improved.

This output focuses on **mobilizing domestic financing**, and will:

- Improve national policies, plans and distribution mechanisms related to social protection programs and fiscal decentralization to ensure that they are responsive to climate change concerns. By strengthening such in-country systems for the distribution of public funds the CRPP will support the channeling of public funds for adaptation measures that can be targeted locally and meet local adaptation needs, especially the needs of poor women and men.
- Support the enhancement of national climate information services (CIS) to meet the information needs of decision-makers. National systems for the distribution of public funds for local adaptation such as adaptive social protection and climate risk informed decentralized governance processes will need to be informed by climate risk information, and thus require improved climate information services infrastructure. This intervention directly supports overcoming the current barrier of lack of suitable, locally relevant, and accessible climate information to inform resilience investments targeted at the poor.

It is expected that the interventions described above will result in increased investment from international financial institutions in local level pro-poor climate change adaptation in CRPP countries, and increased investment from governments in local level pro-poor climate change adaptation. This will support transformative change by (i) creating sustainable and predictable financing through improved mechanisms for the delivery of public funds (ii) reach scale by benefitting a large number of poor and vulnerable people, and (iii) create systemic change through improving the participation of women and men in adaptation related decision-making (climate risk informed decentralisation). Outcome 1 and 2 are closely linked and interrelated as we would expect to see that with an improving enabling environment for financing and implementing pro-poor local adaptation measures at scale in CRPP countries, the increasing political will and institutional capacity results in stepping up the mobilisation of both domestic and international finance at scale.

Supporting assumptions are that (i) a sufficient number of appropriate large-scale investment projects for implementing adaptation measures at the local level are identified and, (ii) improving CIS systems and frameworks will result in their increased use in pro-poor climate financing decision-making frameworks such as social protection systems and decentralisation processes. The key potential risk in this pathway is that COVID-19 impact further restricts already tight governments budgets and spending capabilities that results in reduced availability of domestic and international financing for adaptation.

Pathway 3: Implementing large-scale pro-poor adaptation solutions to build long-term capacity to deal with climate shocks and stresses. The objective of this pathway is to implement local adaptation measures that specifically aim to build the capacity of poor men, women and children in areas of high climate risk to adapt to multiple, long-term and future climate risks and to learn and adjust in the aftermath of extreme events. This pathway will help to overcome the barrier of limited access to adaptation finance for implementing local solutions at scale. The pathway leads to the achievement of two of the program's outcomes; **Outcome 3:** Food security of poor and vulnerable communities is increased; and **Outcome 4:** Exposure of poor and vulnerable people to climate shocks and stresses is reduced. The path towards achieving this outcome and overcoming the above stated barriers includes implementing the following activities under two outputs.

Output 5: Climate resilient pro-poor livelihoods investments implemented.

- Implement large-scale investment projects that support community-based adaptation actions that are targeted at the poor and vulnerable population that improve agroecological systems and agriculture supply chains and information technology and risk management services, making their livelihoods more climate resilient.

Output 6: Climate adaptation pro-poor infrastructure implemented.

- Implement large-scale investments projects that support community-based adaptation actions that are targeted at the poor and vulnerable population to reduce the likelihood of the occurrence and/or the magnitude of climate-related hazards via the installation of natural or man-made barriers or ecosystem-based infrastructure, that block or absorb rapid-onset hazards such as floods and creeping hazards such as sea level rise.

It is expected that by implementing climate resilient pro-poor livelihoods investments, poor and vulnerable people will have access to the financing required to apply and adopt new resilient livelihood techniques and skills that will improve their access to safe and nutritious food despite the impacts of climate change. This will result in improved food security. It is also expected that by implementing climate adaptation pro-poor infrastructure investments, governments and communities themselves will have access to financing for the installation/construction of measures that reduce the extent and severity of the impact of climate hazards, thus reducing the exposure of poor and vulnerable people to climate shocks and stresses. By using new or existing large-scale government programs as vehicles for the delivery of these adaptation solutions to the local level, the CRPP can contribute to transformational change because it (i) builds on existing national programs that meet the needs of poor and vulnerable population and are thus pertinent and have a high buy-in from DMCs and can be easily replicated; (ii) they are large scale in nature, but individual activities are tailored to local circumstances; and (iii) they can facilitate deep systemic change by addressing underlying drivers of vulnerability, and promoting equity and justice; and (iv) ensure sustainability by strengthening local public and private institutions and creating policy incentives, and financing structures as part of a holistic implementation approach.

The two outcomes under this pathway are closely linked as the focus on nature-based solutions for reducing exposure to climate hazards can also contribute to improved food security outcomes through the protection and provisioning services that they provide. Outcomes 1 and 2 can also be viewed as precursors to the achievement outcomes 3 and 4 and they provide the political will, institutional capacity and financing for the investments that are ultimately expected to be implemented under pathway 3. However, by capitalising on existing opportunities, pathway 3 can start independently of the progress made on outcomes 1 and 2.

Supporting assumptions are that (i) the new techniques and skills introduced for improved climate resilient livelihood meet the context specific needs of the participating population (ii) there is a sufficient skilled and non-skilled labor force available to meet installation needs of green adaptive infrastructure, and new infrastructures are built to adequate standards, and (iii) new/improved infrastructures are well maintained. The two primary potential risks associated with the pathway are (i) low capacity and willingness of end users to apply and adopt new techniques and skills, and (ii) low capacity and willingness of end users to maintain and operate new infrastructure.

Box 3: The CRPP and Systems Transitions for Climate Resilient Development

Climate Resilient Development (CRD) has emerged as one of the guiding principles for climate policy and is identified by the IPCC in its 6th Assessment Report as a core requirement to pursue sustainable development. CRD requires the adoption of adaptation measures to meet basic needs for each human being, eliminate poverty and enable equitable, just and sustainable development.²¹⁰

System transitions have been highlighted as necessary processes for large-scale transformations and a core enabler of CRD, requiring social, institutional and technological change within societal sub-systems.²¹¹ The CRPP has been designed to contribute to progress in core system transitions that catalyze broader societal transformations that steer development in a climate resilient direction.

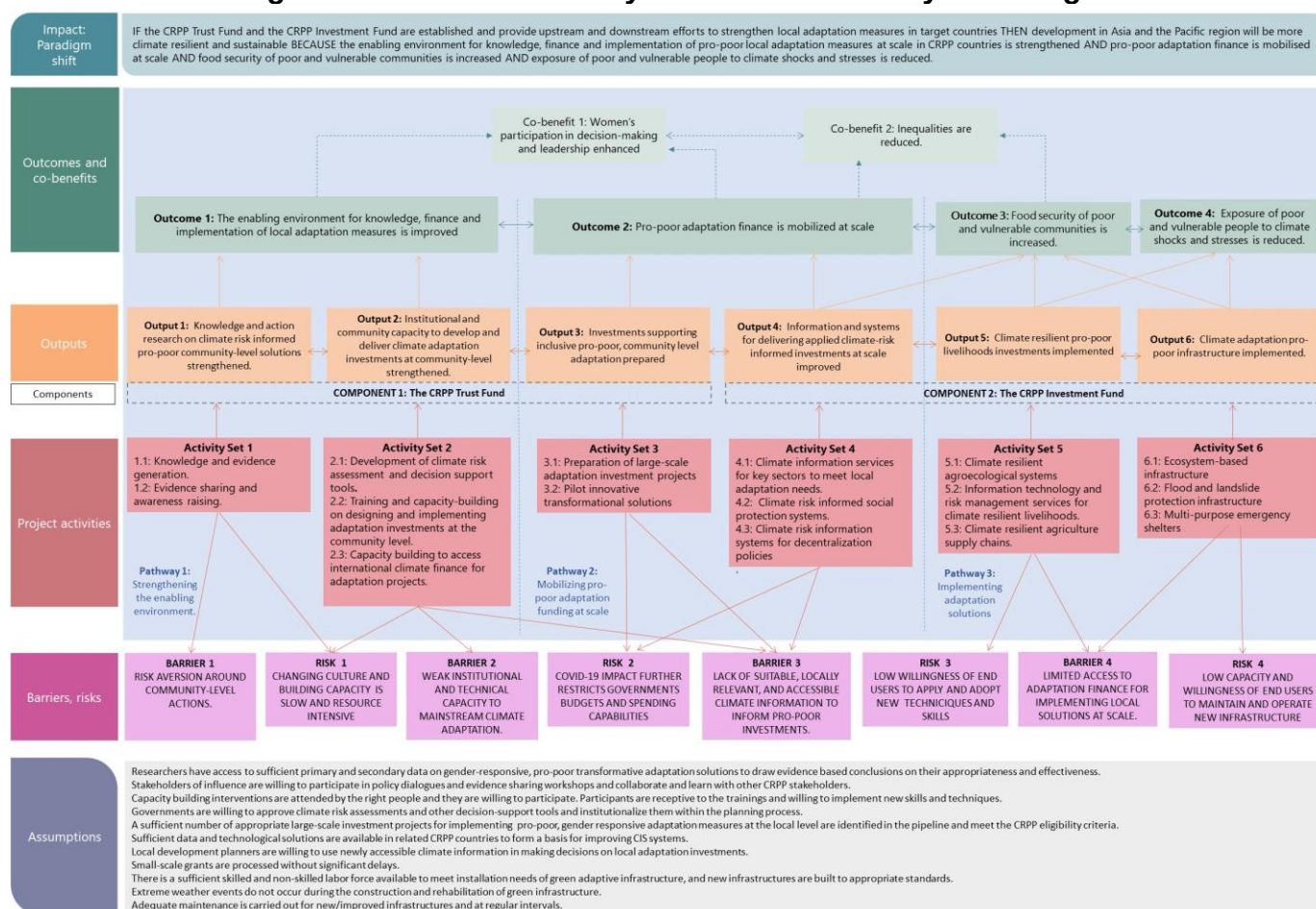
By including adaptation options (**detailed in section B3**) that are targeted at the poor and vulnerable population, and have been identified by the IPCC as offering the potential for change within the key systems, as part of investments in new or existing large-scale government poverty reduction programs, the CRPP can support faster transitions that contribute to more fundamental system transformations.

Co-benefits: The CRPP will have environmental, economic, social and gender co-benefits and will deliver against a number of the SDGs, including 1 no poverty, 3 good health and wellbeing, 11 sustainable cities and communities and 13 climate action. Some of the interventions supported by the CRPP are also likely to have mitigation co-benefits (as highlighted in section b.3) However, two specific co-benefits are considered to be central to the philosophy of the program.

- 1. Women's participation in decision-making and leadership:** Promoting gender equality is a core thrust of the CRPP as is creating and/or improving national and local policies and planning frameworks, so that there are policy incentives and procedures in place for the inclusion of local people in adaptation decision-making. By emphasizing women's participation in the strengthening of such governance structures, women will have the opportunity to participate in a meaningful way in decision-making on local adaptation measures. The explicit targeting of women through women focused investments and resource flows will help to break generations of gender inequalities in economic and human capital endowments and voice and agency that shape their climate adaptive capacity.
- 2. Reduced inequalities.** This social co-benefit is highlighted because of the explicit focus on low income and disadvantaged groups, and the focus on poverty reduction and addressing existing inequalities associated with climate change. The interventions will have positive distributional benefits by targeting these groups, increasing the flows of finance and responsibility to them.

The outputs and activities are described in the next section. **Figure 2** provides a visual summary of the theory of change.

Figure 2: A visual summary of the CRPP Theory of Change



B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MAR 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MAR 1 Energy generation and access	MAR 2 Low-emission transport	MAR 3 Building, cities, industries, appliances	MAR 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☐	☒	☒	☒	☒
Outcome 2	☐	☐	☐	☐	☒	☒	☒	☒
Outcome 3	☐	☐	☐	☐	☐	☒	☒	☒
Outcome 4	☐	☐	☐	☐	☒	☐	☒	☒

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1	☐	☐	☐	☒	☐	☐
Co-benefit 2	☐	☒	☐	☐	☐	☐

B.3. Project/programme description

Components, Outputs and Activities.

As outlined in section B.2, the CRPP comprises two separate but interlinked components; the CRPP Trust Fund (TF), which focuses on upstream support to strengthen the enabling environment required for financing and implementing local adaptation measures at scale; and the CRPP Investment Fund (IF) which focuses on the efficient roll-out of local adaptation measures through downstream investment program. CRPPF is not a separate legal entity, but a facility administered within and by ADB.

CRPP TF. The TF, established in August 2021, is financed by development partners and administered by ADB. The trust fund provides technical assistance and grant resources for improving knowledge and understanding, and strengthening institutions, processes, financial architecture, and capacity needed for building local resilience. This would ultimately lead to a suite of investments which can be implemented by governments with support from financial institutions. Some of these investments will be financed by the CRPP IF. The TF will support implementation of outputs 1, 2 and 3 of the wider CRPP program (described below). The technical assistance projects financed by CRPP TF will primarily be implemented by ADB and the two key CRPP TF partners, including the International Institute for Environment and Development (IIED) and the Huairou Commission (HC). Under the CRPP TF, ADB will engage IIED to implement some of the activities related to action research and knowledge management under output 1, and HC to implement some of the activities related to capacity building under output 2. Detailed [implementation guidelines](#) have been developed as part of the process of operationalising the CRPP TF. Recognizing that longer-term, flexible, and predictable funding is needed for supporting transformational adaptation at the local level, the TF will be implemented over a period of 10 years, between 2021–2030. Implementation will be carried out in a phased manner with phase 1 (2021-2025) with a total amount of USD75 million. To date, the Government of United Kingdom, the Nordic Development Fund, and the French Development Agency has confirmed financial support for the TF, among to a total of ~ USD68 million. Discussions are ongoing with other bilateral donors and foundations to support the TF. All developing member countries of ADB are eligible to receive support from the CRPP TF. However, due to limited availability of funding, the CRPP TF will support selected countries. This will include all seven countries that are part of the CRPP IF plus selected additional countries. Thus, the USD 75million from the CRPP TF will support countries included in the CRPP IF plus additional countries. The CRPP TF will **not** be financed by GCF.

Gender window under CRPP TF: In order to promote investments in women-focused resilience building, the CRPP TF has a dedicated “gender window”, amounting to approximately 15% of the TF. This window will provide technical assistance grants to (i) undertake policy dialogue and analytical studies which make the case for dedicated women-focused investments in adaptation at local level; (ii) undertake project feasibility studies for such investments; and (iii) build capacity of women machineries in country and civil society organizations, especially grassroots women’s organizations to identify, prioritize and implement local adaptation solutions. The window will also provide grant resources to grassroots women’s organizations to implement small-scale innovative adaptation solutions and pilot projects as part of the lesson learning and evidence generating process. Outside of the Gender Window, gender considerations will be mainstreamed in all activities supported by the CRPP, including projects being financed by CRPP IF.

CRPP IF. The IF will be financed by GCF and administered by ADB. It will be a program to provide critical grant and loan resources to implement local adaptation measures as part of ADB financed investments (projects) in seven countries that are targeted at the poorest population. Without GCF financing, these projects will not implement local adaptation measures aiming for transformational changes. Recognizing the importance of providing longer-term financing, the IF will have a time period of 2023–2030. Projects seeking financing from IF will need to be likely approved by ADB Board between 2024 and 2027 and complete implementation by 2030. The IF will have a total of USD120 million (USD100 grant and USD20 loan) from GCF for seven countries and will leverage at least USD555 million from ADB’s resources and USD75 million from bilateral donors and foundations through CRPP TF to support at least 10-15 projects. Under CRPP IF, each of the seven countries will have at least one project and will receive between USD10 million and USD40 million. . The IF will provide grant and loan resources to implement outputs 4, 5 and 6 of the wider CRPP program (described below). The allocation and use of resources under the IF will be aligned with CRPP

theory of change, results framework, project principles and eligibility criteria (described later in this section and **Annex 2.4**) and administrative arrangements. The eligibility criteria and project assessment process ensures that the allocation of resources will take into account the principles for demonstrating the impact potential for adaptation activities as set out in the GCF policy as adopted by the Board in decision B.33/12, paragraph (e). For more details see **section B4** on governance structure.

Interaction between the CRPP TF and CRPP IF. Some of the ways in which the CRPP TF and CRPP IF will interact are as follows: (i) all upstream support related to action research, capacity building and knowledge management provided through the CRPP TF will benefit the countries included in the CRPP IF; (ii) in some cases CRPP TF may support the development of the climate rationale and project preparation of a project, implementation of which will be financed by the CRPP IF; (iii) in some cases the CRPP TF may support implementation of a pilot, which if found to be successful, will be scaled up through the CRPP IF; (iv) in some cases a project may receive support from both CRPP TF and CRPP IF for implementation; (v) projects that are financed by the CRPP IF but may not have received direct project preparation-related support from the CRPP TF, but nevertheless lessons from such projects will be shared with the wider set of countries that are part of the CRPP TF to advance knowledge on local adaptation; and (vi) monitoring, evaluation and learning of the CRPP TF and CRPP IF will be undertaken together using a common results framework. Thus, by combining upstream actions and downstream program, the two components of the CRPP – TF and IF, will contribute to achieving the outcome defined in the previous section.

The design of the CRPP has been significantly influenced by the lessons from similar programs, especially in Africa, funded by the Pilot Programme for Climate Resilience under the Climate Investment Fund and as described in Box 4 below.

Box 4: Lessons from Implementation of the Pilot Programme for Climate Resilience in African Countries

High level coordination across multiple sectors: The PPCR implementation has found that coordination across multiple sectors supported at the highest levels of government was the most effective approach for shaping a program of resilient investments and is promising for implementation effectiveness and anticipated scaling up. The CRPP program design has factored this lesson by prioritizing the outputs under CRPP TF which has a large focus on strengthening institutional coordination. Moreover, ADB has direct relationship with ministries of finance and planning in the countries which play a key role in strengthening coordination across sectors.

Linking and leveraging investments: The PPCR experience highlights that the expectation from countries for linked and leveraged funds at scale through formal MDB collaboration was pivotal for country buy-in. Both linked investments and leveraged investments were instrumental in advancing and catalyzing countries' resilience pathways. This lesson has been factored in the CRPP program design which aims to provide grant resources from GCF in combination with concessional loans and grants from ADB, thereby supporting countries to access funds for adaptation at scale.

A responsive country-based approach to resilience planning: The programmatic approach of PPCR allows for consistency in goal and outcome within a flexible and responsive framework that considers country capacities, political structures, and overall development regimes. This lesson has led to the design of the CRPP as a program (and not simply a multi-country project) with multiple strongly interlinked, but also independently robust, outputs which will provide flexibility to the countries to develop projects while ensuring an overarching vision and common objective of strengthening community resilience. So too, not all outputs will be implemented in all countries, thereby ensuring the GCF resources are used to fill critical gaps, while being guided by a wider common program framework.

Facilitating private sector engagement: Upfront technical assistance and targeted advisory services have been critical for overcoming barriers to engaging the private sector on climate resilience. This includes piloting new modalities of climate adaptation, validating their commercial viability, and creating

an enabling environment for successful investments. This lesson has been factored in the design of the CRPP TF which will provide technical assistance to overcome barriers for scaling up climate adaptation solutions at the local level, including through private sector engagement.

Outputs under the CRPP TF: The CRPP TF supports activities under outputs 1, 2 and 3. Activities under these outputs will not be financed by GCF, but as they underpin and enable the achievement of the overall programme objectives, they are described below.

Output 1: Knowledge and action research on climate risk informed pro-poor community-level solutions strengthened. Knowledge and support technical assistance will be provided to strengthen knowledge, awareness and capacity of national and local institutions and communities for pursuing climate risk informed pro-poor development, especially for improving inclusive decision-making in the context of adaptation and resilience at the local level. Grant resources will also be provided to implement small-scale innovative adaptation solutions and pilot projects as part of the lesson learning and evidence generating process. This output will address issues related to risk perception in support of overcoming barriers related to risk aversion for community adaptation solutions.

Activities. Activities will include supporting governments and communities to (i) conduct research and analysis for identifying a suite of transformational adaptation investments that meet local needs and are aligned with national climate priorities; (ii) organize workshops, seminars or other evidence sharing events on pro-poor adaptation solutions to share experiences of using innovative financing options for adaptation, including private sector financing; (iii) organise a series of policy dialogues to bring together a diverse range of key stakeholders (both nationally and internationally) to identify solutions to overcome key barriers in scaling up transformational adaptation that benefits poor and vulnerable populations and explore opportunities to enhance the effectiveness and equitableness of adaptation finance delivery mechanisms; and (iv) document and share lessons, including lessons emerging from COVID-19 response, which are critical for building future resilience, and for sharing experiences of using innovative financing options for adaptation, including private sector financing. Recognizing the importance of adopting adaptive management principles, the output will include a monitoring and learning system to gather lessons on what works and what needs to be improved.

Output 2: Institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened. This output will strengthen institutions and build national and local capacity to develop and deliver community-level adaptation investments at scale, that meet the needs of the poor and vulnerable population. In doing so, this output will support overcoming the barriers related to weak institutional and technical capacity for delivering adaptation at the local level.

Activities. Activities will support (i) the development and institutionalisation of procedures and mechanisms for the devolution of responsibilities for adaptation planning and implementation, ensuring inclusiveness and participation of poor and vulnerable people; (ii) the implementation of training and capacity building workshops for government officials on applying climate risk information in the formation of plans, policies, and programs; (iii) creating mechanisms and building the capacity of local governments and civil society organisations for monitoring and evaluating adaptation investments; (iv) building capacity within national and local institutions and civil society organisations, including the private sector, to access international climate finance for adaptation projects, including linkage with Enhanced Direct Access pilot, where appropriate. This capacity building will also build on existing GCF readiness and National Adaptation Plan related activities.

Output 3: Inclusive and pro-poor adaptation investment projects identified and prepared. This output supports governments in developing large-scale bankable adaptation projects, that are aligned with national adaptation priorities and address the nexus between climate and poverty, with special emphasis on women. As such, this output supports the mobilisation of large-scale domestic and international financing for local level adaptation and contributes to overcoming the barrier of limited access to finance for adaptation at the local level. Selected projects prepared under this output will be implemented through the CRPP IF outputs 4, 5 and 6.

Activities. Activities under this output will; (i) support governments to prepare (feasibility studies) for adaptation investments identified and prioritized under outputs 1 and 2; and (ii) explore the potential of accessing concessional development and climate finance for implementation of such projects by engaging with potential financing partners (national development banks, multi-lateral development banks, climate finance institutions, bilateral donors, and the private sector) to mobilize adaptation finance, including strengthening linkages with the GCF's Enhanced Direct Access pilot; (iii) provide grant financing (on a very selective basis) to test innovative solutions which can later be scaled up through CRPP IF. Project identification and preparation will be undertaken by the TF, which does not involve GCF finance, and thus there is no need for GCF Project Preparation Facility funding. However, the activities will be closely aligned with the National Adaptation Plans (NAP) and GCF readiness funding and country work programs.

Gender Window: The CRPP Gender Window will support the identification, development, and implementation of investments that have the explicit objective of strengthening the resilience of women to climate change. As such, support will be provided through (i) technical assistance for creating an enabling environment for financing women-focused adaptation measures, identifying investment opportunities and preparing investment projects; and (ii) grant financing, on a selective basis, for implementing investments identified and prepared under the technical assistance. Examples of how the Gender Window can be leveraged to enhance gender equality outcomes of the Investment Fund in CRPP countries, include; (i) generating the necessary political will, and commitment from governments and communities to invest in women-focused adaptation measures that help tackle climate change; (ii) creating policy incentives and procedures for the inclusion of women in adaptation decision-making at the local level and building their capacity to use participatory tools and procedures for climate risk informed development planning and implementation which can empower local women to act on adaptation; and (iii) place women at the centre of the design of large-scale investments that explicitly strengthen the climate resilience of women, particularly poor women, and/or adaptation investments that are led by women.

When approached together, output 1, 2 and 3 will strengthen the enabling environment in the countries for scaling up adaptation solutions that directly benefits the poor and vulnerable populations, including women.

Outputs under CRPP IF: The CRPP IF will support activities under outputs 4, 5 and 6. These outputs will be co-financed by ADB and GCF. In some cases, the outputs may also receive grant co-financing from bilateral donors through the CRPP TF and/or other partner financing institutions. Selected projects emerging from the TF will be delivered using the IF. The focus will be to deliver transformational adaptation solutions through large-scale investments explicitly targeted at the poor and vulnerable population. The CRPP IF proposes to combine USD555 million of ADB finance with USD120 million of financing from the GCF to support implementation of local adaptation measures as part of ADB financed investments (projects) in seven countries. These are also the type of investments that can support green, inclusive and resilient COVID-19 recovery, thereby, ensuring CRPP's alignment with COVID-19 recovery. **Annex 2.2** describes the importance of supporting these outputs in the context of increasing climate risk and barriers in scaling up adaptation, and provides a detailed analysis of the linkages of activities with international standards and frameworks.

Output 4: Information and systems for delivering applied climate-risk informed investments at scale improved: Enhanced climate risk information services that can provide reliable, tailored information based on the past, present and potential future climate will be critical for better understanding of potential risks over various timescales, and to manage current and future climate risks across sectors. Climate information services have also been identified by the IPCC as a key adaptation action for enabling systems transitions. It is critically important that such information captures physical and socioeconomic information of the local area, through engagement with local stakeholders, including women. While there has been improvement in weather and climate forecasting and provision of downscaled climate change projections, existing climate information services are not well focused on the user needs (sectors and at various levels) and also do not reach "the last mile". Often the poor and vulnerable communities whose livelihoods are climate sensitive do not have access to usable and timely information, nor do the local governments for which such information is fundamental to planning and financing adaptation solutions in their constituencies. Global analysis on

climate information services reveals that capacities to deliver climate-risk and reliable early warning information services varies across countries and the functional capacities on monitoring and evaluation of results and benefits of the use of climate services remain weak.²¹² Slow progress in improvements and weak capacities to adapt such practices can be a major barrier to deal with dynamic climate risks. The State of Climate Services, 2019 report notes that “while investments have substantially increased over the past decade, more and better investments are still needed to ensure the provision of high-quality climate information and services for climate-resilient action.”²¹³

In addition, the World Meteorological Organization has identified that there is a need to invest specifically in enhancing countries’ capacities when it comes to impact-based forecasting. Alerts need to go beyond producing accurate forecasts and timely warnings, to better understand and anticipate the likely human and economic impacts due to severe weather and explain what a forecast means for each given location, specifically the types of impacts associated with the forecast to more effectively trigger early action based on the warnings.²¹⁴ Such understanding of impact is critical for informing policy and for ensuring the benefits reach the end users such as farmers.²¹⁵ This output will support countries in developing, maintaining and using climate risk information services for pro-poor decision-making purposes for resilience, thereby overcoming barriers related to lack of suitable, locally relevant and accessible climate information and understanding of impacts. It will also help the large-scale mobilisation of domestic resources for pro-poor adaptation through integrating climate risk information into decision-making frameworks for social protection and decentralised investment planning, thus contributing to overcoming the barrier of limited access to adaptation finance for implementing local solutions at scale. It will support the following types of activities that are closely linked to adaptation options for advancing systems transitions, particularly Land, Oceans, Ecosystems and Societal systems, as identified by the IPCC WG2.²¹⁶ These activities will be selected based on assessment of ongoing initiatives in respective countries in order to ensure complementarity and avoid duplication. The activities are also closely aligned with the four pillars of the UN’s newly launched **Early Warnings for All (EW4A) Initiative**, which stipulates that every person in the world should be protected by an early warning system by 2027 and aims to deliver MHEWS in over 100 countries and will contribute to outcomes such as; improved access and quality of risk data; and strengthened capacity for developing procedures for holistic anticipatory action based on early warnings.

Activity Area 4.1. Climate information services for key sectors to meet local adaptation needs. Under this activity area, it will support activities such as (i) development and strengthening of national frameworks for climate services guided by the five pillars of the Global Framework for Climate Services which includes user interface platforms, climate services information systems, observations and monitoring, research, modelling and prediction, and capacity building; (ii) improvements in multi-hazard early warning systems to support impact based forecasting, especially in the context of agriculture and water resources sector – this will be structured around the key elements identified by the WMO on Multi-hazard Impact-based Forecasting and Warning Services, including partnerships, development of information and services, functional requirements, capacity development and validation; and (iii) adapting and scaling up of gender-responsive climate information services and demonstrate its application to manage farm level climate risk.

Activity Area 4.2: Climate risk informed social protection systems. Under this activity area, it will support activities such as (i) development/strengthening of national policy frameworks and regulations on adaptive and shock responsive social protection following the four building blocks identified by the World Bank, namely, programs, data and information, finance, and institutional arrangement and partnerships; (ii) through provision of hardware and software that allows acquisition, analysis, storage and visualization of hazard, exposure and vulnerability data which can be integrated in national social protection data systems, and trainings in undertaking climate risk-informed targeting (including community-based targeting) of social protection program beneficiaries; (iii) through the provision of early warning equipment and trainings on modelling, forecasting and early warning capacity, which can support forecast based financing through social protection programs, based on lessons learned from similar initiatives being undertaken by humanitarian agencies such as the Red Cross and Red Crescent Societies; (iv) through the provision of ICT infrastructure that can be deployed in social protection programs to facilitate transfer of assistance in the aftermath of extreme events in high risk areas; and (v) in providing trainings for social protection field workers to build their capacity on disaster preparedness measures. COVID-19 pandemic has further highlighted the

importance of adaptive and shock responsive social protection and thus many countries are in the process of strengthening social protection systems as part of their inclusive COVID-19 recovery, which directly contributes to outcome related to reducing exposure of poor and vulnerable population to shocks and stresses. Resources from the CRPP IF will not be used to cover cost of transfers (such as cash transfers) provided through social protection programs.

Activity Area 4.3 Climate risk information systems for decentralization policies. This will support national governments in strengthening decentralization policies and frameworks from climate risk perspective based on experiences of the United Nations Capital Development Fund (UNCDF) in introducing performance-based climate resilience grant mechanism under its Local Climate Adaptive Living Facility (LoCAL); specifically, (i) national and local governments in the preparation of climate risk and vulnerability assessments and dissemination of such information in order to inform government programs on decentralization and build capacity of local governments through trainings, workshops and consultations in using the results of climate risk and vulnerability assessments to develop local adaptation plans, identify adaptation investments as part of local development plans, mobilize finances for adaptation, formulate incentives to scale up adaptation, and increase awareness among local stakeholders on adaptation; (ii) local governments in integrating climate adaptation into local development planning and budgeting processes including undertaking capacity building, improvement of methodologies, and strengthening public participation processes, especially to engage local women in decision-making related to resilience building; and (iii) implementation of adaptation measures that have been prioritized in local development plans through the provision of small scale grants (such as grants tied to results or performance) to grassroots women's organizations using existing budgeting and fiscal transfer mechanisms and building capacity for procurement and monitoring of resilience interventions. With the shared goals of increasing awareness of and capacities to respond to climate change at the local level, mainstreaming climate change adaptation into local government planning and budgeting systems and investments and increasing the amount of finance available for local adaptation to climate change adaptation, the CRPP provides a significant opportunity to replicate and scale-up some of the key activities implemented under LoCAL.

Output 5. Climate resilient pro-poor livelihoods investments implemented. Many low-income countries in Asia and the Pacific region have predominantly agrarian societies. In Papua New Guinea for example 87% of the total population live in rural areas, and Cambodia (76%), Vanuatu (74%), Timor-Leste (69%), Lao PDR (64%) and Pakistan (63%) all have majority rural populations. Agriculture accounts for a significant proportion of GDP in many of these countries (Vanuatu 27%, Pakistan 23% and Cambodia 23%) and many rural households in poor communities depend on subsistence agriculture for their livelihoods and food security²¹⁷. In Lao PDR for example, 50% of rural households are subsistence farmers with incomes below USD300 per year and only about 30% are food secure. In Papua New Guinea and Vanuatu, employment in agriculture is more than 50% of total employment. These people's livelihoods are heavily dependent on natural resources, such as rainfall that are directly influenced by changes in weather and climate, and poor communities often inhabit marginal land on hillsides, in deserts and floodplains, and on coastal zones that are highly exposed to floods, droughts and storm surges, which can erode their natural asset base. Subtle changes in rainfall and temperature, as well as increasing extreme events can have significant multiple impacts on their income and food security. This output will therefore support poor and vulnerable communities to build their capacity to adapt to climate change by adopting adaptation measures that help to create climate resilient sustainable livelihoods. New adaptation interventions will be needed at the farm level to transform agroecological systems (e.g., the adoption of climate resilient crop varieties and improved soil and water management), provide farmers in poor communities with access to climate information, early warning and risk management services that meets their specific needs and support adoption of new production practices (e.g., index-based weather insurance), and supply chains from local to national level that are robust and ensure a stable supply of agricultural products and safe and nutritious food despite the impacts of climate change (e.g., improving transport and logistics infrastructure etc.).

Activities under this output will be guided by the Sustainable Livelihoods Framework (SLF), first proposed by the then UK Department for International Development, which identifies five types of assets or capitals upon which livelihoods are built upon, namely; human (set of skills, knowledge, ability to work, and good health), socio-political (quantity and quality of social resources as networks and access to wider institutions

in society), natural (fresh water, soil quality, forests, healthy ecosystems), physical (infrastructure, services, and productive assets), and financial (cash and other liquid resources, such as savings, credit, remittances, pensions). The SLF has been effectively applied for exploring the many complex factors influencing household food security and climate resilience and thus provides a useful conceptual basis for the identification of the activities under this output.²¹⁸ With its focus on capacities, vulnerabilities and livelihood strategies, the SLF helps develop resilience-building strategies that can be effective in increasing the capacity to cope with and adapt to climate-related impacts. This livelihood resilience perspective is particularly useful in the context of the CRPP as it provides a basis for assessing people's capacity for, and differences in, perceiving risk and taking anticipatory actions, either individually or collectively, and places people at the center of the analysis, located within, rather than dominated by, ecosystems, technologies, political contexts, markets and resource networks.

This output will finance the following types of activities. The activities will be selected based on assessment of ongoing initiatives in respective countries in order to ensure complementarity and to avoid unnecessary duplication. The activities are closely aligned to the GCF paradigm shifting investment pathways outlined in the Agriculture and Food Security Sectoral Guide, as well as the adaptation options identified by the IPCC as offering the potential for progressing transitions in Land, Ocean and Ecosystems and Societal systems. Some activities, such as promoting the uptake of climate smart agricultural practices are also anticipated to have mitigation co-benefits.²¹⁹

Activity Area 5.1: Climate resilient agroecological systems. The main focus will be preserving and increasing natural capital, such as water and soil for productive purposes, and building human capital, as identified in the SLF, through supporting knowledge and skills generation in relation to climate resilient production practices. This output is closely aligned with the GCF Agriculture and Food Security Paradigm Shifting Investment Pathway 1: Promoting resilient agroecology. This activity area will support activities such as (i) provision of agriculture extension services, especially to women farmers, for new climate resilient crop varieties and production systems to enhance crop productivity based on the local farming systems, agro-ecological zones and climate risks; ii) supporting the adoption of agro-forestry (which also may have additional mitigation benefits such as sequestration of carbon in biomass and soil) through developing the skills required to establish and manage an agroforestry system (selection of trees and crops, marketing of products, creating an enabling environment); (iii) promotion of effective and efficient use of water by adapting and scaling up water retention, communal water harvesting and storage; iv) supporting integrated land and water management and improvement of soil management techniques and drainage systems such as terracing which refers to the division of hills and mountains into narrow, graduated steps to facilitate crop cultivation, and minimum/no tillage that helps conserve soil moisture, reducing evaporative loss, thus requiring less water than conventionally tilled field; (v) provision of small revolving grants that can support upfront capital for women farmers to adopt climate-resilient practices and vi) supporting national institutions to offer climate resilient employment through improved technical and vocational education and training (TVET), including trainings targeted at women employed in agriculture.

Activity Area 5.2: Information technology and risk management services for climate resilient livelihoods. The focus will be to build human and financial capital, as identified in the SLF, through improving access to digital information services and low-cost risk transfer mechanisms. Building on the climate information generated through the expanded and upgraded climate information services under output 4, this activity area will support activities such as (i) investing in information technology systems to facilitate rapid information flows to small-holder farmers (ii) developing low cost risk transfer mechanisms for small-holder farmers, such as weather indexed based micro-insurance schemes tailored to individual country contexts, to support the adoption of climate resilient production practices, (iii) Design and deliver capacity building programs on financial literacy for insurance consumers (iv) design and deliver trainings and awareness raising for social protection program beneficiaries, especially women, on climate-resilient livelihoods (linked to output 4.2). This activity area is closely aligned with the GCF Agriculture and Food Security Paradigm Shifting Investment Pathway 2: Facilitating climate informed advisory and risk management services. Resources from the CRPP IF will not be used to cover cost of insurance premium for disaster risk financing.

Activity Area 5.3 Climate resilient agriculture supply chains. The focus will be to build human capital and physical capital, as identified in the SLF, to deal with climate impacts to the agricultural supply chain. This activity area is closely aligned with the GCF Agriculture and Food Security Paradigm Shifting Investment Pathway 3: Reconfiguring food systems. This will support activities such as i) improving the transport and logistics infrastructure, especially through the use of ICT solutions, to successfully bring products to market despite the impact of extreme weather events due to climate change; (ii) introduction of protective measures to the siting and storage of seeds, tools, vehicles and energy infrastructure as well as installation of new or improved post-harvest storage and processing facilities to protect against the impact of climate change on productive inputs and marketable produce; (iii) providing technological solutions and capacity building for improved food processing strategies that can help increase the value and durability of perishable foods such as fruits and vegetables. The COVID-19 pandemic has highlighted the importance of resilient agriculture supply chains and are therefore being prioritized by countries as part of COVID-19 recovery and will contribute to outcomes related to increasing food security of poor and vulnerable population.

Output 6. Climate adaptation pro-poor infrastructure implemented. There are urgent needs to scale up investments in adaptation infrastructure, i.e., infrastructure that has a primary purpose of building resilience to climate change (not climate proofing of development infrastructure) and would minimize the exposure and physical vulnerabilities of the poor and vulnerable communities to the impacts of climate change and protect their productive assets. Moreover, when designed in close collaboration with local stakeholders, such infrastructure can provide additional benefits. For example, by favouring green over grey coastal infrastructure to reduce impacts of storm surges, it is possible to deliver additional social and environmental benefits, that help build community resilience, as well as providing economic benefits. This output will support following types of activities. The activities will be selected based on assessment of ongoing initiatives in respective countries in order to ensure complementarity and to avoid duplication. They are also closely aligned to the adaption options identified by the IPCC as supporting Land, Ocean and Ecosystem transitions and urban and infrastructure system transitions. Some activities will also have important mitigation co-benefits, such as rehabilitation of mangrove forests which could potentially mitigate upwards of $-27.6 \pm 1.7 \text{ Mg CO}_2\text{-e per hectare per year}$.²²⁰

Activity Area 6.1 Ecosystem-based infrastructure. This will support activities such as (i) national and local governments in collaboration with national scientific organizations in identifying locally appropriate green infrastructure to reduce exposure and vulnerability of climate impacts on lives and livelihoods of poor and vulnerable communities; (ii) in undertaking trainings for local governments, communities and community-based organizations in planting, rehabilitation and maintaining green infrastructure; (iii) local governments/communities (especially women's groups) by providing small grants for planting and rehabilitation of green infrastructure; and (iv) in developing policy, legal frameworks and financial models for scaling up investments in green infrastructure in partnerships with private sector. The implementation of the activities will be guided by the IUCN Global Standards for Nature-based Solutions, which serve as a mechanism for developing a consistent approach to designing and implementing nature-based solutions, including ecosystem-based adaptation measures, and will thereby guide the CRPP to deploy such investments in a systematic way, and ensure quality in the design and execution.

Activity Area 6.2 Flood and landslide protection infrastructure. This will support activities such (i) undertaking local planning processes to identify and prioritize small scale protective infrastructure and recognize linkages with wider infrastructure systems; (ii) in undertaking civil works for small scale flood protection and/or landslide protection measures, especially measures that use a combination of grey and green measures and are prioritized by local communities; and (iii) in undertaking trainings for local governments and communities, especially women, in maintenance of protective infrastructure.

Activity Area 6.3 Multipurpose emergency shelters. This will support activities such as (i) undertaking local planning, with active participation of women, to identify the need of multipurpose emergency shelters to save the lives and livestock of poor and vulnerable population in times of extreme weather events; (ii) local governments in constructing emergency shelters (with gender inclusive and socially appropriate design features) to reduce the exposure of poor and vulnerable population to extreme weather events; (iii) in undertaking trainings for local governments and communities in planning and maintenance of multipurpose

emergency shelters; and (iii) in undertaking capacity building for local communities on disaster preparedness including evacuation planning.

Outputs 4, 5 and 6 will deliver adaptation activities through the provision of combined grant finance with major investment funding to deliver adaptation at scale at the local level. Since these outputs will be part of wider ADB investments with governments, this will ensure scale, replicability and sustainability.

Ideas on Project Concepts

The following paragraphs provide a summary of ADB investments (projects) that while currently in their very early stages of conceptualization, have the potential to align with the objectives of CRPP (subject to findings of climate risk and adaptation assessments to be undertaken during project preparation). Some of these project concepts are described in more detail in **Annex 2.3**, and demonstrate examples of large-scale interventions that could serve as vehicles for the delivery of the specific local adaptation solutions described above using the CRPP IF resources provided by GCF. Without financing from the IF, activities proposed under these projects that explicitly promote adaptation to climate change at the local level will not be implemented.

Cambodia: The proposed **Climate Resilient Smallholder Development Project** will build on the successful implementation of past and ongoing ADB financed Tonle Sap Poverty Reduction and Smallholder Development Project. The project promoted community-driven development through livelihood improvement groups (LIGs) and provided them with group revolving funds managed by a committee of members. At the end of the first production cycle, LIG members will repay the value of the inputs to the group to establish a group revolving fund from which group members can subsequently borrow to finance future activities. The LIG model has been shown to be a viable mechanism to enable the poor to build up assets and subsequently to access funds from formal rural credit organizations. The LIG model enhanced the income-earning opportunities of its members supported by credit, skills development, and market access. The project promoted community-led smallholder development, while incorporating climate resilience and climate risk reduction mainstreaming in agriculture development activities and project designs. Moving forward the proposed project will build on the LIG model to strengthen community-led adaptation actions. It will integrate four types of adaptation measures related to (i) rural infrastructure resilience by rehabilitating rural infrastructures responding to the climate resilience condition; deploying climate resilience varieties of crops and livestock and bio-engineering to strengthen infrastructure; (ii) community resilience through building social networks focusing on adaptation and disaster risk management (adaptive social protection); (iii) institutional resilience through strengthening capacity of local institutions to integrate adaptation and disaster risks concern in development planning; and (iv) financial resilience by mobilizing green finance and introducing insurance solutions in rural area to support adaptation and disaster risk management efforts. The proposed project directly contributes to the activities under **output 4** and **output 5** of the CRPP.

Lao PDR: The proposed **Building Resilience and Reducing Rural Poverty Project** will build the resilience of rural poor in Lao PDR to climate change by expanding the coverage of the non-contributory social assistance and integrating shock responsive features that will enable it to rapidly expand as necessary in response to the impacts of climate and weather-related hazards, such as floods. The inclusion of shock-responsive features will help deliver vital income support to households in the instance of a shock, such as floods to smooth consumption and minimise negative coping strategies. The project will expand coverage of the existing nutrition-sensitive conditional cash transfer (CCT) program implemented by the Ministry of Agriculture and Forestry (MFA) and the World Bank into new geographic locations that have high levels of climate and disaster risk, strengthen underdeveloped systems for the delivery of cash transfers, and build the capacity of MFA and the Ministry of Planning and Investment (MPI) to implement priority social protection programs. Specific actions will include, (i) integrating vulnerability to climate-risks and shocks into the social registry and selection criteria for geographic locations and households that can receive the assistance based on climate risk; (ii) link the social registry with early warning systems; and (iii) provide operational and technical assistance for day-to-day implementation and project management functions such as environmental and social safeguards management. Thus, support from the CRPP IF will support Laos

government to make their existing social protection system more responsive to climate related shocks and thereby use such system in the future to reach to vulnerable households in the aftermath of disasters. Further to these activities, the program proposes to pilot an economic inclusion programme which will deliver a package of interventions tailored to address climate vulnerabilities in selected households in conjunction with the CCT. These may include; (i) skills training for climate resilient agriculture and non-agriculture income generating opportunities; (ii) savings and credit support through linkages to financial services and digital banking mechanisms; and (iii) life skills training on climate issues such as disaster preparedness and health and livestock management in post-flood situations. This innovative climate-resilient economic inclusion pilot will generate rigorous evidence on impact and cost-effectiveness to inform social protection and climate adaptation policies in Lao PDR, and contribute to the global evidence base on how these programs need to be tailored to build longer-term resilience to climate change. The project directly contributes to the activities under **output 4** and **output 5** of the CRPP.

Lao PDR: The proposed **Flood and Drought Mitigation and Management Project** aims to increase the resilience of rural livelihoods and reduce the economic losses from floods and droughts as a result of increasing extreme weather events caused by climate change in Lao PDR. Flood and drought events already occur and have been increasing in frequency over the last three decades, affecting the large proportion of the population that relies on agricultural activities as the primary source of livelihood, and live on or below the national poverty line. The project will address specific adaptation needs in the country and fill the current gaps in adaptation financing by focusing on managing the risk and impacts of more frequent and intensified floods and droughts on the lives of poor rural households in the three central provinces of Bolikhamxai, Khammouan and Vientiane Capital. The project is expected to harness flood water for usage during dry season and droughts and reduce damages from floods using strategies such as the planting of permanent tree crops and/or natural regeneration of upland areas, balanced with lowland community outreach initiatives, improve water reliability during droughts by adapting and scaling up water retention, communal water harvesting and storage, provide modernized agrometeorological information systems and improve nutrition awareness and facilities. The project contributes to Lao PDR's climate change priorities by aligning to the first target of the country's Nationally Determined Contributions (NDCs) through investing in improved vegetative cover of watersheds as a means of flood control and by promoting crop and nutrition diversification. It will also support post COVID-19 economic recovery and rural resilience by (i) generating rural employment through construction; (ii) increasing incomes through the reduction of flood damages and losses and diversification into high value crops; and (iii) investing in better diets and sanitation. Floods, drought, and epidemics stretch women's resources, further exposing them to vulnerabilities. By targeting the activities at the rural population where women in Lao traditionally take on a high proportion of the household and farm responsibilities, the project will be able to directly support women's gender specific needs.²²¹ The project directly contributes to the activities under **output 5** and **output 6** of the CRPP.

Pakistan: The proposed **Sindh Coastal Resilience Project** will enhance resilience of coastal area livelihoods and infrastructure in the Sindh province of Pakistan through integrated water resources, drainage, and flood risk management schemes; and restored natural protection buffers. It will reforest mangrove species along with other fresh and saline tolerant tree species in strategic areas. In doing so, the project will address the specific adaptation needs of the rural agrarian populations that live in the coastal zone of the Indus Delta that are exposed to present and future climate extremes and slow onset climate related hazards and fill an identifiable funding gap in this regard. The project will support women to play a role in building coastal resilience as decisionmakers, stakeholders, and educators. Specifically, the project will contribute to reducing coastal erosion rate, provide protection from storm surge, and reduce saline intrusion. The pertinence of the project for climate adaptation in Pakistan is also reflected in the fact that both the updated NDC (2021), and Sindh Climate Change Policy (2022) identify the Indus River delta and coastal region as priority areas in need of (i) improved climate risk assessment, planning, and emergency response; (ii) enhanced water resources and coastal zone management; (iii) climate-resilient protective infrastructure; and (iv) restoration of ecosystems. The project directly contributes to the activities under **output 6** of the CRPP.

Papua New Guinea: The proposed **Improved Technical and Vocational Education and Training for Climate Resilient Employment Project** aims to strengthen climate resilience through technical and

vocational education. In PNG, over 85% of the population are classified as rural and depend on small-scale and subsistence farming. Further, many of these rural people live in remote and inaccessible locations. The ongoing efforts of this population to transition out of marginalization and poverty are likely to be undermined and reversed by climate change. In response, the project focuses on the vulnerable, small-scale farming communities in eight target provinces. Each province is exposed to a specific set of climate hazards. The government has prioritized strengthening the technical and vocational education (TVE) system to reduce the poverty of the rural poor and achieve sustainable development. Building on this, the project focuses specifically on training and vocational education for vulnerable people in the agriculture and construction sectors. It includes specific measures to support women and informal workers. Through the TVE system, the project will develop climate resilient agriculture, climate resilient construction processes and will help some vulnerable people to transition out of agriculture into climate resilient livelihoods. The project will mainstream climate change adaptation into the governance, the incentive systems, the management, the curricula, the training modules and the training programs of the TVE system. In addition, it will pilot practices and techniques in college-community partnerships – a learning process. The project directly contributes to the activities under **output 5** of the CRPP.

Timor-Leste: The proposed **Agriculture Value Chains and Water Harvesting Project** aims to enhance the sustainability of rural livelihoods in Timor-Leste by increasing the resilience of poor subsistence farming families to the negative impacts of climate change. Specifically, the project focuses on protecting and improving the productivity of small-scale agriculture by using climate-smart approaches, including investments in water and forest management and conservation. To ensure the sustainability of project investments, the project also aims to enable smallholder producers, especially women, to expand their participation in, contribution to, and benefit from agricultural value chains. The project will support activities in the municipalities of Dili (non-urban), Manatuto, Manufahi, and Oecusse which together represent the country's six agro-ecological zones and associated farming systems and are areas that have been identified as being at high risk from the impacts of climate change and include communities with high levels of poverty. The project is in line with the priorities of the NDC, the National Adaptation Programme of Action to Climate Change, and GCF country program. The project directly contributes to the activities under **output 5 and 6** of the CRPP.

Vanuatu: The proposed **Climate Change and COVID-19: Building Comprehensive Resilience in Poor and Vulnerable Groups in Vanuatu Project** will respond to climate challenges by creating, modifying and enhancing climate responsive social protection systems and enhancing climate resilient livelihood support mechanisms. Existing social protection initiatives provide targeted support payments to the vulnerable. These are temporary, typically raised in response to disasters, and dependent on international partners. Moreover, they do not consider climate change impacts, they are small scale and lack the institutional capacity required to upscale. Given the projected increase in extreme weather events due to climate change, Vanuatu requires social protection systems that are able to effectively protect the vulnerable against climate shocks. Thus, the project will strengthen the capacity of social protection systems to function in advance of and through climate extremes, to respond to the specific challenges of both slow and rapid onset climate disasters, and to meet the needs of the most vulnerable. Building on existing, related initiatives in Vanuatu, the project will also help vulnerable populations to adapt to climate change and become climate resilient by establishing schemes that develop resilient livelihood skills and practices. The project includes a strong focus on women. The project notably contributes to the activities under **output 4** of the CRPP.

Project Eligibility criteria

Projects that seek GCF co-financing under the CRPP IF will be subject to a strict assessment and selection process, that will be based around a framework of criteria that is closely aligned with GCF investment criteria, while also reflecting the core values and priorities of the CRPP. These criteria are also aligned with the selection criteria for CRPP TF. In order to be considered for GCF co-financing, project proposals must also demonstrate close alignment to the IRMF of the GCF and its indicators. The project eligibility criteria will be applied in addition to due diligence conducted by all projects (such as safeguards) and conduct of economic analysis as per ADB's ongoing practice. The eligibility criteria would include the following. See **Annex 2.4** for more details.

- i. **Impact potential:** Following criteria will be used to determine the project's impact potential
 - a. The project must clearly set out the specific climate risks (hazard, exposure and vulnerabilities), under current and future conditions of climate change and socioeconomic development, that the project aims to address, articulating a clear causal narrative between specific activities and their objective of reducing climate risk. Projects that include arguments that are well evidenced and quantified using reliable data sources and information will be prioritized. *[Required]*
 - b. Prioritization will be given to those projects that have targeting intervention (TI) for poor population – (TI-household); (TI-Geographic); or (TI- SDG)²²² *[Strongly Desired]*
 - c. The project must clearly describe the adaptation benefit that is expected to be generated by the implementation of the project activities. Only projects that identify measurable adaptation benefits will be prioritized.
 - d. Projects must align to GCF adaptation result areas and prioritization will be given to those projects that have higher expected final targets for the related indicators.
 - e. Prioritization will be given for projects with a higher expected total number of beneficiaries relative to total population (disaggregated by gender)
 - f. Prioritization will be given for projects with lower cost per beneficiary
- ii. **Paradigm shift potential:** The potential of the sub-project to catalyze impact beyond a one-off project investment will be determined through following criteria:
 - a. The project must clearly describe how the project outcome will be **sustainable** in the long run i.e., through integration into national/local development processes, through securing additional public and/or private financing; and capacity of national/local institutions to implement similar activities without GCF funding. [e.g., climate risk informed national targeting systems for social protection programs] *[Required]*
 - b. Prioritization will be given to projects which demonstrate the potential of being **replicated** elsewhere in the same country or other countries in region. [e.g., successful agroforestry initiatives replicated in different ecological zones of a country] *[Strongly Desired]*
 - c. The project promotes **innovation**, such as, through the development and/or adoption of new technology or approaches which can help address adaptation barriers in the region. [e.g., use of ecosystem-based solutions to improve water management in urban informal settlements] *[Desired]*
 - d. The project promotes sharing of knowledge and learning within and with other CRPP countries *[Required]*
- iii. **Sustainable development potential:** All projects must demonstrate sustainable development potential by identifying at least one **positive co-benefit** from the following and have associated indicators. Prioritization will be given to projects that demonstrate more than one co-benefit. *[Required]*
 - a. Economic co-benefits –explicitly aim at poverty reduction
 - b. Social co-benefits – human health and social development through improved food and nutrition security and reduced loss of life resulting from the impact of extreme weather events
 - c. Environmental co-benefits –improved ecosystem services and biodiversity
- iv. **Gender:** Gender is a standalone criterion that projects must meet independently of the environmental, social and economic co-benefits. The project must either be categorized as **Gender Equity Theme (GEN)** or **Effective Gender Mainstreaming (EGM)** as per Guidelines for Gender Mainstreaming Categories of ADB Projects⁶ *[Required]* A project is assigned “gender equity theme” (GEN) if the project outcome directly addresses gender equality and/or women's empowerment by narrowing gender disparities, including under the five pillars: (i) women's economic empowerment increased, (ii) gender equality in human development enhanced, (iii) gender equality in decision-making and leadership enhanced, (iv) women's time poverty and drudgery reduced, and (v) women's

⁶ The projects of the Asian Development Bank have four gender mainstreaming categories (i) category I: gender equity as a theme (GEN); (ii) category II: effective gender mainstreaming (EGM); (iii) category III: some gender elements (SGE); and (iv) category IV: no gender elements (NGE).

resilience to external shocks strengthened. A project is assigned “effective gender mainstreaming” (EGM) if the project outcome does not explicitly address gender equality or women’s empowerment, but project outputs contribute to addressing gender equality and/or women’s empowerment by narrowing gender disparities, including under the five pillars described above.

- v. **Needs of the Recipient:** Following criteria will be used to determine how far the project meets the need of the recipient.
 - a. The project aims to overcome **barriers** related to technical, institutional and/or financial barriers to adaptation. Prioritization will be given to projects that aim at overcoming more number of barriers. *[Required]*
 - b. The project must strengthen capacity of **local institutions** to implement adaptation projects, enhance coordination and improve governance. *[Required]*
- vi. **Country Ownership:** Following criteria will be used to determine alignment with national and/or local priorities
 - a. All projects must demonstrate alignment with **national climate priorities** as reflected in nationally determined contributions (NDCs), national adaptation plans (NAPs) and/or national /local disaster risk reduction plans. *[Required]*
 - b. All projects must demonstrate **stakeholder engagement** in developing the project, including consultation with local stakeholders – local government and civil society organization. *[Required]*
 - c. All projects must demonstrate engagement with **national designated authorities** (NDAs) in developing the sub-project. *[Required]*
 - d. *All projects will be implemented in countries that have provided a **no-objection letter** to participate in the program [Required]*
- vii. **Efficiency and Effectiveness:** Following criteria will be used to determine eligibility
 - a. Resources from CRPP IF are being used to offset the incremental cost of climate resilient investments, provide targeted climate resilient technical assistance, or for non-revenue generating outputs and non-market elements. *[Required]*
 - b. Resources from CRPP IF are blended with ADB grants and/or loans (to act as ADB co-financing) in order to augment the volume of financing available for adaptation and better tailor concessionality to achieve transformative investments which would otherwise not proceed. The ratio GCF financing: ADB Grant/Loan should at least be 1: 4 *[Required]*
 - c. Projects should ensure that the concessional resources from GCF do not displace or substitute investments that might have taken place anyway using regular ADB borrowing. *[Required]*

In recognition of the importance of the contribution of each and every project to climate adaptation impact for the achievement of the program’s anticipated outcome, the CRPP has developed a dedicated tool for use in the project screening process that will ensure a transparent and consistent approach to the selection of eligible projects related to this criterion. The climate impact project screening tool is included in **annex 2.4.**

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

1. Executing Entity of CRPP

As Executing Entity for the CRPP, ADB will be responsible for the overall Program, particularly the provision of technical assistance support under CRPP TF (Component 1), and the provision of grants and loans for projects under CRPP IF (Component 2).

Subject to GCF Board approval for this funding proposal, ADB and GCF will enter into a program-specific legal agreement (the “Funded Activity Agreement” or “FAA”) in accordance with the AMA, which will set out the specific details concerning GCF’s provision of and ADB’s administration of the GCF funds.

GCF funds for the CRPP will be ring-fenced within an ADB GCF Trust Fund which was created pursuant to the AMA, and as such benefit from the same privileges and immunities as ADB’s resources. Usage of GCF funds will be based on criteria outlined in this proposal which will be included in dedicated **IF Implementation Guidelines** to be developed during inception. These guidelines will complement the program Term Sheet and FAA, and include further guidance on implementation arrangements, project selection and prioritization, monitoring and evaluation, and reporting.

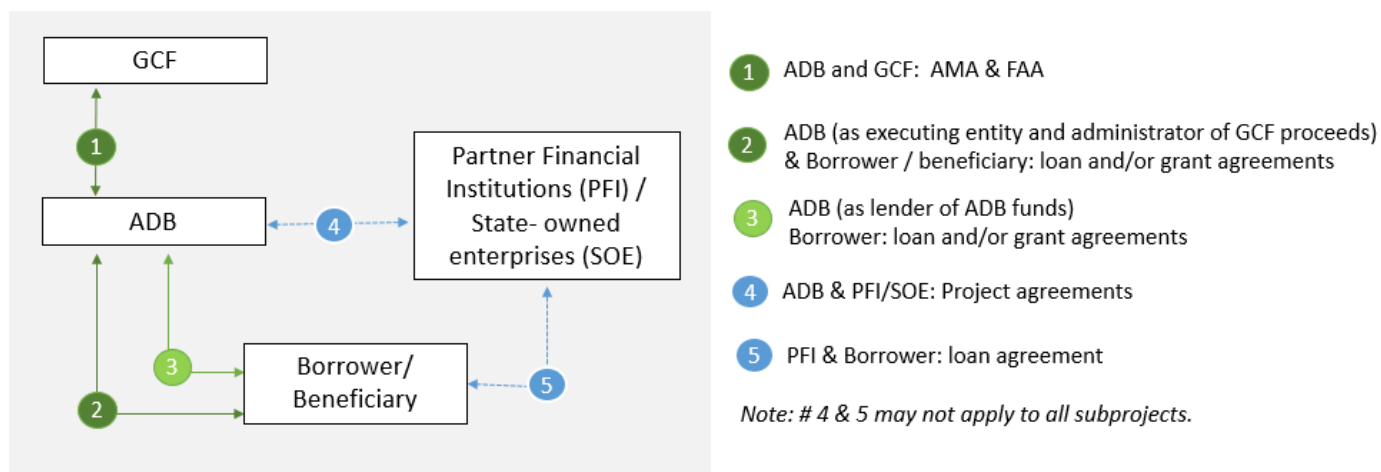
For Projects implemented by an executing agency and/or implementing agency (each as defined by ADB), where such agency will be using or channeling GCF Proceeds for the purposes of the Funded Activity and/or such agency will execute, carry out or implement Funded Activities, the executing agency and/or implementing agency will be the Executing Entity. In such cases, in addition to the implementation arrangements described in this Funding Proposal, ADB will enter into a subsidiary agreement with the relevant Executing Entity.

2. CRPP IF Legal Arrangements

Under CRPP IF, ADB will offer concessional loan and grants to sovereign borrowers to be funded by GCF. For each CRPP IF project, ADB will sign separate loan and/or grant agreements with the sovereign borrowers/beneficiaries for each source of financing (i.e. ADB’s own resources and GCF funds), and through this, each project will be implemented in compliance with ADB policies and CRPP requirements. Depending on the structure of the project, ADB will also enter into project-level agreements with relevant borrower’s agencies for the implementation of projects. Such agencies may include financial intermediaries, such as National Development Banks and potentially, GCF Direct Access Entities, where feasible. If additional co-financing from other partner financial institutions are also included, these will be provided in parallel to the ADB and GCF funded loans and grants. Loans funded by GCF funds are expected to be *pari passu* with the ADB loans, except for the interest rate and tenor applicable to the GCF funds.

For each project, ADB enters into separate legal agreements with borrowers/recipients for each financing source. A loan/grant from ADB’s own resources will be covered by one legal agreement, while the loan/grant from GCF funds will be covered by another legal agreement, with cross-references between the two as appropriate. (Figure 3)

Figure 3: CRPP Investment Fund Overview of Legal Arrangements



GCF loans and grants will be used to support projects implemented by governments, or government-owned entities based on the terms and conditions agreed with ADB in each respective project's loan and grant agreements and project administration manual (PAM). Each potential project will be identified during ADB's country programming process, prepared in accordance with ADB's internal policies and procedures, and evaluated against the criteria presented in this proposal and agreed with GCF, to determine eligibility for funding under the program. CRPP projects will be fully administered by ADB and likewise benefit from ADB's project preparation, due diligence processes, project administration, and monitoring and evaluation systems. Each project will be presented for approval to the ADB Board on a transaction-by-transaction basis, which approval shall include the provision of GCF funds to be administered by ADB.

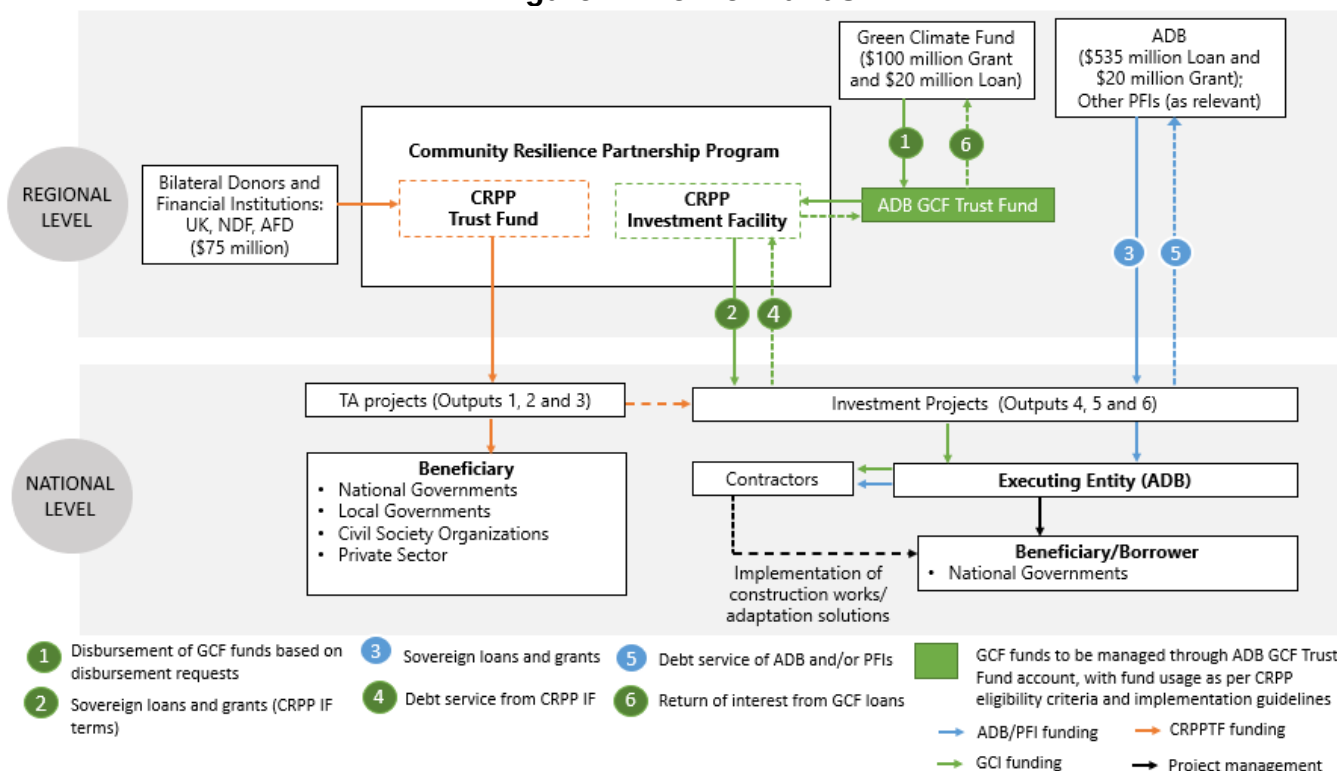
3. CRPP IF Flow of Funds

Figure 4 shows the flow of funds related to CRPP TF and IF, in line with the program description and implementation arrangement described in **section B3 and B4** of this proposal.

The CRPP IF will be implemented from 2023 to 2030. Projects seeking financing from IF will need to be likely approved by ADB Board between 2024 and 2027 and complete implementation by 2030. Projects which will be supported by the program will be identified in accordance with ADB's country programming process and selected using the eligibility criteria presented in this proposal. Disbursement requests from ADB to GCF are expected to be made on an annual basis in accordance with the expected ADB approval dates, and projected disbursement plans and timelines of the projects in the CRPP pipeline. Funds will be disbursed by GCF into existing ADB trust fund that has been established for GCF financed projects, and subsequently deployed to approved projects in accordance with ADB policies and procedures.

Payments of interest and principal for GCF funded projects loans will be repaid back to ADB, into the ADB GCF Trust Fund. Principal, interest payments, and any unused funds or income will be subsequently returned to GCF.

Figure 4: Flow of Funds



4. CRPP TF Legal arrangement and Flow of Funds

The CRPPTF was established in 2021 as a multi-donor trust fund under CRFPF to support TA projects, and any other activities that may be agreed upon between financing partners and ADB. Contributions from bilateral donors will be held and administered by ADB separately from ADB's other resources. Contributions to financial partnerships will be effected through an Instrument of Contribution or through the signing of a trust fund contribution agreement.

5. Role of ADB

As a GCF AE, ADB and GCF have entered into several project-specific legal agreements which are under implementation, delivering climate impacts throughout ADB's developing member countries.

In its capacity as AE and Executing Entity, ADB will administer grants and loans from GCF under the CRPP IF as follows:

- Provide grants and loans as per the CRPP IF terms and requirements, alongside loans using ADB own funds and other ADB-administered parallel financing (as appropriate), applying the same duty of care and due diligence for all sources.
- Carry out such management and administration of the GCF funds for grants and loans in accordance with its policies, procedures and practices, and with at least the same degree of care as it uses in the administration of its own funds or other donor funds, consistent with the AMA.
- Establish a governance structure of the CRFPF will comprise of (i) CRFPF Partnership Group; (ii) CRFPF Steering Committee (iii) CRFPF Working Group; and (iv) CRFPF Secretariat.
- Oversee and manage the dedicated CRPP Project Structuring Team, staffed by full-time and intermittent experts, including an additional experts specifically engaged for the CRPP, with the following responsibilities:
 - originate projects and develop a pipeline for CRPP funding,
 - provide support to structure projects and identify financing partners;

- identify and screen potential projects, review project documentation and ensure the project meets the program's criteria;
- verify and monitor project and facility impacts and prepare annual performance reports to GCF;
- manage and monitor procurement and contracts.
- Work closely with other CRPP financing partners to identify additional parallel financing sources for projects if needed.

6. Governance Structure of CRPP

The CRPP will be operationalised through the **Community Resilience Financing Partnership Facility (CRFPF)**, implemented and managed by ADB, the Accredited Entity. The Facility will aim to mobilize financial and knowledge resources from various development partners to support scaling up of climate adaptation investments at local level. A key objective of the Facility is to help strengthen partnerships in the climate adaptation space, especially in the context of local actions for climate adaptation, among governments, donors, civil society, and the private sector, and influence regional and global directions on community led adaptation and thereby advance elements of locally led adaptation principles.

The governance structure of the CRFPF will comprise of (i) CRFPF Partnership Group; (ii) CRFPF Steering Committee (iii) CRFPF Working Group; and (iv) CRFPF Secretariat. (**Figure 5**).

CRFPF Partnership Group: The CRFPF Partnership Group will be the primary advisory body to ensure the expected outcomes of the CRPP are achieved. The CRFPF Partnership Group will have a Chair and a Co-Chair. A representative from ADB will serve as the Chair. A member from one of the CRPP financing partners will serve as the Co-Chair. Specific roles and responsibilities of the CRFPF Partnership Group will include: (i) selecting the co-chair (on an annual basis); (ii) providing strategic direction to the CRPP in line with the results framework; (iii) review and approve CRPP annual report; (iv) establishing linkage between CRPP and other global and regional resilience programs; (v) providing guidance on mobilization of resources for CRPP TF; (vi) approving the TOR for CRPP-related strategic studies and evaluations to guide the results and assess the impact of the program; and (vi) providing inputs and attend the annual CRPP Partnership Forum.

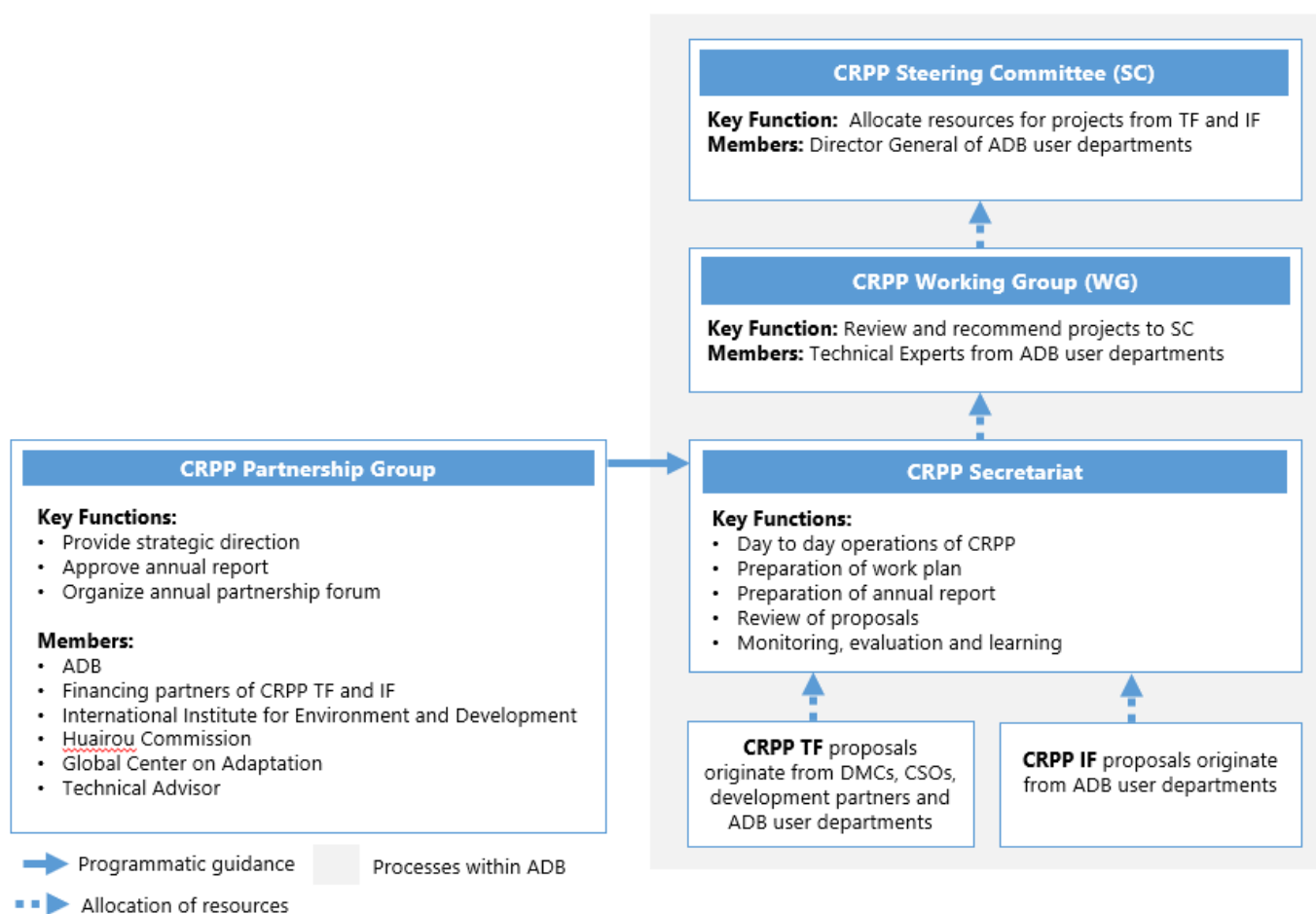
The Members of the Partnership Group will include (i) Asian Development Bank – represented by the CRFPF Facility Manager; (ii) Countries and financial institutions that provide financial contribution to the TF and/or IF; (iii) countries that receive CRPP support; (iv) Non-financing partners - International Institute for Environment and Development and Huairou Commission; and (vii) Independent technical advisors (two positions, on an annual basis). Such an inclusive governance structure will enable successful delivery of the CRPP programmatic approach and be able to influence wider actions in support of community-resilience, thereby enabling larger impact of the program beyond its immediate footprint.

CRFPF Steering Committee: The CRFPF Steering Committee will be responsible for (i) approving CRFPF annual work plan based on endorsement received from financing partners; (ii) approving allocation of resources for specific projects in alignment with the eligibility criteria defined in the CRFPF TF and IF implementation guidelines; and (iii) approving the CRFPF annual report based on endorsement from financing partners. The Steering Committee will be chaired by Director General, Climate Change and Sustainable Development Department (CCSD) of ADB and the members of the Steering Committee will include the Directors General of the user departments, i.e. ADB departments applying for resources from CRPP TF and IF. The Chair of the CRPP Steering Committee will be the designated authority for approving allocation of CRFPF resources for specific projects. After approval by the CRPP Steering Committee for fund allocation, the proposals will be processed in accordance with all relevant policies and procedures for technical assistance and grant approval, including without limitation, consulting services and procurement, social and environmental safeguards, financial management and reporting, and anticorruption and governance.

CRFPF Working Group: The CRFPF Working Group will be responsible for reviewing project proposals and ensure they meet the eligibility criteria outlined in Annex 2.4 and make recommendations to CRFPF Steering Committee for allocation of resources. The Working Group will be chaired by Director Climate Change CCSD, and comprise of ADB staff from relevant departments with technical expertise on social development, gender, climate and disaster resilience, urban development, and agriculture and food security, and governance. Where needed, the Working Group will engage suitable qualified climate risk consultants to review the climate rationale of the project proposals. In particular the Working Group will ensure gender considerations are suitably met by the projects.

CRFPF Secretariat: The secretariat will support the mission of the CRPP and manage its day-to-day operations. The secretariat will be housed at the ADB and operate under ADB's policies and procedures. The secretariat will include a (i) CRPP Fund Manager, (ii) Monitoring and Evaluation and Learning Expert; (iii) Knowledge Expert; and (iv) Gender Expert. The roles and responsibilities of the CRFPF Secretariat will include: (i) supporting the organization and deliberation of CRFPF Partnership Group, TF Committee and IF Committee meetings; (ii) preparing the annual work plan and annual report; (iii) supervising day-to-day implementation of annual work plan; (iv) preparing minutes of working group meetings on evaluation of IF sub—projects and sharing the minutes with GCF; (v) monitoring, evaluation and learning; (vi) fostering partnerships with relevant institutions and programs at global and regional level; and (vii) disseminate the CRPP results and achievements. The role of gender expert will be to ensure gender requirements are being met.

Figure 5: CRPP Governance Structure



7. Allocation of Resources

The administration of different funds under the CRFPF is governed by respective Implementation Guidelines that would be endorsed by the respective CRFPF financing partners and approved by the CRFPF Steering Committee on or before the first fund allocation. The CRPP IF Implementation Guidelines will be endorsed by GCF before approval by CRFPF Steering Committee. The Implementation Guidelines can be amended from time to time, reflecting revisions in CRFPF operations. The Steering Committee will approve any change or amendment after concurrence by the respective financing partner(s), and the Facility Manager will communicate the changes and amendments to the financing partner(s) and relevant ADB user departments.

Procedure for Allocation of Resources from CRPP TF: The procedure for allocation of resources from TF (not funded by GCF) will include the following steps:

- Project proposals for TF can be initiated by countries, development partners, and ADB. In Q1 of each year, the CRPP Secretariat in close consultation with the CRPP Partnership Group will prepare the annual work plan which would include the list of prioritized project proposal. The work plan for the TF will be approved by the financing partners of the CRPP TF.
- ADB user department will prepare project proposal using the application form (to be developed as part of Implementation Guidelines) and draft concept paper in standard ADB concept paper format and will submit them to the CRPP Secretariat, depending on the nature of application. Applications will be reviewed in batches.
- The applications will be reviewed by the CRPP Secretariat to ensure that they comply with the eligibility criteria described in the Implementation Guidelines and priorities in the annual work plan. If the application does not meet the criteria, the CRPP Fund Manager will discuss the issues with the user department for revision or withdrawal. If the application complies, the application will be included in the batch for circulation to the CRPP Working Group, as the case may be. The CRPP Fund Manager will make a recommendation to the CRPP Working Group on each proposal based on three criteria: (i) compliance with the implementation guidelines for the use of the funds, (ii) estimated amount of climate finance, and (iii) the overall responsiveness to the eligibility criteria. The CRPP Fund Manager will also advise the respective working groups on the availability of CRPP TF resources to support the applications.
- The working groups will have seven working days for review, comment, and endorsement. The user department will then have five working days to take the comments into consideration and revise (if necessary) the application. Once completed, the CRPP Fund Manager will forward the applications to the CRPP Steering Committee to approve the allocation of resources from the CRPP TF.
- After approval by the CRPP Steering Committee for fund allocation, the proposals will be processed in accordance with all relevant policies and procedures for TA and grant approval, including without limitation, consulting services and procurement, social and environmental safeguards, financial management and reporting, and anticorruption and governance.

Procedure for Allocation of Resources from CRPP IF: The GCF funds for the CRPP IF will be ring-fenced within ADB GCF Trust Fund which was created pursuant to the AMA. Usage of GCF funds for individual projects will be based on criteria outlined in this proposal which will be included in a dedicated CRPP IF Implementation Guidelines to be developed during inception. The procedure for allocation of resources from CRPP IF (which will blend ADB and GCF finance) will include the following steps:

- Project proposals for CRPP IF can be initiated by ADB in discussion with relevant government. In Q1 of each year, the CRPP Secretariat in close consultation with the CRPP Partnership Group will prepare the annual work plan which would include the list of prioritized project proposal.
- ADB user department will prepare project proposal using the application form (to be developed as part of Implementation Guidelines) and draft project document in standard ADB format and will submit them to the CRPP Secretariat, depending on the nature of application. Applications will be reviewed in batches.
- The applications will be reviewed by the CRPP Secretariat to ensure that they comply with the eligibility criteria (**Annex 2.4**). If the application does not meet the criteria, the CRPP Fund Manager will discuss the issues with the user department for revision or withdrawal. If the application complies, the application will be included in the batch for circulation to the CRPP Working Group, as the case

may be. The CRPP Fund Manager will make a recommendation to the CRPP Working Group on each proposal based on compliance with the eligibility criteria for the use of the funds from CRPP IF. The CRPP Fund Manager will also advise the respective working groups on the availability of CRPP IF resources to support the applications.

- The working groups will have seven working days for review, comment, and endorsement. Where needed, the working group may engage technical climate change experts (consultants) to review the climate rationale of the projects. The user department will then have five working days to take the comments into consideration and revise (if necessary) the application. Once completed, the CRPP Fund Manager will prepare the minutes of the decisions of the working group and forward the applications to the CRPP Steering Committee to approve the allocation of resources from the CRPP IF.
- After approval by the CRPP Steering Committee for fund allocation, the proposals will be processed in accordance with all relevant policies and procedures for grant and loan approval, including without limitation, consulting services and procurement, social and environmental safeguards, financial management and reporting, and anticorruption and governance.
- The CRPP Secretariat will provide semi-annual reporting to the GCF Secretariat so that the GCF Secretariat can continuously assess the compliance of the approved projects with the eligibility criteria, and where appropriate recommend remedial measures. Such reporting will be in addition to annual performance reports.

8. Track Record of ADB

ADB will administer the Community Resilience Financing Partnership Facility, which will include the CRPP TF and the CRPP IF. ADB has vast experience in developing and financing climate adaptation projects and in administering Trust Funds, as described in paragraphs below.

ADB commitments in scaling up in climate adaptation. ADB has an extensive experience in developing, financing and implementing climate adaptation projects in Asia and the Pacific region. ADB has approved its Strategy 2030, which recognizes “resilience” as a key objective for achieving sustainable and inclusive development in Asia and the Pacific. ADB’s Strategy 2030 identifies seven operational priorities, including a priority on “*Tackling climate change, building climate and disaster resilience, and enhancing environmental sustainability.*” The Strategy sets out that in order to achieve its climate objectives, ADB will scale up its climate financing targets to ensure that (i) ‘75% of the number of its committed operations (on a three-year rolling average, including sovereign and non-sovereign operations) will be supporting climate change mitigation and adaptation by 2030’; and (ii) ‘climate finance from ADB’s own resources will reach USD80 billion cumulatively from 2019 to 2030. Further, in 2021, ADB has announced its ambition of increasing total climate finance from USD80 billion to USD100 billion by 2030, out of which around USD34 billion will support climate adaptation. In 2021, ADB has also announced its commitment to align its new operations with the goals of the Paris Agreement starting July 2023 for sovereign operations. Meeting these climate targets will require expanding investments in climate adaptation and ensuring that the benefits of such investments reaches the poorest and vulnerable population. It will also require strengthening country capacity to undertake transformational planning and programming that encourages paradigm shift, closely aligned with NDCs, National Adaptation Plans and National Disaster Risk Reduction Plans, which is directly aligned with GCFs strategic priorities 2020-2023. The CRPP aims to help ADB achieve its ambitious climate objectives and targets and develop and finance a new generation of adaptation projects.

ADB’s priorities on climate adaptation. ADB has been one of the leading MDBs on climate change. It has been implementing a climate risk management framework since 2014. Application of the framework is mandatory in all ADB investments and runs from the early concept phase of a project through to implementation. The framework allows integrating climate risk considerations systematically in ADB financed projects. However, recognizing the importance of increasing climate risk and its huge impact on vulnerable countries and population, ADB aims to scale up investments in projects that have the primary purpose of adaptation and addressing underlying drivers of vulnerability. It is expected that such projects can help steer development in a resilient direction and catalyse transformational planning and programming, which is in line with GCF’s strategic priorities 2020-2023. It is important that such projects advance resilience in four

key aspects: physical resilience (such as community-level infrastructure on cyclone preparedness); social and institutional resilience (such as adaptive social protection to support communities deal with droughts), eco-based resilience (such as promoting nature-based solution to deal with floods), and financial resilience (such as disaster microfinance). The four key aspects allow advancing all four adaptation targets of GCF's strategic priorities 2020-2023, namely, livelihoods and people and communities; health, food and water security; infrastructure and the built environment; and ecosystems and ecosystem services. ADB has capacity in designing and implementing such holistic solutions, but there is a need to do more and scale up. The CRPP aims to help ADB promote such holistic approach towards resilience.

ADB's approach to delivering adaptation at scale through partnerships. Partnerships is an important aspect of ADB Strategy 2030. In the context of adaptation, this becomes even more critical because of the need for understanding local context (both from climate risk perspective as well as underlying social, economic, cultural and political drivers that contribute to vulnerabilities and thereby increasing climate risk). Thus, the CRPP adopts a partnership approach. The CRPP TF will be delivered by ADB in close partnerships with global policy and action research organizations (such as the International Institute for Environment and Development), and coalitions of grassroots organizations and bilateral agencies (such as the Huairou Commission). The CRPP IF is planned to be delivered by ADB in close partnership with GCF.

ADB experience in administering trust funds. ADB administers several trust funds to support many of its projects, programs, technical assistance, and other activities. ADB administers contributions of financing partners as a trustee or administrator. As of 2019, ADB managed a total of 45 trust funds (18 are active single-partner trust funds, 12 are multi-partner trust funds, and 14 trust funds are under financing partnership facilities) which contributed around USD181 million in ADB's sovereign operations. ADB's financing partnership facilities, providing platforms for long-term and multiparter cooperation that links various forms of assistance in a coordinated matter, has five facilities focused on clean energy, health, regional cooperation and integration, urban development, and water. In 2019, ADB's trust funds received replenishments totalling USD297.1 million and contributions for new trust funds amounting to USD28.7 million.²²³

B.5. Justification for GCF funding request

Why GCF financing?

The CRPP IF will use financing from GCF in order to support:

The poorest countries with high exposure to climate risk in implementing transformational adaptation measures. The CRPP IF aims to support adaptation projects in (i) low-income developing countries that confront severe structural impediments to sustainable development and climate change adaptation, i.e. that are highly vulnerable to climate shocks and stresses and have low levels of adaptive capacity (technical, financial and human) to undertake adaptation measures at scale; (ii) small islands developing states that experience high climate related shocks today and face very high future climate change risks, and have a narrow resource bases, thereby with high needs for adaptation; and (iii) in targeted poor communities (rural and urban) in middle-income countries. While these countries have identified adaptation as a climate priority, actions at scale remain limited due to lack of financing to support measures that aim for transformational change. Targeting the poorest and most vulnerable countries is in line with the use of GCF finance to meet the additional costs of climate action that would otherwise not be available for vulnerable groups. The focus on the poorest countries and communities becomes even more important in the post COVID-19 context, which has hit the poor the hardest.

The poorest and most vulnerable communities in enhancing their well-being despite increasing climate impacts. The CRPP has an explicit focus on adaptation solutions at the local level. Thus, CRPP IF will only support local adaptation measures that are targeted at the poorest and most vulnerable communities. While government budgets and supporting international development, finance is helping these groups, there are insufficient funds to address the rising costs of climate impacts, without which, the numbers in poverty are projected to increase. Domestic budgets are not available to meet this shortfall. Part of this shortfall will be addressed by concessional ADB funding with a combination of loan and grant depending on

country status, but the rate of indebtedness of many governments means that not all interventions can be supported through loans, even at concessional rates. However, without access to concessional financing, these communities will not be able to undertake adaptation measures and thereby not escaping climate-induced poverty. The situation has become even worst in the current pandemic. Targeting the poorest and most vulnerable communities is in line with the use of GCF finance to meet the additional costs of climate action that would otherwise not be available for vulnerable groups.

Projects promoting adaptation measures as public goods. Investments in adaptation are often stymied by their public good nature. Such investments may be underfunded because investors do not take account of their positive externalities, such as the benefits to downstream communities in flood risk management. Moreover, as discussed earlier, a key barrier for scaling up adaptation is the fact that adaptation finance is not reaching the local level, where it is needed the most. This requires improving national and local financial systems through climate risk information, improved poverty and vulnerability databases for targeting, digital technology to enable reliable transfer of resources, and local capacity to administer resources. Specifically grant financing from GCF will be critical to help implement adaptation measures that act as public good, especially in the post-COVID-19 context where the financial capacity of countries are already stretched, and thereby enabling the flow of future adaptation finance to the front line.

Projects implementing transformational adaptation measures. The CRPP has a focus on promoting transformational adaptation that address the nexus of climate change, gender and poverty i.e., measures that aims at fundamentally changing systems and that challenges and seeks to dismantle the social, political processes and structures that construct vulnerability, and allow it to accumulate and persist. Transformational adaptation is particularly important when solutions are targeted at the poorest communities, since the barriers faced by these communities are not just restricted to technical and financial, but equally include social and political set-up, perceptions and biases. Thus, transformational measures will allow adopting inclusive approaches, so that poor and vulnerable communities, especially women, have an equal seat at the table and are in a position to interact with different stakeholders, leading to a process where different interests are negotiated. Without financing, especially grant financing from GCF, it will not be possible to promote such transformational measures. The proposed concessional finance from GCF through CRPP will address the barriers to the uptake of transformational adaptation, and thus justifies GCF funding. In turn, this will allow greater leveraging of ADB investment, thus increasing the total level of finance flowing to the most vulnerable groups. While incremental adaptation is equally critical, it will be financed as part of ADB's regular financing.

Project outputs with non-revenue elements. Under-investment in adaptation is reinforced by a typical preference for the use of limited public resources for projects that address short-term development priorities, yield assured near-term benefits, and generate positive streams of direct or indirect income, particularly in poorest countries. With loan financing, governments also prefer projects that support direct capital recovery. However, the benefits of adaptation projects are realized in the form of reduces losses (which may not be reaped for many years), rather than streams of positive income. Thus, grant resources from GCF will be critical to help to reduce the cost of investment sufficiently to offset the incremental costs of climate-resilient investment, especially where this is funding non-market elements (e.g., nature-based solutions, and thus green over grey) or targeting extremely vulnerable groups where there is low potential for revenue generation. Moreover, such measures often generate co-benefits, such as alleviating poverty, enhancing development and other welfare benefits, which need to be better identified and captured to encourage greater investments.

Overcoming risk perceptions of community level adaptation interventions. Implementing adaptation measures that are local yet can be implemented at scale remains a challenge, partly due to the fact that local measures have high transaction cost, are subject to local governance-related challenges and favour for supporting national level projects that align to existing political economy. However, the benefits of implementing adaptation measures at local level through local institutions and communities are many and critical if long-term sustainability is to be ensured. The CRPP IF can demonstrate the benefits of decentralized and inclusive solutions that are designed based on principles of subsidiarity. Thus,

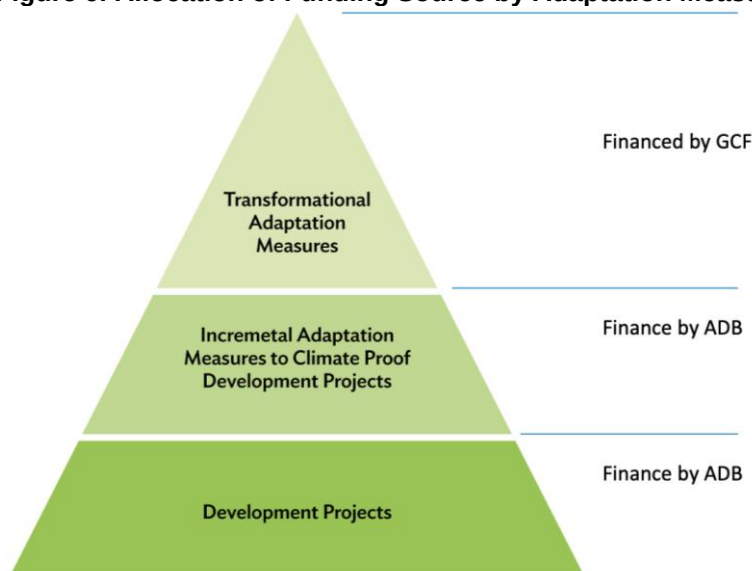
concessional financing will improve the bankability of community resilience projects with national governments, allowing more innovative measures to be taken.

Thus, without GCF support, current opportunities for transformational adaptation for the poor and vulnerable will not be realized. The GCF funds provided will serve the dual purpose of making the community resilience investment more viable, while providing immediate adaptation benefits to local populations. By offering critical resources from GCF in combination with the ADB's finance, the CRPP IF will enable countries to implement local adaptation measures, that they would otherwise not be able to do. GCF contribution of USD120 million will leverage a further USD555 million of ADB finance and USD75 million of financing from other bilateral donors (through the TF), representing more than fivefold leveraging ratio for every USD of GCF finance invested. The GCF's ability to offer grant finance will drive ADB's co-financing into the most transformational and paradigm-shifting investments as shown in **Figure 6**.

Level of concessionality: The CRPP aims to promote transformational adaptation measures that benefits the poorest population. All projects seeking grant financing from CRPP IF will be designed to comply with the use of GCF grant finance and with the principles set out in decision B.09/04. The grant elements will be tailored to

- Cover the incremental cost or the risk premium required to make investments viable and/or overcome risk perception, i.e., to meet the additional costs of ecosystem-based measures, cost of building new livelihoods-related skills, or to create incentives for behavioural change.
- Showcase demonstration effect, i.e., where there is clear demonstration effect in relation to new technology, approach or market, which if needed, is then further supported by TA through CRPP TF to ensure scale-up.
- Cover technical assistance for climate resilient activities and integration.
- Cover the additional costs of climate action that would otherwise not be available, particularly for transformational adaptation activities for poor and vulnerable groups.

Figure 6: Allocation of Funding Source by Adaptation Measures



In all cases, the projects will be designed to address barriers and create the enabling environment for adaptation, to leverage on existing public and private finance, and to ensure there is no distortion or displacement of existing public or private sector funds. The following paragraphs set out the concessional argument for each of three outputs proposed for financing under CRPP IF:

Output 4: The GCF grant finance for climate information for delivering climate-risk informed decisions is targeted to address information failures that prevent climate resilient development and will be largely focused on technical assistance provision (which is eligible for grant finance under B.05/07) to cover the additional activities to strengthen climate services at local level, to integrate climate risk information into social protection programs and decentralizations policies. It is also highlighted that these activities will cover activities that explicitly target the most vulnerable groups.

Output 5: The GCF grant finance for climate resilient pro-poor livelihood investment will focus on addressing risk perceptions for new climate-resilient livelihoods or markets, and to cover the incremental cost or the risk premium required to make these new livelihoods or investments viable. Again, the activities will also be targeted at the most vulnerable groups

Output 6: The GCF grant and loan finance for climate resilient pro-poor infrastructure investment is targeted at the additional costs of green, resilience infrastructure, which are needed to deal with climate risks and that would otherwise not be available, for poor and vulnerable groups. The primary focus is to invest in ecosystem-based adaptation, as this provides a range of public good characteristics, and has benefits in reducing non-market costs (providing environmental and social benefits).

B.6. Exit strategy

The CRPP is designed to promote sustainability through the adoption of the following approaches:

The CRPP will support outputs that are closely aligned with national climate priorities and to address climate impacts that hinders the achievement of poverty reduction. With a key focus on mainstreaming, the CRPP TF will catalyze impacts beyond a one-off project investment and deliver a paradigm shift by ensuring that climate risk informed pro-poor development becomes the new normal. It will do so by strengthening **policies and regulatory frameworks** that promote climate-sensitive development, and through capacity building on use of climate risk information for decision-making, with policy makers, planners, and local government and communities. This will ensure the sustainability of outcomes beyond completion of the project. The GCF financing will be used through the CRPP IF to support rolling out of such climate-sensitive development, especially in the context of localization and decentralization. By **strengthening national and local institutions**, the CRPP IF will ensure that its interventions will contribute to building policy coherence around climate resilience, and is taken up nationwide by relevant agencies. Countries' ownership will be achieved through **stakeholder engagement and capacity building** at various decision-making levels – government, civil society organizations, communities, financial intermediaries, and the private sector. With such a wide-ranging target, the CRPP will contribute to bringing about a paradigm shift.

Scaling up adaptation in the context of pro-poor development. The CRPP is based on the understanding that poverty reduction strategies/programs financed by national and local budgets will not be achieved due to the increasing impacts of climate change and thus need to be designed with a strong climate rationale. Accordingly, the CRPP TF will work with countries to integrate the CRPP activities fully with such opportunities, build capacity, and support in preparing such programs. Projects seeking finances from CRPP IF will have to demonstrate that proposed adaptation outputs are embedded within large programs that are being partly financed by government, and therefore will be continued beyond CRPP life span. For example, it will support integration of climate risk information in national social protection policies and programs that are embedded in national legislation and financed by government budgets.

Promoting adaptation through national curriculum systems. The CRPP recognizes that transformational solutions will require building life skills among the poor and vulnerable communities on addressing climate risk and exploring opportunities. The CRPP IF prioritizes development of new life skills as part of technical and vocational training and curriculum that targets climate resilience in key sectors. The integration of these elements into national curriculum being largely financed by government budget, increases the level of sustainability by ensuring that these life skills are taught and will remain beyond CRPP life span. For example, it will support integration of climate smart agriculture value chain practices in the curriculum of government agriculture training colleges.

Strengthening national and local institutions. A key focus of CRPP is to strengthen the capacity of national and local institutions in order for them to develop and implement adaptation solutions and ensure adaptation finance reaches the poor. While the CRPP TF aims to do so by building capacity of NDAs, all projects seeking financing from CRPP IF will have a key focus on strengthening capacity of national and local institutions. This will enhance climate resilience integration between institutions and to improve local governance so that benefits of adaptation reaches all in a just and equitable manner and continues beyond CRPP life span. For example, it will support building capacity of local governments to use climate risk information in planning and budgeting processes in order to inform capital and recurrent expenditure.

Engagement of communities, especially women. Meaningful participation of communities, especially women, is key for implementing sustainable transformational solutions at the local level. This is especially true for CRPP which has a large focus on demonstrating social, economic and environmental co-benefits of adaptation solutions, which is directly related to livelihoods and well-being of the poor communities. The CRPP TF has a dedicated output on strengthening engagement of communities and building their capacity, especially capacity of grassroots organizations through local policy and planning reforms. The CRPP IF will support projects that demonstrate engagement of national and local stakeholders, including, government and civil society organizations in developing the project and has an explicit role for communities in implementation. Such engagement will increase the buy-in for the proposed adaptation solutions and thereby contribute to its sustainability. For example, co-design of neighborhood scale ecosystem-based water management measures with the women from urban informal settlement, will ensure the social and cultural suitability of the measures and that local communities are involved in its maintenance beyond the project period.

Creating opportunities for scaling up. A key focus of the CRPP TF is to support governments and communities in accessing adaptation finance, including from the private sector. The financing support from CRPP IF will have an impact beyond financing the targeted projects. By supporting local adaptation initiatives in the region, it brings an explicit focus to the need for more such programming nationally, which could better leverage and use government funds. The leveraging of GCF resources alongside ADB will create a critical mass of fund volumes to achieve impact which will then open the way for mainstreamed sector and climate support. For example, projects on ecosystem-based adaptation will explore the potential of strengthening frameworks for private sector involvement to support long term sustainability and explore innovative options for revenue generation such as leveraging ecotourism initiatives.

FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	120			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	USD 20,000,000	Enter years	Enter years	Enter %		
(ii) Subordinated loans	Enter amount	Enter years	Enter years	Enter %		
(iii) Equity	Enter amount	Enter years		Enter % equity return		
(iv) Guarantees	Enter amount					
(v) Reimbursable grants	Enter amount					
(vi) Grants	USD 100,000,000					
(vii) Results-based payments	Enter amount					
(b) Co-financing information	Total amount			Currency		
	630			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Asian Development Bank	Senior Loans	535	million USD (\$)	Varies by country	Varies by project	senior
Asian Development Bank	Grant	20	million USD (\$)	Enter years Enter years	Enter%	Options
Bilateral Donors	Grant	75	million USD (\$)	Enter years Enter years	Enter%	Options
Click here to enter text.	Options	Enter amount	million USD (\$)	Enter years Enter years	Enter%	Options
Total financing = (a)+(b)	Amount			Currency		
	750			million USD (\$)		
(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	Bilateral donors include funding contributions from UK Aid, Nordic Development Fund, Agence française de développement (AFD). In addition, ADB is aiming to mobilize further resources for the Trust Fund from other bilateral donors and foundations. ADB will leverage further resources by strategically partnering with other trust funds of ADB, such as the Urban Resilience Trust Fund (in the context of urban poor), Asia Pacific Climate Finance Fund (in the context of developing pro-poor financial risk management solutions), and proposed ADB initiative/s on healthy oceans (in the context of coastal communities). ADB will further seek to mobilize co-financing or parallel financing from other donors for the projects financed by the CRPP IF.					
Note on ADB Senior Loans Tenor & Grace: For Group A countries Concessional-Assistance only (Cambodia, Lao PDR, Vanuatu): 32 years tenor, 8 years grace period. For Group B OCR Blend Country (Pakistan, Papua New Guinea, and Timor Leste): Concessional loan carries 25 years tenor, including 5 years grace period and equal amortization while for Regular OCR loan tenor, and grace period varies. For Group C Regular OCR (Ordinary Capital Resources) Countries (Indonesia): (i) Project loans consist tenor and grace period that varies by projects, subject to maximum average loan maturity limit of 19 years. Average loan maturity is calculated based on loan term, grace period and amortization type; (ii) Policy-based loan, consist 15						

years tenor including 3 years of grace period, subject to maximum average loan maturity limit of 19 years. See <https://www.adb.org/what-we-do/public-sector-financing/lending-policies-rates>.

Note on ADB Senior Loans Pricing: For Group A countries Concessional-Assistance only (Cambodia, Lao PDR, Vanuatu): 1% during grace period, 1.5% during amortization period. For Group B countries OCR Blend Country (Pakistan, Papua New Guinea, and Timor Leste): Concessional OCR Loan 2% interest fixed for the life of the loan; Regular OCR; under the Flexible Loan Product facility, interest rate is calculated based on outstanding balance comprised of 6-month SOFR compounded in arrears + 50 bps, maturity premium of 0-20 bps (if applicable), (Rebate)/Surcharge and commitment fee of 15 bps calculated based on undisbursed balance. For Group C Regular OCR Countries (Indonesia): under the Flexible Loan Product facility, interest rate is calculated based on outstanding balance comprised of 6-month SOFR compounded in arrears + 50 bps, maturity premium of 0-50 bps (if applicable), (Rebate)/Surcharge and commitment fee of 15 bps calculated based on undisbursed balance. See <https://www.adb.org/sites/default/files/flp-overview.pdf>.

Country groupings are updated from time to time. Information can be accessed here: <https://www.adb.org/what-we-do/public-sector-financing/lending-policies-rates>

C.2. Financing by component

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
Component 1: CRPP Trust Fund	1. Knowledge and action research on climate risk informed pro-poor community-level solutions strengthened.	14,000,000	NA	NA	14,000,000	Grants	UK Aid, Nordic Development Fund, Agence française de développement (AFD)
	2. Institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened.	12,000,000	NA	NA	12,000,000	Grants	UK Aid, Nordic Development Fund, Agence française de développement (AFD)
	3. Inclusive and pro-poor adaptation investment projects identified and prepared.	45,000,000	NA	NA	45,000,000	Grants	UK Aid, Nordic Development Fund, Agence française de développement (AFD)
Component 2: CRPP Investment Fund	4. Information and systems for delivering applied climate-risk informed	118,000,000	20,000,000	Grants	5,000,000	Grants	ADB
			NA	Loans	93,000,000	Loans	

	investments at scale improved						
	5. Climate resilient pro-poor livelihoods investments implemented	238,000,000	40,000,000 NA	Grants Loans	5,000,000 193,000,000	Grants Loans	ADB ADB
	6. Climate adaptation pro-poor infrastructure investment	312,500,000	40,000,000 20,000,000	Grants Loans	10,000,000 242,500,000	Grants Loans	ADB ADB
Component 3: Program Implementation and M&E and Learning Support	7. Program Implementation and M&E and Learning	10,500,000			4,000,000	Grant	UK Aid, Nordic Development Fund, Agence française de développement (AFD)
					6,500,000	Loans	ADB
Indicative total cost (USD)		750,000,000	120,000,000		630,000,000		

Annex 17 provides **indicative** GCF financing for each country covered by the program. Under Component 2 CRPP IF, each country will receive between USD10 million and USD40 million. Among the seven countries, only Pakistan will take GCF loan.

C.3 Capacity building and technology development/transfer

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☐ No ☒

Supporting capacity development is a key focus of all outputs under the CRPP IF. This includes building capacity of national and local institutions on use of climate risk information for decision-making purposes; and strengthening capacity of local communities in implementing adaptation solutions in the context of livelihood diversification, climate smart land and water management, ecosystem-based infrastructure, and operations and maintenance of climate adaptation infrastructure. Special emphasis is on building capacity of poor and vulnerable women in adaptation planning and implementing adaptation measures. All capacity building activities will adopt approaches that promote institutionalization and long-term sustainability by partnering with local technical organization and civil society organizations, especially grassroots women's organizations. Such approaches will also help in replication and scaling up of adaptation solutions in other sectors or parts of the country. GCF financing will be critical for undertaking such capacity building activities.

EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

D.1. Impact potential

The impact of the CRPP will be contributing to a paradigm shift towards climate resilient sustainable development in Asia and the Pacific region. The program will achieve this impact by strengthening the resilience of poor and vulnerable people to the impacts of climate change in selected countries in the Asia and Pacific region through implementing outputs related to (i) strengthening knowledge and action research on climate risk informed pro-poor community-level solutions; (ii) strengthening institutional and community capacity to develop and deliver climate adaptation investments at community-level; (iii) identification and preparation of inclusive and pro-poor adaptation investment projects; (iv) improving information and systems for delivering applied climate-risk informed investments at scale, (v) implementing pro-poor climate resilient livelihood strategies and practices and (vi) implementing climate adaptation pro-poor infrastructure. In doing so, the CRPP will directly address the specific risks and vulnerabilities to the poorest and most vulnerable people and their livelihoods posed by increased weather extremes and slow-onset climate related hazards that result from climate change. This will support transitions in key systems that are necessary for large-scale transformations and a core enabler of climate resilient development outcomes and sustainable poverty reduction, in line with the objectives of the Paris Agreement and the key recommendations from the WG2 contributions to the IPCC 6th Assessment Report.

The pathway towards the achievement of improved resilience includes the implementation of local climate change adaptation measures delivered as part of **at least 10-15 large-scale projects**, co-financed by the ADB. On completion of these projects, it is expected that a **minimum of 3,219,400 of the poorest and most vulnerable people** will directly benefit and thus be more resilient to climate change as a result. Direct beneficiaries are considered to be those people that are specifically targeted by the intervention and participate in, or are recipients of, goods (e.g. agricultural extension), services (e.g. early warning/climate information) or new knowledge or improved practices (e.g. water harvesting, agroforestry) related to the three IF output areas, during the lifetime of the projects *and* are expected to have a measurable adaptation benefit from the intervention. The following table provides a breakdown of minimum expected direct beneficiaries per CRPP country.

Table 3: Direct beneficiaries disaggregated by country		
Country	No. of Direct Beneficiaries	% of direct beneficiaries of total population
Cambodia	1,300,000	8.13%
Indonesia	1,200,000	0.45%
Lao PDR	141,000	2.04%
Pakistan	488,000	0.23%
Papua New Guinea	10,000	0.11%
Timor-Leste	60,000	4.62%
Vanuatu	20,40	7.02%

In addition to those directly benefiting from the activities under each project, there will be a number of people that indirectly benefit. It is expected that a **minimum of 4,465,000 people** will indirectly benefit from the program. Indirect beneficiaries are those people that are not specifically engaged by the project activities and thus do not participate in, or receive the goods, services or knowledge of practices, but are located in the coverage area and are perceived to have indirect access or exposure to them. This includes those that live in areas where the installation of protective green infrastructure is considered to have reduced exposure. For example, these may be people that live within the catchment area of an improved climate adapted integrated water management system but have not been directly engaged in training or awareness raising, or climate vulnerable people eligible for potential social protection payments based on upgraded targeting criteria, but do not receive the payments during the project time-frame. The indirect beneficiaries will be estimated for each project during the detailed design phase and reported on in the annual reporting to GCF. The below table shows the minimum expected number of indirect beneficiaries per CRPP country. It should

be noted that the direct total beneficiaries is higher than indirect due to the social protection program related projects that will result in a large number of direct beneficiaries.

Table 4: Indirect beneficiaries disaggregated by country		
Country	No. of Indirect beneficiaries	% of Indirect beneficiaries of total population
Cambodia	1,365,000	8.53%
Indonesia	450,000	0.17%
Lao PDR	1,143,000	16.57%
Pakistan	1,200,000	0.57%
Papua New Guinea	25,000	0.27%
Timor-Leste	241,5000	18.58%
Vanuatu	40,000	13.76%

The number of beneficiaries presented in this section is an approximation based on current sub-project concepts. Due to the diversity of country contexts and the range of activities that the CRPP is able to deploy under each project, both the total direct and indirect beneficiaries as well as in terms of total population will vary widely and will be confirmed as part of the on-going project designs and reported on in annual reporting. In populous countries such as Pakistan and Indonesia, targeted interventions such as the installation of green infrastructure may have a relative beneficiary population of 1-2% of the total province or city population. However, national level social protection and decentralization in those same counties may reach 5% total beneficiaries. Similarly, improved climate information services in Vanuatu have the scope to reach 80-100% of the population, and livelihoods improvements in Lao PDR and Cambodia could realistically reach 10-20% of the population in terms of total beneficiaries in the selected provinces of project implementation, depending on the final intervention.

In all cases, **women will make up at least 50% of the beneficiaries**. However, this will be a minimum requirement for all projects and it is anticipated that in many projects, women will make up more than 50% of the total direct beneficiaries.

D.2. Paradigm shift potential

It is expected that the CRPP will catalyze following impact that will help achieve the paradigm shift of *Increased Climate Resilient Sustainable Development in Asia and the Pacific*.

Scaling-up and replication. As described earlier, finance for local climate adaptation remains limited, especially finances that targets the poorest communities. The CRPP TF aims to support countries in identifying and developing local adaptation solutions that ensure no-one is left behind. The CRPP IF will only support projects that demonstrate the potential of being **replicated** elsewhere in the same country or other countries in the region. With GCF funding, these projects will be able to address relevant barriers to climate adaptation, be they market failures, information failures, policy failures, or behavioural barriers. By empowering government and communities with the knowledge and skills to take appropriate action to adapt, improving systems for delivering climate-resilient investments, and proving their effectiveness through pilots, the project will build the foundations for replication, support transitions in key systems and thereby drive a paradigm shift towards climate resilient sustainable development. GCF funding will be used together with the ADB investment to enable the project to deliver at **scale**. The CRPP through its TF and IF aims at overcoming this barrier by mobilizing resources close around USD750 million for adaptation solutions targeted at the poor and vulnerable communities. This means the GCF investment can spark a major scaling up that is many times the original investment. Further, with a strong focus on partnership the CRPP aims at leveraging and influencing further resources through national budgets, private sector, bi-lateral development partners and civil society organizations, thereby multiplying the delivery of adaptation solutions.

Creation of enabling environment (and sustainability of outcomes). The CRPP will support outputs and outcomes that are closely aligned with national climate priorities. The CRPP TF will therefore catalyze

impacts beyond a one-off project investment and deliver a paradigm shift by ensuring that climate risk informed development becomes the new normal. It will do so by strengthening policies and regulatory frameworks that promote climate-sensitive development, and through capacity building and use of climate risk information for decision-making, with policy makers, planners, and local government and communities. This will ensure the sustainability of outcomes beyond completion of the project. The GCF financing will be used through the CRPP IF to support rolling out of such climate-sensitive development, especially in the context of localization and decentralization. By strengthening national and local institutions, the CRPP IF will ensure that its interventions will contribute to building policy coherence around climate resilience, and is taken up nationwide by relevant agencies. Countries' ownership will be achieved through stakeholder engagement and capacity building at various decision-making levels – government, civil society organizations, communities, financial intermediaries, and the private sector. With such a wide-ranging target, the CRPP will contribute to bringing about a paradigm shift.

Contribution to the regulatory framework and policies. With a key focus on mainstreaming, the CRPP will promote the integration of climate change considerations into policies and decision-making processes. It has an explicit objective to shift to adaptive programming under output 4, increasing the use of climate information in national social protection systems and building capacity of national agencies, local government and civil society organizations. It also will improve climate risk information systems and integrate them into decentralized decision-making processes. The CRPP IF programming will capitalize on on-going decentralization processes to support governments build robust, science-based local climate assessment frameworks that can inform planning processes, equipping decision-makers with the data, tools, guidance, and information needed to inform local planning and investment. The focus on such actions is expected to lead to paradigm shift towards climate resilient development as they are explicitly linked to actions identified by the IPCC AR6 as highly likely to support systems transitions and contribute to transformational change²²⁴.

Innovative solutions and approaches. The CRPP builds on the lessons learned from the implementation of large-scale resilience programs, which identify the need for innovative measures and approaches. These lessons include: (i) resilience building is not just determined by what is supported but how it is supported, including longer-term commitment; (ii) the need for adopting flexible programming approaches, including longer-lead time for design of interventions; and (iii) the need for making resources available for facilitating testing of ideas, learning, and evaluations. Accordingly, the CRPP is planned for a 10-year implementation period. The CRPP TF will support three interrelated outputs (described as in **section B2**) including support to development of investment ideas; and making resources available through a combination of technical assistance and grant financing for pump priming of ideas and facilitating knowledge and learning. The CRPP IF will support financing of adaptation projects. In particular, GCF financing will be used to finance activities that can help overcome information, finance, institutional and gender-related barriers in an innovative way. As a result, it is anticipated that the GCF financed CRPP IF, through the size and the type of investments, will deliver transformational change.

Sharing of knowledge and learning. The CRPP TF has dedicated activities on knowledge and learning which will contribute significantly to generation and sharing of knowledge on climate risk and resilience. It will include a monitoring, evaluation and learning component, which will build collective learning processes, and provide the information and mechanisms for sharing lessons. Learning is critical to operationalize adaptive management principles at a program level. The CRPP TF will ensure that knowledge emerging from GCF financed projects under the CRPP IF are shared with wider stakeholders. The multi-country nature of the program means there will be very large potential for learning and replication, with the transfer of experience between programmes.

Table 5 shows how the proposed GCF CRPP financed outputs (described in section B.3) will complement past and ongoing similar initiatives in the CRPP IF countries that are financed by UNFCCC-related funds and contribute to the paradigm shift. It should be noted that the review of current and past projects is just a first step in the identification of specific interventions under the CRPP, the final projects that receive funding for each country will result from comprehensive consultation with government counterparts, community groups and other stakeholders to determine the specific opportunities for replicability and scaling-up of other interventions based on lessons learned and national and local priorities.

Table 5: Value addition of the CRPP

Country	Relevant adaptation investments	Value addition of the CRPP
Output 4: Information and systems for delivering applied climate-risk informed investments at scale improved		
Cambodia	GEF: Reducing the Vulnerability of Cambodian Rural Livelihoods through Enhanced sub-national Climate Change Planning and Execution of Priority Actions (UNDP, approved on 24 Oct 2013) GEF: Strengthening Climate Information and Early Warning Systems in Cambodia to Support Climate Resilient Development and Adaptation to Climate Change (UNDP, approved on 2 May 2013) GEF: Promoting Climate-Resilient Water Management and Agricultural Practices (UNDP, approved on 19 Dec 2007, completed)	Scaling up successful interventions. There is a wide disparity in terms of investment in Climate Information Services (CIS) in CRPP countries. For example, the GEF has invested heavily in improving CIS in Cambodia for over a decade, improving monitoring and forecasting hardware and software and building capacity for its sectoral application. This has largely been done in the context of strengthening the mechanisms for local adaptation financing and mainstreaming adaptation into local development planning, including undertaking strategic vulnerability and adaptation assessments as part of the planning process. These investments laid the ground work for the interventions under the UNCDF LoCAL initiative which has scaled up local government led adaptation planning and decentralized financing mechanisms. In this case, the CRPP is an opportunity to inject additional resources and capacity to capitalize on GEF investments through supporting the scaling-up of LoCAL work in Cambodia, and support the national roll-out of the local adaptive living facility mechanism. As an MDB, ADB can add significant value to this work through its ability to work within government institutions, national financial institutions and development partners to leverage policy changes and crowd-in funding for the sustainability of such interventions.
Lao PDR	GEF: Strengthening Agro-climatic Monitoring and Information Systems to Improve Adaptation to Climate Change and Food Security in Lao PDR (FAO, approved on 01 Jan 2014)	Replication of effective interventions. CRPP output 4 is also an opportunity to replicate effective interventions through the transfer of knowledge and skills and apply best practice between and within participating countries on the development and use of applied CIS to increase the scale, efficiency and effectiveness of climate change adaptation financing at the local level. For example, the LoCAL program in Lao PDR is at a more nascent stage than Cambodia but is capitalizing on the fiscal decentralization and delegation of authority and responsibility from the central government to local administrations in line with the government's Sam Sang (3 Builds) decentralization policy, under which local administrations have been given a greater role and responsibility in exercising discretionary decision-making power and authority. The CRPP can build on the investments of GEF in agrometeorological services and provide further resources to support the first phases of LoCAL in Lao PDR and foster close collaboration and partnership between the two neighboring countries.
Pakistan	Adaptation Fund: Reducing Risks and Vulnerabilities from Glacier Lake Outburst Floods in Northern Pakistan (UNDP, approved December 2010) GCF: Scaling-up of Glacial Lake Outburst Flood (GLOF) risk reduction in Northern Pakistan (UNDP)	
Timor Leste	GCF: Enhancing Early Warning Systems to build greater resilience to hydro-meteorological hazards in Timor-Leste (UNEP, approved October 2021) GEF: Strengthening Community Resilience to Climate-induced Disasters in the Dili to Ainaro Road Development Corridor, Timor	

	Leste (UNDP, approved May 2013) GEF: Strengthening the Resilience of Small Scale Rural Infrastructure and Local Government Systems to Climatic Variability and Risk (UNDP, approved Dec 2011) (Component 1)	adaptation goals. For example, the CRPP would bring together countries with more advanced CIS capacity and infrastructure such as Indonesia and Vanuatu, together with countries such as Timor-Leste and Papua New Guinea under one common program, with common goals, objectives and approach, fostering peer to peer knowledge and information generation and sharing and exchanges between the relevant agencies as well as development partners to work directly towards the promotion of knowledge and skills related to climate information services and catalyze future investments in national climate information service frameworks.
Vanuatu	GCF: Climate Information Services for Resilient Development in Vanuatu (SPREP, approved December 2016)	Promoting innovative interventions that help fill critical gaps. The CRPP is also an opportunity to capitalise on previous and current investments in CIS infrastructure and expand its application to innovative projects that can improve the mechanisms for channelling adaptation finance into the hands of the poor and most vulnerable, and thereby fill a critical gap. For example, the current landscape of adaptation investment activities in Vanuatu presents a significant opportunity to extend the application of the enhanced CIS mechanisms and products from on-going GCF funded activities and complement the more infrastructure and strategic framework focused investments of GEF, by developing national level mechanisms that can support the delivery of targeted and timely adaptation finance directly to the hands of the poorest and most vulnerable populations through climate informed adaptive social protection systems and forecast-based financing. Pakistan is another example where dedicated funding for CIS from GEF and GCF for EWS on Glacial Lake Outburst Floods and a current 210 million USD investment from the World Bank on improving hydrometeorological and climate services presents a significant opportunity to link the modernising CIS infrastructure with the well-established social protection mechanisms to make them climate adaptive. Such a systemic approach to adaptation will enable those people that rely on fragile livelihoods that are frequently and increasingly disrupted by climate related shocks and stresses, to take steps both before and after the impact of extreme events to ensure minimal negative impacts to their sources of income and food security and take advantage of new opportunities specific to their individual needs.
Output 5: Climate resilient pro-poor livelihoods investments implemented		
Cambodia	GCF: Public-Social-Private Partnerships for Ecologically-Sound Agriculture and Resilient Livelihood in Northern Tonle Sap Basin (PEARL) (FAO, approved March 2023) GCF: Climate-Friendly Agribusiness Value Chains Sector Project (ADB, approved March 2018) Adaptation Fund: Enhancing Climate	Scaling up successful interventions. Climate resilient rural livelihoods have received significant UNFCCC backed investment in some of the CRPP countries and little or no investment in others. For example, Cambodia, Lao PDR and Pakistan have received investment in climate resilient agricultural value chains, climate smart agriculture, and watershed and land management, particularly from the GEF but also from GCF, and the Adaptation Fund has invested in livelihood adaptation highly targeted towards the most vulnerable populations in Indonesia. The activity areas proposed under output 5 of the CRPP are highly complementary to the past and on-going resilient livelihood initiatives in these countries and thus present a significant opportunity to channel fresh resources into scaling-up and replicating selected interventions both within countries and

	Resilience of Rural Communities Living in Protected Areas of Cambodia (UNEP, approved June 2012) GEF: Promoting Climate-Resilient Livelihoods in Rice-Based Communities in the Tonle Sap Region (FAO, approved June 2019) GEF: Building Adaptive Capacity through the Scaling-up of Renewable Energy Technologies in Rural Cambodia (S-RET) (IFAD, approved June 2015) GEF: Strengthening the Adaptive Capacity and Resilience of Rural Communities Using Micro Watershed Approaches to Climate Change and Variability to Attain Sustainable Food Security (FAO, approved Sep 2011, completed) GEF: Climate Adaptation and Resilience in Cambodia's Coastal Fishery Dependent Communities (FAO, approved June 2018) GEF: Vulnerability Assessment and Adaptation Programme for Climate Change in the Coastal Zone of Cambodia Considering Livelihood Improvement and Ecosystems (UNEP, approved July 2009, completed) (cross cutting)	between countries and build on the lessons learned from previous projects. The specific added value of the CRPP in this regard comes from its unique design that will combine GCF grant financing with large co-financing from ADB, which would result in a significant scale-up (both geographically and financially speaking) from previous and current funding promoting transformational change. Replication of effective interventions in geographical areas with gaps. Within countries, the CRPP is an opportunity to capitalize on the nascent national frameworks and knowledge base regarding the fore-mentioned intervention areas, and to transfer, replicate and scale up similar initiatives to other geographies with high climate risk. For example, existing activities and investments related to livelihoods in Lao PDR are currently focused in the rural northern highlands and as such there is the potential of neglecting large swaths of important agricultural communities in the central provinces that are experiencing devastating impacts from floods and droughts, which are likely to worsen due to climate change, and where the poverty head count rate is some of the highest in the country and is showing the slowest rate of reduction.²²⁵ The CRPP can capitalise on the knowledge, lessons learned and take advantage of the related newly strengthened institutional frameworks and fill the financing gap in the to the lesser funded, central provinces that are highly exposed and vulnerable to the impacts of climate change. Similarly, there is an identifiable gap in large scale climate adaptation investments in Pakistan that relates to the coastal zones. Much of the focus of investment from large-scale adaptation funds in Pakistan has been in the inland districts of the Indus River Basin and the northern provinces. By focusing investment on livelihood adaptation in the coastal zones of the Indus River Delta, interventions in Pakistan will not only fill a funding gap in terms of geographical distribution, but also directly complement existing investments by GCF that focus on the provision of CIS for livelihood decision-making (from federal to farm level). Balancing the geographical distribution of adaptation financing in such a way is crucial as Pakistan's 990km coastline is located in the provinces of Sindh and Balochistan, which have the highest rural poverty rates among Pakistan's provinces and the highest urban-rural poverty gaps.²²⁶ Promoting targeted gender-focused interventions. In addition, the CRPP will, where appropriate, direct resources towards activities in this area that have an explicit and targeted focus on women's empowerment, which would be a particular asset in a country such as Cambodia where the transition of employment away from agriculture has been greater for men than for women, but this is not necessarily reflected adaptation interventions where existing initiative may not have such an explicit focus on women. Help fill critical adaptation gaps. Other countries, such as Papua-New Guinea have only received relatively recent and small-scale adaptation funding on climate resilient livelihoods from UNFCCC mechanisms. The programmatic, large-scale approach proposed by the CRPP presents an important opportunity to channel GCF's adaptation resources into a
Indonesia	Adaptation Fund: Community Adaptation for Forest-Food Based Management in Saddang Watershed Ecosystem (Partnership for Governance Reform (Kemitraan) of Indonesia, approved July 2020) Adaptation Fund: Adapting to Climate Change through Sustainable Integrated Watershed Governance in	

	Indigenous People of Ammatoa Kajang Customary Area in Bulukumba Regency, South Sulawesi Province, Indonesia (Partnership for Governance Reform (Kemitraan) of Indonesia, approved September 2020) Adaptation Fund: Community Adaptation for Forest-Food Based Management in Saddang Watershed Ecosystem (Partnership for Governance Reform (Kemitraan) of Indonesia, 8 July 2020) GEF: Seventh Operational Phase of the GEF Small Grants Programme in Indonesia (UNDP, approved June 2020) GEF: Strategic Planning and Action to Strengthen Climate Resilience of Rural Communities in Nusa Tenggara Timor Province (SPARC) (UNDP, approved March 2011)	country that is considered high climate risk, and where over 85% of the population live in rural areas where access to markets, services and income-generating opportunities is limited. Channelling new adaptation resources to the country as part of large-scale poverty reduction projects will provide the possibility for delivering resilient livelihood interventions at-scale. The programmatic approach will also facilitate the replicated implementation of climate resilient livelihood projects that has been demonstrated to be effective in supporting transformative change in other CRPP countries, such as in Indonesia and Timor-Leste thus reducing risk and maximizing administrative efficiencies. This would be an important and necessary step forward for building the adaptive capacity of subsistence and small-scale farmers that make up over 70% of the PNG population. Furthermore, it is self-evident that with 37.5% of the population living below the national poverty line (USD1.90 PPP/day) gender inequality at extremely high levels, with the country ranking 159 out of 160 countries on the Gender Inequality Index, and one of the most culturally and geographically diverse nations on earth, Papua New Guinea, requires carefully considered and specific investment in adaptation solutions that are explicitly pro-poor and gender empowering and that are driven by a local approach that promotes context specificity. Indonesia is also yet to receive adaptation funding from GCF and has only received relatively small grants from GEF for building climate resilient livelihoods, this is in the world's fourth most populous country where nearly 5% of the population have a power purchasing parity of USD1.9 a day or less. The poorest and most vulnerable populations in Indonesia are being left behind in terms of adaptation financing, and thus there is a need for investment in projects that have the capacity to reach a significant number of people living with high climate risk.
Lao PDR	GEF: Integrated Water Resource Management and Ecosystem-based Adaptation (EbA) in the Xe Bang Hieng River Basin and Luang Prabang City (UNDP, approved June 2020) GEF: Climate Smart Agriculture alternatives for upland production systems in Lao PDR (FAO, approved June 2019) GEF: Climate Adaptation in Wetlands Areas (CAWA) in Lao PDR (FAO, approved on 01 Jan 2014) GEF: Improving the Resilience of the Agriculture Sector in Lao PDR to Climate Change Impacts (UNDP, approved September 2009, completed)	Support complementary interventions. Finally, there are some countries where there has been a much greater emphasis on climate adaptive infrastructure and further investment in adapting rural livelihoods is required to complement and balance out the investment portfolio. In Timor-Leste increasing investment in agricultural diversification, and climate resilient value chain development would balance the portfolio of investments and complement the more infrastructure focused interventions to date and address the high risk posed by climate change to the countries important agriculture sector of which more than 80% of the population rely as a source of income and for their food security. Investments in Vanuatu also represent large contributions to financing of climate risk management activities but place a strong emphasis on the installation of protective infrastructure along with strengthening national and local government frameworks for climate risk reduction, and so there is a need for scaling-up investment in livelihood adaptation activities.
Pakistan	GCF: Transforming the Indus Basin with Climate Resilient	

	Agriculture and Water Management (UN FAO) GEF: Sixth Operational Phase of the GEF Small Grants Programme in Pakistan (UNDP, approved April 2016) GEF: Mainstreaming Climate Change Adaptation through Water Resource Management in Leather Industrial Zone Development (UNIDO, approved March 2014)	
Papua Guinea	New	Adaptation Fund: Adaptation of Small-Scale Agriculture for Improved Food Security of Resilient Communities in Papua New Guinea (ASSA) (The Pacific Community (SPC), approved March 2023)
Timor Leste		GEF: Adapting to climate change and enabling sustainable land management through productive rural communities in Timor-Leste (UNEP, approved 2023) GEF: IKAN Adapt: Strengthening the adaptive capacity, resilience and biodiversity conservation ability of fisheries and aquaculture-dependent livelihoods in Timor-Leste (FAO, approved June 2019)
Vanuatu		GCF: Vanuatu community-based climate resilience project (VCCRP) (Save the Children Australia, approved May 2022) GEF: Adaptation to Climate Change in the Coastal Zone in Vanuatu (UNDP, approved February 2013)
Output 6: Climate adaptation pro-poor infrastructure implemented		
Cambodia		Adaptation Fund: Climate Change Adaptation through Protective Small-Scale Infrastructure Interventions in Coastal Settlement of Cambodia (UN Habitat, approved January 2021) Support transfer of knowledge between geographies and sectors. The CRPP can complement previous and current investments in climate adaptive infrastructure in participating countries by supporting the transfer of knowledge and skills between geographies and sectors, capitalizing on progress made in terms of the establishment of related policies and strengthened institutions, and pooling resources to build

	GEF: Vulnerability Assessment and Adaptation Programme for Climate Change in the Coastal Zone of Cambodia Considering Livelihood Improvement and Ecosystems (UNEP, approved July 2009, completed)	coordinated and transformative change through the replication and scaling up of activities. For example, in Indonesia the Pekalongan City Resilience Project, which seeks to address urban flood risk, presents an opportunity for collaboration and coordination via the transfer of knowledge and capacity to other cities on the Island of Java. Replicate interventions in new areas. In Lao PDR, current climate change financing for adaptation infrastructure and nature-based solutions is mainly targeted towards activities which focus on urban and peri-urban settings. Rural areas in Lao PDR account for nearly 90% of the poor population, and rural poverty is almost three times higher than in urban areas. The CRPP can bridge this urban-rural gap, by expanding on existing activities that have a more urban focus, and thus addressing the pressing adaptation infrastructure needs of the rural populations. Specifically, there is a clear opportunity to capitalize on the nascent national frameworks and knowledge base regarding Integrated Water Resource Management and Ecosystem-based Adaptation. Help fill critical adaptation gaps. In Pakistan, there is an identifiable gap in terms of large-scale climate adaptation infrastructure investments, and particularly those that relate to the coastal zones and protecting lives and rural livelihoods from coastal flooding and the impacts of sea level rise. The one UNFCCC backed investment project from the Adaptation Fund that has adaptation infrastructure at its center is focused on water management infrastructure for flood and drought resilience. Papua New Guinea is another CRPP country where investment in adaptation infrastructure has been minimal, with only one project from the Adaptation Fund addressing climate change-related flood risk in the North Coast and Islands Region leaving a large gap and thus opportunity for the CRPP to direct fresh resources for this output in the country and bring together neighboring countries such as Vanuatu and Timor-Leste which have invested more explicitly in local, small-scale and nature-based adaptation infrastructure solutions.
Indonesia	Adaptation Fund: Safekeeping-Surviving-Sustaining towards Resilience: 3S Approach to Build Coastal City Resilience to Climate Change Impacts and Natural Disasters in Pekalongan City, Central Java Province (Partnership for Governance Reform (Kemitraan) of Indonesia, approved 23 September 2020)	
Lao PDR	GCF: Building resilience of urban populations with ecosystem-based solutions in Lao PDR (UNEP) Adaptation Fund: Building climate and disaster resilience capacities of vulnerable small towns in Lao PDR (UN-Habitat, approved July 2019) Adaptation Fund: Enhancing the climate and disaster resilience of the most vulnerable rural and emerging urban human settlements in Lao PDR (UN Habitat, approved October 2016) GEF: Integrated Water Resource Management and Ecosystem-based Adaptation (EbA) in the Xe Bang Hieng River Basin and Luang Prabang City (UNDP, approved June 2020) GEF: Effective Governance for Small Scale Rural Infrastructure and Disaster Preparedness in a Changing Climate (UNDP, approved August 2011)	

Pakistan		Adaptation Fund: Enhance community, local and national-level urban climate change resilience to water scarcity, caused by floods and droughts in Rawalpindi and Nowshera, Pakistan (UN-Habitat, approved June 2020)
Papua Guinea	New	Adaptation Fund: Enhancing adaptive capacity of communities to climate change-related floods in the North Coast and Islands Region of Papua New Guinea (UNDP, approved 2012, completed)
Timor Leste		GCF: Safeguarding rural communities and their physical assets from climate induced disasters in Timor-Leste (UNDP) GEF: Enhancing Climate Resilience of the Urban Services Sector in Timor Leste (ADB, approved March 2014) GEF: Upscaling Climate-Proofing in the Transport Sector in Timor-Leste: Sector Wide Approaches (ADB, approved, March 2014) GEF: Building Shoreline Resilience of Timor-Leste to Protect Local Communities and their Livelihoods (UNDP, approved on 30 Jul 2014) GEF: Strengthening the Resilience of Small-Scale Rural Infrastructure and Local Government Systems to Climatic Variability and Risk (UNDP, approved on 16 Dec 2011)
Vanuatu		GCF: Enhancing Adaptation and Community Resilience by Improving Water Security in Vanuatu (Pacific Community, approved October 2022) GEF: Adaptation to Climate Change in the Coastal Zone in Vanuatu – Phase II (VCAP II) (UNDP, approved December 2019)

	GEF: Protecting Urban Areas Against the Impacts of Climate Change in Vanuatu (ADB, approved March 2014) GEF: R2R: Integrated Sustainable Land and Coastal Management (FAO, approved November 2013)	
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D.3. Sustainable development

Through investments in local level adaptation, the CRPP directly targets the pillars of sustainable development - environment, social and economic.

Environmental co-benefits. The CRPP TF will support countries to increase awareness and build capacity to develop and implement large-scale adaptation projects, including projects that promote ecosystem-based adaptation measures such as mangrove protection, management of urban waterways, and urban greening projects. The CRPP IF will support implementation of such measures in the context of poverty reduction programs in rural and urban areas, thereby promoting environmental co-benefits. Such measures become critical in the context of community resilience, because often the livelihoods and well-being of poor and marginalized populations are dependent on natural assets. Since such livelihoods are often impacted by climate-related shocks and stresses, it would be critical to ensure natural-based solutions are adopted, where appropriate (guided by decision on scale and subsidiarity) to strengthen resilience of poor population. It is expected that CRPP TF and IF will promote nature based solutions through a wide range of investments, such as (i) community infrastructure - adopting natural assets in place of/in mix with grey infrastructure to build resilience of local communities; (ii) livelihoods – promoting nature-based solutions as part of climate smart agricultural practices; (iii) well-being – promoting protection of natural assets as part of social protection programs, such as public works programs; (iv) governance and decentralization – increasing role of women's group in protecting natural resources etc.

Social co-benefits. All interventions supported by CRPP TF and IF will have strong social benefits, because of the explicit focus on low income and disadvantaged groups, and the focus on poverty reduction and addressing existing inequalities associated with climate change. The interventions will have positive distributional benefits by targeting these groups, increasing the flows of finance and responsibility to them, and providing opportunities to participate collectively and create social networks, thus will lead to high gains in social capital. The explicit targeting of women through a gender window will also lead to additional social benefits. It is expected that CRPP TF and IF will promote a wide range of adaptation investments with social co-benefits, including (i) adaptive and shock responsive social protection; (ii) community-driven development in rural and urban areas; and (iii) climate resilient livelihoods support.

Economic co-benefits. There is a strong economic element to the investments to be promoted by CRPP TF and supported by CRPP IF, which is captured through the triple dividend of investing in resilience.²²⁷ This includes i) the economic activity stimulated by reduced impacts, forward-looking planning, long-term capital investments and entrepreneurship as well as macro-economic benefits, and ii) development co-benefits from resilience investments, including positive spillovers as well as non-market benefits. Moreover, with a focus on inclusive governance, the CRPP will promote bottom-up processes, which are critical for inclusive sustainable development.

Gender-responsive development impact. The CRPP seeks to address the gender-climate vulnerability-poverty nexus, through supporting an integrated set of activities across both the Trust Fund and The Investment Fund that combine to address structural drivers of women's inequality, women's disproportionate vulnerability to climate change and multidimensional poverty. All projects supported by the CRPP Investment Fund must either be categorized as Gender Equity Theme (GEN) or Effective Gender Mainstreaming (EGM), meaning that specific targeted activities that address issues of women's inequality must be identified in

project Gender Action Plans (GAPs), with associated budget allocation and monitoring and reporting plans. For GEN projects, the project's results framework must include outcome indicators that are specific to promoting gender equality, thus gender becomes a fundamental part of the project's design. The CRPP Trust Fund has dedicated resources for undertaking upstream analytical and capacity building work that will support the identification of the types of appropriate strategies and interventions that can be effective in addressing issues of climate risk, gender inequality hand in hand with poverty reduction interventions. These strategies are articulated at a higher level in the CRPP's Gender Action Plan (GAP) presented in annex 8.2.

Promoting gender equality is a core thrust of the CRPP and which will be promoted through all projects supported by the CRPP TF and CRPP IF. The focus on gender equality extends from existing evidence that when grassroots women's groups are involved in leading the identification and implementation of adaptation measures the benefits are manifold. First, adaptation investments identified and prioritized by women are not focused exclusively on women but aim to improve the lives of their families and communities. Second, the strategies adopted by women highlight their leadership role in claiming resources and public recognition, thereby bringing a shift in viewing women's status from victims and passive beneficiaries to stakeholders and drivers of resilient development.²²⁸ Such examples are also visible in the current pandemic response where grassroots women's groups have been instrumental in ensuring that government support reaches the last mile. It is thus important that such women's leadership roles are promoted in different investments thereby boosting women's adaptive capacity to deal with climate shocks and stresses. At the same time, women-focused investments and resource flows are also necessary to break generations of gender inequalities in economic and human capital endowments and voice and agency that shape their climate adaptive capacity. An annex on Gender assessment and program-level action plan is included as an **Annex 8**.

Thus, the CRPP will deliver against several of the SDGs, including 1 no poverty, 3 good health and wellbeing, 5 gender equality, 10 reduced inequalities, 11 sustainable cities and communities and 13 climate action.

D.4. Needs of recipient

Addressing resilience needs of the poor and vulnerable population. The recipients of the program are the poor and vulnerable, who by definition have high adaptation needs. These groups are likely to be the most affected by climate change, and do not have the finance or capacity to anticipate, absorb, adapt to and recovery from climate shocks and stresses, and need support to strengthen their assets, livelihoods and well-being. Thus, the CRPP will directly meet the needs of these groups by supporting resilience investments, as well as strengthening institutions and systems, to allow climate finance to reach the poor and support local adaptation. Without external finance, it will not be possible to deliver the scale of climate adaptation needed. In particular, GCF resources will help finance transformational change in areas that other financial institutions tend to ignore. These resources also provide greater flexibility, through the provision of grant finance alongside ADB financing, which is likely to be needed for community resilience investments. This will allow tailored packages of community investment, allowing the combination of finance to deliver a more transformative overall effect, because it allows the targeting of multiple barriers. Thus, there is a high recipient need for GCF support. This will apply to the least developed countries, the small island developing states, and the poorest in middle income countries.

Cambodia is an LDC in Southeast Asia with a population of 16.3 million in 2018, 76 percent of its population lives in rural areas and the urban concentration is located in the most fertile river valleys and flood plains. About 90% of the poor live in the countryside and around 4.5 million people remain near-poor, vulnerable to falling back into poverty when exposed to economic and other external shocks.²²⁹ Cambodia's population relies heavily on agriculture, fisheries, and natural resources extraction providing 25% of GDP and employing 49% of the country's labor force. Climate change shocks are projected to victimise 61% of the rural Cambodian population; in particular, about 7 million smallholder farmers and 25,000 workers in construction industry. Women are similarly more disadvantaged, especially in decision-making and income disparity.²³⁰ Still as an LDC, Cambodia's fiscal management in solving local development problems is compounded by climate shocks and the COVID-19 pandemic, such the constraints on the public budget management requires outsourced assistance especially in dealing with uplifting conditions of the rural poor.

Indonesia has the largest economy in Southeast Asia. With high economic growth, the poverty rates in the country in recent decades have decreased from 24% in 1999 to 10% in 2020. Nonetheless, poverty remains high in absolute numbers, because of the large population size, and thus numbers tens of millions of people. Poverty rates are higher particularly in remote areas and among certain groups.²³¹ While Indonesia has urbanized rapidly, not everyone has equally benefitted from it, with nearly 7% of urban population living under the poverty line and roughly equal percentage being near poor. Prior to COVID-19, an estimated 61.8 million people, or 21.5% of the population, were considered poor or to have a high risk of falling into poverty, making them vulnerable to shocks.²³² With COVID-19, it is estimated that those below the national poverty line has increased and may reach 12.8% by the end of 2020.²³³ Inequality also remains high, with the bottom 40% of the population accounting for 17.7% of consumption, whereas the top 20% consumes 45.6%.²³⁴ Urban areas have higher inequality than rural areas.²³⁵ The absolute number of urban poor will continue to increase as the country urbanizes and thus is an increasingly important target group in Indonesia's poverty reduction policies.²³⁶ To reduce poverty and stimulate economic inclusion, Indonesia has prioritized poverty reduction programs, including social protection programs and urban informal settlements upgrade. These programs do not consider long-term changes in climate and thus the impacts generated by these programs will be undermined in the future due to increase in climate shocks and stresses. It is therefore critical that such poverty reduction programs integrate climate resilience considerations to build adaptive capacity of local institutions and communities.

Lao PDR is a landlocked country in Southeast Asia and is one of the LDCs. It is a highly vulnerable country with low readiness. While the country's economy has been growing, delicately balanced macroeconomic outlook, weak human resources and infrastructure, and the recurrence of climate hazards continue to pose challenges to the Lao PDR's development.²³⁷ A large percentage of population remain close to the poverty line: data from 2019 show that the poverty rate in rural areas was 23.8% compared to 7.0% in urban areas.²³⁸ Agriculture accounts for 16% of GDP and engages, to various degrees, over 65% of the population (of whom more than half are women). The large section of the rural population engaged in agriculture are particularly vulnerable to the impacts of natural hazards due to their large dependence on subsistence agriculture production and inadequate social protection systems. With projected increase in temperature and intensified precipitation, livelihoods and agricultural yields will be greatly impacted due to damage to rural infrastructure by floods, increased crop evapotranspiration in the dry season with associated higher demand for water, more frequent land inundation and consequently crop losses during floods, loss of livestock due to water shortage or by destructive floods. Investments will be needed to support the rural poor in climate resilient livelihood diversification, in improving land water management practices in response to change climate, and in strengthening climate services for improved decision-making. As an LDC, Lao PDR lacks the financial resources to address climate threats, especially for the most vulnerable, and the constraints on the public budget mean Government cannot fill the adaptation funding gap. Moreover, the COVID-19 crisis is expected to result in loss of jobs for almost half a million of people roughly 383,000 people are expected to be pushed deeper, or back into poverty. Such situation further exacerbates pre-existing vulnerabilities, for example, the food insecurity in the northern rural provinces, being already a critical issue as seven months rice availability deficit has been recorded even before the Covid-19 outbreak; and the 1.7 million school children having their learning setback.²³⁹

Pakistan is characterized by diverse topography, ecosystems, and climate zones, and faces multiple climate hazards: the country is one of the most vulnerable to climate shocks in the world today. As of 2015 approximately 24% of the population still lived below the national poverty line and 20% of the population remained undernourished. However, the International Monetary Fund (IMF) projects due to COVID-19 crisis up to 40 percent of population will be living below the poverty line. Agriculture is still the largest employer, occupying 42% of the workforce and contributes 21% to GDP, making potential climate impacts and adaptation needs in the sector a high priority, especially given the high proportion of subsistence farmers. However, it is expected that agricultural growth will lag if lockdowns continue and disrupt needed transportation, logistical support, labor, and access to inputs for the next planting season.²⁴⁰ Pakistan is also a highly water stressed country, ranked 14th globally in terms of water stress. The target area for the project is the coastal areas, which are already affected by coastal and river floods, which are projected to increase significantly with climate change. Poverty rates in these coastal areas are among the highest in the country. These

coastal communities do not have the resources to adapt, and the local government budgets are not sufficient to deal with the rising climate-related impacts, and these areas have very high needs, especially the poorest and most vulnerable which this program will target.

Papua New Guinea. Located in the South West Pacific, PNG comprises the eastern half of New Guinea island, four additional islands (Manus, New Ireland, New Britain, and Bougainville), and around 600 smaller islets and atolls. PNG is a highly climate vulnerable country²⁴¹ and is also considered to be a Small Island Developing State (SIDS). PNG has consistently ranked among the lowest 30 countries in the Human Development Index, and has poverty rates between 30-40% and an adult literacy rate of around 50%. A large proportion of its population is rural (87%). Agriculture, fishing, community forestry, and small-scale mining are primary livelihood activities in rural areas. Estimate suggest that 70% of households in PNG remain dependent on subsistence agriculture.²⁴² The combined issues of widespread poverty, poor infrastructure and dependence on climate sensitive livelihoods all heighten the vulnerability of the local population. 83% of food consumed is produced in country, thus, any disruption to local food production (from climate or other shocks) has an immediate and significant impact on food security.²⁴³ Climate change is likely to increase hazards, and without adaptation measures, the exposure of the rural communities will further increase. As an SIDS, PNG lacks the financial resources to address the very large climate threats, especially for the most vulnerable, and the constraints on the public budget mean Government cannot fill the adaptation funding gap. The COVID-19 crisis is having negative impacts on unemployment (31% of household heads lost their jobs) and poverty (increase by 4.5 percentage points). The 2020 fiscal balance is estimated to worsen from -5% of GDP pre-COVID-19 to -8.2% of GDP post-COVID-19. Public sector debt is expected to reach 49.8% of GDP in 2020, compared with 39.8% in 2019.

Timor-Leste occupies the eastern half of the island of Timor, which it shares with the Indonesian province of East Nussa Tenggara. The island has an extensive coastline stretching 700km, while inland the topography is mountainous. The country is one of the poorest countries in Asia and is a LDC. Data for 2016 estimate that 30.7% of the population were living on less than the internationally recognised poverty line of USD1.90/day, and 45.8% were classified as living in multi-dimensional poverty.²⁴⁴ Such conditions make the population more vulnerable to shocks and stresses. Agriculture accounts for 9% of GDP, and 41% of employment, however, excluding oil revenue plays a much more significant role in the economy.²⁴⁵ Most agriculture is subsistence farming, with coffee the only major crop grown for export. Rural areas suffer from food insecurity, due to a combination of low productivity, high population growth and unpredictable rainfall coupled with extreme climate events.²⁴⁶ Gender analysis shows that for women, these circumstances are felt even more acutely.²⁴⁷ Most farming households have minimal engagement with agricultural markets and can be characterized as low-input, low-output producers. Although the Timor-Leste people have some capacity to adapt to these complex changes, such coping strategies are still reliant on climate-sensitive natural resources, aggravated by their limited access to alternative livelihoods. Traditional gender roles exacerbate the risks for women in a changing climate, while minimal access to weather and climate forecasting hinders adaptive actions. With agricultural systems, including the livelihood conditions and food insecurity issues, expected to be severally affected by both by increases in extreme events, as well as changes in temperature and rainfall³⁶ solutions in adaptation are urgently needed. As an LDC, Timor-Leste lacks the financial resources to address climate threats, especially for the most vulnerable, and the constraints on the public budget mean Government cannot fill the adaptation funding gap. The COVID-19 crisis is expected to worsen the economic situation and result in large scale loss in livelihood and increase in food insecurity.

Vanuatu is a small island developing state in the Pacific that is characterized by low-lying areas that have high levels of climate risk, especially to tropical cyclones. Vanuatu was ranked as the country most at risk from disasters in the annual World Risk Report (2019). As a series of islands and atolls lying in the tropics, it faces both slow and rapid onset meteorological hazards. Disasters resulting from these climate hazards are a very significant contributor to issues of social deprivation and impede economic development in Vanuatu; these events cost on average an estimated 6% of GDP every year. As an island nation, the country is particularly at risk from sea-level rise. Vanuatu lacks the financial resources to address climate threats, especially for the most vulnerable, and the constraints on the public budget mean Government cannot fill the adaptation funding gap. The country's GDP is likely to contract by 9.8% in 2020 due to reduced tourism,

remittances, and exports due to COVID-19, exacerbated by damages from Cyclone Harold. It is estimated that the population living below USD1.90/day will rise from 13.9% to 18.3% in 2020 if consumption expenditure declines by 10%.

D.5. Country ownership

The CRPP IF will be implemented in seven countries. Consultations have been undertaken with stakeholders from Cambodia, Lao PDR, Indonesia, Pakistan, PNG, Timor Leste, and Vanuatu. For these countries, ADB has investments in the pipeline that have the potential to meet the objectives of the CRPP. In-depth consultations were also organized with selected development partners, civil society organizations and grassroots women's organizations, in order to understand local needs and most importantly the type of transformational approaches needed at the local level.

In order to ensure scale, replicability and sustainability, implementation of the CRPP will be country-led and tailored to fit country and local context, recognizing the broader political economy context. Thus the outputs identified under CRPP IF are closely aligned with national climate priorities (as outlined in NDCs and/or NAPs) and national poverty reduction priorities (as outlined in national development plans). For more information, please refer to **Annex 2.2**.

Cambodia has prioritized climate adaptation as a priority and to reduce its vulnerability. It has identified adaptation priorities in its National Adaptation Programme of Action (NAPA) and Climate Change Strategic Plan (CCSP), 2014-2023, and has embedding adaptation in the National Sustainable Development Plan (NSDP) and the sectoral development plans. The CCSP has 8 strategic objectives and several strategies that provide entry points for a coherent approach to integrate with the national development processes. Within the framework, sectoral plans have identified 171 actions of which 93% are adaptation actions. To support the implementation of the adaptation actions the National Adaptation Plan (NAP) has been formulated and to overcome the financing barrier the Government endorsed the NAP Financing Framework and Implementation Plan in 2017. The Financing plan has identified 40 priority actions with an estimated budget of USD 530 million covering key sectors including infrastructure and agriculture.²⁴⁸ Priority actions in agriculture sector includes promoting and scaling up climate smart farming systems and promotion of post-harvest technology. In 2020, Cambodia has developed its Updated Nationally Determined Contribution with identified priority adaptation actions in agriculture and water resources, forestry, coastal zones, human health, energy, industry, infrastructure, livelihoods and tourism. The government has identified the need to improve the climate risk management into the provincial level policy, planning and budgeting process and also the opportunity to integrate as part of the National Programme for Sub National Democratic Development (NPSNDD) process. Budget availability, knowledge on climate change impacts and implications, lack of technical capacity across ministries have been identified as key barriers for effective adaptation.²⁴⁹

Indonesia has put in place ambitious climate adaptation actions outlined in National Action Plan on Climate Change Adaptation (RAN-API) and its Climate Resilient Development Policy 2020-2045, which prioritizes four sectors– water, marine and coastal, agriculture, and health, each of which is critical for strengthening resilience of the urban and rural poor population. It further identifies infrastructure, governance, technology and capacity building measures for adaptation. In infrastructure, green infrastructure is recognized as a key priority, especially for the low-income people and vulnerable groups. In governance, the RAN-API prioritizes strengthening of institutions and information systems in order to ensure benefits of adaptation are reaching the poor and vulnerable population. Further, the recently approved Rencana Pembangunan Jangka Menengah Nasional (RPJMN) (National Medium-Term Development Plan) 2020-2024 has identified one of its seven development agenda as development of living environment to increase resilience to disasters and climate change. The Government of Indonesia has also recognized the role of social protection in resilience building and the government has developed the Adaptive Social Protection Roadmap.

Lao PDR has prioritized increasing climate resilience in agriculture, forest and land-use and strengthening management of water resources as top priorities of its NDC. This includes promoting climate resilience of farming systems, strengthening resilience of infrastructure, and enhancing water resources information system. Additionally, the NDC recognizes building capacity will be crucial for effective implementation of

adaptation measures, including capacity for risk-informed decision-making at all levels from national to local. Further, the Agricultural Development Strategy to 2025 and Vision to 2030, have prioritized water resources management to support the production of both wet and dry season crops.²⁵⁰ The government prioritizes interventions in agriculture and rural development by: (i) improving agro-met information, and coverage to reach target users; (ii) increasing forest and vegetative land cover; (iii) developing water management infrastructure for flood control and irrigation; and (iv) modernizing irrigation technologies and practices.²⁵¹

Timor Leste in the Dilli Declaration on Climate Change Adaptation recognizes that strong interdependence of climate adaptation and development target, and that the national development priorities can only be sustainably achieved by considering climate adaptation. The NDC identifies food security as a key adaptation priority in order to reduce vulnerability of farmers to drought and flood by improving their capacity to plan for and respond to future climate conditions. In particular, the NDC identifies agroforestry and water shed management as key measures for strengthening food security. Priorities related to water resources management includes enhancing government and community strategies to respond to drought exacerbated by climate change; and enhancing water harvesting model, water distribution system and management system at all levels to avoid water shortages due to climate change. Promoting subnational capacity development for improved adaptation planning and implementation. In 2021, Timor Leste has developed its National Adaptation Plan which has identified seven priority areas: agriculture, biodiversity and ecosystem, disaster risk reduction, health, infrastructure, tourism, and water.

Pakistan in its updated NDC (2021) adopts a long-term vision of mainstreaming climate resilience considerations in the economically and socially vulnerable sectors of the economy. In particular, it identifies agriculture, water resources management, and infrastructure as key adaptation-related priorities. The National Climate Change Policy (2012) spells out the policy measures for adaption in vulnerable ecosystems including coastal and marine ecosystems. The framework for implementation of climate change policy (2014-2030) provides a detailed list of actions with timeframe for implementation, including priorities related to water sector, agriculture and livestock, marine and coastal ecosystems, and disaster preparedness. The framework emphasizes the importance of implementation adaptation measures that focuses on women's needs.

PNG in its NDC has identified food security caused by crop failures a priority for adaptation purposes. In its Second National Communication, PNG emphasizes the importance of agriculture, capacity building, technology transfer, education and training to adaptation. The National Climate Compatible Development Management Policy prioritizes provision of education and training on resilient agriculture and resilient construction to poor and marginalized people. Further, the long-term development strategy: PNG Vision 2050 identifies environmental sustainability and climate change as one its seven strategic focus areas and stresses the importance of engaging with communities in achieving inclusive sustainable development. PNG has developed its National Adaptation Plan in 2023 which has prioritized four key sectors that include agriculture, infrastructure, transport and health.

Vanuatu: The objectives of the CRPP are aligned with NAPA (2007), the Vanuatu Climate Change and Disaster Risk Reduction Policy (2016-2030) and the Vanuatu Recovery Strategy, 2020 – 2023 (VRS, developed in early 2020 in response to Typhoon Harold and to the Covid Pandemic). The Climate Change and Disaster Risk Reduction Policy (2016-2030) has identified community focus is one of its principles of implementing climate adaptation interventions, with an objective to empower communities at the local level through bottom up approaches, drawing on local skills, values and traditional knowledge, and enhancing decentralization in planning, programs and projects.

Consultation with NDAs. In the course of development of this proposal ADB has officially coordinated with the NDAs, including having consultations with NDAs and relevant sector ministry to explain the program proposal, understand their priorities and reflect such needs in the proposal. The NDAs were provided with copy of the draft program proposal and no-objection letter has been obtained from 6 of the 7 countries.

D.6. Efficiency and effectiveness

The projects funded under the CRPP will be subject to ADB's economic appraisal criteria and will have to meet an economic internal rate of return (EIRR) threshold. This means that in present value terms, the total economic benefits will exceed the total economic costs (when analysed in accordance with ADB's 2017 Guidelines for the Economic Analysis of Projects).

The projects will also be subject to ADB's requirements for financial management and analysis. For each project, financial analysis will be prepared in accordance with ADB's Financial Management and Analysis of Projects guidelines. In particular (i) the estimated investment and operations costs, as well as cash inflows, will be clearly presented and will be reasonable; (ii) the financial internal rate of return (FIRR) calculated on a real basis consistent with the ADB Financial Management and Analysis of Projects guidelines will be greater than the weighted average cost of capital; and (iii) the FIRR will be robust under various sensitivity scenarios.

The CRPP programmatic approach proposed will deliver very high efficiency and effectiveness, because the GCF funds will be leveraged against large volumes of ADB finance. At the CRPP IF level, the program will use USD120 million from GCF and leverage USD 630 million from ADB and other TF donors. This will result in a ratio of co-financing of more than 5:1, which is high for adaptation, especially when targeting the most vulnerable. This will mean the GCF investments will benefit from existing ADB funded management and implementation activities. The marginal transaction costs for implementing GCF funds will therefore be extremely low (and highly efficient).

The use of GCF finance to fund the climate additionality on top of existing ADB funds will also be extremely effective, because it will work with existing financial, governance and implementation activities, ensuring high conversion of inputs to outputs and outcomes. This means the GCF funds will deliver a very large number of beneficiaries/ USD invested.

The projects that are put forward for financing under CRPP IF will be scrutinized against strict economic and financial viability prior to funding, and only those that pass these criteria will be taken forward. The exact EIRR and FIRR will vary with the projects funded, and it is not possible at this stage to provide values for individual pipeline projects. However, a review of the economic benefits of community level adaptation, as well as the three main proposed outcome areas where GCF funds will be used, was undertaken. The economic justification and rates of return are summarized below.

Community level adaptation: There are a wide set of interventions that can hence community-based resilience, including local and community-based adaptation planning and financing. Evaluations of these projects has found good benefit to cost ratios (BCR) in Asia^{252, 253} and there is good evidence from an ex-post analysis of 23 field tested pilots.²⁵⁴ It is also highlighted that interventions that target the most vulnerable are likely to have very positive distributional benefits. In theory these distributional issues can be taken into account in economic appraisal (in cost-benefit analysis) by using equity (or distributional) weights to adjust present values. Very few studies apply equity weights, though such an approach would further increase the benefits of the proposed interventions below.

Output 4: Improved information and systems for delivering applied climate-risk informed investments at scale. These activities are primarily focused on integrating climate risk information into decision-making processes. Social protection programs have positive benefit to cost ratios in themselves and have been recommended²⁵⁵ as a form of climate sensitive development for small and frequent shocks in their own right. Economic analysis of social protection programs²⁵⁶ show high benefit to cost ratios, e.g., one review study looking in detail at 8 studies (4 ex ante, 4 ex post) found BCRs that were as high as 6:1. The GCF funds for this output will be focused on enhancing the climate targeting of funded social protection activities. Numerous studies have identified the advantages and benefits of making social protection programs climate smart,²⁵⁷ to explicitly target climate extremes and to ensure climate shocks do not undermine social protection program effectiveness. The intervention will include the use of climate information, including weather and climate services (including early warning), for social protection. Analysis

of such applications find high benefit to cost ratios (BCRs)^{258,259,260,261} and recent reviews identify on average a benefit to cost ratio of 9:1.²⁶² Benefits increase with increasing climate change (and shocks) and thus these interventions are highly efficient as well as effective.

Output 5: Climate resilient pro-poor livelihoods investment. These activities are focused on additional activities that climate smart livelihoods program investments. These include a wide range of types of interventions but are often focused on agriculture. They can include activities that build the resilience of current livelihoods, with weather and climate services, climate smart extension services, capacity building, and climate smart agriculture, and these interventions have been found to have high benefit to cost ratios.^{263,264} Economic review of Adaptation for Smallholder Agriculture Programme reported positive benefit cost ratios across 32 country-level projects, up to 6:1 (and EIRRs of between 15% and 35%).²⁶⁵ The interventions will also include some activities to help communities' transition to new livelihoods due to climate change threats. This may include moving to new resilient varieties, or developing new climate resilient livelihoods, which are also found to generate positive benefits to cost ratios.^{266,267,268}

Output 6. Climate adaptation pro-poor infrastructure investment. This activity is focused on climate adaption infrastructure. Such interventions have been demonstrated to have very positive economic returns. A systematic review²⁶⁹ of flood risk management appraisal (27 studies, mix ex ante and ex post) found a BCR of 5 to 1 for flood related risks. Further review work, looking at 110 observations on investments/projects from 32 studies and databases in 16 countries²⁷⁰ found a mean BCR of 6. These economic benefits have been found for multiple hazards, which include sea level rise²⁷¹ and floods.²⁷² One particular focus of the GCF funds will be the targeting of green infrastructure for resilience (ecosystem-based adaptation). These provide alternatives to hard engineered protection, such as mangroves and wetlands, river catchment management, and even small-scale local schemes. The review of the literature finds these have high benefit to cost ratios, for example, coastal EBA was found to have among the highest BCR for coastal resilience^{273,274}. Importantly, these schemes have high co-benefits, which include environmental benefits (including carbon sequestration), and these can be particularly high for local ecosystem-based community-based adaptation.²⁷⁵

LOGICAL FRAMEWORK				
E.1. Project/Programme Focus				
☐ Reduced emissions (mitigation) ☒ Increased resilience (adaptation)				
E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)				
Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Large proportions of the poorest and most vulnerable population living in areas with high climate risk in CRPP countries are highly vulnerable to the impacts of climate change, but they are not able to adopt measures to increase their climate resilience because finance for local climate adaptation is limited and is not targeted at the poorest communities.	<u>Low</u>	Paradigm shift would involve the mobilization of large-scale domestic and international financing targeted at the local level to support large numbers of poor and vulnerable people in Asia and the Pacific region to adapt to climate change. Paradigm shift depends on using this large-scale investment to transform the lives and livelihoods of the poorest and most vulnerable communities towards a climate resilient and sustainable development pathway and catalysing investments and transformative adaptation beyond the program's initial investments.	The CRPP is projected to mobilize US\$750 million in domestic and international climate finance by 2030 to improve the climate resilience of at least 5 million poor and vulnerable people across 7 countries in the Asia and Pacific region. This means that GCF investments will spark major scaling up that is many times the original investment and will be able to catalyse adaptation impact on a scale that is beyond the scope of similar investment amounts. Further, as a regional program with a strong focus on partnership the CRPP aims at leveraging and influencing further resources through national budgets, private sector, bi-lateral development partners and civil society organizations, thereby multiplying the delivery of adaptation solutions.
Replicability	Investments in climate information systems, climate resilient livelihoods, and climate adaptation infrastructure that target poor and vulnerable communities have been made to various degrees in CRPP countries. However, information	<u>Low</u>	If knowledge and evidence on the benefits and successful implementation of local level adaptation measures that build the resilience of the poor are systematically documented and shared there is potential for overcoming information failures, policy failures and behavioural failures, which will facilitate the exportation of key structural elements of successful projects within countries,	The programmatic approach of the CRPP and the knowledge strengthening aspects of the CRPP TF component will allow for the systematic generation and dissemination of lessons learned, best practice, and evidence, building knowledge and capacity for further investments and action elsewhere within similar sectors, across different country districts as well as across different countries in the region. It will also generate the political will and

	failures, market failures, policy failures and behavioural failures in the target countries are significant barriers to replication of promising local level adaptation measures that build the resilience of poor and vulnerable people, in different geographies or sectoral settings.		across different sectoral settings, and across the region.	behavioural change among decision-makers required to prioritize the types of investments implemented under the CRPP and strengthen the policies and regulatory frameworks that promote climate resilient and sustainable development, further facilitating replication and scaling.
Sustainability	In most cases, policies, programs and decision-making frameworks in CRPP countries do not prioritise climate sensitive development and the channelling of adaptation resources to the local level for targeted adaptation interventions for the poorest and most vulnerable communities. Weak institutional and community capacity results in the inability to plan and implement adaptation measures at the local level that build resilient livelihoods in poor and vulnerable communities and protect them from increasing climate shocks and stresses.	<u>Low</u>	Paradigm shift would involve climate risk informed development becoming the new normal, with climate adaptation at the local level prioritised in poverty reduction and development policies and programs. It will also involve improved capacity of communities and local institutions to design and implement adaptation measures to make livelihoods climate resilient and minimize the negative impacts of climate shocks and stresses.	Through its investments in climate information services, the CRPP will promote the integration of climate change considerations into development policies, programs and decision-making processes. It will use climate information in national social protection systems and capitalize on on-going decentralization processes to support governments to build robust, science-based local climate assessment frameworks that can inform planning processes, equipping decision-makers with the data, tools, guidance, and information needed to inform local adaptation planning and investment. By strengthening national and local institutions, the CRPP IF will ensure that its interventions will contribute to building policy coherence around climate resilience, and is taken up nationwide by relevant agencies. Countries' ownership will be achieved through stakeholder engagement and capacity building at various decision-making levels – government, civil society organizations, communities, financial intermediaries, and the private sector.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
<u>Total number of direct and indirect beneficiaries</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Detailed guidance will be provided to user departments for project-level reporting against this indicator as it will vary according to the project level results framework. It will include at least one primary source of verification which is not project generated, such as survey data.	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0)	Direct: 1,691,520 (Male: 845,760 Female: 845,760)	Direct: 3,219,400 (Male: 1,609,700 Female: 1,609,700)	There are sufficient pipeline of appropriate sub-projects that meet the CRPP eligibility criteria.
				Cambodia: 884,000 (Male: 442,000 Female: 442,000) Indonesia: 480,000 (Male: 240,000 Female: 240,000) Lao PDR 72,800 (Male: 36,400 Female: 36,400) Pakistan 195,200 (Male: 97,600 Female: 97,600) Papua New Guinea: 6,800 (Male: 3,400 Female: 3,400)	Cambodia: 1,300,000 (Male: 650,000 Female: 650,000) Indonesia: 1,200,000 (Male: 600,000 Female: 600,000) Lao PDR: 141,000 (Male: 70,500 Female: 70,500) Pakistan: 488,000 (Male: 244,000 Female: 244,000) Papua New Guinea: 10,000 (Male: 5,000)	There are no significant delays with the approval of CRPP program sub-projects. The methodology and assumptions for estimating beneficiary numbers under Core Indicator 2 for each ARA are presented in annex 2.4, Table 2.

⁷ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
				Timor-Leste 44,400 (Male:22,200 Female:22,200) Vanuatu: 8,320 (Male: 4,160 Female: 4,160) Indirect: 1,642,000 (Male: 821,000 Female: 821,000) Cambodia 546,000 (Male: 273,000 Female: 273,000) Indonesia 36,000 (Male: 18,000 Female: 18,000) Lao PDR 457,200 (Male: 228,600 Female: 228,600) Pakistan 480,000 (Male: 240,000 Female: 240,000)	Female: 5,000) Timor-Leste: 60,000 (Male: 30,000 Female: 30,000 Vanuatu: 20,400 (Male:10,200 Female:10,200) Indirect: 4,465,000 (Male: 2,232,500 Female: 2,232,500) Cambodia 1,365,000 (Male: 682,500 Female: 682,500) Indonesia 450,000 (Male: 225,000 Female: 225,000) Lao PDR 1,143,000 (Male: 571,500 Female: 571,500)	

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
				Papua New Guinea 10,200 (Male: 5,100 Female: 5,100) Timor-Leste 96,600 (Male: 48,300 Female: 48,300) Vanuatu 16,000 (Male: 8,000 Female: 8,000)	Pakistan 1,200,000 (Male: 600,000 Female: 600,000) Papua New Guinea 25,500 (Male: 12,750 Female: 12,750) Timor-Leste 241,500 (Male: 120,750 Female: 120,750) Vanuatu 40,000 (Male: 20,000 Female: 20,000)	
<u>ARA1 Most vulnerable people and communities</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Detailed guidance will be provided to user departments for project-level reporting against this indicator as it will vary according to the project results framework. It will include at least one primary source of verification which is not project generated, such as survey data.	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0)	Direct: 1,691,520 (Male: 845,760 Female: 845,760) Cambodia: 884,000 (Male: 442,000 Female: 442,000) Indonesia: 480,000 (Male: 240,000 Female: 240,000)	Direct: 3,219,400 (Male: 1,609,700 Female: 1,609,700) Cambodia: 1,300,000 (Male: 650,000 Female: 650,000) Indonesia: 1,200,000 (Male: 600,000 Female: 600,000)	Governments are willing to integrate climate information systems and related decision support tools into their decision-making processes. Detailed baseline and targets will be provided for each sub-project as part of detailed preparation activities and aggregated into the program reporting framework. A common methodology is used for estimating direct and indirect beneficiaries.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
				Lao PDR 72,800 (Male: 36,400 Female: 36,400) Pakistan 195,200 (Male: 97,600 Female:97,600) Papua New Guinea: 6,800 (Male: 3,400 Female: 3,400) Timor-Leste 44,400 (Male:22,200 Female:22,200) Vanuatu: 8,320 (Male: 4,160 Female: 4,160) Indirect: 850,000 (Male: 425,000 Female: 425,000) Cambodia: 546,000 (Male: 273,000 Female: 273,000)	Female: 600,000) Lao PDR: 141,000 (Male: 70,500 Female: 70,500) Pakistan: 488,000 (Male: 244,000 Female:244,000) Papua New Guinea: 10,000 (Male: 5,000 Female: 5,000) Timor-Leste: 60,000 (Male: 30,000 Female: 30,000) Vanuatu: 20,400 (Male:10,200 Female:10,200) Indirect: 2,316,500 (Male: 1,158,250 Female: 1,158,250) Cambodia 1,365,000 (Male: 682,500	

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
				Indonesia 36,000 (Male: 18,000 Female: 18,000)	Female: 682,500 Indonesia 90,000 (Male: 45,000 Female: 45,000)	
				Lao PDR 124,600 (Male: 62,300 Female: 62,300)	Lao PDR 311,500 (Male: 155,750 Female: 155,750)	
				Pakistan 168,000 (Male: 97,600 Female: 97,600)	Pakistan 420,000 (Male: 210,000 Female: 210,000)	
				Papua New Guinea 6,000 (Male: 3,000 Female: 3,000)	Papua New Guinea 15,000 (Male: 7,500 Female: 7,500)	
				Timor-Leste 36,000 (Male: 18,000 Female: 18,000)	Timor-Leste 90,000 (Male: 45,000 Female: 45,000)	
				Vanuatu 10,000 (Male: 5,000 Female: 5,000)	Vanuatu 25,000 (Male: 12,500 Female: 12,500)	
<u>ARA1 Most vulnerable people and communities</u>	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new</u>	The specific methodology to measure this indicator will be project-specific given the	0 (Male: 0 Female: 0)	548,000 (Male: 274,000)	1,370,000 (Male: 685,000)	Smallholder farmers and other poor and vulnerable people

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
	<u>climate-resilient livelihood options</u>	variety in livelihood options supported by the CRPP IF. It will include at least one primary source of verification which is not project generated, such as survey data.		Female: 274,000) Cambodia 520,000 (Male: 260,000 Female: 260,000) Lao PDR 12,000 (Male: 6,000 Female 6,000) Timor-Leste 12,000 (Male:6,000 Female: 6,000) Papua New Guinea 4,000 (Male: 2,000 Female: 2,000)	Female: 685,000) Cambodia 1,300,000 (Male: 650,000 Female: 650,000) Lao PDR 30,000 (Male: 15,000 Female 15,000) Timor-Leste 30,000 (Male:15,000 Female: 15,000) Papua New Guinea 10,000 (Male: 5,000 Female: 5,000)	targeted by the program are willing to participate and adopt climate resilient livelihoods strategies. 40% of individuals targeted adopt the new skills/technologies over the course of the program implementation. Only beneficiaries who have received GCF-funded support and are applying improved or new livelihood options will be counted. Each beneficiary will be counted only once unless a same individual adopted more than one improved or new climate-resilient livelihoods options.
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Detailed guidance will be provided to user departments for project-level reporting against this indicator as it will vary according to the project results framework. It will include at least one primary source of verification which is not project	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0)	Direct: 383,600 (Male: 191,800 Female: 191,800) Cambodia 364,000 (Male: 182,000 Female: 182,000)	Direct: 959,000 (Male: 479,500 Female: 479,500) Cambodia 910,000 (Male: 455,000 Female: 455,000)	There are sufficient pipeline of appropriate sub-projects that meet the CRPP eligibility criteria and contribute to ARA2. There are no significant delays with the approval of CRPP program sub-projects.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
		generated, such as survey data.		Lao PDR 8,400 (Male: 4,200 Female: 4,200)	Lao PDR 21,000 (Male: 10,500 Female: 10,500)	A common methodology for identifying beneficiaries across CRPP sub-projects and indicators is included in the CRPP's Monitoring, Evaluation and Learning (MEL) Plan.
				Papua New Guinea 2,800 (Male: 1,400 Female: 1,400)	Papua New Guinea 7,000 (Male: 3,500 Female: 3,500)	
				Timor-Leste 8,400 (Male: 4,200 Female: 4,200)	Timor-Leste 21,000 (Male: 10,500 Female: 10,500)	
				Indirect: 575,400 (Male: 287,700 Female: 287,700)	Indirect: 1,438,500 (Male: 719,000 Female: 719,000)	
				Cambodia 546,000 (Male: 273,000 Female: 273,000)	Cambodia 1,365,000 (Male: 682,500 Female: 682,500)	
				Lao PDR 12,600 (Male: 6,300 Female: 6,300)	Lao PDR 31,500 (Male: 15,750 Female: 15,750)	
				Papua New Guinea 4,200 (Male: 2,100 Female: 2,100)	Papua New Guinea 10,500 (Male: Female:)	

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
				Timor-Leste 12,600 (Male: 6,300 Female: 6,300)	Timor-Leste 31,500 (Male: 15,750 Female: 15,750)	
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>	Measured through the Food Insecurity Experience Scale (FIES) methodology by undertaking a (representative sample) household survey.	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0) (Baseline value will be generated based on FIES survey results at the start of related projects.)	383,600 (Male: 191,800 Female: 191,800)	959,000 (Male: 479,500 Female: 479,500)	Rural communities and smallholder farmers are willing to apply and adopt new skills, technologies and practices that increase availability and access to safe and nutritious food, despite the impacts of climate change.
				Cambodia 364,000 (Male: 182,000 Female: 182,000) Lao PDR 8,400 (Male: 4,200 Female: 4,200) Papua New Guinea 2,800 (Male: 1,400 Female: 1,400) Timor-Leste 8,400 (Male: 4,200 Female: 4,200)	Cambodia 910,000 (Male: 455,000 Female: 455,000) Lao PDR 21,000 (Male: 10,500 Female: 10,500) Papua New Guinea 7,000 (Male: 3,500 Female: 3,500) Timor-Leste 21,000 (Male: 10,500 Female: 10,500)	Beneficiary households are targeted based on their food security status.
<u>ARA3 Infrastructure and built environment</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Detailed guidance will be provided to user departments for project-level reporting against this indicator as it will vary according to the project results framework. It will include at least one primary	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0)	Direct: 72,160 (Male: 36,080 Female: 36,080) Indonesia 60,000 (Male: 30,000 Female: 30,000)	Direct: 180,400 (Male: 90,200 Female: 90,200) Indonesia 150,000 (Male: 75,000 Female: 75,000)	There are sufficient pipeline of appropriate sub-projects that meet the CRPP eligibility criteria and align with ARA3. There are no significant delays with the approval

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
		source of verification which is not project generated, such as survey data.		Timor-Leste 12,000 (Male: 6,000 Female: 6,000) Vanuatu 160 (Male: 80 Female: 80) Indirect: 234,000 (Male: 117,000 Female: 117,000) Indonesia 180,000 (Male: 90,000 Female: 90,000) Timor-Leste 48,000 (Male: 24,000 Female: 24,000) Vanuatu 6,000 (Male: 3,000 Female: 3,000)	Timor-Leste 30,000 (Male: 15,000 Female: 15,000) Vanuatu 400 (Male: 200 Female: 200) Indirect: 585,000 (Male: 292,500 Female: 292,500) Indonesia 450,000 (Male: 225,000 Female: 225,000) Timor-Leste 120,000 (Male: 60,000 Female: 60,000) Vanuatu 15,000 (Male: 7,500 Female: 7,500)	of CRPP program sub-projects.
<u>ARA3 Infrastructure and built environment</u>	<u>Core 3: Value of physical assets made more resilient to the effects of climate change and/or more</u>	Actual costs incurred presented in project procurement reports.	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0	\$5 million – Flood and landslide protection	\$12 million – Flood and landslide protection	Skilled workers and appropriate materials are available. There are no significant procurement delays.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
	<u>able to reduce GHG emissions</u>		Female: 0)	\$1 million - multi-purpose emergency shelters	\$3 million - multi-purpose emergency shelters	Favourable climatic conditions support construction, impact of any extreme events is not severe.
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Detailed guidance will be provided to user departments for project-level reporting against this indicator as it will vary according to the project results framework. It will include at least one primary source of verification which is not project generated, such as survey data.	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0)	Direct: 203,200 (Male: 101,600 Female: 101,600) Lao PDR 8,000 (Male: 4,000 Female: 4,000) Pakistan 195,000 (Male: 97,500 Female: 97,500) Indirect: 800,000 (Male: 400,000 Female: 400,000) Lao PDR 320,000 (Male: 160,000 Female: 160,000) Pakistan 480,000 (Male: 240,000 Female: 240,000)	Direct: 508,000 (Male: 254,000 Female: 254,000) Lao PDR 20,000 (Male: 10,000 Female: 10,000) Pakistan 488,000 (Male: 244,000 Female: 244,000) Indirect: 2,000,000 (Male: 1,000,000 Female: 1,000,000) Lao PDR 800,000 Male: 472.500 Female: 472.500 Pakistan 1,200,000 (Male: 600,000 Female: 600,000)	There are sufficient pipeline of appropriate sub-projects that meet the CRPP eligibility criteria and align with ARA4. There are no significant delays with the approval of CRPP program sub-projects.

E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁷	
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Primary data from custom tracking tool to be designed under the program/ / primary data source from site surveys and remote sensing data.	Direct: 0 (Male: 0 Female: 0) Indirect: 0 (Male: 0 Female: 0)	50,000 Ha	106,000 Ha	A sufficient number of relevant sub-projects in the sector of climate resilient agroecological systems that meet the programs investment criteria are identified and approved. Households are willing to adopt CSA and agroforestry techniques.
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 4.1: Hectares of terrestrial forest, terrestrial non-forest, freshwater and coastal marine areas brought under resoration and/or improved ecosystems</u>	Primary data from custom tracking tool to be designed under the program/ primary data source from site surveys and remote sensing data.	0 ha	5,000 ha (Mangrove forest regeneration)	12,000 ha (Mangrove forest regeneration)	A sufficient number of relevant sub-projects with ecosystem-based adaptation that meet the programs investment criteria are identified and approved. Favourable climatic conditions support restoration, impact of extreme events is not severe. Ecosystem restoration techniques chosen are successful, with good operation and maintenance approaches.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	The institutional and regulatory framework for climate resilient development varies in CRPP countries and across key sectors within these countries. However, in general, there are currently very limited and weak institutional and regulatory frameworks in place that support financing of adaptation measures at the local level in CRPP countries, that are targeted at the poor , are responsive to their needs, and build their capacity to steer development in a resilient direction.	<u>low</u>	Institutional and regulatory frameworks in place in at least 3 CRPP countries that incentivize and facilitate the channelling of domestic and international climate finance for adaptation to the local level that explicitly targets poor and vulnerable people and is responsive to their needs. Institutional and regulatory frameworks in key sectors such as agriculture, social protection, urban planning, and natural resources and environment, incentivize the financing and implementation of locally lead, community level adaptation measures in at least 4 CRPP countries.	The CRPP IF component will invest in improving national climate information systems that can be used to inform government policy and programs such as decentralization processes and social protection to make them responsive to climate change and facilitate the channelling of domestic resources for adaptation at the local level. The CRPP TF can provide policy linked TA, leveraging loans from the CRPP IF to bring about institutional and regulatory improvements in key sectors to facilitate local level adaptation that is responsive to the needs of poor and vulnerable people, and builds their capacity for climate resilient development.	<u>Multi-countries</u>
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Knowledge and lessons learned on designing and implementing local level adaptation measures that are targeted at poor and vulnerable communities are not systematically documented and shared and institutional learning in this regard does not take place in CRPP countries.	<u>low</u>	Knowledge and lessons learned from the implementation of local level adaptation measures under the program, that explicitly benefit poor and vulnerable people, are systematically documented and used to inform the design and implementation of measures across	The CRPP multi-country programmatic approach and knowledge and learning strategy will facilitate the systematic documentation of lessons learned from the implementation of GCF investments and sharing of these within participating countries, across sectors and Asia and the Pacific region.	<u>Multi-countries</u>

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)					
Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
	In many CRPP countries, climate information is not systematically used in government programming, the formation of national and sectoral policies or development planning decision-making at the national or local level. Poor and vulnerable people, especially women, are often left out of formal decision-making processes related to adaptation and climate resilient sustainable development. As a result investments in climate adaptation are less likely to meet their specific needs and contribute to climate resilient sustainable development.		different districts, sectors and countries in the region. Communities and governments have the capacity to access, interpret and use climate risk information to inform policy, programming and planning decisions at the national and local level and to design and prioritize adaptation measures as part of climate resilient development processes. Poor people living in areas of high climate risk participate in adaptation decision-making processes.	GCF investments in climate information services under the CRPP IF component combined with institutional capacity building for decision-makers and the improvement of planning processes and methodologies will increase the availability and use of climate information in decision-making. Investments from the CRPP IF will target poor and vulnerable people and include activates to build their human capital and capacity to participate in local level decisions on adaptation and climate resilient development.	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Outcome 1: Improved enabling environment for knowledge, finance and implementation of local adaptation measures.	No. of CRPP countries that have new/improved national and local policies, plans, programs that integrate climate information for financing community-level adaptation.	Drafts of new/improved policies, plans, or programs.	0	2	5	Critical Assumptions Decision-makers are willing to make changes to policies, plans and programs based on climate information.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	No. of CRPP countries where poor and vulnerable people (women and men) report improved meaningful participation in adaptation-related decision-making (disaggregated by age, sex, physical/mental impairment, rural-urban)	Primary data from survey and focus groups.	0	2	7	Critical Assumptions Decision-makers appropriately use the tools and processes developed under the program to engage community members in prioritizing adaptation measures. Methodologies and minimum quality assurance criteria for countries to be counted under such indicators will be included in the CRPP MEL Plan.
Outcome 2: Pro-poor adaptation finance mobilized at scale.	At least \$668.5 million mobilized for pro-poor, gender-responsive community level investments in participating countries.	Government/ADB project pipeline list /CRFPF Annual Work Plans/Annual Government Expenditure Plans in Participating countries.	0	USD 267.4 million	USD 668.5 million	Critical Assumptions A sufficient number of appropriate large-scale investment projects for implementing adaptation measures at the local level are identified in the pipeline.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	At least 10 investment projects that promote building climate resilience at the community level in participating DMCs are identified.	Government/ADB project pipeline list /CRFPF Annual Work Plans/Annual Government Expenditure Plans in Participating Countries	0	5	10	
Output 1: Knowledge and action research on climate risk-informed pro-poor community level solutions strengthened.	1.1 No. of studies on gender-responsive, pro-poor transformative adaptation solutions carried out (disaggregated by theme women-focused/nature-based solutions, etc.)	1a. Project documents / Annual progress reports.	0	5	10	Critical Assumptions Stakeholders of influence are willing to participate in policy dialogues and evidence sharing workshops and collaborate and learn with other CRPP stakeholders. Researchers have access to sufficient primary and secondary data on gender-responsive, pro-poor transformative adaptation solutions.
	1.2 No. of participants from key stakeholder groups in CRPP countries who attend evidence-sharing events that have increased awareness on pro-poor adaptation solutions (at least 50% women participants)	1b. Project reports; annual progress reports	0	200	400	
	1.3 Number of high-level national and regional policy dialogue forums attended by key CRPP stakeholders	1c. Project reports; annual progress reports	0	6	10	
						Critical risks Facilitating communication and collaboration between key stakeholders within

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	(governments, civil society, donors, private sector) (2021 baseline = 0)					participating DMCs is time consuming and delays progress.
Output 2. Institutional and community capacity to develop and deliver climate adaptation investments at community level strengthened.	2.1 No. of participants from key stakeholder groups in CRPP countries, of which at least 50% are women, report improved skills or knowledge on developing, financing, implementing, and monitoring and evaluating adaptation investments (2021 baseline = 0)	2a. Project reports; annual progress reports	0	300	700	Critical Assumptions Capacity building interventions are attended by the right people and they are willing to participate. Climate risk assessments and other decision-support tools are approved by governments and become institutionalized within the planning process.
	2.2 No. of participating DMCs that have new/improved and operational participatory and gender-responsive climate risk assessment tools for community-level adaptation planning.	2b. Project reports; annual progress reports	0	2	4	Critical Risks The process of formal approval of new tools delays their institutionalization.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	2.3 Number of government officials, of which at least 50% are women, report improved skills or knowledge to apply climate information into plans, policies, and programs.	2c. Project reports; annual progress reports	0	100	300	
Output 3: Investments supporting inclusive, pro-poor community-level adaptation prepared	3.1 Number of climate adaptation investments approved (disaggregated by livelihood, green infrastructure, and climate information services focus).	Project reports; annual progress reports	0	5	15	Critical Assumption A sufficient number of appropriate large-scale investment projects for implementing adaptation measures at the local level are identified in the pipeline.
	3.2 % of projects that engage with communities in decision-making on supported adaptation projects.	Project reports; annual progress reports, (including scorecards)	0	100%	100%	Critical Risk Borrower governments fiscal burden may not allow for external borrowing, and / or governments may choose to invest in non-resilience projects with external financing.

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	3.3 Number of initiatives funded that include gender-responsive community resilience-building measures implemented.	Project reports; annual progress reports	0	5	15	
Output 4: Information and systems for delivering applied climate-risk informed investments at-scale improved	4.1 No. of CRPP IF sub-projects approved and/or implemented that include core activities identified under Output 4.	Document review/project reports; Project DMFs	0	2	4	Critical Assumptions A sufficient number of appropriate large-scale investment projects are identified that are suitable for integrating Output 4 core activities. There is political will to integrate CIS into social protection systems decision-making processes. Sufficient data and technological solutions
	4.2 Volume (USD) of investment allocated to Output 4.	Document review/project reports; Project DMFs; Project Budgets	0	47,200,000 USD (GCF Grant 8,000,000 USD ADB Grant 2,000,000 USD ADB Loan 37,200,000 USD)	118,000,000 USD (GCF Grant 20,000,000 USD, ADB Grant 5,000,000 USD ADB Loan 93,000,000 USD)	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
	4.3 Number of national frameworks for CIS set-up or improved, and that are gender responsive	Document Review	0	2	4	are available in related CRPP countries. Local development planners are willing to use newly accessible climate information in making decisions on local adaptation investments.
	4.4 No. of social protection systems in CRPP countries that are climate change responsive.	Document review	0	1	2	
	4.5 No. of CRPP countries where local development plans are formulated using participatory climate risk and vulnerability assessment information generated from the program.	Document review	0	1	2	
Output 5: Pro-poor livelihood adaptation investments implemented.	5.1 No. of CRPP IF sub-projects approved and implemented that include core activities identified under Output 5.	Document review/project reports; Project DMFs	0	2	6	A sufficient number of appropriate large-scale investment projects are identified that are suitable for integrating Output 5 core activities. Relevant investment projects are approved and implemented without significant delays.
	5.2 Volume (USD) of investment allocated to Output 5.	Document review/project reports; Project DMFs; Project Budgets	0	95,200,000 USD (GCF Grant 16,000,000 USD, ADB Grant 2,000,000 USD)	238,000,000 USD (GCF Grant 40,000,000 USD)	

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
				ADB Loan 77,200,000 USD)	ADB Grant 5,000,000 USD ADB Loan 193,000,000 USD)	
Output 6: Pro-poor climate adaptive-infrastructure investments implemented.	6.1 No. of CRPP IF sub-projects approved and implemented that include core activities identified under Output 6.	Document review/project reports; Project DMFs	0	2	5	A sufficient number of appropriate large-scale investment projects are identified that are suitable for integrating Output 6 core activities.
	6.2 Volume (USD) of investment committed to core activities under Output 6 as a result of the CRPP.	Document review/project reports; Project DMFs; Project Budgets	0	124,800,000 USD (GCF Grant 18,000,000 USD, GCF Loan 6,000,000 USD ADB Grant 4,000,000 USD ADB Loan 97,000,000 USD).	312,500,000 USD (GCF Grant 40,000,000 USD, GCF Loan 20,000,000 USD ADB Grant 10,000,000 USD ADB Loan 242,500,000 USD).	Relevant investment projects are approved and implemented without significant delays.
Project/programme co-benefit indicators						
Co-benefit 1: Women's participation in decision-making and leadership enhanced.	Number of CRPP countries with new/improved institutional mechanisms in place for meaningful participation of women in adaptation-related decision-making.	Secondary data collection / government policy and planning documents	0	1	3	Participating governments are willing for TA for policy reform to be attached to loans. Changes to planning and decision-making processes are accepted

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
						and adopted by government institutions.
	Proportion of entities (public, private or social) implementing CRPP country-level activities that are women-led.	Secondary data collection / procurement documents, partnership agreements etc.	0	10%	15%	There are a sufficient number of women-led entities applying in a competitive bidding process.
Co-benefit 2: Inequalities are reduced.	No. of people with increased access to social protection as a result of the CRPP implementation.	Secondary data collection / Government social protection database	0	1,500,000	2,800,000	Integrating climate information into social protection programming and decision-making results in the expansion of such programs reaching new beneficiaries.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Output 1. Knowledge and action research on climate risk-informed pro-poor local solutions strengthened.			
Activity 1.1 Knowledge and evidence generation.	This activity will focus on carrying out research studies and analysis to build an evidence base for pursuing climate risk-informed poverty reduction strategies and programs in education, health, social protection, financial inclusion, and livelihoods-related sectors. It will do so in the context of transformational adaptation investments that meet local needs and are aligned with national climate priorities.	Sub-activity 1.1.1: Undertake macro level climate risk and vulnerability assessments in the context of poverty reduction strategies and programs. Sub-activity 1.1.2. Carry out analysis to identify gaps and barriers in designing and implementing climate risk informed national poverty reduction policies and programs that responds to the resilience-related needs of the poor and vulnerable.	– 4 macro level climate risk and vulnerability assessment reports. – 10 research reports documenting findings from research studies.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
		Sub-activity 1.1.3. Identify and document transformational community-level climate adaptation measures implemented under major government programs and by civil society organizations. Sub-activity 1.1.4. Document lessons, including lessons emerging from the COVID-19 response, which are critical for building future resilience, and for sharing experiences of using innovative financing options for adaptation, including private sector financing.	
Activity 1.2 Evidence sharing and awareness raising	This activity will focus on disseminating the findings from activity 1.1 and using this evidence base to generate political buy in for investments in local level pro-poor climate adaptation. It will also bring together different stakeholders in dialogue to share their experiences and build collaboration.	Sub-activity 1.2.1. Organize workshops, seminars, or other evidence-sharing events on pro-poor adaptation solutions. Sub-activity 1.2.2. Conduct national- and regional-level policy dialogues to influence the wider enabling environment for planning and programming to support investments in local adaptation solutions, including through participation of the private sector. Sub-activity 1.2.3 Engage with potential financing partners (national development banks, multilateral development banks, climate finance institutions, bilateral donors, and private sector) to mobilize adaptation	– 10 reports from workshops, seminars, or other evidence-sharing events including participant evaluations. – 10 evaluation reports from policy dialogues.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
		finance, including strengthening linkages with the Green Climate Fund's Enhancing Direct Access pilot.	
Output 2: Institutional and community capacity to develop and deliver climate adaptation investments strengthened.			
Activity 2.1 Development of climate risk assessment and decision support tools.	This activity focuses on developing participatory climate risk and gender, poverty, and other related assessment tools, and supporting their institutionalisation in local adaptation decision-making processes, ensuring inclusiveness and participation of poor and vulnerable people.	Activity 2.1.1 Design relevant tools using participatory and community driven processes in 3 CRPP countries. Activity 2.1.2 Pilot tools in 3 CRPP countries. Activity 2.1.3 Integrate the tools into local development planning process. Activity 2.1.4 Support government officials to develop local development plans.	– Climate risk assessment and decision support tools developed in at least 2 countries. – Local development plans informed by climate risk assessments in at least 2 CRPP countries.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 2.2. Training and capacity-building on designing and implementing adaptation investments at the community level.	This activity focuses on implementing training and capacity-building workshops on developing, financing, implementing, and monitoring and evaluating adaptation investments at the community level, among local government and civil society organizations, as well as at the national level and with the private sector.	Activity 2.2.1 Implement training and capacity building for government officials on applying climate risk information in the formation of plans, policies, and programs. Activity 2.2.2 Implement training and capacity building for local government and civil society organisations for monitoring and evaluating adaptation investments	– 5 Capacity building material packages for training workshops. – 20 training workshop reports presenting outcomes and evaluation of the trainings.
Activity 2.3 Capacity building to access international climate finance for adaptation projects.	This activity will focus on building capacity within national and local institutions and civil society organisations, including the private sector, to access international climate finance for adaptation projects, including linkage with Enhanced Direct Access pilot, where appropriate. This capacity building will also build on existing GCF readiness and National Adaptation Plan related activities.	Activity 2.3.1 Implement trainings and capacity building for government officials, civil society organisations, and the private sector, to access international climate finance for adaptation projects.	– 5 Capacity building material packages for training workshops. – – 20 training workshop reports presenting outcomes and evaluation of the trainings.
Output 3: Inclusive and pro-poor adaptation investment projects identified and prepared.			

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 3.1. Preparation of large-scale adaptation investment projects	This activity will focus on providing technical assistance to governments for the design and preparation of large-scale bankable adaptation projects, identified and prioritized under outputs 1 and 2, that are aligned with national adaptation priorities and address the nexus between climate and poverty, with special emphasis on women.	3.1.1 Carry out feasibility studies and other preparatory work including detailed climate risk assessments. 3.1.2 Engage with potential financing partners (national development banks, multi-lateral development banks, climate finance institutions, bilateral donors, and the private sector) to mobilize adaptation finance.	– At least 15 detailed feasibility studies including climate risk assessments reports. – At least 15 consultation reports from engagements with potential financing partners. – At least 15 approved investment projects.
Activity 3.2. Pilot innovative transformational solutions	This activity will focus on providing grant financing and implementation support to governments and communities on a selective basis to test innovative adaptation solutions which can later be scaled up through large-scale investment projects.	Activity 3.2.1 Conduct needs assessment and feasibility analysis to identify potential solutions and localities (link with Output 1). Activity 3.2.2 Implementation of grants Activity 3.2.3 Evaluation of pilot projects and recommendations for scaling-up	– 6 technical reports documenting the implementation and evaluation of innovative climate adaptation solutions and practices implemented through small grants which can be scaled up through national government programs on poverty reduction.
Output 4: Information and systems for delivering applied climate-risk informed investments at-scale improved			

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 4.1. Develop and strengthen climate information services for key sectors to meet local adaptation needs.	This activity will focus on developing climate information services and early warning systems in selected CRPP countries to ensure that they meet local adaptation needs and are gender responsive. Sub-activities will be implemented in line with global standards and internationally recognized frameworks.	Sub-activity 4.1.1: Develop and strengthen national frameworks for climate services. Sub-activity 4.1.2: Improve multi-hazard early warning systems for the water and agriculture sector to support impact-based forecasting, especially in the context of agriculture and water resources sectors. Sub-activity 4.1.3: Adapt and scale up of gender-responsive climate information services and demonstrate its application to manage farm level climate risk.	– National frameworks for CIS set-up or improved in at least 1 CRPP country. – Multi-hazard early warning systems set-up or improved to include impact-based forecasting in agriculture and water sector in at least 1 CRPP country. – Gender-responsive climate information mainstreamed into climate information and early warning mechanisms in at least 1 CRPP country.
Activity 4.2 Climate risk informed social protection systems.	Make social protection climate change responsive by improving early warning systems and integrating them into adaptive and shock responsive social protection systems. A holistic approach to adaptive social protection will be taken following the four building blocks identified by the World Bank, namely, programs, data and information, finance, and institutional arrangement and partnerships.	Sub-activity 4.2.1 Develop/strengthen national policy frameworks and regulations on adaptive and shock responsive social protection. Sub-activity 4.2.2 Installation of hardware and software for hazard, exposure and vulnerability data collection and analysis and integrate it in national social protection data systems. Sub-activity 4.2.3 Provide trainings in undertaking climate risk-informed targeting (including community-based	– National policy frameworks and regulations on adaptive and shock responsive social protection established/strengthened in at least 1 CRPP country. – National level climate risk information databases and data gathering services installed and integrated into national social protection systems in at least 1 CRPP country. – Trainings for targeting Social Protection Program beneficiaries using climate risk information

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
		targeting) of social protection program beneficiaries	implemented in at least 1 CRPP country.
		Sub-activity 4.2.4 Installation of early warning equipment (hardware and software)	– Early warning systems hardware and software installed as appropriate in at least 1 CRPP country.
		Sub-activity 4.2.5 Trainings on modelling, forecasting and early warning capacity, which can support forecast based financing through social protection programs	– Training of officials in forecast-based financing through social protection systems in at least 1 CRPP country.
		Sub-activity 4.2.6 Provision of ICT infrastructure that can be deployed in social protection programs to facilitate transfer of assistance in the aftermath of extreme events in high-risk areas;	– ICT infrastructure that supports the transfer of financial assistance in extreme climate events installed in at least 1 CRPP country.
		Sub-activity 4.2.7 Provide trainings for social protection field workers to build their capacity on disaster preparedness measures;	– Trainings for social protection field workers to build their capacity on disaster preparedness measures implemented in at least 1 CRPP country.
			– Trainings and awareness raising for social protection program beneficiaries, especially women, on climate-resilient livelihoods implemented in at least 1 CRPP country.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 4.3 Climate risk information systems for decentralization policy.	This activity area will support national governments in the development of the policy, decision-making framework (protocols and processes) and tools, for the integration of climate risk information into decentralized development finance allocation, strengthening decentralization policies and frameworks from climate risk perspective based on experiences of the United Nations Capital Development Fund (UNCDF) in introducing performance-based climate resilience grant mechanism under its Local Climate Adaptive Living Facility (LoCAL).	Sub-activity 4.3.1 Prepare climate risk and vulnerability assessments. Sub-activity 4.3.2 Conduct trainings, workshops and consultations to build the capacity of local governments in using the results of climate risk and vulnerability assessments for local adaptation planning and financing. Sub-activity 4.3.3 Improve methodologies and processes for local development planning and budgeting so that they include climate risk considerations. Sub-activity 4.3.4 Strengthen public participation processes, especially to engage local women in decision-making related to resilience building; Sub-activity 4.3.5 Implement adaptation measures that have been prioritized in local adaptation/development plans through the provision of small-scale grants to grassroots women's organizations.	– Participatory climate risk and vulnerability assessments undertaken in at least 1 CRPP country. – Trainings/workshops implemented for local governments in climate risk informed planning and budgeting in at least 1 CRPP country. – Planning guidelines and written protocols and policies include climate risk considerations in at least 1 CRPP country. – Local planning guidelines and policies in at least 1 CRPP country include provisions for local consultations and engagement of women. – Local adaptation measures implemented by grass roots women's organizations in at least 1 CRPP country.
Output 5: Pro-poor livelihood adaptation investments implemented.			

Activity 5.1 Climate resilient agroecological systems.	This activity area will focus on preserving and increasing natural capital and building human capital, as identified in the SLF, through supporting knowledge and skills generation in relation to improved climate resilient agricultural production practices.	Sub-activity 5.1.1 Provide agriculture extension services, especially to women farmers, for new climate resilient crop varieties and production systems Sub-activity 5.1.2 Support the adoption of agro-forestry through developing the skills required to establish and manage an agroforestry system (selection of trees and crops, marketing of products, creating an enabling environment); Sub-activity 5.1.3 Support national institutions to offer climate resilient employment through improved technical and vocational education and training (TVET), including trainings targeted at women employed in agriculture. Sub-activity 5.1.4 Transfer of small grants for adapting and scaling up water retention, communal water harvesting and storage infrastructure; Sub-activity 5.1.5 Transfer of small grants for improvement of land and soil management techniques such as terracing and minimum/no tillage. Sub-activity 5.1.6 Design and implement skills development and training courses on integrated water management and land and soil management practices.	– Agriculture extension services for new climate resilient crop varieties and production systems provided in at least 2 CRPP countries. – Households receive small grants for implementing new diversified cultivation practices in at least 2 CRPP countries. – Skills development program for agro-forestry implemented in at least 1 CRPP country. – TVET training programs in at least 1 CRPP country offer courses, particularly targeted at women, that support climate resilient employment. – Small-scale community-based water management systems installed in at least 1 CRPP country. – New, climate resilient, soil management techniques adopted in at least 2 CRPP countries. – Trainings on new integrated land and water management practices implemented in at least 2 CRPP countries.
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E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 5.2 Information technology and risk management services for climate resilient livelihoods.	This activity area will focus on building human and financial capital, as identified in the SLF, through improving access to digital information services and low-cost risk transfer mechanisms.	Sub-activity 5.2.1 Invest in information technology systems to facilitate rapid information flows to small-holder farmers. Sub-activity 5.2.2 Develop low cost risk transfer markets and mechanisms for small-holder farmers. Sub-activity 5.2.3 Design and deliver capacity building programs on financial literacy for insurance consumers Sub-activity 5.2.4 Design and deliver trainings and awareness raising for social protection program beneficiaries, especially women, on climate-resilient livelihoods.	– New/improved ICT systems installed in at least 1 CRPP country. – New markets for low-cost weather indexed micro-insurance created in at least 1 CRPP country. – Financial literacy trainings carried out in at least 1 CRPP country. – Training for social protection beneficiaries carried out in at least 1 CRPP country.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 5.3 Climate resilient agriculture supply chains.	This activity area will build human capital and physical capital, as identified in the SLF, to deal with climate impacts along agricultural supply chains.	Sub-activity 5.3.1 Install protective measures to the siting and storage of seeds, tools, vehicles and energy infrastructure as well as installation of new or improved post-harvest storage and processing facilities. Sub-activity 5.3.2 Install and/or improve transport and logistics infrastructure that ensures products reach markets despite the impact of extreme weather events due to climate change; Sub-activity 5.3.3 Install technological solutions for improved food processing strategies that can help increase the value and durability of perishable foods such as fruits and vegetables.	– Protective measures installed in at least 1 CRPP country. – Improved, climate risk informed, transport and logistics infrastructure installed in at least 1 CRPP country. – Improved food processing hardware and software installed in at least 1 CRPP country. – Capacity building activities carried out in at least 1 CRPP country.
Output 6: Pro-poor climate adaptive-infrastructure investments implemented			

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 6.1: Ecosystem-based infrastructure	This activity area will support the identification and installation of locally appropriate green infrastructure to reduce exposure and vulnerability of climate impacts on lives and livelihoods of poor and vulnerable communities. The implementation of the activities will be guided by the IUCN Global Standards for Nature-based Solutions.	Sub-activity 6.1.1 Identify locally appropriate green infrastructure in collaboration with national and local governments and national scientific organizations. Sub-activity 6.1.2 Undertake trainings for local governments, communities and community-based organizations in planting, rehabilitation and maintaining green infrastructure; Sub-activity 6.1.3 Provide small grants to local governments/communities (especially women's groups) for planting and rehabilitation of green infrastructure; Sub-activity 6.1.4 Develop policy, legal frameworks and financial models for scaling up investments in green infrastructure in partnerships with private sector.	– Consultations/workshops on locally appropriate green infrastructure undertaken in at least 3 CRPP countries. – Trainings for local governments, communities and community-based organizations in planting, rehabilitation and maintaining green infrastructure undertaken in at least 3 CRPP countries. – Small grants provided to local governments/communities for green infrastructure in at least 3 CRPP countries – Policy, legal frameworks and financial models for scaling up investments in green infrastructure in partnerships with private sector developed in at least 3 CRPP countries.

E.6. Project/programme activities and deliverables			
Activities	Description	Sub-activities	Deliverables
Activity 6.2 Flood and landslide protection infrastructure	This activity area will support local governments to identify and implement small scale flood and landslide protective infrastructure, especially measures that use a combination of grey and green infrastructure and are prioritized by local communities.	Sub-activity 6.2.1 Identify and prioritize small-scale protective infrastructure as part of local government planning processes. Sub-activity 6.2.2 Undertake civil works for small scale flood protection and/or landslide protection measures, especially measures that use a combination of grey and green infrastructure. Sub-activity 6.2.3 Undertake trainings for local governments and communities, especially women, in maintenance of protective infrastructure.	– Small scale protective infrastructure installed in at least 2 CRPP countries. – Trainings for local governments and communities, especially women, in maintenance of protective infrastructure implemented in at least 2 CRPP countries.

E.6. Project/programme activities and deliverables

Activities	Description	Sub-activities	Deliverables
Activity 6.3 Multi-purpose emergency shelters	This activity area will support local governments and communities to identify and construct multi-purpose emergency shelters with gender inclusive and socially appropriate design features complemented by community-based disaster preparedness training.	Sub-activity 6.3.1 Identify the need for multipurpose emergency shelters through local participatory planning processes. Sub-activity 6.3.2 Construct emergency shelters with gender inclusive and socially appropriate design features Sub-activity 6.3.3 Undertake trainings for local governments and communities in planning and maintenance of multipurpose emergency shelters; Sub-activity 6.3.4 Undertake capacity building for local communities on disaster preparedness including evacuation planning.	– Multi-purpose emergency shelters constructed in at least 1 CRPP countries. – Trainings on the maintenance of the multi-purpose emergency shelters implemented in at least 1 CRPP countries. – Community disaster preparedness trainings implemented in at least 1 CRPP countries. – Community evacuation plans formulated in at least 1 CRPP countries.

E.7. Monitoring, reporting and evaluation arrangements

Monitoring, Evaluation and Learning (MEL) are considered cornerstones of the CRPP, and the final detailed MEL framework and implementation plan will reflect this commitment, not only to closely monitoring the progress of the individual sub-projects and the program as a whole according to the results framework, but to using the CRPP as an opportunity to advance the collective understanding of how to design and implement effective pro-poor and gender sensitive adaptation solutions. The principals that have informed the general approach to MEL under the CRPP and will guide the development of the detailed MEL framework and implementation plan are based on a detailed study undertaken as part of the CRPP scoping activities. A detailed review of large-scale climate adaptation financing mechanisms was carried out, and best practice and lessons learned were consolidated into a set of general recommendations to guide MEL activities under the CRPP. The arrangement for monitoring, reporting and evaluation are presented in more detail in **Annex 11**.

Guiding principles: The key principles that will guide the MEL activities include;

Participation: The procedures for monitoring will involve communities and local stakeholders, including civil society organizations, at all stages of the project/program cycle from the beginning. This will not only build the capacity of local actors in monitoring and reporting techniques, but create buy-in to project outputs (thus supporting sustainability) and foster learning and awareness raising in terms of climate change adaptation. Evaluations will be planned and conducted in ways that enhance the likely utilization of both the findings and of the process itself to inform decisions and improve performance. Key to promoting the maximum utility of the evaluation outputs, will be the engagement of the primary user groups. Under the CRPP IF, the primary user groups are the Executing Entity (EE), the National Designated Authority (NDA), ADB and GCF.

Accountability vs Learning: MEL activities under the CRPP must strike an appropriate balance between reporting against quantitative indicators that tell us what milestones are being achieved, and a more qualitative and reflective approach that seeks to answer questions about how and why these milestones are / are not being achieved.

Practicality: The CRPP MEL activities will have to balance and overcome multiple and competing demands, risks and constraints of monitoring and evaluation of such a large program. The usual trade-offs between cost, reporting burden, and a desire for methodological rigor and comprehensive learning are acknowledged, as is the need to accept reality on the ground in the project countries. These factors are to be considered in the development of the detailed MEL plan and managed accordingly.

MEL Approach: The following points identify two of the key pillars of the CRPP MEL approach (further discussion on the CRPP MEL approach is included in annex 11).

A multi-level approach: Monitoring and evaluation under the CRPP will be carried out at two levels; project level, and program level. Despite the differences in scope of M&E at different levels, in order to achieve a sufficient level of evidence generation for reflection and learning for the benefit of both projects and the program, the overarching fund level M&E framework and plan will be designed in conjunction with the project-level frameworks and be guided by a common vision of the program across all levels of implementation. This will greatly facilitate the ability to carry out coherent, coordinated MEL activities throughout the program. A multi-level approach which seeks the alignment of results frameworks between the project, CRPP program, and GCF fund level is considered to be especially critical to generating comprehensive and consistent data that can be used to measure the contribution of the program to transformative change and the assessment of key assumptions that provide the logic for activities.

A flexible and adaptive approach: Responding and adapting to changing contexts and the evolution of knowledge and understanding is a key enabling process for building resilience, and thus a learning approach may also involve adjusting the processes, results framework and indicators, of the M&E framework. Experience over time will inform what works and does not work well, and periodic updates to all major components of the M&E system should be expected. Thus, the MEL frameworks need to allow for continual adjustments based on the emerging body of knowledge and experience regarding the monitoring and measuring of resilience. This means that the MEL framework should not be static, rather it should be tailored to the implementation stage by, for example, evolving towards asking broader evaluation questions as the program moves through its lifecycle.

Implementation Arrangements: A CRPP MEL Expert will be part of the CRPP secretariat and will be responsible for the coordination and management of all MEL activities under the CRPP and ensuring independent and impartial review of the IF project activities and the CRPP program. Full roles and responsibilities of the MEL expert will be established as part of the initial role out of the CRPP.

MEL Implementation Guidelines: At the CRPP inception, using the CRPP Trust Fund incubation resources, a MEL inception workshop, involving the CRPP partnership group members, NDAs and EE's, will be held to raise awareness on MEL and promote ownership of the CRPP MEL activities, processes and future outcomes. The discussions from the workshop will feed into the development of the CRPP MEL plan and guidelines which will provide specific and practical steps on how to align CRPP MEL requirements with existing ADB and other CRPP partner's processes and a monitoring system will be established to collect and collate information from projects. This integrated approach will minimize reporting burdens, while ensuring that the CRPP results framework and GCF reporting requirements are adequately covered.

Establishing the baseline: Baselines and targets for the sub-project M&E frameworks will be established. In addition to the related CRPP program indicators, the baseline assessments will include the articulation of qualitative and quantitative baselines and targets for paradigm shift and enabling environment indicators as required for the GCF IRMF. Qualitative baselines will establish a narrative which articulates the current context in which the project will be working and target scenarios hypothesising the paradigm shift potential and change to the enabling environment that the project will support, including how the project will specifically contribute to that change. This will be aligned with the overall program narrative, but with specific details related to the country and project context. The coverage, or unit of analysis, for this baseline exercise will depend on the country and the type of sub-project proposed in the country as per the IRMF guidelines. Quantitative baselines will be developed for each paradigm shift assessment dimension and enabling environment indicators using the three-point scaling rating in the corresponding scorecard.

Monitoring and Reporting Arrangements: Sub-project level monitoring and reporting under the CRPP will combine ADB policies and protocols with GCF criteria to ensure that CRPP results framework is complementary to existing requirements, without adding unnecessary extra reporting burden. ADB project monitoring systems will be followed to track implementation progress. Standard ADB policies, during the loan implementation period, require the executing / implementing agency of individual projects prepare:

1. Semi-annual progress reports which provide:
 - a. progress achieved by output as measured through the indicator's performance targets in the design and monitoring framework
 - b. key implementation issues and solutions and
 - c. updated implementation plan;
2. Consolidated audited and entity financial statements;
3. Semi-annual environmental and social monitoring reports, including subprojects' environmental management plans, gender action plans; and
4. Project Completion Report within 3 months of physical completion of the project.

In line with GCF M&E policy, annual performance reports (APRs) for the CRPP will be undertaken by the project management units under the guidance and supervision of the CRPP MEL Expert. The project level APRs will include a narrative report (with supporting data as needed) on

implementation progress based on the CRPP results framework (which explicitly covers the GCF results framework) and considerations on the ongoing performance of the project/program against the GCF investment framework criteria, including updates on the indicator and a report on ESS as well as gender. Reporting requirements on ESS in the APR will be consistent with the GCF environmental and social management framework. In order to ensure participatory monitoring, an annual participatory review for local stakeholders, notably project affected people and communities, including women and civil society organizations will be organized by the NDA or focal point as part of the annual project reporting process. Based on the project reports, ADB will prepare program level CRPP annual reports detailing a) activities conducted during the year, and priorities for the next year b) potential issues and solutions c) progress against targets and indicators given in this proposal.

Evaluative Monitoring: Evaluative monitoring is an approach that will be used to bridge the gap between traditional monitoring and evaluation timeframes, to understand how and why change is happening. An evaluative approach to monitoring goes beyond the typical focus on tracking progress against milestones and is designed with the aim of understanding how resilience is being built. It is a complementary approach to traditional monitoring, that will be included in order to support annual program-level analysis and synthesis of all projects progress against the ToC pathways. In practice this will require integrating an evaluative lens into monitoring processes so that key evidence, findings and lessons can be generated as projects are implemented.⁸

Evaluation Arrangements

The evaluation arrangements will consist of mid-term reviews and final evaluation reports for each sub-project with investment from GCF as well as the CRPP program as a whole. As with monitoring and reporting, the CRPP evaluation plan will align GCF policies, especially the GCF Evaluation Policy, with ADB policies and protocols to ensure that there is a coordinated and coherent approach to evaluation, that maximizes learning and accountability, but is efficient in its allocation of resources, and does not overburden project management and other key stakeholders.

Mid-term review (projects): The CRPP MEL Expert will be responsible for overseeing the Mid-Term Review (MTR) and the Terminal Evaluation (TE) of the GCF funded activities. The MEL Expert, in coordination with the project management units, will oversee the process of hiring external, independent entities or consultants to carry out the MTR, which will provide an assessment of project performance at the project's mid-point. This will be a formative exercise and will include analyzing whether the project is on track, what problems and challenges the project is encountering, and what corrective actions are required so that the project can achieve its intended outcomes by project completion in the most efficient and sustainable way.

Mid-term review (Program): ADB will conduct a mid-term review of the program in the 5th year of the program implementation to take stock of results and implementation of the first phase of investments under the program. This program-level evaluation will be undertaken by an independent entity and coordinated by the MEL Expert, which will assess the performance of the program against the GCF evaluation criteria, including financial/economic performances as part of the program efficiency and effectiveness criterion. The evaluation will determine progress being made toward the achievement of outcomes and outputs and will identify course correction if needed. It will include a range of appropriate quantitative and

⁸ Evaluative Monitoring is one of a number of innovative approaches to M&E that was used extensively in the UK Government funded Building Resilience and Adaptation to Climate Extremes and Disasters (BRACED) program and offers significant transferable potential to the CRPP M&E framework.

qualitative research methods such as surveys, and key informant interviews, to evaluate the program's impacts on resilience, human well-being, and other key indicators relevant to the result framework at the program levels. Recommendations of the mid-term review will be incorporated into the design of the program to secure maximum results and sustainability during subsequent program implementation.

Terminal Evaluations (projects): A Terminal Evaluation (TE) will take place at the end of each sub-project implementation period, according to ADB and GCF established policies. ADB's Independent Evaluation Office will be responsible for assuring the quality and independence of the TE at the end of project implementation, which is a summative evaluation, and will liaise with the CRPP MEL Expert throughout the process. An independent assessment of project performance against standard evaluation criteria (e.g. strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability) will be made based on documentary evidence, stakeholder interviews and, in most cases, a field mission. . The draft TE report will be sent to project stakeholders during a commenting process managed by the Evaluation Office and the CRPP MEL Expert. Formal comments on the report will be shared in an open and transparent manner. The evaluation reports will be publicly disclosed and will be followed by a recommendation compliance process.

Terminal Evaluation (Program): An independent Terminal Evaluation (TE) will take place at the end of programme implementation according to ADB and GCF established policies, especially the Evaluation Policy. The TE will be an assessment of programme performance against standard evaluation criteria (e.g., strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability). The TE will use the paradigm shift and enabling environment scorecards to report against the targets identified in the baseline. The draft TE report will be shared with programme stakeholders for comment and subsequently disclosed publicly.

Knowledge management

The implementation of systematic MEL activities in the CRPP will generate information that will advance understanding, not only of how countries can respond to the challenge of climate change, but also organizational performance and the relevance, efficiency and effectiveness of the approach taken for the management and implementation of a large-scale climate adaptation fund. The CRPP will systematically record and disseminate lessons learned in terms of its programmatic approach towards MEL, with their reflection papers and critical analysis providing a wealth of information that can be used to continuously improve MEL practice in the context of resilience. In order to ensure that lessons learned from the CRPP have maximum utility and contribute to inward and outward learning on climate resilience programming, a knowledge management strategy will be developed as part of the roll-out of the CRPP and will include a clearly defined work program and accompanying budget.

RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

The level of key risks that will affect the CRPP performance are generally low to medium and expected to be mitigated with the ADB's established rules and procedures. Key risks identified are related to implementation delays due to COVID-19, changing government priorities, compliance to GCF and ADB procurement rules and policies and environmental and social risks.

The CRPP builds on over 11 years of adaptation programming at scale, including 8 years of implementing a systematic climate risk management framework in ADB, and over 50 years of development finance operations. ADB will use existing institutional arrangements, thus there a high confidence on the project delivery and success. For these reasons, the risks to the project implementation and execution are considered low. ADB has existing and highly rigorous protocols and guidance for financial and operational risk and these will be applied in the project. ADB's Safeguards Policy Statement of 2009 that sets out the policy objectives for key safeguards areas (namely environmental safeguards, involuntary resettlement safeguards, and Indigenous Peoples safeguards) is in place and its effectiveness continually being assessed and enhanced to better manage the environmental and social risks arising from projects and in ensuring their social and environmental sustainability.

Selected Risk Factor 1 Implementation delays due to COVID-19

Category	Probability	Impact
<u>Technical and operational</u>	<u>Medium</u>	<u>Medium</u>

Description

Projects face implementation delays due to COVID-19 related lockdowns and travel restrictions

Mitigation Measure(s)

COVID 19 has caused delays in the preparation and implementation of many ADB projects. It has affected schedules for feasibility studies, implementation of detailed designs, and initiating construction. In order to keep the operations moving, ADB has adjusted its approaches and strategies to respond to the situation such as use of online consultation meetings, and maximize engagement of available local experts to do sections of feasibility studies that require ground work.

Selected Risk Factor 2 Lack of prioritization by governments for projects

Category	Probability	Impact
<u>Governance</u>	<u>Low</u>	<u>Medium</u>

Description

Borrower governments fiscal burden may not allow for external borrowing, and / or governments may choose to invest in non-resilience projects with external financing.

Mitigation Measure(s)

The program will work closely with the governments to promote country ownership and help them understand and prioritize resilience projects. This can be done through supporting policy dialogues to influence for transformational planning and programming and investing in local adaptation, and targeted technical assistance which aims to strengthen institutions and capacity to access funding, prepare and implement adaptation projects. The potential and opportunities for resilience financing for COVID-19 recovery will also be raised with governments during ADB country programming discussions, as part of project origination and project identification processes.

Selected Risk Factor 3 Procurement (including local project procurement)

Category	Probability	Impact
<u>Technical and operational</u>	<u>Low</u>	<u>Low</u>

Description		
Lack of understanding and compliance with ADB procurement rules; Limited capacity in resolving procurement related issues		
Mitigation Measure(s)		
All procurement to be financed under the program must be carried out in accordance with ADB's procurement policy and regulations. As required, and on a project-by-project basis, ADB will support procurement processes and offer training from procurement planning to contract management. Support will also be provided to implementing agencies in the completion and monitoring of procurement related documents such as procurement plans, bidding documents, requests for quotation, contract monitoring forms, among others. Such measures will lower the probability of the occurrence of procurement related risks.		
Selected Risk Factor 4 Project level: ESS and GESI alignment		
Category	Probability	Impact
Technical and operational	Low	Low
Description		
Beneficiaries' limited capacity to comply with national regulations and/or ADB and GCF environmental and social policy requirements and gender and inclusion requirements		
Mitigation Measure(s)		
To ensure that environmental and social risks and impacts of CRPP IF supported applications are managed, an Environmental and Social Management Framework (ESMF) is in place that will apply to any project that utilizes GCF funds through the program. The ESMF will be implemented and maintained by CRPP IF as part of its risk assessment process and environmental and social safeguards compliance undertaking in order to meet ADB safeguards requirements as per ADB SPS, GCF's Environmental and Social Policy, Good International Industry Practices and national and local laws of recipient governments. The policies, set of procedures and designated staff responsible for implementation detailed in the ESMF will guide CRPP-IF in managing its overall exposure to environmental and social risks of its current and future portfolio of adaptation investments. The set of procedures integrated in the ESMF include screening and categorization, conduct of due diligence, monitoring and managing non-compliance. Furthermore, both ADB SPS and GCF's ESS standards recognize stakeholder engagement and disclosure, among others, as compliance benchmarks on ensuring broad multi-stakeholder support and participation throughout project lifecycle and strengthening beneficiaries' capacity for compliance with national and local regulations and ADB and GCF environmental and social policy requirements and gender and inclusion requirements which will be integrated in the CRPP IF ESMF.		
Selected Risk Factor 5 Financial		
Category	Probability	Impact
Credit	Low	Low
Description		
Lack of ability of borrowers to meet repayment of debts from the program		
Mitigation Measure(s)		
ADB's comprehensive due diligence and technical support will mitigate the risk that the borrower may have related to inability to service the loan due to insufficient net operating cash flow to meet the debt service. In addition, all loans from the program will be aligned with ADB sovereign loans and thus be covered by the similar sovereign guarantees. As such the credit or loan default risk on repayment of program loans is low and related only to risks of the sovereign itself.		

Selected Risk Factor 6 Money laundering, terrorist financing and prohibited practices		
Category	Probability	Impact
Prohibited practices	Low	Low
Description		
GCF financing through CRPP IF results in money laundering, terrorist financing or other prohibited practices.		
Mitigation Measure(s)		
GCF financing will be provided as loans and/or grants to sovereign entities. As GCF funds will be provided alongside co-financing from ADB, ADB anticorruption and integrity policies and required due diligence will apply at project level, which will mitigate this risk. ADB conducts integrity due diligence (IDD) on nongovernment counterparts or recipients of ADB financing, to assess any previous involvement in integrity-related or money-laundering issues. In addition, ADB conducts proactive integrity reviews on ongoing sovereign projects to mitigate integrity risks. ADB's Office of Anticorruption and Integrity (OAI) through its Due Diligence Unit, is tasked to (i) advance awareness of ADB's anticorruption and AML/CFT policies in collaboration with other departments; (ii) act as focal point for and provide independent advice to Management and concerned departments on significant integrity or money laundering or financing of terrorism (ML/FT) concerns, including investigations as required to facilitate ADB's ability to assess significant integrity or ML/FT concerns; and (iii) provide oversight and guidance to concerned departments on integrity and ML/FT risks, including AML/CFT internal controls, and integrity due diligence, which encompasses ML/FT checks. Furthermore, all projects must comply with ADB's Anticorruption Policy (1998, as amended to date) and acknowledge that ADB reserves the right to investigate directly, or through its agents, any alleged corrupt, fraudulent, collusive or coercive practice relating to the projects; and recipients of the grant/loan shall cooperate with any such investigation and extend all necessary assistance for satisfactory completion of such investigation. In terms of financial management assessment, in accordance with ADB's Charter, all projects must take necessary measures to ensure that the proceeds of any loan made, guaranteed or participated in by ADB are used only for the purposes for which the loan was granted and with due attention to considerations of economy and efficiency.²⁷⁶ Accordingly, ADB undertakes financial due diligence and financial management assessment (FMA) of the executing agency and/or implementing agency of ADB-funded projects. The FMA is a review of the entity's systems, to assess gaps and mitigation measures for financial and management accounting, reporting, auditing, and internal controls. It also involves an assessment of the entity's disbursement and cash flow management arrangements, and governance and anticorruption measures.²⁷⁷ ADB defines "executing agency" as the government agency with overall responsibility and oversight for the carrying out of a loan, grant or a TA-grant funded project, and it will be specifically identified in the financing agreement for the project. A project may also include one or more "implementing agencies", which are governmental or other bodies responsible for implementing specific activities under a project. The proceeds for projects under the program will be disbursed in accordance with ADB's Loan Disbursement Handbook (2017, as amended from time to time), which sets out detailed parameters and procedures for the release of funds to the borrower. Public disclosure of the audited project's financial statements, including the auditor's opinion on the project financial statements, will be guided by ADB's Access to Information Policy. ADB's OAI serves as the initial point of contact for all allegations of integrity violations in projects and other activities of ADB. Allegations are independently and objectively assessed and, where necessary, investigated pursuant to ADB's Anticorruption Policy. OAI registers and reviews all complaints to determine whether they require further investigation.²⁷⁸ Under ADB's Administrative Order 2.10: Whistleblower and		

Witness Protection, ADB protects, to the extent possible, the identities of whistleblowers and witnesses acting in good faith, to ensure they are not subjected to retaliation.

Pursuant to ADB's procurement rules, individuals and entities listed in the United Nations Security Council (UNSC) sanctions list will not be eligible to participate in, or to be awarded, a contract financed under the CRPP. ADB also screens implementing agencies and payees against the UNSC sanctions list and undertakes the necessary risk assessment and management. Moreover, none of the ADB's project activities will be undertaken in any jurisdiction which is subject to UNSC sanctions.

GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment

CRPP IF was screened in accordance with GCF's Environmental and Social Policy (2018)²⁷⁹, Revised Environmental and Social Policy (2021)²⁸⁰ and ADB's Safeguards Policy Statement (2009)²⁸¹. While both policies are generally substantively equivalent, the categorization is slightly different particularly involving investments through financial intermediaries (this, including other emerging concerns that are not common to both policies, will be addressed as compliance measures in the CRPP IF). The CRPP IF proposes the establishment of an investment facility that will provide grant financing (proposed through GCF) and ADB donor financing (and potentially other multi donor co-financing) to adaptation investments that will implement climate risk information, climate resilient livelihoods and infrastructure and facilitate transformation adaptation for the poor and vulnerable groups. ADB through SDCC, will host and manage the CRPP IF operations, acting as financial intermediary for providing GCF funds alongside ADB funds to support investments in selected countries. The CRPP IF is categorized as Category FI (financial intermediary) in line with ADB SPS Safeguard requirements (SR 4: Special requirements for different finance modalities) that include, among others, the need to establish an environmental and social management framework (ESMF). The ESMF should have the following components: a) an indication of relevant environmental and social policies, and applicable requirements; b) guidelines and procedures for conducting due diligence of proposed investments, including screening of proposed projects to be financed; c) identification of organization capacity and competency; d) guidelines for safeguards monitoring; and e) guidelines for managing non-compliances. Screening and categorization of projects will be undertaken in accordance with the ESMF to ensure compliance with ADB's requirements specified in Safeguard Requirements (SR) 1 on Environmental Safeguards; SR 2 on Involuntary Resettlement Safeguards; and SR 3 on Indigenous Peoples Safeguards.

The GCF's environmental and social risk category determines the extent of the due diligence required, the scope of the mitigation measures and the resources necessary to implement environment and social management plans, stakeholder engagement and information disclosure. Based on the screening and categorization for investments through financial intermediation, the CRPP IF is rated as Category I-2 or Medium level of intermediation. This is when an intermediary's existing or proposed portfolio includes, or is expected to include, financial exposure to category B activities. GCF defines category B activities as moderate risk activities that include those with risks and impacts "that are less adverse than Category A activities in terms of magnitude and likelihood of Occurrence" and where risks and impacts are considered limited, and the magnitude is expected to be low to moderate. The safeguards' risks and impacts of Category B investments are few, contained within the footprint of the activities, largely reversible, and readily mitigated through generally accepted mitigation measures and good international industry practices.

Category I-2 is proposed for CRPP-IF on the basis of the investments that it will support, that falls under the following: a) adaptation investments that support implementation of Information and systems for delivering applied climate-risk informed investments at scale (directly linked to attainment of Output 4); b) climate resilient pro-poor livelihoods investments (directly linked to attainment of Output 5; and c) Investments on climate adaptation pro-poor infrastructure (directly linked to the attainment of Output 6). Category I-2 is expected to include financial exposure to category B activities as defined above. An ESMF is required for the CRPP IF where specific investments and their location have yet to be clearly defined.

When project details are available, appropriate environmental and social due diligence should be prepared. The CRPP IF ESMF will guide and provide the framework for managing environmental and social risks associated with sub investments, and guide the risks evaluation at appraisal and monitoring stages. This is to ensure that financial intermediation operations result in projects that are environmentally and socially sound and in compliance with relevant national and local laws on environmental and social regulations, ADB SPS and GCF's ESP, and Good International Industry Practices.

The establishment of the ESMF is in line with ADB SPS and GCF revised ESP and will be maintained by the CRPP IF to guide the identification, assessment, management and monitoring of the environmental and social risks, including risks of sexual exploitation, abuse and harassment (GCF's Revised Environmental and Social Policy (2021)). This process is critical to support decision-making on whether the level of risk associated with investment is acceptable to CRPP-IF, to minimize environmental and social risks and impacts and to identify potential co-benefits associated with CRPP-IF investments.

The CRPP IF ESMF includes the following processes:

- a) Screening and categorization. Screening is undertaken to determine if project falls on the list of prohibited investment activities list (SPS, Appendix 5). The project will not be pursued if it is an excluded activity. ADB adopts a classification system that assigns one of the following categories to reflect the significance of a project's potential environmental and social risks and impacts:
 - Category A for projects likely to have significant adverse environmental impacts, significant adverse involuntary resettlement impacts and significant impacts on indigenous peoples that are irreversible, diverse and unprecedented. Safeguard requirements include the preparation of environmental impact assessment and development of resettlement plan and indigenous peoples plan;
 - Category B for proposed projects with potential limited adverse environmental impacts, involuntary resettlement impacts that are not deemed significant and limited impacts on indigenous peoples. Safeguard requirements include the preparation of an initial environmental examination, resettlement plan and indigenous peoples plan;
 - Category C rating for activities with minimal or no adverse environmental risks or impacts, no involuntary resettlement impacts and not expected to have impacts on indigenous peoples require no further action; and
 - Category FI when a proposed project involves the investment of ADB funds to or through a financial intermediary with safeguard requirement include the need for the FI to develop an environmental and social management system.
- b) Due diligence and review including systematic identification, quantification and assessment of environmental and social risks associated with sub projects will be undertaken, and the application of the mitigation hierarchy (avoidance, minimization and mitigation, and eventually offset of residual impacts) will be applied to the risks identified. The scope and extent of due diligence is defined by the risk category identified in the screening and categorization procedures.
- c) Information disclosure. In line with ADB's Public Communications Policy (2018, as amended from time to time), safeguards documents will be disclosed in the ADB website to make stakeholders be aware of and understand the risks and impacts of the project. Full disclosure of the safeguard reports to GCF Board and Active Observers will be coordinated with the GCF secretariat in line with the GCF Information Disclosure Policy.
- d) Consultation and participation. ADB requires borrowers and clients to undertake meaningful consultations by engaging with communities, people affected by proposed projects (including persons in vulnerable positions and situations), through consultation and informed participation.
- e) Monitoring and reporting. Procedures to monitor ongoing compliance with and the progress of implementation of safeguards plans will be integrated in the ESMF, including procedures for the implementation of corrective actions to address non-compliance and unanticipated environmental and social risks; and
- f) Grievance redress mechanism (GRM). ADB requires the borrowers/clients to establish and maintain a GRM to receive and facilitate resolution of affected people's concerns and grievances. In case of occurrence of Sexual Exploitation and Abuse and Sexual Harassment (SEAH), ensure that there are established accessible and inclusive survivor-centred and gender-responsive grievance redress mechanisms in place and procedures to address SEAH. The ADB Accountability Mechanism is also in place to provide an independent forum and process whereby

people adversely affected by ADB-financed projects can voice, and seek a resolution of their problems, as well as report alleged violations of ADB's operational policies and procedures.

G.2. Gender assessment and action plan

Women have a critical role to play in advancing investment in building resilience to climate change, especially at the local level, and must be recognized as vital drivers of resilient development. However, social and political inequalities, as well as patriarchal power structures, remain deeply entrenched in CRPP countries, often limiting women's leadership and participation in decision-making, and contributing to disproportionate levels of vulnerability and low adaptive capacity compared to men. This is particularly the case in poor communities in low-income countries as multidimensional poverty significantly limits women's ability to confront climate change. The following are examples of factors identified in the CRPP program level Gender Assessment that contribute to women's increased vulnerability and lower adaptive capacity compared to men in CRPP countries. They are closely related to socially created gender functions, systemic discriminations, and the power imbalances between women and men, and include;

- Women typically have less access to all types of information including climate information and early warning.²⁸² This is related to gender-based differences in literacy rates and access to education that are present in most CRPP countries, with differences often more pronounced in rural than urban areas and in some countries more than others. In **Pakistan** for example, less than half of women are literate (46%), compared with 71% of men, and in **Cambodia**, the gender gaps in education are most prominently seen in rural areas.
- Significant gaps remain in social protection programs reaching women. In **Indonesia**, insufficient use of up to date sex-disaggregated data for monitoring and targeting of beneficiaries, complicated information about protective measures and limited assistance available from official sources, and inadequate support to complete enrolment procedures have been identified as important issues²⁸³. In **Pakistan**, large gender gaps in mobile phone ownership and national ID possession, has meant that women are at risk of being disproportionately excluded from the Ehsaas Emergency Cash Program, which was a key part of the government's Covid-19 response²⁸⁴.
- Women in many CRPP countries are largely excluded from decision-making and are absent from leadership positions. This is particularly true at the local level. This is often due to discriminatory customary norms and cultural and structural barriers that prevent women from participating, such as in **Vanuatu** where women are excluded from direct roles in the village decision-making body, and in some cases women are invited to attend but decisions are already made by the men, or in **Timor-Leste** where 95% of the village chiefs are men, despite having a constitution that affirms the right of equal participation of women and men in political life.
- Gender gaps in access to agricultural extension and rural advisory services are prevalent in many CRPP countries, such as in **Cambodia**, where women receive only 10% of agricultural extensions services despite comprising 51% of the primary workforce in subsistence agriculture and 57% of the workforce in market-oriented agriculture²⁸⁵. This is despite evidence that suggests women are more likely than men to use the information they receive²⁸⁶.
- Social, cultural and legal barriers prevent women from accessing financial services in many CRPP countries, such as **Pakistan, PNG, Timor-Leste, and Vanuatu**. Women have limited ownership of productive assets, such as agricultural land and property and the consequent lack of collateral represents a barrier for women's access to formal financial services. Other barriers include; lack of collateral and ID documents that hinder the access to formal financial services; low digital literacy and mobile phone ownership, that deter women's outreach through digital financial services.
 - There is a significant disparity between men and women in earnings, the quality of jobs, and voice and representation in leadership positions in many CRPP countries, and women's economic

activities often involve low value-added activities. In **Papua New Guinea**, for example, there is a strong gender gap in employment with only 14% of women as waged workers compared to 33% of men, and 85% of women are in vulnerable employment compared with 65% of men²⁸⁷. In **Lao PDR**, 95.5% of women are informally employed and dominate small-scale retail and vending enterprises²⁸⁸. Under-employed workers are also more likely to be women as they juggle employment and family responsibilities. In **Lao PDR**, women spend significantly more time on unpaid domestic labour (4 hours per day on average) compared to men (30 minutes per day)²⁸⁹.

- Women's adaptive capacity is also limited by the fact that they often face barriers to accessing and control over natural resources. Women are underrepresented in environmental decision-making and natural resources management, such as forest user groups, water user associations, and environmental policy-making. Due to these disadvantages, women can be less likely to benefit from projects that promote nature-based solutions.
- Women are often reluctant to use emergency shelters, and there are frequent examples of women refusing to use emergency shelters due to either design concerns or because of their bad experiences from the past²⁹⁰. This is often because basic gender needs like separation toilet facilities, dedicated space for women, and private space for pregnant woman and nursing mothers are rarely factored into the design and management of emergency shelters. Evidence shows that women and girls are at greater risk of gender-based violence (GBV), including sexual harassment and domestic violence during emergency situations and in the aftermath of disasters in part because of poor emergency-shelter design and inadequate coordination by disaster responders²⁹¹.

Building women's resilience to climate change and empowering women through transformational investments in climate adaptation at the community level are core pillars of the CRPP, reflecting ADBs commitment to gender equality. The approach toward implementing investment activities that build women's resilience to climate change will emphasize women's role in resilience as "active stakeholders" and "game changers" rather than "vulnerable" and "victims." The CRPP also understands that structural gender inequalities are driving women's climate risk in CRPP countries, and so a gender transformative approach that aims to change structural inequalities and address issues of climate justice²⁹², will be a central element of CRPP investments. CRPP investments in climate resilience therefore represent an opportunity to promote a positive change towards gender equality and challenge socially created gender functions, systemic discriminations and the power imbalances between women and men, which intensify the social and economic barriers faced by women to adapt to climate change²⁹³. As such, the CRPP will go beyond mainstreaming gender considerations into programming, delivering specific outputs for strengthening women's adaptive capacity, and challenging the structural sociocultural norms that drive their vulnerability through transformational investments.

Examples of gender interventions that are identified in the CRPP Program Level Action Plan, and address some of the specific issues identified above include;

- Undertaking action research studies that address gaps in understanding around women and resilience in CRPP countries and make the case for women-focused adaptation investments.
- Support innovative adaptation initiatives and pilot activities at the community level to identify and test women-focused adaptation measures that could be scaled up through larger investments in the future.
- Carry out policy dialogues and forums for exchange at the regional, national, and local levels that bring together grassroots women's organizations, government ministries from various sectors, the private sector, academic institutions, and other civil society organizations, providing a platform for exchange and collaboration between actors that may not traditionally work closely together.

- Creating policy incentives and procedures for the inclusion of women in adaptation decision-making at the local level and building their capacity to use participatory tools and procedures for climate risk informed development planning and implementation to empower local women to act on adaptation.
- The CRPP will seek to tackle the limitations of existing social protection schemes in terms of reaching women, ensuring that social protection programs include specific identification of, and outreach to, women, using sex-disaggregated data for targeting beneficiaries.
- Recognizing that financial inclusion is key to strengthening women's resilience to climate related shocks and stresses, the CRPP will seek to expand women's access to instruments such as indexed-based microinsurance, through targeted financial products and services as part of its investment projects that can be paired with capacity building on savings and investments, supporting preparedness to shocks while empowering women financially, socially and politically.
- CRPP investments will seek to address gender-related factors that influence women's ability to participate in value chains, increase access to and control over income, promote women's engagement in agricultural entrepreneurship orientated activities, deepen women's chances to increase value of their agricultural activities, as well as diversify their income-earning opportunities, and provide women with leadership roles.
- Investments will also integrate initiatives that reduce women's time poverty and burden of care in ways that reduce and redistribute labor and time commitments to benefit them directly.
- The CRPP will integrate gender considerations in initiatives related natural resource management that can make them more effective and sustainable²⁹⁴[\[xvii\]](#), as well as ensuring participation of women in meetings where public works are designed and selected, including capacity-building activities that specifically develop women's technical and leadership skills.
- CRPP supported investments in safe shelter will ensure that shelter design includes features that respond to specific needs of women, and increases their access, ensuring that women of all ages, including those from marginalized groups, are properly engaged to understand their concerns and needs and that designs respond accordingly. Women will also be equally represented in governance structures such as shelter or community committees, to support empowerment through participation and addressing gender norms in terms of decision-making.

The CRPP Gender Assessment and Action Plan are presented in detail in **Annex 8**. The Gender Assessment provides a program level overview of some of the key gender issues that contribute to the vulnerability of women to climate change in CRPP countries and that cause women's lower capacity to adapt than men. Detailed Gender Assessments will be undertaken as part of the detailed design and preparation phase of each project and will be specific to the target area and context of the project. Similarly, the CRPP Gender Action Plan presented in Annex 8 will provide an overarching framework for CRPP implementation, and will be complemented by project specific Gender Action Plans to be developed during detailed preparation phase.

G.3. Financial management and procurement

The CRPP supported projects will follow ADB systems for financial management, disbursement, audit and procurement. Implementation of the program and projects must conform to all applicable ADB policies including those concerning anticorruption measures, anti-money laundering, counter terrorist financing, safeguards, gender, procurement, consulting services, and disbursement.

Financial Management. All projects' financial accounting, disbursement procedures and auditing requirements will be developed during project preparation and will be included their respective grant

implementation manuals. ADB shall ensure that these are in accordance with ADB's policies, procedures and approach on Financial Management and Analysis of Projects.

Disbursement. Project disbursements will follow ADB's Loan Disbursement Handbook (2017, as amended from time to time) and detailed arrangements agreed upon between the government and ADB.

Audit. Public disclosure of the audited project's financial statements, including the auditor's opinion on the project financial statements, will be guided by ADB's Public Communications Policy 2011.

Procurement. All procurement for CRPP supported projects will be done in accordance with ADB's Procurement Guidelines and Procurement Regulations for ADB Borrowers (2017, as amended from time to time), and will be done by the Implementing Agencies for each project. ADB staff will provide support procurement processes and offer training, as needed, to the executing agencies in the completion and monitoring of procurement-related documents. Procurement plans for the respective projects shall be in accordance with their respective grant implementation manuals. A detailed procurement plan for use of GCF is included in **Annex 10**.

G.4. Disclosure of funding proposal

- ☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.
- ☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:
- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
 - redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☐ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁹
- ☐ Annex X Other references

* Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.

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KINGDOM OF CAMBODIA
Nation Religion King



National Council for Sustainable Development

No: 008 NCS

Phnom Penh 09, March ... 2021

Mr. Yannick Glemarec

Executive Director

Green Climate Fund Secretariat

Songdo International Business District

G-Tower, 175 Art Center-daero, Yeonsu-gu, Incheon 22004

Republic of Korea

Re: Funding Proposal for the Green Climate Fund (GCF) Project Preparation Facility by the Asian Development Bank regarding the Community Resilience Partnership Program (CRPP)

Dear Mr. Glemarec,

We refer to the Community Resilience Partnership Program (CRPP) covering multiple countries in Asia and the Pacific for the Green Climate Fund (GCF) to include Cambodia in the program which was submitted by the Asian Development Bank (ADB) to us on 17 February 2021.

The undersigned is the duly authorized representative of the Ministry of Environment and the National Designated Authority and the focal point of the Kingdom of Cambodia.

Pursuant to the GCF decisions B.08/10 and B.13/21, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the CRPP funding Proposal.

By communicating our no-objection, it is implied that:

- (a) The Royal Government of Cambodia has no-objection to the program as included in the CRPP funding Proposal;
- (b) The program as included in the CRPP funding Proposal is in conformity with Cambodia's national priorities, strategies, and plans; and
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the CRPP funding Proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the program as included in the CRPP funding Proposal has been duly followed.

We also confirm that our no-objection applies to all GCF-financed activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website. *RF [signature]*

Sincerely yours,



SAY Samal

Chair of the National Council for
Sustainable Development
Minister of Environment



MINISTRY OF FINANCE OF THE REPUBLIC OF INDONESIA
FISCAL POLICY AGENCY

R.M. NOTOHAMIPRODJO BUILDING 2ND FLOOR JALAN DR. WAHIDIN RAYA NOMOR 1 JAKARTA 10710
TELEPHONE (+62 21) 3441484; FACSIMILE (+62 21) 3848049; WEBSITE www.fiskal.kemenkeu.go.id

Ref. : S-36/KF/2021

12 April 2021

Mr. Yannick Glemarec
Executive Director
Secretariat of the Green Climate Fund
175, Art center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Subject : Funding Proposal for the GCF by Asian Development Bank regarding Climate Resilience Partnership Program (CRPP)

Dear Mr. Glemarec,

We refer to the program of Climate Resilience Partnership Program (CRPP) as included in the funding proposal submitted by Asian Development Bank (ADB) to us on 8 March 2021.

The undersigned is the Chairman of Fiscal Policy Agency, Ministry of Finance as the National Designated Authority of Indonesia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the program as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Indonesia has no-objection to the program as included in the funding proposal;
- (b) The program as included in the funding proposal is in conformity with Indonesia's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the program as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to program as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the program.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours faithfully,



Ditandatangani secara elektronik
Febrio Nathan Kacaribu
Chairman





Lao People's Democratic Republic
Peace Independence Democracy Unity Prosperity

Ministry of Natural Resources and Environment
Department of Planning and Finance

1345
No:/DPF
Vientiane, dated: 24. March 2021

To: The Green Climate Fund (GCF)

Subject: No-Objection Letter - Funding proposal for the GCF by the Asian Development Bank regarding the Community Resilience Partnership Program

Dear Madam/Sir,

We refer to the Community Resilience Partnership Program (CRPP) covering multiple countries in Asia and the Pacific, which was submitted by the Asian Development Bank (ADB) to the Ministry of Natural Resources and Environment (MONRE) on 23 November 2020.

The undersigned is the duly authorized representative of the Ministry of Natural Resources and Environment (MONRE), the National Designated Authority of Lao PDR. Pursuant to GCF decision 8.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our non-objection to the project as included in the funding proposal.

By communicating our non-objection, it is implied that:

- a) The government of Lao PDR has no objection to the project as included in the funding proposal
- b) The project as included in the funding proposal conforms with Lao PDR's national priorities, strategies and plans
- c) In accordance with the GCF's environmental and social safeguards, the funding proposal's project conforms with relevant national policy and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed. We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,



Mr. Phouvang Luangxaysana,
National Designated Authority of Lao PDR



Sr. JS (IC)

F.No1 (2) 2021 ADB-GCF
Government of Pakistan
Ministry of Climate Change
LG & RD Complex, Sector G-5/2, Islamabad

Islamabad, the 4th August, 2022

To: **Green Climate Fund (GCF) Secretariat**
Song Do,
South Korea

Subject: **Funding proposal for the GCF by Asian Development Bank regarding Community Resilience Partnership Program**

We refer to the programme, Community Resilience Partnership Program, multiple countries in Asia and Pacific, as included in the funding proposal submitted by Asian Development Bank to us on 27th July, 2022.

2. The undersigned is the duly authorized representative of the Ministry of Climate Change, the National Designated Authority/Focal Point of Pakistan. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed; we hereby communicate our No-Objection to the programme as included in the funding proposal.

3. By communicating our No-Objection, it is implied that:

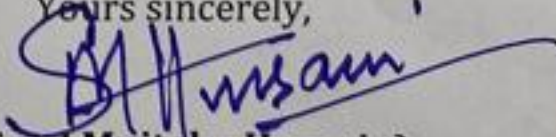
- a) The National Designated Authority (NDA) of GCF-Pakistan has no-objection to the programme as included in funding proposal;
- b) The programme as included in the funding proposal is in conformity with Pakistan's national priorities, strategies and plans;
- c) In accordance with the GCF's environmental and social safeguards, the programme as included in funding proposal is in conformity with relevant national laws and regulations.

4. We confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

5. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

6. We acknowledge that this letter will be made publicly available on the GCF website.

Yours sincerely,


(Syed Mujtaba Hussain)
Sr. JS (IC)/ GCF FP
051-9245585

Copy to:

Country Director,
Asian Development Bank,
Pakistan.

CLIMATE CHANGE & DEVELOPMENT AUTHORITY

OFFICE OF THE MANAGING DIRECTOR



1st Floor, Dynasty Tower, Savannah Heights,
Waigani Drive, WAIGANI, Port Moresby

P O Box 4017, Boroko 111, NCD.
Papua New Guinea



Telephone: (+675) 7411 2493, Facsimile: (+675) 3257620, Email: info@ccda.gov.pg Website: www.ccda.gov.pg

The Green Climate Fund

Songdo Business District
175 Art Center-daero
Yeonsu-gu, Incheon 22004
Republic of Korea

Port Moresby, Papua New Guinea, 02nd March 2021

RE: GCF FUNDING PROPOSAL TITLED “COMMUNITY RESILIENCE PARTNERSHIP PROGRAM” FOR THE ASIAN DEVELOPMENT BANK (ADB) AS THE ACCREDITED ENTITY

Dear Madam, Sir,

We refer to the project (Community Resilience Partnership Program) for countries in the Asia-Pacific region with Papua New Guinea (PNG) included. The PNG National Designated Authority is in receipt of the full funding proposal submitted by the Asian Development Bank on December 22, 2020.

The undersigned is the duly authorized representative of the Climate Change and Development Authority, the National Designated Authority of Papua New Guinea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Papua New Guinea has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Papua New Guinea's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Yours Sincerely,

MR. RUEL YAMUNA, LLB
MANAGING DIRECTOR

“Building a climate-resilient and carbon-neutral pathway through sustainable economic development”



MINISTRO DO COORDENADOR DOS
ASSUNTOS ECONOMICOS (MCAE)



National
Designated
Authority



DEMOCRATIC REPUBLIC OF TIMOR-LESTE SECRETARIA DE ESTADO DO AMBIENTE
AUTORIDADE NACIONAL DESIGNADA (AND)

Edifício do Fomento, Rua: Dom Boa Ventura, Nu. 16, Mandarin-Dili.

executive.secretary@nda.gov.tl Tlf: +67078380222

No. Ref.: 02 AND/SEA/MCAE/II/2021

Dili, 10 Februry 2021

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Asian Development Bank regarding Community Resilience Partnership Program

Dear Madam, Sir,

We refer to the programme, Community Resilience Partnership Program, multiple countries in Asia and Pacific, as included in the funding proposal submitted by Asian Development Bank to us on 07 February 2021.

The undersigned is the duly authorized representative of Office of the Secretariat of State for the Environment, Dr. Julião dos Reis, the National Designated Authority of Timor-Leste.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Timor-Leste has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Timor-Leste's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Dr. Julião dos Reis
Acting Executive Secretary for Timor-Leste NDA for the Green Climate Fund & Principal Adviser
for Climate Finance and Resource Mobilization
Office of the Secretariat of State for the Environment



TEL : (678) 22068

FAX : (678) 22068

Ref: PV/MoCCA/PROJ/ADB-GCF/4. B.2

10th May 2021

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by Asian Development Bank regarding Community Resilience Partnership Program

Dear Madam, Sir,

We refer to the programme, Community Resilience Partnership Program, multiple countries in Asia and Pacific, as included in the funding proposal submitted by Asian Development Bank to us on 19th of Feb 2021.

The undersigned is the duly authorized representative of the Ministry of Climate Change Adaptation, Meteorology, Geo-Hazards, Environment, Energy and Disaster Management (the National Designated Authority of Vanuatu).

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Vanuatu has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with Vanuatu's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

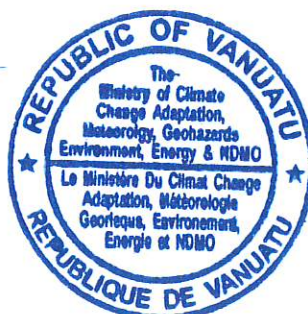
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mrs Esline Garaebiti,
Director General,



Ministry of Climate Change Adaptation, Meteorology, Geo-Hazards, Environment, Energy and Disaster
Management

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Community Resilience Partnership Program
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	Asian Development Bank (ADB)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	Cambodia, Indonesia, Lao PDR, Pakistan, Papua New Guinea, Timor-Leste, and Vanuatu
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Friday, September 22, 2023
Language(s) of disclosure	English, Khmer, Bahasa Indonesia, Lao, Urdu, Pidgin (Tok Pisin), Tetum, and Bislama
Explanation on language	These are the official languages of the target countries.
Link to disclosure	English: https://www.adb.org/documents/annex-6-environmental-social-management-framework Khmer*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-km.pdf Bahasa Indonesia*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-id.pdf

	Lao*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-lo.pdf Urdu*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-ur.pdf Pidgin (Tok Pisin)*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-tpi.pdf Tetum*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-tet.pdf Bislama*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-bi.pdf
Other link(s)	https://www.adb.org/what-we-do/funds/community-resilience-financing-partnership-facility
Remarks	An ESMS consistent with the requirements for a Category I-2 project is contained in the “Annex 6 on Environment and Social Management Framework.”
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report/disclosure on accredited entity's website	Indigenous Peoples Planning Framework
Language(s) of disclosure	English, Khmer, Bahasa Indonesia, Lao, Urdu, Pidgin (Tok Pisin), Tetum, and Bislama
Explanation on language	These are the official languages of the target countries.
Link to disclosure	English: https://www.adb.org/documents/annex-6-environmental-social-management-framework Khmer*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-km.pdf Bahasa Indonesia*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-id.pdf

	Lao*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-lo.pdf Urdu*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-ur.pdf Pidgin (Tok Pisin)*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-tpi.pdf Tetum*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-tet.pdf Bislama*: https://www.adb.org/sites/default/files/institutional-document/910616/annex-6-environmental-social-management-framework-bi.pdf
Other link(s)	https://www.adb.org/what-we-do/funds/community-resilience-financing-partnership-facility
Remarks	In respect of the Indigenous Peoples Planning Framework, an 'Indigenous Peoples Impact Categorization' and 'Outline of an Indigenous Peoples Plan' are included as Annexes to the ESMF. The 'Outline of Resettlement Framework Plan' and 'Outline of a Resettlement Plan' are likewise referenced as Annexes to the ESMF.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Friday, September 22, 2023
Place	Hard copies have been made available in ADB Resident Missions that are currently operating: Cambodia Resident Mission (CARM)** 29 Preah Suramarit Blvd (268), Phnom Penh 12207, Cambodia <i>Contact Information:</i> Mr. Sambath Kim, Senior Social Development Officer, samkim@adb.org Tel +855 23 973 200 Indonesia Resident Mission (IRM)** The Plaza Office Tower, 11th Floor, Jl. MH Thamrin 28-30, Central Jakarta, Indonesia 10350 <i>Contact Information:</i>

	Ms. Naning Mardiniah, Senior Safeguards Officer (Resettlement), nmardiniah@adb.org Ms. Binafeda Harimundarti, Associate Environmental Safeguards Officer, bharimundarti@adb.org Tel +62 21 2992 7388 Lao Resident Mission (LRM)** Corner of Lanexang Avenue & Samsenthai Road P.O. Box 9724 Vientiane, Lao People's Democratic Republic <i>Contact Information:</i> Mr. Vongphet Soukhavongsa, Senior Safeguards Officer, vsoukhavongsa@adb.org Ms. Vinanda Siphonesay, Associate Social Safeguards Officer, vsiphonesay@adb.org Tel +856 21 250444 Pakistan Resident Mission (PRM)** Level 8, North Wing, Serena Business Complex, Khayaban-e-Suhrawardy, G-5, Islamabad, Pakistan <i>Contact Information:</i> Ashfaq Ahmed Khokhar, Senior Safeguards Officer, akhokhar@adb.org Tel +92 51 2600351 to 69, 2087300 Papua New Guinea Resident Mission (PNRM)** Level 2 – Burns Philp Haus, Corner of Musgrave Street & Champion Parade, PO Box 1992, Port Moresby, National Capital District, Papua New Guinea <i>Contact Information:</i> Mr. Jack Stanley, Senior Safeguards Officer, jstanley@adb.org Mr. Paraka Newman, Safeguards Officer, pnewman@adb.org; Tel +657 3221700 Timor-Leste Resident Mission (TLRM)** Rua Cabo Verde No 16, Posto Administrativo, Vera Cruz, Suco Motael, Aldeia Halibur, Dili, Timor-Leste <i>Contact Information:</i> Chandrasekaran Velayutham, Safeguards Specialist (Resettlement), cvelayutham@adb.org Tel +855 23 973 200 Pacific Liaison and Coordination Office (PLCO)** Level 20, 45 Clarence Street, Sydney, NSW Australia 2000 <i>Contact Information:</i> Michelle Dooley, Senior Safeguards Specialist, mdooley@adb.org Tel +678 23610
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Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, October 23, 2023

* Subsequent to the disclosure of this form to the Board and active observers on 22 September 2023, the following updates have been made: The Environmental and Social Management Framework (ESMF) has been disclosed in the following languages on 29 September 2023: Khmer, Bahasa Indonesia, Lao, Urdu, Pidgin (Tok Pisin), Tetum, and Bislama.

** Subsequent to the disclosure of this form to the Board and active observers on 22 September 2023, the following updates have been made: Contact information of the respective ADB Resident Missions, where hard copies of the Environmental and Social Management Framework (ESMF) have been made available, have been included on 29 September 2023.

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP215

Proposal name:	Community Resilience Partnership Program
Accredited entity:	Asian Development Bank
Country/(ies):	Cambodia, Indonesia, the Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste and Vanuatu
Project/programme size:	Large

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The proposal presents a unique and comprehensive approach to scaling up locally led adaptation measures, tailored to the needs of the poor and vulnerable populations in seven countries across Asia and the Pacific region. This locally driven approach is well aligned with updated Strategic Plan-2 priorities (through the individual projects in the target countries).	Limited private sector involvement is foreseen in this programme as it focuses on the nexus of poverty, climate and gender. This aspect will be explored on a case-by-case basis at the individual project level.
The ability of GCF to offer grant finance will drive the Asian Development Bank (ADB) co-financing into the most transformational and paradigm-shifting investments at large scale, targeting the poor and vulnerable populations . GCF contribution of USD 120 million will leverage USD 555 million of ADB finance and USD 75 million of finance from other bilateral donors (through the Community Resilience Partnership Program Trust Fund), representing more than fivefold (5.25) leveraging ratio for every United States dollar of GCF finance invested.	
The accredited and executing entity (ADB) has strong expertise and a proven track record in the region in implementing these activities and producing evidence-based results for effective climate change adaptation.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum XXI, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Project background

3. Countries in Asia and the Pacific region are highly exposed to the damaging impacts of climate change. Between 2000 and 2019, the region experienced 3,068 disaster events accounting for 42 per cent of global events. Adverse impacts of climate change will be largely manifested at the local level. Poor and vulnerable populations face the brunt of climate impacts, with women typically lacking equal preventive and coping capacity due to pre-existing gender inequalities. This requires a large-scale intervention in order to address these issues at a regional level.

4. In responding to the financial gap and needs, the proposed project (the Community Resilience Partnership Program (CRPP)), a regional programme of the Asian Development Bank (ADB), aims to scale up adaptation measures by mobilizing climate investment at large scale at the community level. In particular, it will support measures that address the nexus between climate change, poverty and gender inequality across seven countries (Cambodia, Indonesia, the Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste and Vanuatu) in Asia and the Pacific region.

5. This proposal is a unique and important programmatic concept for scaling up local adaptation investments in the region, building on the lessons learned from similar programmes, especially in Africa, funded by the Pilot Programme for Climate Resilience under the Climate Investment Fund. The goal of this multi-country programme is to improve (i) the enabling environment for knowledge, finance and implementation of local adaptation measures; (ii) the mobilization of pro-poor adaptation finance at scale; (iii) food security of poor and vulnerable communities; and (iv) mitigation of the exposure of poor and vulnerable people to climate shocks and stresses.

6. The CRPP comprises two separate but interlinked components; the CRPP Trust Fund (TF) focuses on upstream support to strengthen the enabling environment required for implementing local adaptation measures at scale, funded by bilateral donors;¹ and the CRPP Investment Fund (IF) focuses on the actions for rolling out local adaptation measures through downstream investment (subprojects), which will be co-financed by ADB and GCF. The CRPP will be operationalized through a special vehicle established by the ADB (Community Resilience Financing Partnership Facility).

7. The CRPP IF will provide grant and loan resources to implement outputs 4, 5 and 6 of the wider CRPP programme (described below).

- (a) Output 4: Information and systems for delivering applied climate risk informed investments at scale improved;
- (b) Output 5: Climate-resilient pro-poor livelihoods investments implemented; and
- (c) Output 6: Climate adaptation pro-poor infrastructure implemented.

8. The CRPP IF will have a total of USD 120 million (USD 100 million grant and USD 20 million loan) from GCF for seven countries (with each country receiving between USD 10 million and USD 40 million) and will leverage at least USD 555 million from ADB's resources and USD 75 million from bilateral donors to support at least 10–15 subprojects (at least one project in each country) from 2023 to 2030. Subprojects seeking financing from the IF will need to be approved by the ADB Board between 2024 and 2027 and complete implementation by 2030.

¹ The Trust Fund will be fully capitalized by bilateral donors and financial institutions, including UK Aid, Nordic Development Fund and Agence française de développement. This total cost excludes the potential contribution from respective governments that will be determined at the individual project level.

9. The combined effect of project interventions will result in strengthened locally led pro-poor adaptation measures and knowledge transfer, enhanced finance mobilization and food security, and reduced climate shocks and stress of poor and vulnerable people in seven countries. The project will support 3,219,400 direct beneficiaries and will also indirectly benefit 4,465,000 people. Its coordinated and programmatic approach is expected to allow for scalability, replication and sustainability of the programme between countries and within countries across Asia and the Pacific region.

10. The CRPP will have environmental, economic, social and gender co-benefits, with strong emphasis especially on women's participation in decision-making and leadership and reduced inequalities. It is expected to deliver against many Sustainable Development Goals (SDGs), including SDG 1: no poverty; SDG 3: good health and well-being; SDG 5: Gender Equality; SDG 10: Reduced Inequalities; SDG 11: sustainable cities and communities; and SDG 13: climate action. There are also potential mitigation co-benefits anticipated through some of the interventions.

2.2 Component-by-component analysis

11. The goal of the proposal will be to deliver transformational adaptation solutions through large-scale investments explicitly targeted at the poor and vulnerable population. There are two main and complementary components to deliver its objective.

Component 1: CRPP TF – supporting countries to develop knowledge, build capacity, and structure adaptation projects through an upstream programme (total cost: USD 72,500,000 grant from UK Aid, Nordic Development Fund, Agence française de développement; GCF cost: 0 or 0 per cent of GCF budget)

12. The first component will be financed through the CRPP TF and focus on the three outputs: (i) strengthening knowledge and action research on climate risk informed pro-poor community-level solutions (USD 15,500,000); (ii) strengthening institutional and community capacity to develop and deliver climate adaptation investments at community level (USD 12,000,000); and (iii) Identifying and preparing inclusive and pro-poor adaptation investment projects (USD 45,000,000).

13. Activities will focus on undertaking the analytical work that complements the design and implementation of projects to be supported by the IF. It will develop climate risk assessment and decision support tools, conduct training and capacity-building on designing and implementing adaptation investments at the community level, prepare large-scale adaptation investment projects, and pilot innovative transformational solutions. These activities will generate knowledge and evidence to be shared.

14. The activities under component 1 are also complementary to the activities planned under component 2. CRPP TF supports the development of the climate rationale and project preparation, the implementation of which will be financed and scaled up by component 2.

Component 2: CRPP Investment Fund (IF) – supporting implementation of local adaptation measures (information systems, resilient pro-poor livelihoods, adaptation infrastructure) through a downstream investment programme (Total cost: USD 668,500,000; GCF cost: 120,000,000 or 100 per cent of GCF budget for the programme)

15. Component 2 will be financed through the CRPP IF, supporting activities under outputs 4, 5 and 6. The three outputs will deliver adaptation activities through the provision of combined grant finance with major investment funding to deliver adaptation at scale at the local level.

16. These outputs will be co-financed by the ADB and GCF. The focus will be to deliver: (iv) improvement of information and systems for delivering applied climate risk informed

investments at scale (total cost: USD 118,000,000; GCF grants USD 20,000,000); (v) implementation of climate-resilient pro-poor livelihoods investments (total cost: USD 238,000,000; GCF grants USD 40,000,000); and (vi) implementation of climate-resilient pro-poor infrastructure investment (total cost: USD 312,500,000; GCF grants USD 40,000,000 and loans USD 20,000,000).

- (a) **Output 4. Information and systems for delivering applied climate risk informed investments at scale improved.** The GCF grant finance for climate information for delivering climate risk informed decisions is targeted to address information failures that prevent climate-resilient development and will be largely focused on technical assistance provision (which is eligible for grant finance under B.05/07) to cover the additional activities to strengthen climate services at local level, to integrate climate risk information into social protection programmes and decentralization policies. It is also highlighted that these activities will cover activities that explicitly target the most vulnerable groups.
 - (i) Activity area 4.1. Climate information services for key sectors to meet local adaptation needs;
 - (ii) Activity area 4.2. Climate risk informed social protection systems; and
 - (iii) Activity area 4.3. Climate risk information systems for decentralisation policies.
- (b) **Output 5. Climate-resilient pro-poor livelihoods investments implemented.** The GCF grant and loan finance for climate-resilient pro-poor livelihood investment will focus on addressing risk perceptions for new climate-resilient livelihoods or markets, and covering the incremental cost or the risk premium required to make these new livelihoods or investments viable. Again, the activities will also be targeted at the most vulnerable groups.
 - (i) Activity area 5.1. Climate-resilient agroecological systems;
 - (ii) Activity area 5.2. Information technology and risk management services for climate-resilient livelihoods; and
 - (iii) Activity area 5.3. Climate-resilient agriculture supply chains.
- (c) **Output 6. Climate adaptation pro-poor infrastructure implemented.** The GCF grant and loan finance for climate-resilient pro-poor infrastructure investment is targeted at the additional costs of green, resilient infrastructure, necessary to deal with climate risks but requiring finance that would otherwise not be available, for poor and vulnerable groups. The primary focus is to invest in ecosystem-based adaptation, as this provides a range of public good characteristics, and has potential benefits in reducing non-market costs (providing environmental and social benefits).
 - (i) Activity area 6.1. Ecosystem-based infrastructure;
 - (ii) Activity area 6.2. Flood and landslide protection infrastructure; and
 - (iii) Activity area 6.3. Multipurpose emergency shelters.

17. The estimated impacts are considered significant and entail substantial adaptation benefits as well as support the resilience of poor and vulnerable populations to climate change. The proposed activities are assessed as being adequate and reasonable to support local communities while contributing potential emissions reduction from activities related to climate-resilient agriculture supply chains and ecosystem-based infrastructure.

18. The activities are also closely aligned with the four pillars of the United Nations' newly launched Early Warnings for All (EW4A) Initiative, which stipulates that every person in the world should be protected by an early warning system by 2027 and aims to deliver multi-hazard early warning systems (MHEWS) in over 100 countries. The proposed programme will

contribute to outcomes such as improved access and quality of risk data, and strengthened capacity for developing procedures for holistic anticipatory action based on early warnings, which has high potential for replication in Asia and the Pacific region.

19. The individual projects will align with the objectives of the CRPP and contribute to specific outputs related to climate resilience, poverty reduction and adaptation measures. They will be financed by the ADB and GCF, with GCF resources specifically supporting local-level adaptation interventions. A list of potential idea notes of individual projects in each country is summarized below with details annexed to the funding proposal.

- (a) **Cambodia:** The Climate Resilient Smallholder Development Project aims to strengthen community-led adaptation actions by taking a community-driven development approach to promote climate-smart agricultural practices, rehabilitate and climate-proof rural infrastructure, strengthen the capacity of local institutions to respond to climate change, and develop financial risk management mechanisms such as crop insurance.
- (b) **The Lao People's Democratic Republic:** The Building Resilience and Reducing Rural Poverty Project aims to enhance the resilience of rural poor communities by expanding non-contributory social assistance programmes to cover vulnerable people living in areas of high climate risk, integrating early warning systems with the social registry to expand coverage in response to climate shocks and stresses, and implementing an economic inclusion programme tailored to address climate vulnerabilities through skills training and financial services provision linked to conditional cash transfers.
- (c) **Pakistan:** The Sindh Coastal Resilience Project aims to enhance the resilience of coastal area livelihoods and infrastructure through integrated water resources and flood-risk management schemes, as well as restored coastal defences using community-led natural solutions. The project also emphasizes the role of women in building coastal resilience. The project directly contributes to the activities under output 6 of the CRPP.
- (d) **Papua New Guinea:** The proposed Improved Technical and Vocational Education and Training for Climate Resilient Employment Project aims to strengthen climate resilience through technical and vocational education. It includes specific measures to support women and informal workers. Through the technical and vocational education system, the project will develop climate-resilient agriculture, climate-resilient construction processes and will help some vulnerable people to transition out of agriculture into climate-resilient livelihoods. The project directly contributes to the activities under output 5 of the CRPP.
- (e) **Timor-Leste:** The Agriculture Value Chains and Water Harvesting Project aims to enhance the sustainability of rural livelihoods by increasing the resilience of small-scale agriculture. It focuses on community-led climate-smart production practices, water, forest management, and enabling smallholder producers to participate in agricultural value chains.
- (f) **Vanuatu:** The Climate Change and COVID-19 Project aims to build comprehensive resilience in poor and vulnerable groups. It focuses on creating climate-responsive social protection systems and enhancing climate-resilient livelihood support mechanisms, particularly for women. The project notably contributes to the activities under output 4 of the CRPP.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

20. The project has medium-high impact potential as an adaptation project, with approximately 3.2 million poor and vulnerable direct beneficiaries and 4.7 million indirect beneficiaries over the 20 years of the programme's lifespan. The CRPP will contribute to locally led sustainable development in the region by strengthening the resilience of nearly five million people and communities in seven countries across Asia and the Pacific region.

21. In all cases, women will make up at least 50 per cent of the beneficiaries. However, this will be a minimum requirement for all individual projects, and it is anticipated that women will make up more than 50 per cent of the total direct beneficiaries.

22. In addition, potential mitigation co-benefits are anticipated through planned activities, such as promoting the uptake of climate-smart agricultural practices and forestation activities. For example, rehabilitation of mangrove forests could lead to potential mitigation of upwards of -27.6 ± 1.7 megagramme of carbon dioxide equivalent (CO₂-eq) per hectare per year.²

Table 1. Expected adaptation outcomes

Country	No. of direct beneficiaries	% of direct beneficiaries of total population	No. of indirect beneficiaries	% of indirect beneficiaries of total population
Cambodia	1,300,000	8.1	1,365,000	8.53
Indonesia	1,200,000	0.45%	450,000	0.17
The Lao People's Democratic Republic	141,000	2.04	1,143,000	16.57
Pakistan	488,000	0.23	1,200,000	0.57
Papua New Guinea	10,000	0.11	25,000	0.27
Timor-Leste	60,000	4.62	241,5000	18.58
Vanuatu	20,40	7.02	40,000	13.76
Total	A minimum of 3,219,400 direct beneficiaries		A minimum of 4,465,000 indirect beneficiaries	

3.2 Paradigm shift potential

Scale: Medium to high

23. The core paradigm shift potential of this project centres on the enhanced mobilization of climate investment at scale in the region. It will allow scaling up and replication, creation of an enabling environment, contribution to the regulatory framework and policies, innovative solutions and approaches, and, lastly, sharing knowledge and lessons.

24. Scaling up and replication: The CRPP IF will only support individual projects that demonstrate the potential of being replicated elsewhere in the same country or other countries in the region. By empowering government and communities with the knowledge and skills to take appropriate action to adapt, improving systems for delivering climate-resilient investments, and proving their effectiveness through pilots, the individual project will build the foundations for replication, support transitions in key systems and thereby drive a paradigm shift towards climate-resilient sustainable development. GCF funding will be used together with the ADB investment to enable the individual project to deliver at scale.

² C Cameron, LB Hutley, DA Friess and B Brown. 2019. High greenhouse gas emissions mitigation benefits from mangrove rehabilitation in Sulawesi, Indonesia. *Ecosystem Services*. Vol 40. Available at <https://doi.org/10.1016/j.ecoser.2019.101035>.

25. Creation of an enabling environment (and sustainability of outcomes): The CRPP will support outputs and outcomes that are closely aligned with national climate priorities. The CRPP TF will therefore catalyse impacts beyond a one-off project investment and deliver a paradigm shift by ensuring that climate risk informed development becomes the new normal. The GCF financing will be used through the CRPP IF to support rolling out of such climate-sensitive development, especially in the context of localization and decentralization. By strengthening national and local institutions, the CRPP IF will ensure that its interventions will contribute to building policy coherence around climate resilience and is taken up nationwide by relevant agencies.
26. Contribution to the regulatory framework and policies: With a key focus on mainstreaming, the CRPP will promote the integration of climate change considerations into policies and decision-making processes. The CRPP IF programming will capitalize on ongoing decentralization processes to support governments build robust, science-based local climate assessment frameworks that can inform planning processes, equipping decision-makers with the data, tools, guidance and information needed to inform local planning and investment.
27. Innovative solutions and approaches: The CRPP builds on the lessons learned from the implementation of large-scale resilience programmes, which identify the need for innovative measures and approaches. The CRPP IF will support financing of adaptation projects. In particular, GCF financing will be used to finance activities that can help overcome information, finance, institutional and gender-related barriers in an innovative way.
28. Sharing of knowledge and learning: Learning is critical to operationalize adaptive management principles at a programme level. The CRPP TF will ensure that knowledge emerging from GCF-financed individual projects under the CRPP IF are shared with wider stakeholders. The multi-country nature of the programme means there will be very large potential for learning and replication, with the transfer of experience between programmes.
29. In addition, the proposed GCF CRPP-financed outputs will complement similar past and ongoing initiatives in the countries that are financed by United Nations Framework Convention on Climate Change -related funds and contribute to the paradigm shift in the region.
30. Global Framework for Climate Services (GFCS) has defined five priority sectors for effective national development planning, comprising agriculture and food security, water, health, disaster risk reduction, and energy. Depending on response capability of communities to hazard types, the potential for benefit-cost of return on investment could be up to 500:1 for densely-populate flood-prone areas, and 2500:1 for heat waves in populated urban centres. By targeting different sectors, the programme will make a significant contribution to the GCF's support for building resilience of multi-sectoral livelihoods of most vulnerable communities.
31. Although the specificities of the social, economic and environmental co-benefits of early warning systems are context-specific, the proposal would benefit from exploring these in a multi-hazard context for a range of priority hazards. ADB should explore the use of the regional, national and subnational Climate Outlook Forum user-interfaces to co-design and co-produce impact forecasts for priority sectors of the communities. This can be done during the implementation of the project.

3.3 Sustainable development potential

Scale: High

32. Through investments in local level adaptation, the proposal directly targets the pillars of sustainable development – environmental, social, economic and gender. Thus, the CRPP will deliver against several of the SDGs, including SDG 1: no poverty; SDG 3: good health and well-being; SDG 5: gender equality; SDG 10: reduced inequalities; SDG 11: sustainable cities and communities; and SDG 13: climate action.

- (a) Environmental co-benefits: It is expected that CRPP TF and IF will promote nature-based solutions through a wide range of investments, such as (i) community infrastructure – adopting natural assets in place of/in mix with grey infrastructure to build resilience of local communities; (ii) livelihoods – promoting nature-based solutions as part of climate-smart agricultural practices; (iii) well-being – promoting protection of natural assets as part of social protection programmes, such as public works programmes; (iv) governance and decentralization – increasing the role of women’s groups in protecting natural resources etc.
- (b) Social co-benefits: All interventions supported by CRPP TF and IF will have strong social benefits, because of the explicit focus on low-income and disadvantaged groups, and the focus on poverty reduction and addressing existing inequalities associated with climate change.
- (c) Economic co-benefits: There is a strong economic element to the investments to be promoted by CRPP TF and supported by CRPP IF, which is captured through the triple dividend of investing in resilience. This includes (i) the economic activity stimulated by reduced impacts, forward-looking planning, long-term capital investments and entrepreneurship as well as macroeconomic benefits, and (ii) development co-benefits from resilience investments, including positive spillovers as well as non-market benefits.
- (d) Gender-responsive development impact: Promoting gender equality is a core thrust of the CRPP which will be promoted through the gender window of the CRPP TF. The focus on gender equality extends from existing evidence that, when grassroot women’s groups are involved in leading the identification and implementation of adaptation measures, the benefits are manifold.

3.4 Needs of the recipient

Scale: High

33. The poverty and vulnerability of the countries and target beneficiaries and their adaptation needs are well elaborated throughout the proposal with descriptions of the scale and intensity of each country’s exposure to climate risks. It highlights the increased socioeconomic vulnerability resulting from the crisis of the coronavirus disease 2019 pandemic, coupled with the recurrence of climate hazards in these countries. Furthermore, the vulnerability of women, especially woman in rural area, is well acknowledged in the funding proposal.

34. The recurrence of climate hazards continues to pose challenges to these countries. Cambodia is a least developed country (LDC) in Southeast Asia with about 90 per cent of the poor living in the countryside. Indonesia has the largest economy in Southeast Asia, nonetheless, poverty remains high in absolute numbers with high inequality. The Lao People’s Democratic Republic as a landlocked country is one of the LDCs with high vulnerability and low readiness for climate change. A large percentage of the population remains close to the poverty line. Pakistan is one of the countries most vulnerable to climate shocks in the world today with high water stress. Papua New Guinea has consistently ranked among the lowest 30 countries in the human development index with 30–40 per cent of poverty rates. While Timor-Leste ranked as one of the poorest LDCs in Asia, Vanuatu, a small island developing state in the Pacific, is exposed to high levels of climate risks, especially to tropical cyclones.

35. Most of the target countries are LDCs with high levels of vulnerability to climate change and financial gaps to cover and sustain climate change adaptation. The proposal highlights the lack of a pro-poor climate information system, public infrastructure, and social services as the major vulnerability factors in the countries. The proposal also notes the lack of access to finance as a constraint to building capacity of government, local governments, communities and vulnerable groups, especially women.

3.5 Country ownership

Scale: High

36. The proposal demonstrates alignment with national adaptation priorities of participating governments, international commitments, and recommendations, such as the Paris Agreement, the Sendai Framework for Disaster Risk Reduction, and the SDGs.

37. Furthermore, the CRPP will contribute to the AE's climate targets and its ambition of providing significant climate finance for adaptation in the region. It also responds directly to the nationally determined contributions of beneficiary countries and builds on the Intergovernmental Panel on Climate Change's call for increased feasibility and effectiveness of adaptation through integrated, multisectoral solutions that address social inequities, differentiated responses based on climate risk and vulnerabilities across a range of biophysical and socioeconomic systems.

38. By strengthening national and local institutions, the proposal seeks to ensure that investments will contribute to building policy coherence around climate resilience and is taken up nationwide by relevant agencies. Countries' ownership will be achieved through stakeholder engagement and capacity-building at various decision-making levels – government, civil society organizations, communities, financial intermediaries, and the private sector.

39. The ADB has demonstrated its strong track record as a credible AE and EE by working with the target countries in the region to address related challenges. It has been developing and implementing climate adaptation projects for the past decade and has the necessary expertise and experience in implementing such projects in the region, including GCF projects in the past years. The ADB builds local capacity on climate change to ensure national specialists as part of the project team for each subproject.

40. The project was designed on the basis of rigorous consultations with relevant stakeholders in the target countries. A stakeholder engagement plan was submitted as an annex to the funding proposal.

3.6 Efficiency and effectiveness

Scale: Medium to high

41. Of the project's total proposed budget of USD 750 million, GCF funding represents 16 per cent of the estimated project cost, with co-financing at 84 per cent. The GCF financing comprises a USD 100 million grant and USD 20 million senior loan. The ADB will provide USD 535 million in senior loans with variable, per-country loan values based on proposed interventions and country-specific needs and USD 20 million in grants. Additionally, bilateral donors including UK Aid, Nordic Development Fund, Agence française de développement will deliver grants of USD 75 million (for more details, please refer to annex 4).

42. The proposed project will provide numerous interventions in multiple countries; therefore, the full spectrum of economic and financial appraisals of proposed individual projects was not feasible at the ex ante analysis. However, a thorough economic and financial analysis (EFA) of selected pilot projects (annex 3) was pursued. At default, as a part of the internal ADB policy guidelines, all other interventions destined for funding under the CRPP IF will be scrutinized a priori against their financial and economic viability. Only subprojects that obtain proper economic net present value (ENPV) and expected internal rate of return (EIRR) thresholds will be funded by the CRPP IF.

43. The results of EFAs of individual pilot projects (tables 2, 3 and 4 below) were pursued over 30 years (table 2) and 25 years (tables 3 and 4), starting from 2024, with a 6-year implementation period and using the economic discount rate of 9 per cent. Please refer to tables 2, 3 and 4 below for detailed EFA estimates.

Table 2. Economic and financial analysis results of individual pilot projects related to Output 1 (Interventions: 1. The Lao People's Democratic Republic Flood and Drought Mitigation and Management)

Estimates		Country	Project Name
*EIRR	13.40%	The Lao People's Democratic Republic	Nam Houm project of the The Lao People's Democratic Republic Flood and Drought Mitigation and Management Sector Project
ENPV	*LAK 105,954 million		
Estimate		Country	Project Name
EIRR	16.67%	The Lao People's Democratic Republic	Houay Gngang project of the Lao People's Democratic Republic Flood and Drought Mitigation and Management Sector Project
ENPV	LAK 23,485 million		
Estimate		Country	Project Name
EIRR	13.81%	The Lao People's Democratic Republic	Hinboun project of the Lao People's Democratic Republic Flood and Drought Mitigation and Management Sector Project
ENPV	LAK 11,199 million		

*EIRR (Expected Internal Rate of Return)

**ENPV (Economic Net Present Value)

***LAK (Laotian Kip Currency)

Table 3. Economic and financial analysis results of pilot projects related to Output 1 (Interventions: Timor-Leste Water Harvesting and Agriculture Value Chains Improvement)

Estimate		Country	Project Name
EIRR	39.1%	Timor-Leste	Rotutu project of Timor-Leste Water Harvesting and Agriculture Value Chains Improvement Project
ENPV	USD 362,900		
Estimate		Country	Project Name
EIRR	24.9%	Timor-Leste	Taituda project of Timor-Leste Water Harvesting and Agriculture Value Chains Improvement Project
ENPV	USD 188,802		
Estimate		Country	Project Name
EIRR	23.9%	Timor-Leste	Maquili project of Timor-Leste Water Harvesting and Agriculture Value Chains Improvement Project
ENPV	USD 183,502		
Estimate		Country	Project Name
EIRR	33.1%	Timor-Leste	Banafi Cuanobe project of Timor-Leste Water Harvesting and Agriculture Value Chains Improvement Project
ENPV	USD 234,737		

Table 4. Indicative/illustrative economic and financial analysis results of Farming System Improvement, Rural Access Road and Water Harvesting Infrastructure (Output 2 indicative models)

Estimates	Country	Project Name
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		Timor-Leste	Farm household models (low elevation)
ENPV	USD 6,180		
Estimate		Country	Project Name
		Timor-Leste	Farm household models (mid elevation)
ENPV	USD 7,910		
Estimate		Country	Project Name
		Timor-Leste	Farm household models (high elevation)
ENPV	USD 2,030		
Estimate		Country	Project Name
		Timor-Leste	Rural Access Road
EIRR	16.2%		
ENPV	USD 96,200		
Estimate		Country	Project Name
		Timor-Leste	Water Harvesting Infrastructure
EIRR	22.1%		
ENPV	USD 59.1 million		

44. The CRPP IF project's EFAs were carried out in accordance with the GCF EFA guidelines and were conducted in financial and economic terms on a per-project basis. The results of outlined EFAs suggest that the proposed investments are likely to be justified from the economic point of view as ENPV and EIRR measures suggest the economic viability of proposed indicative investments (e.g. ENPVs are positive and EIRRs are higher than a chosen economic discount rate of 9 per cent). Furthermore, the results of the EFA suggest that the proposed indicative investment will benefit the entire economy as the benefits of proposed interventions will outweigh the investment costs. The positive direction of the pilot modelling presented is additionally supported by convincing evidence from the ex-post analysis of 23 field pilot tests that tested a set of interventions related to community-based resilience, including local and community-based adaptation planning.³

45. The proposed structure and programmatic approach are assessed to be adequate and reasonable, considering the level of vulnerability and poverty of the target beneficiaries in the region. It is expected to deliver high efficiency and effectiveness as the GCF funds will be leveraged against large volumes of ADB finance. At the CRPP IF level (where GCF funds will be invested) USD 640 million will be leveraged from ADB and other bilateral donors. This results in a ratio of co-financing of more than 5:1, which is assessed to be highly efficient compared to benchmarks in similar/average investments in similar regional contexts, targeting the most vulnerable.

46. The use of GCF finance to fund the climate additionality on top of existing ADB funds is also assessed as effective. The GCF investment will benefit from the existing financial, governance and implementation of activities, which ensure low risk and high conversion of inputs to outputs and outcomes. Hence, the marginal transaction costs for implementing GCF funds will be extremely low.

IV. Assessment of consistency with GCF safeguards and policies

³ Risk to Resilience Study Team. 2009. *Catalyzing Climate and Disaster Resilience: Processes for Identifying Tangible and Economically Robust Strategies: Final Report of the Risk to Resilience Study*. M Moench, E Fajber, A Dixit, E Caspari, and A Pokhrel (eds.). Kathmandu: ISET. p. 328; D Eucker, P Bolte, and F Rahmadana. 2012. *The Long Road to Resilience: Impact and Cost-Benefit Analysis of Community-Based Disaster Risk Reduction in Bangladesh*. Dhaka: Bangladesh Red Crescent Society.

4.1 Environmental and social safeguards

47. **Project background.** The CRPP is part of a regional partnership facility established by the AE to scale up investments in climate adaptation at the community level explicitly targeting the nexus between climate change, poverty and gender. The programme proposed for the GCF financing aims to provide grants and loans funded through the CRPP Investment Fund (IF) to support the efficient roll-out of local adaptation measures in seven host countries, namely Cambodia, Indonesia, Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste, and Vanuatu. Complementary to the technical assistance and grant resources financed by other development partners under the CRPP TF (not funded by GCF), the environmental and social (E&S) risks and impacts of the programme concerned below arise from three outputs that are proposed for the GCF financing as downstream investments on improving and implementing (a) information and systems for delivering applied climate risk informed local investments at scale; (b) climate-resilient pro-poor livelihoods; and (c) pro-poor climate adaptation infrastructures. Among the E&S co-benefits are the empowerment of women's participation in adaptation decision-making and leadership, the reduction of inequalities and poverty associated with climate change, and increases in resilience among low-income and disadvantaged groups.

48. **E&S risk category.** The AE assigned the programme medium level of intermediation, equivalent to the GCF E&S risk category I2. The Secretariat confirms the proposed E&S risk category considering the proposed portfolio is expected to include financial exposure to moderate-risk activities and that Category A activities of potentially significant adverse E&S risks and impacts will not be considered eligible. The Secretariat also confirms that the assigned risk category is within the accreditation level of the AE.

49. **Safeguards instrument.** The AE provided, under the CRPP IF, an Environment and Social Management Framework (ESMF) to articulate the applicable environmental and social policy and reference framework to be followed by the programme; detail the procedures for addressing environmental, social and sustainability issues; prescribe standards and guidelines; and set monitoring compliance procedures to manage E&S risks and impacts associated with the portfolio of investments pursuant to AE's safeguards policies and the requirements of AE's Safeguard Policy Statement (SPS).

50. **Compliance with the GCF environmental and social safeguards (ESS) standards.** The paragraphs below provide a brief description of the programme's compliance with the ESS standards.

51. **ESS 1: Assessment and management of E&S risks and impacts.** As the specific details of the component subprojects are yet to be determined during implementation, the ESMF outlines the reference framework to be complied by the programme, the E&S management procedures for subproject screening and categorization, and the requirements for E&S impact assessments (ESIA) and E&S management plans (ESMP) to address the potential risks and impacts identified. Further to the assessments per the AE's SPS and safeguards policies, the ESMF presents an assessment on the GCF E&S safeguards standards that are likely to be triggered by the proposed CRPP IF investments and suggests the indicative mitigation measures to be detailed at subproject level during implementation. The ESMF also entails three linked documents that further elaborate the safeguards policy structure and requirements related to the subproject E&S due diligence, including (a) the AE's SPS; (b) rapid environmental assessment checklist, various outlines, and monitoring template; and (c) SPS review and update: policy architecture study. While the ESMF has made it clear that the CRPP IF will not finance high-risk activities, the AE is recommended to substantiate the indications on the degree and significance of the E&S risks based on the screening of the existing indicative pipeline and foreseen portfolio of the CRPP IF, especially in relation to the envisioned climate adaptation infrastructure investments.

52. The AE conducted a preliminary conflict sensitivity assessment as part of the programme's design study to understand the conflict situation in the countries included in the CRPP IF. Recognizing the close linkage between climate risk and conflict, where existing conflict situation can exacerbate climate risks and climate change can drive conflict, the AE requires the climate risk assessments to be undertaken as part of subproject identification and preparation process adequately captures the linkage with conflict. Further conflict assessment will be conducted during the programme implementation with focus on the potential of inducing conflicts in relation to community health and safety, land acquisition, and indigenous peoples. In case of expected adverse impacts, an action plan on human rights due diligence will be required. Such action plan will identify national laws and obligations of the country directly applicable to the activities under relevant international treaties and agreements and describe the mitigation measures that will be taken to comply with those obligations and national laws. The AE will ensure that the risks of conflict will be categorized and managed during implementation.

53. **ESS 2: Labour and working conditions.** Financing subprojects on climate-resilient agroecological systems, climate-resilient agriculture supply chains, and the pro-poor climate adaptation infrastructures under the programme may involve hiring of workers by contractors during construction and the employment of workers to operate and maintain the infrastructures following the construction phase. The ESMF triggered the AE's own safeguards policy principles (SPS Safeguard Requirements 1), which require the borrower/client to provide workers with a safe and healthy working environment, taking into account risks inherent to the particular sector and specific classes of hazards in the borrower's/client's work areas, including physical, chemical, biological and radiological hazards, as well as to take steps to prevent accidents, injury and disease arising from, being associated with or occurring during the course of work. The borrower/client is also required to apply preventive and protective measures consistent with international good practice, as reflected in internationally recognized standards such as the World Bank Group's Environment, Health and Safety Guidelines.⁴ Furthermore, based on the Prohibited Investment Activities Lists (Appendix 5 of the AE's SPS linked to the programme's ESMF), the programme will not finance projects or activities involving harmful or exploitative forms of forced labour or child labour. The ESMF suggests mitigation measures to ensure safe and healthy working conditions, compliance with national employment and labour laws, non-discrimination and equal opportunity, access to a workers' grievance redress mechanism (GRM), and occupational health and safety. The AE is recommended to detail the risk screening and mitigation measures on ESS 2 during the programme implementation at the subproject level to make sure the above-mentioned safeguards requirements can be effectively applied to the CRPP IF investments.

54. **ESS 3: Resource efficiency and pollution control.** The ESMF triggered this safeguard standard following the AE's SPS, which requires the borrower/client to apply pollution prevention and control technologies and practices consistent with international good practice during the design, construction, and operation of the programme. SPS Safeguard Requirements 1 covers requirements for pollution prevention, resource conservation, energy efficiency, wastes, hazardous materials, and pesticide use and management. The AE's Prohibited Investment Activities Lists also include the production of or trade in any product or activity deemed illegal under host country laws or regulations or international conventions and agreements or subject to international phaseouts or bans, such as (a) pharmaceuticals, pesticides, and herbicides; (b) ozone-depleting substances; (c) polychlorinated biphenyls and other hazardous chemicals; (d) wildlife or wildlife products regulated under the Convention on International Trade in Endangered Species of Wild Fauna and Flora; (e) transboundary trade in waste or waste products; and (f), production of, trade in, or use of unbonded asbestos fibres. In addition to the general requirements in the ESMF, the programme will require incorporation of

⁴ Available at <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/157871484635724258/environmental-health-and-safety-general-guidelines>.

relevant mitigation measures, such as conserving raw materials and energy, reducing project-related greenhouse gas emissions, reducing water usage, and putting in place solid and hazardous waste management and integrated pest/vector management, within the ESIA and ESMP. The AE is recommended to detail the risk screening and mitigation measures on ESS 3 during the programme implementation at the subproject level to make sure the above-mentioned safeguards requirements can be effectively applied to the CRPP IF investments.

55. **ESS 4: Community health, safety and security.** The ESMF triggered this safeguard standard and identified the main mitigation measures following the AE's own safeguards policy requirements. As per the AE's SPS, the borrower/client needs to identify and assess the risks to, and potential impacts on, the safety of affected communities during the design, construction, operation and decommissioning of subprojects, and will establish preventive measures and plans to address them in a manner commensurate with the identified risks and impacts. Moreover, all the programme activities will require the establishment of preventive and emergency preparedness and response measures to avoid and, where avoidance is impossible, to minimize adverse impacts and risks to the health and safety of local communities. The infrastructure projects to be funded under the programme may pose risks to health and safety of local communities (during the construction and due to exposure to construction traffic and other hazards at construction sites) as well as risks of diseases, gender-based violence, and increase in criminality associated with the labour influx.

56. **ESS 5: Land acquisition and involuntary resettlement.** The ESMF triggered this standard in recognition of the fact that the programme may involve land acquisition and physical and economic displacement in the case of installation of MHEWS and equipment and information and communications technology infrastructure; financing activities for climate-resilient agriculture supply chains; investment in ecosystem-based infrastructure, flood and landslide protection infrastructure; and build-up of multipurpose emergency shelters. As per the programme's ESMF, the AE will screen all subprojects to determine whether they involve involuntary resettlement. For a project involving involuntary resettlement, a resettlement plan will need to be prepared that is commensurate with the extent and degree of the impacts. The degree of the impacts shall be determined by the scope of the physical and economic displacement, and the vulnerability of the affected persons. The AE's Safeguard Requirements 2 (Involuntary Resettlement) in the SPS provides for the scope, objectives and requirements of involuntary resettlement safeguards. SR2 also provides an outline of the Resettlement Action Plan to be detailed where necessary at the subproject level.

57. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The ESMF triggered this safeguard standard as, for example, the proposed climate adaptation infrastructures may be located in sites near or within areas of natural habitat that may or may not be officially declared protected. The AE's SPS requires the borrower/client to assess the significance of project impacts and risks for biodiversity and natural resources as an integral part of the environmental assessment process. The assessment will focus on the major threats to biodiversity, which include destruction of habitat and introduction of invasive alien species, and on the use of natural resources in an unsustainable manner. The borrower/client will need to identify measures to avoid, minimize or mitigate potentially adverse impacts and risks and, as a last resort, propose compensatory measures, such as biodiversity offsets, to achieve no net loss or a net gain for the affected biodiversity. SPS Safeguard Requirements 1 includes detailed provisions for assessing and mitigating impacts on the natural habitat (including on the modified natural habitat, the critical natural habitat and legally protected areas) and required screening and assessment of potential impacts on biodiversity. The ESMF also required that activities shall avoid impacts on biodiversity and ecosystem services, and, if avoidance of impacts is not possible, measures to minimize impacts and compensate and restore biodiversity and ecosystem services will need to be implemented. For activities that have potential adverse impacts on natural habitats, the preparation of a biodiversity action plan (BAP) will be mandatory. The ESMF provides an outline of the BAP as referenced in Linked

Document 2 of the ESMF. The ESMF also requires the implementation of measures to minimize habitat fragmentation (such as through biological corridors), restore habitats during and/or after operations, and implement biodiversity offsets.

58. **GCF Indigenous Peoples Policy and ESS 7: Indigenous Peoples.** Indigenous Peoples are present in all of the seven selected countries for the project, namely Cambodia, Indonesia, Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste and Vanuatu. The ESMF accordingly provides an Indigenous Peoples Impact Categorization (IPIC) assessment process and the outline of an Indigenous Peoples Plan (IPP Outline), which is required for all projects with impacts on Indigenous Peoples. The IPIC provides a detailed Indigenous Peoples Impact Screening Checklist, which covers the triggers required by the GCF Indigenous Peoples Policy (IPP), such as identification of Indigenous Peoples, identification of Indigenous Peoples without requiring status of vulnerability, identification of potential impacts and identification of special requirements. The IPP Outline provides guidance regarding social impact assessment, information disclosure, consultation and participation, beneficial measures, mitigative measures, GRM, monitoring, reporting and evaluation, institutional arrangements, and budget and financing. While the ESMF meets the basic requirements of the GCF IPP, the following recommendations are provided to cover remaining gaps: the AE will ensure that the broad community support obtained for any subprojects will meet the free prior informed consent requirements of the GCF IPP and that subprojects that only develop natural resources will also require special procedures for consent. In line with its roles and functions, the GCF Indigenous Peoples Advisory Group is available to provide advice to the AE, which is also encouraged to share emerging good practices and success stories with the Group.

59. **ESS 8: Cultural heritage.** The ESMF triggered this safeguard standard. Climate adaptation infrastructure projects to be funded by the programme might be located near and/or affect cultural heritage structures or sites. The AE's SPS defines physical cultural resources as movable or immovable objects, sites, structures, groups of structures, and natural features and landscapes that have archaeological, paleontological, historical, architectural, religious, aesthetic, or other cultural significance. It requires chance finds procedures be included in the ESMP when the proposed location of a subproject is in an area where physical cultural resources are expected to be found as determined during the risk assessment process. The AE's SPS also requires that chance finds shall not be disturbed until an assessment by a competent specialist has been made and actions consistent with these requirements have been identified. The ESMF specifies that the programme will not remove any physical cultural resources unless (a) no alternatives to removal are available; (b) the overall benefits of the subproject substantially outweigh the anticipated cultural heritage loss from removal; and (c) any removal is conducted in accordance with relevant provisions of national and/or local laws, regulations, and protected area management plans and national obligations under international laws, and employs the best available techniques. The detailed provisions for physical cultural resources are discussed in SPS Safeguard Requirements 1, including the requirement to consult with affected communities and the relevant national or local regulatory agencies, and measures to avoid or mitigate impacts as part of the environmental planning process. The ESMF suggests the main mitigation measures including compliance with local cultural heritage regulations or protected area management plans, consultation with relevant stakeholders, and measures to implement additional programmes to promote and enhance protected area conservation.

60. **Sexual exploitation, sexual abuse and sexual harassment (SEAH) safeguarding.** The ESMF triggered SEAH safeguards specifying that the programme will not finance investments that are anticipated to have significant SEAH risks and impacts. The AE will take an approach of zero-tolerance towards SEAH and ensure that the risk category of CRPP IF financing is proportional to the nature, scale and location of activity including in relation to SEAH. In addition, the ESMF specifies that the AE is responsible for incorporating and monitoring SEAH prevention, mitigation and response measures guided by the AE's newly

published Good Practice Note ⁵ to address SEAH risks in its operations focusing on sovereign projects involving civil works. The AE will also ensure that adequate information related to CRPP IF SEAH safeguarding is provided to GCF to enable review. The programme will address SEAH risks and impacts as part of the subproject's due diligence ESIA and ESMP. In the event of SEAH incidents, the programme will establish a survivor-centred and gender-responsive GRM for SEAH-related grievances including confidential reporting, safe and ethical documentation of SEAH cases, and mechanisms for timely services and redress for survivors.

61. **Implementation arrangements.** The programme will be managed by the AE, which also serves as the executing entity (EE). It has in-house safeguards staff in its Climate Change and Sustainable Development Department (CCSD) to manage and operate the programme. At the programme level, the AE will monitor portfolio-level safeguards implementation and performance to ensure the transactions meet the AE's safeguards requirements. The Facility Manager of the CRPP IF's umbrella programme will verify that the AE has conducted the necessary safeguards review, ensuring ongoing compliance monitoring and reporting, and confirming that the project aligns with the relevant safeguard policies of both the AE and the GCF, as outlined in the CRPP IF ESMF. Training will be provided to borrowers/clients and project developers as needed, as part of the AE's broader SPS initiative aimed at strengthening country systems. The AE will engage experts on human rights and SEAH from the roster of consultants to contribute to the attainment of the CRPP IF ESMF objectives.

62. **Stakeholder engagement and information disclosure.** The AE provided a stakeholder engagement plan with the proposal, which summarizes the stakeholders' consultations conducted during the preparation and development of the CRPP concept and the present proposal for the GCF. It also presents an analysis of stakeholders' engagement, a brief explanation of the AE's policies and procedures relevant to stakeholders' participation, and the required processes to be implemented during key phases of the project lifecycle. The AE has conducted stakeholder engagements in the preparation of the project, including consultations with relevant national and local implementation partners in developing the project concepts and with the national designated authorities to seek their guidance on project design and to secure the no-objection letters for the programme. For the subprojects that will be supported under the CRPP IF, the AE's SPS requires the borrower/client to carry out meaningful consultation with affected people and other concerned stakeholders, including civil society, and facilitate their informed participation. Consultation will be carried out in a manner commensurate with the impacts on affected communities. However, since the details of subproject disclosure are yet to be agreed upon with the AE, it is recommended to include in the funded activity agreement (FAA) the disclosure and communications process of ESS documentation for Category B subprojects.

63. **Grievance redress mechanism.** The AE will establish a project-level GRM that will address the affected people's concerns and complaints including on SEAH. The ESMF requires that the GRM address affected people's concerns and complaints promptly, using an understandable and transparent process that is gender responsive, culturally appropriate and readily accessible to all segments of the affected people. The GRM will be accessible by the affected stakeholders throughout the programme cycle. In addition to the project-specific GRM and the GCF's independent redress mechanism, the AE also maintains its own accountability mechanism. The accountability mechanism provides an independent forum and process for people adversely affected by projects financed by the AE, including in cases of allegations of violation of the AE's own operational policies and procedures. In line with the GCF Indigenous Peoples Policy, the GCF indigenous peoples focal point will be available for assistance at any stage, including before a claim has been made.

⁵ Available at: [Good Practice Note on Addressing Sexual Exploitation, Abuse, and Harassment in ADB-Financed Projects with Civil Works | Asian Development Bank](#).

4.2 Gender policy

64. The AE has provided a gender assessment and gender action plan therefore complies with the GCF gender policy. It has also provided a programme-level gender assessment and gender action plan.

65. The gender assessment is undertaken through desk review and stakeholder consultations, the latter being with government experts, community and several women's groups. Further assessments are planned at subproject level with a gender equity theme and taking effective gender mainstreaming into consideration. This will be done to ensure key gender-related issues in the project's sectors of focus are understood and issues are addressed based on country circumstances.

66. The current programme-level gender assessment describes the gender issues at the country level and indicates that all seven countries are committed to gender equality and women's empowerment. The constitutions of all the countries guarantee equality for women and men in the economic, political and social spheres.

67. Many of the countries have ratified international agreements such as the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) while having introduced various policies and national frameworks to promote equality. Many of the countries have national policy environment and supportive legislations such as national climate change policies and financial inclusion strategies (Pakistan); national framework action (Papua New Guinea); and Strategic Development Plan and National Adaptation Programme of Action (Timor-Leste).

68. The assessment also shows that though the seven countries are committed to gender equality and women's empowerment, the ranking on the gender index varies and levels of equality are different. Generally, however, all countries need to invest more to meet their own commitments to gender equality and ensure that, as a population more vulnerable to climate change, women are supported to help mitigate and adapt to climate change.

69. The assessment has also demonstrated that violence against women, inequality in access to education, stark gaps in economic and labour force participation and limited participation of women in the political space, are all factors that hinder progress for women. Countries demonstrate gender-based barriers to women in areas of literacy and education, skills to progress in the professional fields, violence during disasters, and the double burden of domestic and reproductive roles. Yet women play a critical role and contribute in all facets of each country's economy and social well-being. The general gender issues as related to programme sectors have been provided for each country.

70. The programme gender action plan (GAP) provided by the AE includes indicative actions with corresponding indicators, targets, budget and timelines of the activities that will be delivered. Gender expertise will be available to support the implementation of all activities included in the plan and provide support in the supervision, monitoring and evaluation of the GAP.

71. The activities of the GAP will address the challenges for women in accessing resources (e.g. finance) and skills. Specific actions in this regard are: investments in technical assistance and grant resources that will develop the capacities and knowledge of various actors to be targeted by the projects; a trust fund with a gender window specifically targeting and investing in women; building women's knowledge and capacity to identify investment opportunities; and preparation of women-focused investment projects.

72. Further actions include a focus on addressing women's needs in accessing climate information. This will be done by ensuring gender issues are mainstreamed in climate information and in systems to deliver local investments that apply climate-risk information at

scale. Climate-resilient pro-poor livelihoods and adaptation investments will also focus on women's needs.

73. It was noted through the assessment that gender inequality emanates from, and continues to be a barrier through, unequal norms and practices existing in countries. The GAP includes activities that are targeted at addressing structural barriers to gender equality and gender issues through policy reform and revision of regulations.

74. The programme will also ensure that employment and income-generation opportunities are presented to women. The various actions and investments presented in the GAP are expected to empower women, facilitating an increased role in decision-making, increasing access to and control of resources in the agriculture sector, and to technologies. In addition, the GAP is expected to build countries' overall capacity to mainstream gender across sectors. The programme will focus on and highlight women-led measures that could be scaled up through larger investments in the future.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

75. GCF is requested to provide a grant of USD 100 million and a senior debt of USD 20 million for the CRPP covering seven countries. The AE is co-financing a senior debt of USD 535 million and a grant of USD 20 million. Bilateral donors are providing a total of USD 75 million in the form of a grant.

76. The CRPP consists of two separate but interlinked funds; namely CRPP TF providing technical assistance for upstream support, and CRPP Investment Fund (IF) focusing on the local adaptation measures through downstream investment in projects. The GCF funds will be used to finance CRPP IF only. Although projects supported under the TF may be financed by the IF and countries may benefit from the TF to develop IF projects, TF support is not a pre-condition to be eligible for IF investment and there is no direct sequencing between CRPP TF and IF.

4.3.2. Accredited entity/executing entity capability to execute the current programme (low risk)

77. The ADB will be both AE and EE for the programme. The CRPP is operationalized through the Community Resilience Financing Partnership Facility (CRFPF) which has been administered by the ADB since 2021. GCF funds for the CRPP will be ring-fenced within an ADB GCF TF. The AE has an extensive track record of managing more than 40 trust funds to support programmes and technical assistance in the region.

4.3.3. Programme-specific execution risks (medium risk)

78. Demand risk: At least one project per country will be financed under CRPP, however the allocation and use of resources under CRPP IF will be determined at a later stage. It will depend on the AE to source projects that meet the eligibility criteria. In case there is a lack of pipelines, the targeted impact will be reduced and GCF funds might not be fully utilized. Currently there is no limit in terms of the extension of the CRPP.

79. Funding risk: The CRPP targets reaching a size of USD 750 million including USD 675 million for CRPP IF. The AE envisages that the time period of the IF will be from 2023 to 2030, meaning individual projects will be likely to be approved between 2024 and 2027, and completed by 2030. The GCF resource is expected to leverage at least USD 555 million from the AE and USD 75 million from other co-financiers. However, the term sheet does not include any timeline for the target fund size to be reached. Comfort can be derived from the covenants for

proportionate disbursement of GCF and co-financing amounts and the eligibility criteria, which includes the ratio of GCF grant to AE loan/grant, should be at least 1:4. If the AE cannot reach its target size, the unused funds need to be reflowed to GCF to avoid the fund being idle.

80. Credit risk: The borrower of the GCF loan will be sovereign or sovereign entity in Pakistan (Caa3 by Moody's), therefore GCF is exposed to the sovereign credit risk. The country experiences several credit challenges with a weak fiscal position. However, the concessional term from GCF will be conducive for the country's debt affordability. The GCF loan will be aligned with ADB sovereign loans, thus be covered by similar sovereign guarantees based on the AE's due diligence.

81. Project viability and concessionality: Since the CRPP will support a wide range of interventions, the overall CRPP level ex ante economic and financial analyses are not feasible. Instead, the AE carried out analyses in each component (Components 4, 5 and 6) with indicative pilot projects. Components 4 and 6 show a relatively high benefit to cost ratio with a range of 6 to 9. Component 5 results in the EIRR between 15 per cent and 35 per cent. However, the exact EIRR and financial internal rate of return will vary depending on the projects to be funded. All the projects under CRPP IF will have to pass the economic and financial viability criteria to be taken forward.

82. Apart from the loan provided to Pakistan, the remaining 83 per cent of GCF financing is a full grant. The GCF grant will be blended with ADB grants and/or loan to pass down the concessionality. One of the mandatory eligibility criteria is that the concessionality of GCF will not substitute investments that might have taken place anyway using regular AE borrowing.

4.3.4. Compliance risk (medium risk)

83. ADB, as accredited entity (AE), has confirmed that the CRPP IF reveals a low-level risk regarding prohibited practices, money laundering and terrorist financing, with a low probability and impact. The ADB mitigation strategy includes enforcing ADB's rigorous due diligence policies at the subproject level. ADB conducts integrity due diligence, proactive integrity reviews, and offers oversight through its Office of Anticorruption and Integrity (OAI).

84. ADB also confirmed that it enforces compliance with its Anticorruption Policy and financial management assessments. Disbursements follow ADB's Loan Disbursement Handbook, reinforcing financial prudence and ensuring financial transparency. OAI handles integrity violations, offering protection to whistleblowers. ADB confirmed that it strictly prohibits individuals/entities on the United Nations Security Council sanctions list from participating in CRPP contracts and avoids operations in sanctioned jurisdictions.

85. The Secretariat has conducted a review of the project in accordance with relevant GCF Board-approved policies and does not find any material issue or deviation with respect to compliance-related issues. The risk-mitigation measures informed by ADB appear to effectively manage the low-level risk of prohibited practices, money laundering and terrorist financing, ensuring transparency and accountability in CRPP IF activities. Based on available information for this funding proposal, the Secretariat has determined a risk rating of "medium" and has no objection to this proposal proceeding.

4.3.5. GCF portfolio concentration risk (low risk)

86. In case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

87. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall programme	Medium
Accredited entity (AE)/executing entity (EE) capability	Low
Project-specific execution	Medium
GCF portfolio concentration	Low
Compliance	Medium

4.4 Fiduciary

88. The ADB will play the dual role of AE and EE in the programme.

89. As AE, the ADB will manage and operationalize the CRPP through the CRPPF. The ADB will host and manage the CRPP IF operations through the Climate Change and Sustainable Development Department (CCSD), acting as financial intermediary to ADB developing members for providing funds from GCF alongside ADB funds to support investments in selected countries.

90. On the other hand, as the EE for the CRPP, ADB will be responsible for the overall programme, particularly the provision of technical assistance support under the CRPP TF (Component 1), and the provision of grants and loans for subprojects under the CRPP IF (Component 2).

91. GCF funds for the CRPP will be ring-fenced within an ADB GCF TF which was created pursuant to the accreditation master agreement, and as such benefit from the same privileges and immunities as ADB resources. Usage of GCF funds will be based on criteria outlined in the funding proposal which will be included in dedicated programme implementation guidelines to be developed. These guidelines will complement the programme Term Sheet and FAA, and include further guidance on implementation arrangements, project selection and prioritization, monitoring and evaluation, and reporting.

92. The ADB will offer concessional loans and grants to sovereign borrowers to be funded by GCF. For each CRPP IF project, the ADB will sign separate loan and/or grant agreements with the sovereign borrowers/beneficiaries for each source of financing. Each project will be implemented in compliance with ADB policies and CRPP requirements.

93. Disbursement requests from the ADB to GCF are expected to be made on an annual basis in accordance with the expected ADB approval dates, and projected disbursement plans and timelines of the projects in the CRPP pipeline. Payments of interest and principal for GCF-funded subproject loans will be repaid to the ADB GCF TF. Principal, interest payments, and any unused funds or income will be subsequently returned to GCF.

94. The CRPP-supported projects will follow ADB systems for financial management, disbursement, audit and procurement. Implementation of the programme and projects will also conform to all applicable ADB policies. In particular, all subprojects financial accounting, disbursement procedures and auditing requirements will be developed during project preparation and will be included in their respective grant implementation manuals. The ADB shall ensure that these are in accordance with ADB policies, procedures and approach on financial management and analysis of projects. Project disbursements will follow ADB's Loan Disbursement Handbook (2017, as amended from time to time) and detailed arrangements agreed upon between the government and the ADB. Public disclosure of the audited project's financial statements, including the auditor's opinion on the project financial statements, will be guided by the ADB Public Communications Policy of 2011.

95. As at the date of this assessment, it is recommended that a Condition Precedent relating to provision of capacity assessment regarding EEs is reflected in the Term Sheet and FAA. This recommendation is subject to change.

4.5 Results monitoring and reporting

96. As an adaptation initiative, the programme aims at generating diverse adaptation results across all four GCF adaptation result areas. A minimum of 3,219,400 direct beneficiaries and 4,465,000 indirect beneficiaries are estimated to receive adaptation benefits, which account for 0.63 per cent and 0.87 per cent of the total population of seven target countries respectively. Definitions and broad methodologies have been provided by the AE to estimate the adaptation beneficiaries. Acknowledging the difficulties of beneficiary estimation at this stage without detailed information about individual projects, it was recommended for the AE to develop and submit a Monitoring, Evaluation and Learning Plan within 12 months of the programme commencement.

97. The theory of change (TOC) diagram adequately captures different levels of expected changes as well as logical linkages between them from the goal statement to the proposed activities. It is helpful that the TOC diagram identifies co-benefits which can be achieved along the way towards climate results and captures the causal relationships between outcomes and co-benefits. The barriers and risks that may hinder the achievement of these changes are identified and linked to pertinent activities which will tackle them accordingly. Project components are well embedded into the project TOC.

98. The logical framework has been developed in accordance with the GCF integrated results management framework. The programme is expected to have huge paradigm shift potential across all seven target countries in all three dimensions: scale, replicability and sustainability.

99. It will contribute to all four GCF adaptation result areas with the CRPP TF and IF which will be established with the programme, and which will finance various individual projects. Four outcomes are identified with six outputs with clear logical linkages. Adequate GCF core and supplementary indicators are chosen to monitor the results of the programme, however it is highly recommended for the AE to provide the Secretariat with concrete methodologies and targets for each indicator once the detailed Monitoring, Evaluation and Learning Plan will be developed. At the GCF outcome level of “Enabling environment”, two GCF core indicators have been identified: institutional and knowledge generation, capture and learning. Given the relatively low or medium baseline of each indicator, the programme is expected to build a comprehensive enabling environment at multi-country level.

100. The programme has identified relevant project-specific indicators for project outcomes and outputs. It is useful that the programme will measure co-benefits with specific indicators: this will increase the level of understanding of the project and will give an opportunity for the Secretariat to capture non-climate results that are worth monitoring and reporting.

4.6 Legal assessment

101. The original Accreditation Master Agreement was signed with the Accredited Entity on 17 August 2017, and it became effective on 6 September 2017. The Accredited Entity’s first term of accreditation expired on 5 September 2022. The Accredited Entity was reaccredited by the Board pursuant to Decision B.33/10. The first amendment and restatement agreement amending and restating the accreditation master agreement was signed on 13 July 2023 and the amended and restated accreditation master agreement (the “AMA”) became effective on 20 September 2023.

102. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme.

103. The proposed programme will be implemented in seven (7) countries: the Kingdom of Cambodia, the Republic of Indonesia, the Lao People's Democratic Republic, the Islamic Republic of Pakistan, the Independent State of Papua New Guinea, the Democratic Republic of Timor-Leste and the Republic of Vanuatu.

(a) The GCF has signed a bilateral agreement on privileges and immunities agreement with two (2) of the seven (7) countries:

- (i) **Independent State of Papua New Guinea:** The GCF has signed a bilateral agreement on privileges and immunities with the Government of the Independent State of Papua New Guinea. It was signed and became effective on 21 September 2016.
- (ii) **Republic of Vanuatu:** The GCF has signed a bilateral agreement on privileges and immunities with the Government of the Republic of Vanuatu. It was signed and became effective on 9 August 2017.

(b) GCF is not provided with privileges and immunities in the five (5) countries listed below in which the proposed programme will be implemented. This means that, among other things, GCF is not protected against litigation or expropriation in these countries, the risks of which need to be further assessed. The Secretariat has engaged with the relevant countries as follows:

- (i) **Kingdom of Cambodia:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Kingdom of Cambodia on 19 April 2017. The latest communication was sent by the GCF Secretariat to the Government of the Kingdom of Cambodia on 5 October 2018.
- (ii) **Republic of Indonesia:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Republic of Indonesia on 7 December 2015. The latest communication took place on 2 April 2019 whereby the GCF Secretariat sent the updated draft agreement to the Government of the Republic of Indonesia.
- (iii) **Lao People's Democratic Republic:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Lao People's Democratic Republic on 21 February 2019. The latest communication took place on 23 March 2022 whereby the GCF Secretariat sent the updated draft agreement to the Government of the Lao People's Democratic Republic. Negotiations on the agreement are ongoing.
- (iv) **Islamic Republic of Pakistan:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Islamic Republic of Pakistan on 15 September 2015. The latest communication took place on 3 March 2023 whereby the GCF Secretariat sent the updated draft agreement to the Government of the Islamic Republic of Pakistan. Negotiations on the agreement are ongoing.
- (v) **Democratic Republic of Timor-Leste:** The GCF Secretariat provided a draft agreement on privileges and immunities and a background note to the Government of the Democratic Republic of Timor-Leste on 7 April 2016. The latest communication took place on 1 January 2021 whereby the GCF Secretariat

sent the updated draft agreement to the Government of the Democratic Republic of Timor-Leste. Negotiations on the agreement are ongoing.

104. The Heads of the Independent Redress Mechanism (IRM) and Independent Integrity Unit (IIU) have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where the GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by the GCF are made only after the GCF has obtained satisfactory protection against litigation and expropriation in the country(ies), or has been provided with appropriate privileges and immunities.

105. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP215

Proposal name:	Community Resilience Partnership Program
Accredited entity:	Asian Development Bank
Country/(ies):	Cambodia, Indonesia, the Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste and Vanuatu
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1. This is the first submission of the "Community Resilience Partnership Program (CRPP)" to the independent Technical Advisory Panel (iTAP) under the standard Proposal Approval Process (PAP). The accredited entity (AE) is the Asian Development Bank (ADB). The funding proposal covers seven countries: Cambodia, Indonesia, the Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste, and Vanuatu.
2. The CRPP is aimed at helping countries and communities in Asia and the Pacific region scale up investments in adaptation to climate change, especially investments at the community level, and targeting the climate change-poverty-gender nexus.
3. The funding proposal has been prepared as a response to three challenges: (i) the increasing climate shocks and stresses in Asia and the Pacific region; (ii) poor and already vulnerable populations in the participating countries that are already suffering climate impacts and require targeted adaptation solutions; and (iii) the need for climate finance, which is critical for facilitating adaptation action by these highly vulnerable social groups.
4. ADB expects to reach a minimum of 3,219,400 direct and 4,465,00 indirect beneficiaries, and to contribute to all four adaptation result areas as follows: most vulnerable people and communities (30 per cent); health and well-being, and food and water security (30 per cent); infrastructure and built environment (20 per cent); and ecosystems and ecosystem services (20 per cent).
5. The CRPP will be operationalized through the Community Resilience Financing Partnership Facility (CRFPF)¹ which was established by ADB in August 2021 and comprises two separate but interlinked components: the CRPP Trust Fund (TF) and the CRPP Investment Fund (IF). CRFPF is not a separate legal entity, but a facility administered within and by ADB.
6. Component 1, the CRPP Trust Fund (TF), focuses on upstream support to strengthen the enabling environment required for implementing local adaptation measures at scale. Component 2, the CRPP Investment Fund (IF) focuses on the efficient roll-out of local adaptation measures through a downstream investment programme. According to its budget, the funding proposal includes a third, smaller component dedicated to programme implementation, monitoring and evaluation and learning (MEL) support.
7. The expected outputs from the CRPP are: (1) knowledge and action research on climate risk-informed pro-poor community-level solutions strengthened; (2) institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened; (3) investments supporting inclusive pro-poor, community-level adaptation

¹ Community Resilience Financing Partnership Facility (CRFPF).

measures prepared; (4) information and systems for delivering applied climate risk-informed investments at scale improved; (5) climate resilient pro-poor livelihoods investments implemented; and (6) pro-poor climate adaptation infrastructure investments implemented.

8. The first three outputs will be funded through the CRPP TF (USD 75 million), supported by bilateral donors including the Government of the United Kingdom, the Nordic Development Fund, and the French Development Agency. Outputs four to six are proposed to be funded through the IF (USD 675 million), which will be financed by GCF (USD 120 million) and the Asian Development Bank (USD 555 million). Thus, funding from GCF, consisting of USD 100 million in grants and USD 20 million in loans, will be channeled solely through the IF.

9. For the projects² financed by the IF and implemented in host countries, the government of the host country, acting through its relevant ministry or department, may be the executing entity and/or implementing agency and will follow ADB procedures.

10. In its assessment the iTAP considered all documents as submitted on 8 August 2023, the follow-up answers both in written form and during the meeting between the iTAP review team, representatives from ADB and members from the Secretariat task team, and additional documents sent to the iTAP as agreed during the question and answer session.

1.1 Impact potential

Scale: High

11. This funding proposal does not consider a mitigation impact potential.

Adaptation impact potential

12. The ADB expects to reach a minimum of 3,219,400 direct beneficiaries and 4,465,00 indirect (see table 1 below for details)

Table 1: Expected beneficiaries of the CRPP per country

Country	Expected minimum number of direct beneficiaries	Percentage of the total population	Expected minimum number of indirect beneficiaries	Percentage of the total population
Cambodia	1,300,000	8.13%	1,365,000	8.53%
Indonesia	1,200,000	0.45%	450,000	0.17%
Lao People's Democratic Republic	141,000	2.04%	1,143,000	16.57%
Pakistan	488,000	0.23%	1,200,000	0.57%
Papua New Guinea	10,000	0.11%	25,500	0.27%
Timor-Leste	60,000	4.62%	241,000	18.58%
Vanuatu	20,400	7.02%	40,000	13.76%
Total	3,219,400	0.63%	4,465,000	0.87%

Source: Funding Proposal. Main document and annex 17

13. Direct beneficiaries are considered to be those who are specifically targeted by the intervention and participate in, or are recipients of, goods (e.g. agricultural extension), services (e.g. early warning/climate information), and/or new knowledge or improved practices (e.g. water harvesting, agroforestry). Indirect beneficiaries are those located in the coverage area and hence likely to receive a measurable adaptation benefit from CRPP projects' intervention. Due to the diversity of country contexts and the range of activities that the CRPP is able to

² In the documents received by the iTAP, the terms "subprojects" and "projects" under the CRPP programme were used interchangeably and for referring to the individual projects financed by CRPP in each participating country. In this assessment we use the term "projects" as it is the term used by ADB in the corrected version of the funding proposal (v.11, 29.August 2023).

deploy under each project, both direct and indirect beneficiaries will vary widely and be confirmed as part of annual reporting.

14. The proposal includes two levels, a programmatic and a project-specific level. The current assessment was done differentiating the two levels.

15. Programmatic level: the funding proposal includes a climate profile for each of the participating countries (annex 2.1). The profiles are based on the information from the World Bank's Climate Change Knowledge Portal³ for presenting the evolution of climate hazards, exposure and vulnerability at the country level. Each profile includes climatological information (e.g. average monthly temperature and rainfall), documented observations about past evolution of climate variables (temperature and rain patterns), as well as future climate projections using four representative concentration pathways.

16. The funding proposal further presents specific socioeconomic conditions of the participating countries including a preliminary analysis of the link between social conflicts and climate change. Based on the climate change-related information and considering the socioeconomic context, the profiles introduce the climate risk in each country. In addition, the profiles present the evolution of climate policy in each of the participating countries.

17. As a response, the proposal introduces indicative examples of activities aimed at reducing climate risks per country as well as potential complementarity and synergies with other ongoing projects that have been financed by the Adaptation Fund, GCF or the Global Environment Facility.

18. The funding proposal explains existing barriers to scaling up adaptation at the livelihood level and how the proposed CRPP will contribute to overcoming these (for more detail see sections 1.2 "Paradigm shift potential" and 1.4 "Needs of the recipient").

19. In terms of monitoring and evaluation, the funding proposal includes a preliminary version of the CRPP monitoring, evaluation and learning (MEL) system, based on the principles of participation, focus on learning and practicality (annex 11). The MEL will combine a multilevel monitoring and evaluation approach (project and programme level) with a theory-based approach and a dynamic approach to monitoring and evaluation. The funding proposal states that a baseline for MEL will be prepared at the beginning of the implementation phase. However, the proposal is not yet clear about how to assess the outcomes of adaptation activities or to quantify the adaptation beneficiaries.

20. Project level: The funding proposal includes a well-structured framework for selecting projects and allocating resources, which comprises five overarching principles, eligibility criteria, project selection process and assessment tools, as well as the governance structure that clarifies the roles and responsibilities in selecting projects in host countries (annex 2.4).

21. The five principles are:

- (a) Respond to current and future climate risks in the context of changing socio-economic development;
- (b) Promote adaptation measures that address underlying drivers of poverty and gender inequality;
- (c) Promote selection of adaptation measures based on the principle of subsidiarity;
- (d) Promote adaptation measures in the context of physical, ecosystem-based, financial, and/or social and institutional resilience; and
- (e) Promote cross-sector linkages and mainstreaming into development.

³ <https://climateknowledgeportal.worldbank.org/>

22. The eligibility criteria for projects in the CRPP mirrors the investment criteria of the GCF, plus one additional eligibility criterion on gender as a stand-alone criterion by which each project must fall under the ADB's Gender Equity Theme or under Effective Gender Mainstreaming. This criterion is important since climate vulnerability is produced to a great extent by underlying socioeconomic inequality and vulnerability. Hence, the CRPP's explicit target on the nexus between climate change, poverty and gender.

23. Component 2, the Investment Fund (Outputs 4, 5 and 6) will deliver adaptation activities through the provision of combined grant finance with major investment funding to deliver adaptation at scale. Indicative ideas for specific national projects to be funded under the CRPP IF are included in Table 2 below.

Table 2: Ideas on project concepts for the CRPP Investment Fund

Country	Name of the project	Areas of adaptation measures	Sector
Cambodia	Climate-resilient Smallholder Development Project	Rural infrastructure Community resilience and Disaster Risk Reduction Institutional resilience Financial resilience through insurance	Agriculture, Natural Resources and Rural Development
Lao People's Democratic Republic	Building Resilience and Reducing Rural Poverty Project	Pro-poor social registry for attending (climate) disaster Link the social registry with a climate early warning system Technical assistance	Agriculture, Natural Resources and Rural Development
Lao People's Democratic Republic	Flood and Drought Mitigation and Management Project	Rural employment through construction Reduction of flood damages and diversification of crops Investments in better diets and sanitation	Agriculture, Natural Resources and Rural Development
Pakistan	Sindh Coastal Resilience Project	Climate risk assessment and emergency response Enhanced water resources and coastal zone management Climate resilient protective infrastructure Ecosystems restoration	Multisector
Papua New Guinea	Improved Technical and Vocational Education and Training for Climate Resilient Employment Project	Including climate-smart agriculture and resilient construction in training and vocational education for vulnerable people, especially women	Education

Timor-Leste	Agriculture Value-Chains and Water Harvesting Project	Using CSA approaches for small-scale agriculture including forest management and conservation	Agriculture, Natural Resources and Rural Development
Vanuatu	Climate Change and COVID-19: Building Comprehensive Resilience in Poor and Vulnerable Groups in Vanuatu	Climate responsive social protection systems and enhancing climate resilient livelihood support mechanisms	Public Sector Management

Source: CRPP funding proposal, main document and CRPP annex 3.

Note: There is no idea/proposal for Indonesia.

24. The funding proposal economic and financial analysis (annex 3) develops three ideas on project concepts based on projects from the ADB: (i) the Lao People's Democratic Republic: Flood and Drought Mitigation and Management Project; (ii) Pakistan: Sindh Coastal Resilience Project; and (iii) Timor-Leste: Water Harvesting and Agriculture Value Chains Improvement Sector Project (annex 2.3). While currently in their very early stages of conceptualization, these are examples of large-scale interventions that could serve as vehicles for the delivery of specific local adaptation solutions promoted by the CRPP, using IF resources provided by GCF.

25. The iTAP acknowledges the completeness of the information regarding the climate adaptation potential in this funding proposal as well as the alignment with three of the four principles for demonstrating adaptation potential, as specified in GCF decision B.33/12: identification, response and alignment. However, the iTAP assesses that the principle of monitoring and evaluating adaptation outcomes is not yet fulfilled in the extent required by the GCF Board Decision B.33/12. That said, the iTAP recognizes that this gap can be resolved during the first year of implementation of the project.

26. Further, the iTAP acknowledges that due to the programmatic approach in this proposal it is difficult to calculate the exact number of beneficiaries, and noted the clarification provided by the AE for arriving at the figures in Table 1. However, the iTAP sees a possibility for over or underestimating the number of beneficiaries at the present stage.

27. The iTAP assesses the adaptation impact potential as high and includes the improvement of the MEL system during the first year of project implementation as a condition for endorsement. Beyond the aspects already outlined in annex 11, the MEL system will include:

- (a) A complete monitoring and evaluation system that allows assessment of adaptation outcomes, including:
 - (i) The corresponding methods for the steps of data gathering, analysis and interpretation, and the frequency of each step;
 - (ii) The roles and responsibilities for the steps above;
 - (iii) A method to assess progress in addressing the climate-gender-poverty nexus that has been included as a key element of this funding proposal; and
 - (iv) A clear method for quantifying the number of direct and indirect beneficiaries; and
- (b) A strategy for linking the MEL of the CRPP's national projects to the national systems for monitoring and reporting progress under the United Nations Framework Convention on Climate Change (UNFCCC).

1.2 Paradigm shift potential

Scale: Medium

28. The funding proposal claims a “contribution to transformational change by: (i) mobilizing large-scale public investments that support community-level adaptation of poor and vulnerable people; (ii) developing national and local policies, plans, and programs that promote financing for community-led adaptation; and (iii) increasing the meaningful participation of poor women and men in resilience-related decision-making” (annex 11).

29. The funding proposal identifies four barriers to strengthening the climate resilience of poor and vulnerable people in Asia and the Pacific Region:

- (a) Lack of suitable, locally relevant and accessible climate information to inform resilience investments targeted at the poor;
- (b) Limited access to adaptation finance for implementing local solutions at scale;
- (c) Weak institutional and technical capacity to mainstream climate adaptation in policy, planning and implementing adaptation solutions; and
- (d) Negative perceptions and risk aversion around community-level actions.

30. As a response to these barriers, the funding proposal proposes three pathways of change. The first is strengthening the enabling environment for financing and implementing pro-poor local adaptation measures at scale in the participating countries (Outputs 1 and 2). The second pathway is mobilizing pro-poor adaptation finance at scale (Outputs 3 and 4) and the third pathway for change is implementing large-scale pro-poor adaptation solutions to build long-term capacity to deal with climate change shocks and stresses (Outputs 5 and 6).

31. According to the funding proposal, the components of the project are aimed at operationalizing these pathways of change. The first component, the funding proposal CRPP Trust Fund (no GCF contribution) will finance the three outputs looking at improving the enabling conditions in the participating countries and securing a regulatory framework and policies that facilitate adaptation to climate change. These three outputs are:

- (a) Knowledge and action research on climate risk-informed pro-poor community-level solutions strengthened;
- (b) Institutional and community capacity to develop and deliver climate adaptation investments at community level strengthened; and
- (c) Inclusive and pro-poor adaptation investment projects identified and prepared.

32. The budget allocates USD 72.5 million for activities related to these outputs. Even if the amount per country will not cover all costs for preparing and enforcing an institutional framework that fully enables adaptation to climate change at the national, subnational and local levels, the iTAP considers that the CRPP will contribute to building up and/or improving the institutional framework at the national level. However, the iTAP is less confident that this will lead in all countries to the establishment or improvement of institutional frameworks at the subnational and local levels, where the second component of the CRPP will invest.

33. The second component of the funding proposal is the CRPP Investment Fund, where the GCF funding will be placed. This component will finance the implementation projects promoting:

- (a) Information and systems for delivering applied climate-risk informed investments at scale improved;
- (b) Climate resilient pro-poor livelihoods investments implemented; and
- (c) Climate adaptation pro-poor infrastructure implemented.

34. Almost 90 per cent of the total budget is allocated to the IF (USD 668.5 million out of a total of USD 750 million).
35. It is promising that one of the eligibility criteria for proposals in the CRPP IF is a paradigm shift potential defined as “the potential of the project to catalyse impact beyond a one-off project investment” (annex 2.4 – eligibility criteria), and that, in line with this definition, the project proposals will have to demonstrate sustainability in the long run, replicability and innovation potential.
36. Further, because these projects will be implemented by the corresponding ministries in the host countries, the iTAP sees a potential for replication and upscaling. However, this potential is reduced by the low allocation of resources to training and learning (2.5 per cent of the total amount for training, workshops and conferences, and 1.2 per cent for monitoring, evaluation and learning) and the low number of beneficiaries in percentage of the total population in most participating countries (see table 1).
37. The iTAP assesses that other factors will further reduce the paradigm shift potential. First, the lack of a strategy for generating long-term adaptation finance at a programmatic level, which is one of the key barriers identified in the theory of change. The iTAP recognizes that the funding proposal mentions potential participation of the private sector and the need for securing long-term financing sources (private or public), but there is no dedicated activity on preparing and/or testing financing mechanisms for adaptation in the participating countries that could contribute to long-term funding for those practices identified as resilient to climate change.
38. Second, the iTAP noticed that the project concepts presented in the funding proposal and in annex 2.3 are based on existing ADB projects, which implies a certain level of continuation of business-as-usual for activities in a development bank. Although this is deemed as a synergy between multilateral organizations and does not challenge the quality of the projects, at the same time it reduces their innovative character.
39. Due to the above-mentioned reasons, the iTAP assesses the paradigm shift potential of the CRPP funding proposal as medium.

1.3 Sustainable development potential

Scale: High

40. Funded in 1966, the ADB has a long tradition of promoting sustainable development in its member countries. The current funding proposal is based on and uses both lessons learned and competences resulting from this experience. This is clearly reflected in the design of the funding proposal as addressing the interface between the Paris Agreement, the Agenda 2030 and the Sendai Framework for Disaster Risk Reduction.

Programme level

41. Based on the suggested project concepts included in the funding proposal, the iTAP assesses that implementation of the proposed programme can contribute to achieving several Sustainable Development Goals (SDGs) including, SDG 1: no poverty; SDG 5: gender equality; SDG 10: reduced inequalities; and SDG 13: climate action. Contribution to further SDGs can be achieved in the specific projects, for instance the concept for Papua New Guinea will address SDG 4: quality education.
42. Economic co-benefits: The funding proposal aims at facilitating access to finance for poor people in order to be trained for and/or to participate in climate resilient economic activities including construction or agriculture. Therefore, the CRPP will secure income to vulnerable households during implementation of the projects. According to the exit strategy of the specific projects it is possible that such income may continue in the long term.

43. **Social co-benefits:** The CRPP focuses on reducing the vulnerability of people affected by poverty, with special attention to vulnerable women. Depending on the characteristics of the specific projects, the CRPP can promote additional social co-benefits including access to decision-making, improvements in social protection systems, access to education and training, or reduction of disaster risks and, last but not least, securing life through access to multipurpose emergency shelters.
44. **Environmental co-benefits:** Several activities included in this proposal can have positive environmental co-benefits including biodiversity conservation, soil protection or promoting and securing other environmental services. Activities like the promotion of climate resilient agroecological systems (Output 5) or the use of ecosystem-based infrastructure and nature-based adaptation measures (Output 6) are examples of such. In addition, and depending on the specific design of the projects, flood and landslide protection infrastructure can contribute to protect important ecosystems.
45. **Gender-related co-benefits:** The promotion of gender equality is reflected in the funding proposal through its focus on the climate-poverty-gender nexus. Consequently, a stand-alone eligibility criterion on gender has been included for assessing the projects to be funded through the CRPP IF, which will benefit from the GCF contribution. Furthermore, projects within the CRPP have to be categorized as Gender Equity Theme or Effective Gender Mainstreaming. At the programmatic level a Gender Assessment (annex 8.1) and Gender Plan (annex 8.2) are included. The Gender Assessment includes a detailed analysis of gender issues per country, providing a good information basis that justifies considering gender issues as a central element in the proposal. It is highly recommended to strengthen the assessment, the gender criterion for project selection, and to secure the inclusion of gender-related indicators in the MEL system, so that progress in addressing gender-related inequalities can be perceived throughout programme implementation.
46. The iTAP deems the climate-poverty-gender nexus as a promising approach for addressing climate change adaptation and vulnerability.

Portfolio level

47. According to the funding proposal, the CRPP IF “will establish and maintain an effective environmental and social management system to identify, assess, manage, and monitor the potential environmental and social impacts and risks associated with its portfolio of investments on climate adaptation” (annex 6, Environmental and Social Management Framework). This framework is built on the ADB Safeguard Policy Statement and follows the GCF revised Environmental and Social Policy. The annex further states a list of environmental and social safeguard requirements and clarifies that during implementation of the CRPP IF environmental and social due diligence (including sexual exploitation, abuse and harassment [SEAH] due diligence) will be undertaken by the specific project’s entities.
48. As part of the Environmental and Social Management Framework, the CRPP IF will respect the ADB Prohibited Investment List, a generic list applied by the bank. The Environmental and Social Management Framework includes a list of templates and checklists that should facilitate due diligence and monitoring of environmental and social impacts in a consistent manner across the projects in the CRPP portfolio.
49. Considering the above, the iTAP assesses the potential contribution of the CRPP to sustainable development as high, as long as the recommendation from the iTAP with regard to monitoring, reporting and evaluation of gender issues at the programmatic level is followed.

1.4 Needs of the recipient

Scale: High

50. Table 3 summarizes comparable information about the climate vulnerability and socioeconomic development of the countries participating in the CRPP. Although all countries are vulnerable to climate change, the iTAP noticed that both vulnerability and socioeconomic development are at uneven stages across the participating countries.

Table 3: Summary of the climate vulnerability and socioeconomic index of CRPP countries

Country	ND-GAIN* (2021)		Poverty level	Gini index	Gender Inequality Index (2021)		Human Development Index (2021-22)	
	Rank (out of 186)	Score			Value	Rank	Index	Rank
Cambodia	144	40.1	17.7%	n/a	0.46	116	0.59	146
Indonesia	99	47.6	9.5%	39.9	0.44	110	0.7	114
Lao People's Democratic Republic	121	43.8	18.3%	38.8	0.48	120	0.61	140
Pakistan	151	39.6	21.9%	29.6	0.53	135	0.54	161
Papua New Guinea	167	36.8	39.9%	41.9	0.73	169	0.56	156
Timor-Leste	123	43.7	41.8%	28.7	0.38	89	0.61	140
Vanuatu	140	41.4	15.9%	32.3	n/a	n/a	0.61	140

* Notre Dame Global Adaptation Initiative

Sources:

ND-GAIN: [Notre Dame Adaptation Initiative](#)

Poverty level as the headcount ratio at national levels (% of population): [World Bank Poverty](#)

Gini Index: [World Bank – Gini index](#)

Human Development Index: [UNDP HDI](#)

51. With its focus on the climate-poverty-gender nexus, the CRPP IF is well designed for contributing to key climate and development concerns in the participating countries. Furthermore, in the examples of possible projects at the country level included in the funding proposal, the ADB shows how these projects can address the specific characteristics of the nexus at the national level.

52. The proposal identifies limited access to adaptation finance for implementing local solutions at scale as one of the three key barriers to climate change adaptation. With an investment of over 50 per cent of the total budget in implementing adaptation measures (27.5 per cent civil works and 23.5 per cent equipment), the CRPP will certainly facilitate financial means for adaptation measures during its implementation period.

53. The inclusion of “needs of the recipient” as an eligibility criterion formalizes the assessment of the potential to overcome barriers and to strengthen the capacity of local institutions at the project/portfolio level.

54. Further, several activities in the CRPP are aimed at strengthening institutions and implementation capacity, especially those under Outputs 1 and 2 (CRPP Trust Fund). These activities contribute to satisfying the needs of the recipient, even though these are not co-funded by the GCF contribution.

55. Based on the analysis above, the iTAP assesses the response of the CRPP to the needs of the recipients as high.

1.5 Country ownership

Scale: Medium

56. The iTAP received no-objection letters from the seven participating countries with the funding proposal submission.

57. The iTAP found a good correspondence between the project ideas included in the proposal and the existing climate and/or adaptation policies at the national level and as included in the registers managed by the UNFCCC secretariat (see table 4).

Table 4: Overview of the national programmes on adaptation by CRPP countries as submitted to the UNFCCC secretariat

Country	Nationally Determined Contribution (NDC)	National Communications (NC)	National Adaptation Plans (NAPs)	National Adaptation Programmes of Action (NAPAs)
Cambodia	First submission updated (2020)	NC3 – Sept. 2022	7 July 2021	March 2007
Indonesia	Enhanced version - 3 (2022)	Revised NC3 – Feb. 2018	n/a	n/a
Lao People's Democratic Republic	First submission updated (2021)	NC2 – June 2013	n/a	May 2009
Pakistan	First submission updated (2021)	NC2 – Aug. 2019	15 Aug 2023	n/a
Papua New Guinea	Second submission (2020)	NC2 – Dec. 2015	13 April 2023	n/a
Timor-Leste	Updated NDC (2022-30)	NC2 – Nov. 2020	31 March 2021	Sept. 2011
Vanuatu	NDC revised and enhanced (3 version, 2022)	NC3 – March 2021	n/a	Dec. 2007

Sources:

Nationally Determined Contributions: <https://unfccc.int/NDCREG>

National Communications: <https://unfccc.int/non-annex-I-NCs>

National Adaptation Plans: <https://napcentral.org/submitted-naps>

National Adaptation Programmes of Action: <https://unfccc.int/topics/resilience/workstreams/national-adaptation-programmes-of-action/napas-received>

58. In the climate policies of all seven countries there is an explicit or implicit mention of the climate-poverty-gender nexus and the need to reduce climate risks of the population affected by poverty, with special consideration of women (and youth). Other, more specific examples of correspondence between the national policies and the project ideas were identified by the iTAP, for instance:

- (a) In its Nationally Determined Contribution (NDC) of 2020, Papua New Guinea identifies the need to promote both the understanding of climate risks and to facilitate training around the value of ecosystem-based adaptation as promising adaptation actions;
- (b) In its National Adaptation Programme of 2021, Cambodia specifically looks at gender and climate change and mentions rural infrastructure, institutional resilience or insurance schemes for reducing climate and disaster risks; and
- (c) The National Adaptation Programme of Pakistan (2023) mentions disaster risk management, ecosystem restoration and coastal and marine protection as priority adaptation areas.

59. The ADB has currently 12 projects approved by GCF, with a volume of funded activity financing of USD 947,481,020 and in different stages of implementation. These figures demonstrate the implementation capacity of the AE.

60. However, the iTAP could not assess the capacity of executing entities because these have been identified in the funding proposal only in a generic manner and no assessment of the capacity of possible executive entities for the proposed ideas has been submitted. The funding proposal states that “for Projects implemented in Host Countries, the government of the Host Country, acting through its relevant ministry or department, may be the executing agency and/or implementing agency”. It is noted that among the participating countries uneven capacity at the ministerial level is to be expected. Where the capacity of the executing or implementing entities is a matter of concern, the iTAP recommends that capacity development or other support should be added to project implementation, as appropriate.

61. A further reason for concern is the level of engagement with the civil society organizations (CSOs) and other relevant stakeholders in each country. Although the iTAP recognizes the consultations related to the proposal preparation and the stakeholder mapping and engagement plan (annex 7), three further reasons for concern were observed:

- (a) According to information in the proposal, most of the consultation and dialogue meetings focused on national government representatives, international donors and international non-governmental organizations. National consultations were also conducted in some countries with the participation of representatives of the national level, but it remains unclear if and how the subnational and local levels participated and if they had any influence on the project design or will have in future. The iTAP also notes that engagement with representatives of CSOs for preparation of this funding proposal has been limited;
- (b) The language relating to the Stakeholder Engagement Plan (annex 7) is very vague and would apply to any development initiative. For instance, it is not clear how a legitimate representation of key stakeholders will be secured and their participation financed during the design of the projects or how/which subnational authorities will be included, or how competences and expertise from the subnational and local levels will be used in the design and implementation phases; and
- (c) The main clients of the CRPP Investment Fund will be the ADB User Departments. This poses the risk that local community representatives and organizations will have less access to prepare and/or contribute to project proposals to be submitted to the CRPP Investment Fund.

62. Considering the above, the iTAP assesses the country ownership as medium, and recommends that the envisaged future stakeholder consultations during the design of the countries' projects will be given high priority.

1.6 Efficiency and effectiveness

Scale: Medium to High

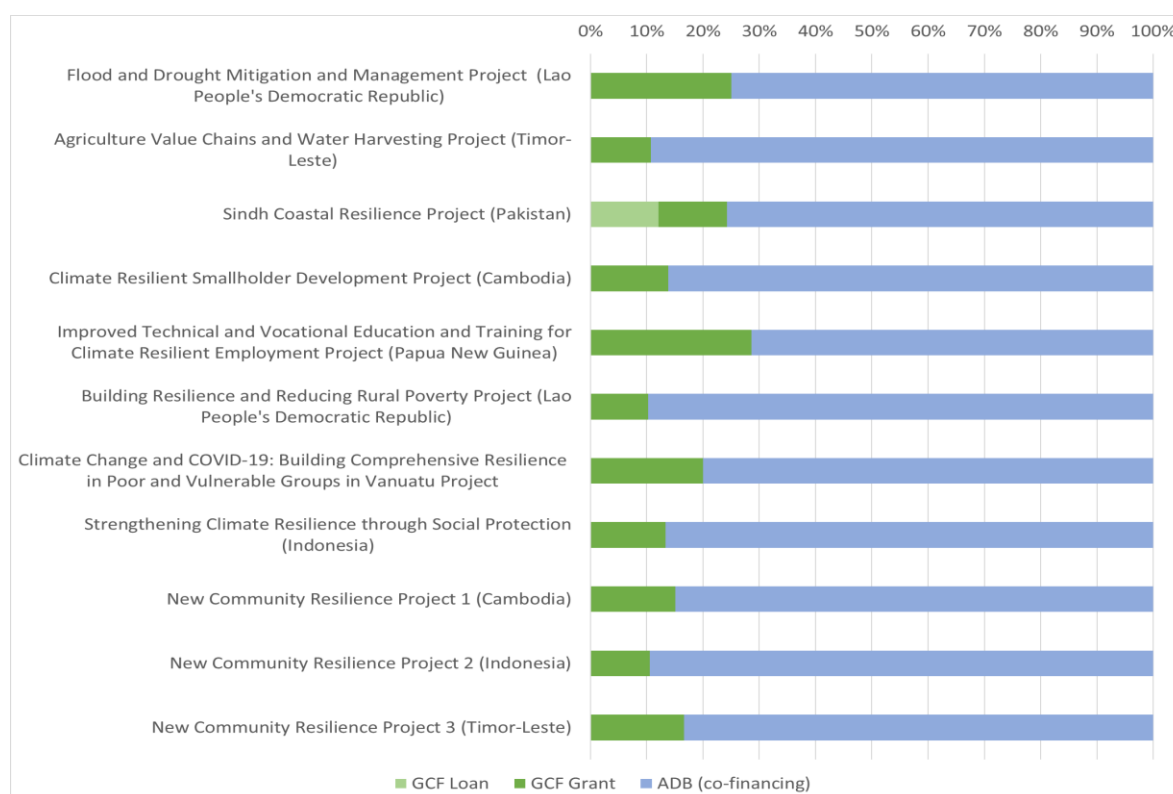
63. Out of the USD 120 million GCF funds, USD 20 million will be lent in United States dollars to the Government of Pakistan (loan component). The GCF loan will be disbursed in line with the project implementation. For the Pakistan project, the project implementation is assumed to develop over four years as follows: 20 per cent in year 1, 25 per cent each in years 2 and 3 and 30 per cent in year 4. The GCF loan would be for a period of 40 years, including a grace period of 10 years. The GCF loan will be priced at a fixed interest rate of zero per cent for Pakistan (high concessionality).

64. USD 100 million from GCF funds will be provided as grants in United States dollars to the Governments of Cambodia, Indonesia, the Lao People's Democratic Republic, Pakistan, Papua New Guinea, Timor-Leste and Vanuatu.

65. ADB will manage GCF funds through a Trust Fund. All interest income against GCF investments will flow through GCF and all surplus funds of GCF in the ADB Trust Fund (after the first-phase repayments commence) will be invested in approved treasury investment vehicles at an annual interest rate of 1 per cent.

66. The ratio of co-finance in the CRPP IF is USD 1:4.60 for GCF and USD 1:5.25 in the whole CRPP (Trust Fund and Investment Fund). This high co-funding can also be seen in the expected distribution of funds for the projects in the pipeline (see figure 1).

Figure 1: CRPP co-financing per project idea

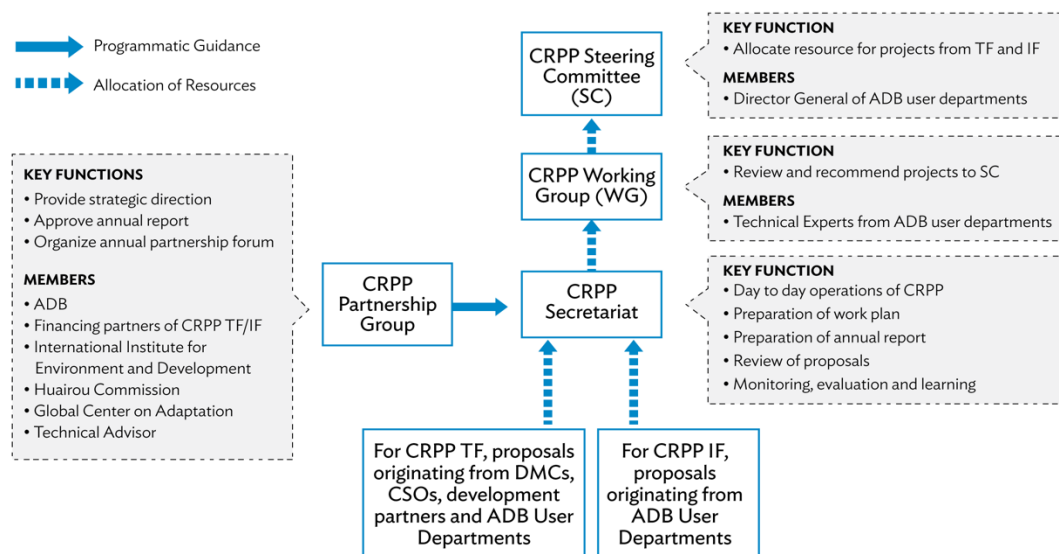


Source: Annex 4 – Budget 2.4

67. The CRPP IF will be implemented from 2023 to 2030. Projects seeking financing from IF will probably need to be approved by the ADB Board between 2024 and 2027 and complete implementation by 2030. Projects which will be supported by the programme will be identified in accordance with ADB's country programming process and selected using the eligibility criteria presented in this proposal.

68. The funding proposal does not include a co-financing letter from ADB. For every project with GCF co-financing, the CRPP secretariat/screening committee will ensure that the eligibility criteria have been met to enable the project to access CRPP funding and ADB will secure the co-investment ratio is maintained. Figure 2 shows the governance structure of the CRPP including the roles and responsibilities in selecting projects in the CRPP IF.

Figure 2: CRPP governance structure



Note: ADB stays for Asian Development Bank, CRPP stays for Community Resilience Partnership Programme, IF stays for Investment Fund, TF stays for Trust Fund, SC stays for Steering Committee.

Source: Annex 2.4 CRPP Design Study – Eligibility Criteria 3, Aug 2023

69. The iTAP is aware of the financial record of the ADB as a major advantage for increasing the viability of the CRPP. The iTAP already expressed its concern about the long-term sustainability of the programme, due to the fact that no long-term financing mechanisms are foreseen. The iTAP assesses the efficiency and effectiveness of this funding proposal as medium to high.

II. Overall remarks from the independent Technical Advisory Panel

70. The iTAP notes that the CRPP mirrors the GCF investment criteria as their own eligibility criteria for the projects to be funded through the Investment Fund. The iTAP highlights that the level of stringency used when applying and assessing these criteria prior to funding projects will determine the success of the CRPP, and therefore recommends to the AE to include indicators related to the GCF sub-criteria, as well as on the additional criterion on gender, when preparing the guidance for project preparation.

71. The iTAP recognizes that stakeholder consultation is an important part of the envisaged programme activities, and that the lack of stakeholder engagement constitutes a barrier in the implementation of climate change adaptation measures. Therefore, the iTAP recommends that the forthcoming stakeholder consultations and engagement in the respective countries ensure to include civil society and private sector actors, and to institutionalise stakeholder involvement processes throughout programme implementation.

72. Further, the iTAP noticed the shared concern with the secretariat with regard to the monitoring and evaluation plan, as expressed in the draft term sheet that was included in the FP package submitted to iTAP.

73. The iTAP endorses this full proposal *Community Resilience Partnership Program (CRPP)* for approval by the GCF Board, with one condition:

Condition precedent to second disbursement:

74. The Accredited Entity shall design an updated Monitoring, Evaluation and Learning (MEL) System and provide to the Fund, in a form and substance satisfactory to the GCF Secretariat, evidence of the adoption of this system. The system shall include the following aspects, in addition to the requirements of the MEL system submitted with the Funding Proposal:

- (a) a methodology for checking compliance with progress against the eligibility criteria of the CRPP.
- (b) a complete monitoring and evaluation system that allows the Accredited Entity to assess the adaptation outcomes, including:
 - (i) the corresponding methods for the steps of data gathering, analysis and interpretation, and the frequency of the performance of each step;
 - (ii) the roles and responsibilities for performing the steps above;
 - (iii) a method to assess progress in addressing the climate-gender-poverty nexus as described in the Funding Proposal; and
 - (iv) a clear method for quantifying the number of direct and indirect beneficiaries.
- (c) a strategy for linking the MEL system of the CRPP's national projects to the national systems for monitoring, evaluation and reporting progress under the UNFCCC in the relevant Host Countries.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP215)

Proposal name:	Community Resilience Partnership Program
Accredited entity:	Asian Development Bank
Country/(ies):	Cambodia, Indonesia, the Lao People's Democratic Republic, Papua New Guinea, Pakistan, Timor Leste and Vanuatu
Project/programme size:	Large

Impact potential
The AE appreciates that the iTAP recognizes the impact potential of the program to be high and that the monitoring, evaluation and learning systems of the program can be finalized in the first year of the program implementation.
Paradigm shift potential
The AE appreciates that the iTAP considers that the program will contribute to strengthening institutional framework at the national level which is essential for paradigm shift and takes note of iTAPs views on the need for similar frameworks at the subnational/local level and the need for new generation of investment projects by development bank's such as the AE in concern.
Sustainable development potential
The AE appreciates that the iTAP recognizes that the sustainable development potential of the program is high and the program's approach of dealing with climate-poverty-gender nexus as a promising approach for addressing climate risk and vulnerability.
Needs of the recipient
The AE appreciates that the iTAP recognizes that program's high responses to the needs of the recipient countries.
Country ownership
The AE appreciates the iTAP suggestions for the program to strengthen capacity development of the executing agencies and stakeholder engagement at the local level at the level of individual projects.
Efficiency and effectiveness
The AE appreciates iTAP views on the efficiency and effectiveness of the program.

Overall remarks from the independent Technical Advisory Panel:

The AE highly appreciates iTAP endorsement of the program and notes the condition to update the monitoring, evaluation and learning system of the program precedent to the second disbursement.



Annex 8.1 Gender Assessment

31 May 2023

Contents

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1. ABOUT THE ANNEX

This annex presents the gender assessment for the Community Resilience Partnership Program (CRPP). The Gender Assessment describes the CRPPs guiding principles and approach for responding to the specific needs of women through its investments in climate change adaptation before presenting an overview of the gender context in CRPP countries. This provides backdrop for a description of critical gender issues related to the types of investments that are likely to be supported by the CRPP, and how such investments can address specific issues that contribute to women's vulnerability and help to build their resilience to climate change. These sections provide the basis for the Gender Action Plan, which identifies specific actions, design features, and monitoring indicators and targets that will be implemented through the CRPP as part of its supported investments.

The CRPP program level gender assessment presented in this annex has been informed by extensive desk-based research carried out as part of the CRPP scoping studies, as well as consultations with governments and communities and expert insights from ADB and CRPP implementation partners such as the Huairou Commission, a women-led social movement of grassroots groups from poor urban, rural and indigenous communities in 45+ countries. As the CRPP is a large program that will consist of a number of sub-projects that are still at concept stage, the aim of the assessment presented in this annex is to provide an overview of key issues related to the sectors and areas of activity of the CRPP. This will serve as a foundation for developing more detailed and project specific gender assessments as part of the sub-project design and preparation.

2. INTRODUCTION

Women have a critical role to play in advancing investment in building resilience to climate change, especially at the local level, and must be recognized as vital drivers of resilient development. However, social and political inequalities, as well as patriarchal power structures, remain deeply entrenched, often limiting women's leadership and participation in decision-making, and contributing to disproportionate levels of vulnerability and low adaptive capacity compared to men.

This is particularly the case in poor communities in low income countries as multidimensional poverty significantly limits women's ability to confront climate change. In Asia, women account for 80% of all people living on less than \$2 per day¹, and their economic activities are often highly concentrated in sectors that are dependent on natural resources such as agriculture. For example, women represent 73% of the agricultural labor force in Pakistan, and 70% in Papua New Guinea and the Lao People's Democratic Republic². Women also tend to be overly burdened with household work and caring for children, the sick, and the elderly; women do not always have the same rights as men to land; women often have less access to financial services, technology, information, and early warning systems;^a and women, particularly rural women, are often held back from fully participating in their community and in formal decision-making processes. All of these factors contribute to increased vulnerability and lower adaptive capacity, and are closely related to socially created gender functions, systemic discriminations, and the power imbalances between women and men.

^a For further details on the specific vulnerability context of women, see Habtezion (2016); Miles and Wiedmaier-Pfister (2018); and Neumayer and Plümper (2007).

While it is crucial to recognize and address women's vulnerabilities to climate change and understand the structural causes for them, it is fundamental to refrain from categorizing women as innately weak, passive receivers of protection and support. In the past, this approach has prevented women from being recognized as active agents in resilience-building programs. The adoption of programs designed on the assumption that women are powerless further exposes them to climate change shocks and disasters, as well as hinders the realization of women's rights to full and equal participation and decision-making³. Women's unique knowledge, skills, and perspectives, mean that they can contribute positively to the identification of appropriate adaptation measures and ensure that they are implemented effectively and efficiently. Women can also play a leadership role in accessing public and private resources for adaptation, which in turn can give them recognition within the community and promote their position in the family and wider society, creating a positive feedback loop for women's empowerment. There are increasing examples of grassroots women organizations closely collaborating with local governments to make use of their knowledge and capacity to assess, prioritize, negotiate, and influence resource allocation for resilience investments targeted at vulnerable communities⁴.

3. ADB's GENDER POLICY FRAMEWORK

Accelerating progress in gender equality in Asia and the Pacific is one of the seven operational priorities in ADB's Strategy 2030. ADB recognizes that gender equality is critical in its own right as well as for helping realize socio-economic development. Strategy 2030 holds that, to achieve a prosperous, inclusive, resilient, and sustainable region, it is imperative that ADB contributes to the efforts of accelerating gender equality outcomes in the region in five areas: economic empowerment, human development, decision-making and leadership, time poverty reduction, and resilience to external shocks. As such, the CRPP contributes towards the gender operational priorities of ADB's Strategy 2030, and achieving the following target - "by 2030, at least 75% of the number of ADB's committed operations (on a 3-year rolling average, including both sovereign and non-sovereign operations) will promote gender equality."

ADB has established an institutional framework on gender and development—consisting of its Gender and Development Policy and Gender Operational Plan (OP2). The Gender Operational Plan, 2013–2020 is the latest action plan to operationalize the GAD policy. The plan adopts a twin approach of gender mainstreaming and targeted investments in women and girls.

4. CRPP AND GENDER EQUALITY

4.1 The CRPP's Approach to Gender Equality

Building women's resilience to climate change and empowering women through transformational investments in climate adaptation at the community level are core pillars of the CRPP, reflecting ADB's commitment to gender equality. This is demonstrated in the CRPP program architecture. The CRPP is operationalized through the Community Resilience Financing Partnership Facility (CRFPF) which comprises two core components; (i) a multidonor Trust Fund, which provides technical assistance and grant resources for improving knowledge and understanding, and strengthening institutions, processes, financial architecture, and capacity needed for building local resilience; and (ii) the proposed Investment Fund, which will be co-financed by GCF and administered by ADB that will provide critical grant and loan resources to implement local adaptation measures as part of ADB financed investments (sub-projects) in seven countries. The CRPP Trust Fund includes a dedicated Gender Window that will support the identification, development, and implementation of investments that have the explicit objective of strengthening the resilience of women to climate change. As such, support will be provided through (i)

knowledge and support technical assistance for identifying investment opportunities for building women's resilience and creating an enabling environment for financing women-focused adaptation measures; (ii) transactional technical assistance for preparing women focused investment projects; and (iii) grant financing, on a selective basis, for implementing investments identified and prepared under the technical assistance^b.

Through these two funds, CRPP will be able to invest in climate adaptation and resilience measures that build on the existing capacities of poor women and implement specific women-centred adaptation initiatives that are designed through inclusive processes where women are engaged at all levels of decision-making.

The approach toward implementing investment activities that build women's resilience to climate change will emphasize women's role in resilience as "active stakeholders" and "game changers" rather than "vulnerable" and "victims." The CRPP also understands that structural gender inequalities are driving women's climate risk in CRPP countries, and so a gender transformative approach that aims to change structural inequalities and address issues of climate justice⁵, will be a central element of CRPP investments. CRPP investments in climate resilience therefore represent an opportunity to promote a positive change towards gender equality and challenge socially created gender functions, systemic discriminations and the power imbalances between women and men, which intensify the social and economic barriers faced by women to adapt to climate change⁶. To do this, CRPP investments will include interventions that are related to policy reforms (e.g., revising the land tenure system); revising regulations (e.g., the need to provide collateral for accessing housing microfinance); information management (e.g., ensuring that women have access to climate information); and capacity building. As such, the CRPP will go beyond mainstreaming gender considerations into programming, delivering specific outputs for strengthening women's adaptive capacity, and challenging the structural sociocultural norms that drive their vulnerability through transformational investments.

The CRPP's approach to gender is closely aligned with both the GCF's gender policy and ADB's gender mainstreaming framework. This is reflected in the following principles for gender that have been identified to guide CRPP Investments, and are reflected in the CRPP Gender Action Plan.

All CRPP sub-projects will:

- Ensure women's equitable participation in projects, including public consultations;
- Promote increased participation of women in decision-making and leadership roles;
- Include capacity-building components that are either gender responsive or specifically target women;
- Integrate gender-considerations in all parts of the project management, including in the monitoring and evaluation;
- Document lessons learned specifically relating to gender to support future learning and replication.
- Follow the principles of an intersectional approach.

The portfolio of CRPP sub-projects will include outputs that:

- Support employment and income generation opportunities for women;
- Promote women's access and control over resources;
- Increase women's access to relevant technologies;

^b More information on the CRPP's Gender Window and the types of activities it supports can be found [here](#).

- Build the capacity of different actors to mainstream gender;

ADB uses four gender categories— gender equity theme (GEN), effective gender mainstreaming (EGM), some gender elements (SGE), and no gender elements (NGE). GEN projects have outcomes that directly address gender equality and women's empowerment by narrowing gender disparities. By contrast, EGM projects need to have several outputs that contribute to gender equality and women's empowerment. All CRPP supported projects will be required to be GEN or EGM gender mainstreaming category (as described in CRPP IF eligibility criteria in **Annex 2.4**) and follow ADB guidelines on gender mainstreaming. GEN and EGM projects are required to have gender action plans (GAPs), and so, as per ADB policy, each project under the CRPP will conduct a project related gender analysis to set evidence- based gender targets in project GAPs, allocate resources for the implementation of the GAP, and regularly monitor and report on GAP implementation.^c

4.2 Methodology for the preparation of Sub-project Gender Assessments

Project specific gender assessments will be undertaken as part of detailed sub-project design stage in line with ADB policy related to GEN and EGM projects. The assessments will include field visits and the use of surveys, and/or focus groups for consultations with women, and stakeholders representing women. The specific groups that will be consulted will vary according to the sub-project but will include poor and vulnerable women in the target communities (e.g women farmers), community women's groups, women leaders at the local and national level and within relevant sector ministries, as well as women in business and private enterprise. Consultations as part of the gender assessment will also include consultations that include other vulnerable groups such as ethnic minority groups and persons with disabilities. These assessments will be used to understand the different roles, interests and priorities of women, men and vulnerable groups in the sub-project implementation areas and in the context of the sub-project's main objectives (e.g climate resilient livelihoods), and inform the design of sub-project activities and the design of project specific Gender Action Plans.

The CRPP program level gender assessment is considered a living document and will be updated and evolve overtime as the CRPP sub-project gender assessments are carried out, giving more specific details about target areas within CRPP countries and in the context of sub-project outcomes.

5. THE GENDER CONTEXT IN CRPP COUNTRIES

5.1. Cambodia

Cambodia is the second poorest country in ASEAN and among the poorest in Asia with nearly 4.5 million Cambodians living close to the poverty line. In 2017, Cambodia ranked 116 out of 160 countries on the Gender Inequality Index; 146/189 on the Human Development Index.

The Government adopted its Gender and Climate Change Action Plan 2014-2018 and identified climate change mitigation and green growth as one of its strategic focus for women's economic empowerment. The Ministry of Agriculture, Fisheries and Forestry's Gender Mainstreaming Policy and Strategic Framework in Agriculture 2016-2020 identifies literacy; representation in agriculture administrations; access to agricultural extensions services; access to farming equipment and technologies; and access to financial services, land and markets as key challenges facing women in

^c An intersectional approach will be a core element of the activities supported by the CRPP's Gender Window that are discussed in the subsequent sections.

the agriculture and natural resource sector. Women's predominance in the agriculture sector, their reliance on income earned in this sector, and their more limited access to education and information, however, make them more vulnerable to the negative impacts of climate change.

Education. The largest gender gaps in education are seen in rural areas (over 10%) with females in rural areas between the ages of 6-24 years the least likely to attend school (at 55.5%). Inequality in education at 27.3% is more than double that of the region (13.1%). Girls have closed the gap in completion rates for upper secondary levels (over 21% for girls and 19% for boys in SY 2016/17.), however, gross enrolment rates in upper secondary for girls remains low at nearly 27%⁷ and gaps remain between regions⁸. 5.5% of women in the labour force had post-secondary education in 2017; and only 7% of women in the labour force had completed upper secondary school. Nearly 15% of women had none or only some education; and only 25% of working women had completed primary school.

Women earn substantially less than men in Cambodia, but the earnings gap narrows with the attainment of upper secondary and tertiary education. The gender earnings gap in 2014 was as high as 30% among those with low education and drops to 7% among college graduates. Within the same sector female workers have a higher share in lower-paid positions and disparities in education levels, particularly of older women, perpetuate the gender earnings gap. Participation of females in public TVET is increasing, and with a large jump in female enrolment in the 2017-2018 academic year, females are now enrolled in equal proportions to men. However, women are more likely to be enrolled in the shorter, lower-level and less-technical courses and in more gender-stereotyped courses.

Gender based violence. The National Survey on Women's Health and Life Experiences in Cambodia, conducted by the National Institute of Statistics of the Ministry of Planning and Ministry of Women's Affairs in 2015, revealed that 20% of Cambodian women reported experiencing physical or sexual violence from an intimate partner and while 90% of these women were hurt badly enough to require healthcare, only 53% ever accessed it. The Domestic Violence Law, enacted in 2005, criminalizes physical and psychological abuse against dependent household members, as well as forms of sexual aggression, including sexual harassment and violent sex. However, the Domestic Violence Law fails to define what constitutes domestic violence and who can be a victim⁹.

Cambodia has high levels of economic participation of women with over 80% of working age women^d working in 2017 (nearly 90% for men)¹⁰ the highest proportion in the region. Of these, 42% were in paid positions of which nearly half were in the garment industry. Women, through their wages and the remittances they send home, are supporting families and communities, with women sending a higher proportion (20%) of their salaries back to their families than men¹¹.

Significant gender disparities can be seen in earnings, the quality of jobs, and voice and representation in leadership positions. Nearly 55% of women work in vulnerable employment¹² and nearly twice as many women than men of working age are not in the labour force. The definition of employment in Cambodia includes subsistence farming and unpaid family members. These activities often involve low value-added activities and the International Labour Organization estimates that 36% of workers are underemployed, significantly higher than official figures. Underemployment is higher in rural areas with many under-employed likely to be women as they juggle employment and family responsibilities.

Nearly 80% of the Cambodian population lives in rural areas and a large majority of them are involved in farming activities¹³. Despite widespread gender inequalities in access to productive resources, services and markets, rural women are responsible for 80% of Cambodia's food production¹⁴. In 2017, nearly 40% of working women (35% of working men) were employed in the agriculture sector¹⁵ and agriculture is the largest sector of employment for both men and women.

^d Working age is defined as between the ages of 15-64 years.

5.2. Indonesia

The population of Indonesia is slightly higher than 273.5 million, about 50.1% of which (134.4 million) are women^{16,17}. According to the World Economic Forum (2020), Indonesia ranks 85th on the Global Gender Gap Index. Both the Asian Development Bank (2015) and the World Economic Forum (2020) reported that Indonesia made relevant progress in reducing gender differences and this effort resulted in a growing participation of women in economics and in a consistent increase in access to both health and education.

Indonesia has ratified the Convention on the Elimination of All Forms of Discrimination Against Women. It has also promulgated Presidential Instruction No. 9/2000 regarding Gender Mainstreaming into National Development, which is aimed at reducing the gap between Indonesian women and men in accessing and obtaining development benefits, as well as to increase participation in and control over the development process. This has been followed up by Ministerial Decrees in most Ministries. Gender priorities highlighted in the National Medium-Term Development Plan For 2020-2024, include: (i) gender equality and justice in education, (ii) life expectancy, (iii) health, (iv) workforce, (v) economy, (vi) political representation in parliament, (vii) eradication of domestic violence, and (viii) tackling climate related challenges (access to water, energy and sanitation and capacity to face food scarcity and disaster). Strategies to be implemented include: (i) strengthening policy and regulation, (ii) accelerating gender mainstreaming, (iii) awareness raising, (iv) enhancing role and participation of women in development, and (v) improving network and coordination among stakeholders, all of which are part of 'National Strategy for Acceleration of Gender Mainstreaming Implementation'. The Long-Term Strategy for Low Carbon and Climate Resilience 2050 mentions that it will use these strategies as a basis for addressing gender issues in climate change adaptation¹⁸. Indonesia's Nationally Determined Contribution (NDC) also highlights how mapping gender issues in climate change in all development sectors will be crucial in implementing the policy and that enhancing the role of women in development and strengthening women's capacity and leadership in climate change is critical¹⁹.

Despite the commitments presented in these policy documents, the country is still facing significant challenges in a number of areas: the economic gap between males and females persists; rates for gender-based violence and maternity mortality are extremely high; female access to political participation is exceptionally low^{20/21}.

Indonesia ranks 36th in the 2016 World Risk Report²² and, with a score of 10.24%, is considered a country at "high risk". The situation of Indonesia is extremely precarious since it is categorized as a country at "high risk" or "very high risk" in all the dimensions that the index in question takes into account (exposure, vulnerability, susceptibility, lack of coping capacities and lack of adaptive capacities).

Indonesia is prone to disasters and women are more impacted by natural disasters than men. For example, in the case of the Indian Ocean Earthquake and Tsunami which led to the loss of many lives in Indonesia, it was reported that the percentage of women deceased was three to four times higher than the percentage of men deceased²³. According to The International Federation of Red Cross and Red Crescent Societies²⁴, Indonesia has a solid legal framework that protects women and vulnerable groups. The 2007 Law on Disaster Management and the Regulation 21/2008 on Disaster Management recognises the need to protect "vulnerable groups" in emergency response, including women.

Health and Gender-Based Violence. According to the 2020 Global Gender Gap Report, the health gender gaps have nearly closed (score of 97.4%²⁵), showing a significant improvement in the last decade. Nevertheless, Indonesia is still facing significant challenges, like maternity mortality²⁶, gender-based violence and child marriage. Indonesia is the 8th country for number of child marriages in the world, and the 2018 Socio-economic National Survey (Susenas), reports that 11.1% of women aged from 20 to 24 married underage²⁷.

In the last decade, the country saw a substantial growth in reported cases of violence against women. In 2010 105 103 cases were reported and in 2013 even 279 760²⁸. The 2016 Indonesian National Women's Life Experience Survey (SPHPN) reports that one third of the women interviewed has been victim of physical and/or sexual violence²⁹. The situation has become even worse during the COVID-19 pandemic, as gender abuses have raised by 12%³⁰. In most cases violence against women relates to domestic violence. In both 2010 and 2013, more than 95% of the cases were of domestic violence (96% in 2010 and 98% in 2013)³¹.

Education. The 2020 Global Gender Gap Report shows the existence of a narrow gap in terms of literacy rates: overall, 94% for females and 97% for males³². However, there remains substantial gender gap in levels of access to advanced education with 59% of PhD graduates are men and 41% women³³.

Political participation. In 2008, the government approved a set of electoral laws that introduced a 30% minimum quota of female candidates in the electoral lists of every party. In a study on women representation in Indonesia, Hage (2014) suggested that the reasons behind the low rates of female elected officials lie in the lack of "political experience, networking and financial support". Furthermore, Hage (2014) reported that the preference of the electorate towards male candidates is dependent on a culture that depicts women as having less capacity than men³⁴.

Economic participation and labor. Indonesia is among the five most improved countries in terms of economic participation and opportunity and is ranked 68th in the respective index³⁵. According to the 2020 Global Gender Gap Report, the last year saw the country improving by more than 5% and jumping 28 positions up. Currently, 55% of leadership and senior positions are held by female professionals³⁶. Not only is Indonesia one of the six countries in the world where women hold more leadership and senior positions than men, but it also detains the highest share of women in such roles³⁷.

Despite these outstanding improvements, Indonesia is still facing relevant challenges. Specifically, only 54% of women has a job³⁸ and about half of them work in the informal sector³⁹. In the last 6 years the share of working women increased by about 4%^{40,41}. Most of women work in small family businesses are poorly paid or unpaid⁴². Accordingly, only 33% of women works in the non-agricultural sector⁴³ and on average, they get paid 50% less than men^{44,45}. Millions of Indonesian women work in foreign countries in the domestic sector, and in many countries, they risk "psychological, physical, and sexual abuse, situations of forced labor, and inhuman conditions"⁴⁶. 35% of small and medium businesses are owned by female entrepreneurs, but mostly in the labor-intensive informal sector⁴⁷. It is hard for women to grow their business, as they have to face a significant lack in knowledge of business management, poor access to financial institutions and a significant likelihood to be discriminated by government officials⁴⁸. Overall, the 2015 Indonesia's Gender Analysis reports that the most relevant challenges that stand among women and economic participation depend on the difficulty to get trained, on the discrimination in the labor market, and on the workload derived from family and child care⁴⁹.

The outbreak of COVID-19 pandemic can have disastrous consequences on women. Women, who make up the majority of caregivers and health workers in Indonesia, are saving millions of lives while putting themselves at higher risk of becoming infected during COVID-19 pandemic.

McKinsey and Company (2018) provided a strategy aimed at advancing women's equality in Asia Pacific and specifically focusing on Indonesia. The strategy was based on four main points. First, it advised to increased infrastructure and services (childcare and care), as well as expand the transportation infrastructure in light of the needs of working women. Secondly, the strategy outlined the necessity to expand training programmes for women such as business training for entrepreneurs and digital training in secondary school. Third, it recommended Indonesia to work on public-awareness aimed at changing attitudes towards females and their roles in the society. Fourth, the strategy reported the need for more legislation aimed at allowing women to actually work (e.g. part-time work; parental leave; discrimination in hiring process) and for more protection for women from abuses and discrimination⁵⁰.

5.3. Lao People's Democratic Republic (Lao PDR)

Lao PDR is a landlocked country with a population of 7.4 million of which women accounted for 50.1%⁵¹ and 15% of households are headed by women⁵². The majority of the population live in rural areas and 35.7 % of the population is urban⁵³. Changes in rainfall and temperature have consequences for ecosystems, communities and livelihoods that depend on them⁵⁴. The loss of different crops and livestock due to environmental disasters affect women, men and ethnic minority groups in a different way. Female-headed households are likely to be particularly impacted by disasters, given that they were already disadvantaged. Female headed households in Lao have average land size 16% smaller than male-headed households⁵⁵. A high proportion of food-insecure households have low education and literacy rates. In rural households, women are responsible for raising livestock, which are crucial for women's livelihoods, especially in disasters and emergencies. Unpredictable floods regularly destroy crops, severely impact women's income opportunities. In turn floods, drought, and other natural disasters damage women's resources, and women need to adopt extreme coping mechanisms, which include migration and child marriage.

The Constitution of Lao PDR guarantees equality between men and women in politics, economy, culture, society and in the family⁵⁶. In the voluntary review to the 2030 Agenda for Sustainable Development the government identified women's empowerment as a crosscutting strategy to achieve its national development goals⁵⁷. The Government of Lao PDR is prioritizing the following strategies in the implementation of its sustainable development plan: "(1) Improving the implementation and monitoring of national gender equality laws, policies, and instruments; (2) Strengthening national capacity to generate gender statistics and enable gender-sensitive planning; (3) Developing measures to combat violence against women; (4) Harnessing the full potential of Lao women's participation in the market economy as entrepreneurs, employees, and consumers; and (5) Addressing the issues that prevent adolescent girls from reaching their full potential"⁵⁸. The Lao National Adaptation Program of Action does not explicitly address gender challenges or opportunities in relation to climate change^{59,60}. The same it is valid for the national Strategy on Climate Change⁶¹.

Gender gaps remain, particularly among rural households, poor households and ethnic minorities, especially in the highland areas where a large number of ethnic minorities live. Women in Lao play important roles in agriculture, small-scale businesses, manufacturing and provision of basic services. Laotian women are very restricted in their mobility due to social norms, domestic duties, and costs of transportation. In rural areas and among ethnic communities, men exercise control

over the social life of their family which impacts on women's freedom of movement, access to information and access to financial services^{62,63}.

With 49 ethnic groups, Lao PDR is one of the most ethnically diverse countries in the world. Poverty rates are especially high among rural and ethnic communities. Ethnic women suffer from lack of opportunities in education and income generation, low mobility, time poverty, lack of access to land and productive assets, and gender-based violence.

Women's Participation in Decision Making. According to the World Economic Forum (2020), Lao is among the 10 best performers on the Economic Participation and Opportunity subindex, with 27.52% of women elected in the national parliament, which is above the world average⁶⁴. However, at the provincial and district level the participation of women is low, and there are no women governors⁶⁵. At the village level only 1.7% of village chief are women⁶⁶.

Education. The gender gap in enrollment in primary and secondary education in Lao is closing: in primary education, 89.9% of boys and 89.4% of girls are enrolled, in secondary education 37.6% of boys and 38.6% of girls⁶⁷. However, girls still face barrier to access education, especially in the rural and poorest areas of the country, due to household labour, financial constraints, language barriers and insufficient school infrastructure. Furthermore, many girls are not able to study due to safety concerns such as sexual harassment and gender-based violence, mobility constraints and child marriage⁶⁸. Data also suggests that families prefer to educate boys, instead of girls, particularly among the poorest families and ethnic communities⁶⁹. This is ultimately reflected in gender differentials in literacy rates: while the male literacy rate is 89.96%, only 79.39% of women are literate⁷⁰.

Health and Gender-Based Violence. Gender based violence is common in Lao, and 30% of women reported that a husband is justified to beat his wife⁷¹. Lao has one of the highest rates of child marriage in the region, with 39% of women married before age 18 in rural areas⁷², including through the practice of "bride kidnappings"⁷³. It was reported that child marriage increases following a disaster. Gender-based violence in Lao increases after disasters: for example, following flooding in Oudamxay in 2016, 27% of the survey respondents had heard of incidences of rape and 43% reported knowing someone injured by sexual and gender-based violence⁷⁴. The Lao PDR Government (2018b) noted that the social welfare and legal systems need to be strengthened in protect order to women and children from violence⁷⁵.

Women's access to healthcare is limited due to financial barriers, language, remote access, domestic responsibilities and restricted mobility, especially for women living in rural areas⁷⁶. In the review to the 2030 Agenda for Sustainable Development, the Government noted a poor coverage and quality of health services, highlighting that anemia rates, low contraceptive use and high fertility contribute to a high incidence of maternal mortality. The level of adolescent pregnancy is one of the highest in the region⁷⁷.

Economy. Women's labor force participation in Lao PDR is amongst the highest in the region at 79.5% for women and 81% for men, however the division of labor is strongly gendered and wage gaps persists. Women spend more time on unpaid domestic labour (4 hours per day on average) compared to men (30 minutes per day)⁷⁸. This suggests that women have also less time to engage in paid productive work.

Women are more likely to be informally employed (95.5%) than men (91.7%)⁷⁹, and unpaid family workers (64% of women compared to 27% of men⁸⁰. Women dominate small-scale retail enterprises. The most common sectors for women's micro and small enterprises are retail,

vending, guesthouse and restaurants⁸¹. These enterprises are more likely to be home-based. The Government recognized that promoting SMEs will be especially beneficial for women⁸². It adds that “the legal framework for doing business does not have gender discriminatory elements, but in practice, household responsibilities and lack of mobility due to personal duties makes it more difficult to start a business”⁸³.

The largest sector in Lao is agriculture, which employs 69% of women and 66% of men⁸⁴. Women in agriculture mainly undertake tasks such as planting, harvesting and tending to smaller livestock. Women are also largely employed in the service sector and in garment factories, which employs more than 100000 people in Lao, among whom 85% are women aged 17-25. Normally, they are hired into unskilled or semiskilled roles⁸⁵.

Although women and men have equal status for land tenure, property ownership and inheritance rights, and women recently have benefited from increasing rates of female and joint land title, customary laws and discriminatory norms prevent women’s access to land⁸⁶. Joint entitlement are not common in rural areas. Women tends to have more access to financial services. For example, more women own a bank account: 31.9% of women compared to 26.9% of men⁸⁷.

5.4. Pakistan

Pakistan is the fifth most populated country in the world, and women make up 48.8 per cent of the Pakistan’s population⁸⁸. Women suffer from pervasive gender-based violence, domestic abuse, honor killings, sexual violence, institutional discrimination, lack of access to sexual and reproductive rights, lower economic empowerment and political participation, among other issues⁸⁹. The International Monetary Fund estimates that, by addressing the gender parity, Pakistan would gain USD 91.5 billion per year⁹⁰.

The Government stated its priorities related to SDG 5 on gender equality: (1) Monitoring progress on the implementation of gender-responsive policy frameworks; (2) Create structures to support women in the labour market; (3) Establish systems and structures to develop, implement, monitor and evaluate behavioral change interventions to promote gender parity at all levels; (4) Encourage women entrepreneurs; and (5) strengthen structures to combat violence against women⁹¹. Similarly, one of the priorities to meet SDG 8 on sustainable growth is to “increase women’s labour force participation by investing in skills development programmes, girls’ education and creating a conducive environment for women’s employment”⁹².

The National Climate Change Policy developed in 2012 stated that one of its goals is to “promote pro-poor and gender-sensitive adaptation measures”, and it identified policies to address cross-cutting areas, such as “disaster preparedness, poverty, gender, capacity building, technology transfer, and international cooperation”⁹³. The Framework for Implementation of Climate Change Policy 2014-2030 included a paragraph on the gender dimensions of climate risk and resilience, which notes that gender differentials are critical to the success of any adaptation program and recognizes that “women in rural areas depend on climate-sensitive resources for survival and their livelihoods, they do not enjoy equal authority, decision-making power and resources needed to adapt to climate change”. Moreover, it acknowledges that women are more vulnerable than men to climate change, due to their limited access and control over decision-making and economic resources, and states that the government “is determined to support gender integration into various efforts to address climate change”⁹⁴.

The National Sustainable Development Strategy recognizes the important role that women play in agriculture, including livestock raising, small cottage industries and feeding their households. It

remarks women's participation at all levels of the crop production process, such as sowing, transplanting, harvesting and post-harvest operations. However, women do not have equal rights to land ownership and development.

Political Participation. According to the Global Gender Gap Report 2020 Pakistan ranks 93 out of 152 countries for political empowerment of women⁹⁵. Quotas are used to ensure a minimal presence of women in elected bodies: 60 seats in the National Assembly and 17 seats in the Senate are reserved for women. It was reported that women participate actively as political party members, however they are not able to secure leadership positions within parties, with the exception of women's wings⁹⁶.

Health and Gender Based Violence. Violence against women and girls, including domestic violence, rape, so-called "honor" killings, acid attacks, forced marriage and frequent is pervasive. 21% percent of women in Pakistan marry as children, and UNFPA estimated that 32% of women have experienced physical violence and 40% per cent of ever-married women have suffered from spousal abuse⁹⁷. Gender based violence occurs widely in communities even before emergencies, and it is exacerbated by humanitarian crisis or environmental disasters, as a more insecure physical environment adds further risks for women. Oxfam estimated that levels of threats, harassment, community-level disputes and domestic violence in Pakistan all increased during emergencies⁹⁸. Furthermore, Pakistan has one of Asia's highest rates of maternal mortality, and malnutrition severely affected the most vulnerable, including children and women of reproductive age⁹⁹.

Education. Pakistan has a gender gap in education of almost 20%. There are striking gender disparities in education, across the entire country. Less than half of women are literate (46%), compared with 71% of men, while the share of women enrolled is systematically lower than the share of men across primary, secondary and tertiary education^{100,101}.

Economic participation and labour. The levels of women's financial inclusion in Pakistan is low, the market is underserved and there would be significant potential to expand women's access through targeted financial products and services¹⁰². In 2015, the Government launched its National Financial Inclusion Strategy that sets targets to pursuit universal financial access, it aims at providing access to formal accounts to 50 percent of adult population by 2020¹⁰³.

Only one-quarter of women participate in the labour force compared with 85% of men. It is estimated that only 18% of Pakistan's labour income goes to women¹⁰⁴. 92.1% of women are informally employed versus only 79.6% of men¹⁰⁵. Often women do not accept paid employment due to security concerns or mobility restrictions generated by cultural and social norms. Almost 40% of women who are not working reported that the main reason for this is that male family members do not permit them to work outside the home¹⁰⁶. Those who work, are mainly employed in farms¹⁰⁷.

Women have less access than men to livelihood and productive assets such as land, employment and livestock and lower control over these resources. In emergencies, access to and control of livelihood become even more limited.

5.5. Papua New Guinea

Papua New Guinea (PNG) population is 8,5 million, 51% men and 49% women¹⁰⁸. Notably, the population is among the most diverse in the world due to the mountainous geography of the country and remoteness of many islands, which have kept communities separate. Its social

composition is extremely complex, with many ethnic groups and over 800 distinct languages still spoken¹⁰⁹. While over 50% of men can communicate in English, only 30% of women can, limiting women's ability to participate in the political and economic life of the country¹¹⁰. The scattered geographical reflects into relevant government costs for both logistic and supply management. Many are excluded from accessing basic services, including health, education, sanitation and safe water in the rural regions, without connectivity and infrastructure¹¹¹.

Gender equality and women empowerment are significant challenges in PNG, and systemic violations of women's rights exist throughout the country. The country scores low in all the major gender indicators: it occupies the 161th position (out of 162 countries) in the 2018 Gender Inequality Index¹¹² and the 127th position (out of 153 countries) in the Global Gender Gap Index¹¹³.

Violence against women is extremely high and PNG's social norms often exclude women from leadership and decision-making roles. PNG is one of the two countries in the world (the other is Vanuatu) where there aren't women in the parliament. Deep-rooted discriminatory norms reinforce patriarchal systems and biases: women in PNG are subjected to harmful practices, such as child marriage and gender-biased sex selection due to son preference.

The National Framework of Action 2017-2030, acknowledges the importance of women's participation, calling for an inclusive engagement without discriminations and for the promotion of women leadership. The plan notes that women's participation is pivotal for disaster risk reductions policies, plans and programmes and mentions that "adequate capacity building measures need to be taken to empower women for preparedness as well as build their capacity for alternate livelihood means in post-disaster situations"¹¹⁴. However, women's perspectives are often not effectively incorporated into the climate change adaptation and disaster management regulations and decision-making processes at all levels.

Health and gender-based violence. Gender-based violence including domestic violence is pervasive, 67% of women in PNG had been hit by their husbands¹¹⁵. PNG has one of the highest rates of maternal death in the world, only slightly more than 50% of women and girls give birth in a health facility or with the support of a skilled birth attendant. For every 100,000 live births, 215.0 women die from pregnancy related causes; and the adolescent birth rate is 52.7 births per 1,000 women aged 15-19¹¹⁶. Although in 2016 the Government adopted a National Strategy to Prevent and Respond to Gender Based Violence 2016-2025., the response to end violence remains inadequate and fragmented¹¹⁷.

Education. More than a third (37%) of the population is illiterate with the majority living in rural areas, where they lack access to education due to the limited number of schools and the physical distance to schools. There is a persistent gender gap in school enrolment, mainly driven by social cultural norms according to which education of girls is not relevant, especially when they are old enough to contribute in the household or get married. 34.6% of women adolescent are out-of-school, compare dot only 18% of men¹¹⁸. With regards to primary education, 71% per cent of women are enrolled, and 76% of men¹¹⁹. Similarly, in secondary education women enrolled are 29% and men 35%¹²⁰. Many girls leave schools after puberty due to the lack of schools, sexual harassment, expectations of doing household chores or preparation for early marriage. Gender inequalities are also present through tertiary and technical and vocational education: university enrollments comprise 61% men and only 39% women¹²¹. Furthermore, in universities there is a strong sex segregation with women being concentrated in the fields of education, nursing, and commerce.

Political participation. Cultural and structural barriers prevent women from participating in political life and currently in PNG there are no women in the parliament. The country scores 127th (out of 127 countries) in the Gender Political Empowerment index¹²². In, 2017, for the first time in 25 years no woman was able secure a seat in the parliament, despite more female candidates running for office than ever before (167 of the total 3,332).

Economic participation and labour. According to the most recent national census conducted in 2011, about 88% of the population lives in rural areas, relying subsistence agricultural and informal cash crop activities¹²³. PNG has the highest poverty rate in the South Pacific, with 40% of its population living below the poverty line¹²⁴. High rates of poverty disproportionately impact women, limiting their access financial services and products as well as their access to livelihoods opportunities and employment. There is a strong gender gap in employment: only 14% of women are waged workers compared to 33% of men, and 85% of women are in vulnerable employment compared with 65% of men¹²⁵. Women are mainly small business owners in the informal sector and engage in crafts, small-scale farming and trading. They face relevant barrier to access markets and small loans to finance their businesses¹²⁶.

Many rural women are subsistent farmers, responsible for the food-crop production for household consumption and the rearing of livestock such as pigs and chickens¹²⁷. In the coastal regions of the country, women use to catch small fish and harvest various shellfish for household consumption and local sale¹²⁸.

The formal financial inclusion gender gap is 29%, the highest in the South Pacific. This is reflected in a relevant gap in the use of financial products, with men owning up to two times more financial products than women¹²⁹. Sociocultural norms limit women's control over their financial resources and productive assets. Women are often solely responsible for household chores and expenses, with husbands or fathers placing further demands on women's finances, leaving a very limited opportunity for them to save money¹³⁰.

5.6. Timor-Leste

The Democratic Republic of Timor-Leste has a population of 1.3 million people, 49% of whom are women¹³¹. Most live in rural areas (60%) and there is a significant socio-demographic divide, with rural women and men having lower education, less employment and more poverty¹³². Rural populations in Timor-Leste have significantly lower levels of access to a range of services, such as water and sanitation, health, education, agriculture and finance¹³³.

Timor Leste Strategic Development Plan 2011-2030 calls for reforms on the areas of human capital, infrastructure, market formation, economy diversification and regulatory development^{134,135}. Furthermore, the plan makes commitment to gender equality and women empowerment, setting the goal of making Timor-Leste a "gender-fair society where human dignity and women's rights are valued and protected"^{136,137}.

The National Adaptation programme of action on climate change includes provisions on gender equality and women's empowerment in the context of climate change. It stresses the importance of improving both institutional and community capacity, including women and vulnerable groups, to increase disaster preparedness and response. Furthermore, it reiterates the importance of integrating gender into water management, in particular by "addressing vulnerable people's rights including women in water related legislation making and its management"¹³⁸. The programme aimed at promoting an active participation of women both at central and district level. The Green Climate Fund (2019) noted that it is crucial to have a gender-focused approach to address and

analyze that gender relations in climate change adaptation and disaster risk reduction taking into account the socio-economic and cultural context of Timor-Leste¹³⁹. Interventions should be designed considering which are the norms and expectations for women and men and how this might affect project outcomes.

Political Participation. The Constitution affirms the right of equal participation of women and men in political life, stating that the “direct and active participation by men and women in political life is a requirement of, and a fundamental instrument for consolidating, the democratic system” and stressing that “the law shall promote equality in the exercise of civil and political rights and non-discrimination on the basis of gender for access to political positions”¹⁴⁰. However, women still face many challenges in attaining political equality, due to lack of family support, limited confidence, geographical isolation, an inability to finance campaigns¹⁴¹. A recent study suggested that in East Timor there is a strong and institutionalized domination of the political arena by men, especially at the local level¹⁴². Women achieve great equity at the national level, thanks to a quota system introduced in 2006, which requires that women comprise at least one-third of listed candidates. Currently there are 40% of women in parliament. However, women have almost no representation at the local level, where 95% of the village chiefs are men^{143,144}. The Law on the Election of the National Parliament ensures women’s representation with a requisite that one in every three candidates elected is a woman. However, this does not guarantee women’s political influence and leadership. In turn, women do not frequently occupy key positions in the government, such as ministers or secretaries of state¹⁴⁵.

Education. Gaps in the education system generated low levels of education and literacy among the adult population. Although the enrollment rates in primary education have increased in the last years, secondary and tertiary enrollment remains low, and the quality of education is scarce¹⁴⁶. There are significant disparities between women and men in Timor-Leste. Only 64.21% of women are literate, compared to 71.87% of men¹⁴⁷. Women are less likely than men to have participated in formal education at all levels¹⁴⁸ and are more likely to have missed out on schooling and less likely to be enrolled post primary education. 58% of women older than 25 have never been to school, compared with 43% of men; and only 16% of women in the same age group have completed secondary or tertiary education, compared with 25% of men¹⁴⁹. Repetition and dropout rates remain high, due to lack of school readiness and language barriers¹⁵⁰. Notably, sexual harassment and violence in schools are critical issue¹⁵¹. Gender segregation is also evident in technical and vocational education. Women (56%) are normally enrolled in administration, finance, and information technology programs, while men are commonly enrolled in technical programs such as construction and auto mechanics¹⁵².

Health and Gender-Based Violence. Gender-based violence remains widespread and is one of the highest in the world with 59 % of ever-partnered women aged 15-49 years experiencing violence at least once in their lifetime¹⁵³. Younger women are particularly at risk of sexual violence and early marriage. The traditional patriarchal system has a negative relevant impact to the empowerment and well-being of women. The idea that women are subordinate “property” originated by the dowry system is one of the main causes of domestic violence. Gender-based violence is exacerbated by limited capacity of the formal criminal justice system, and cases of violence are mainly addressed only at the family level.

Timor-Leste has a high birth rate, accompanied by high maternal and child death rates: maternal mortality is still amongst the highest in the world at 557 death per 100,000 live births¹⁵⁴. 60% of women have reported problems in accessing healthcare¹⁵⁵. Six percent of teenage girls (age 15–19) are already mothers, and those in rural areas are twice as likely to be mothers as their urban

counterparts¹⁵⁶. Hunger and poor nutrition are also a serious concern, with 27% of women aged 15–49 being malnourished¹⁵⁷.

Notably, crisis and emergencies exacerbate gender-based violence. CARE (2020) reported that in Timor-Leste violence worsened during disaster, mainly due to increased pressures from having less work and insufficient food as well as for being displaced¹⁵⁸.

Economic participation. Women in Timor-Leste have limited awareness and resources to control productive assets, such as financial services, land, marketing facilities and information. Although the constitution protects women's equal right to land ownership^e, traditionally land is always in the hands of men and passed down to the men of the family¹⁵⁹. The division of labour in Timor-Leste is strongly gendered due to discriminatory patriarchal values, norms and practices. Women and girls bear responsibilities for household work, reproductive work and caring for the elderly and children¹⁶⁰. Women are normally only employed in subsistence cultivation, informal small-scale trading, and home-based industries. This is due to gender-based barriers such as lower levels of literacy, education and training; lack of professional skills in the formal sector; low support from family members; domestic and reproductive obligations; and mobility constraints due to safety and security concerns^{161,162,163,164}.

Women's participation in the labor force is substantially lower than men's participation: 56% of men are classified as being in the labor force compared with only 27% of women¹⁶⁵. Women are concentrated in wholesale and retail trades (25% of employed women) and in education, health, arts, and others low paid sectors (12%)¹⁶⁶. Women are also more likely to be employed in the informal economy (69%) compared to men (71%)¹⁶⁷. Most people work in the agricultural sector (65%), followed by services and sales (15%)^{168,169}. The activities performed within these sectors are strongly gendered. For example, in agriculture, men undertaking tasks considered labor intensive and paid better, such as working in rice fields, coffee growing and selling livestock. On the contrary, women common activities include planting, weeding, and harvesting¹⁷⁰. Women play an important role in Timor-Leste's agriculture. However, due to gendered barriers in agricultural production and income generation, women produce 15% less per hectare of land compared to men¹⁷¹.

5.7. Vanuatu

The Republic of Vanuatu (Vanuatu) is an island country located in the South Pacific Ocean and inhabited by 298,434 people (50.9% men and 49.1% women)¹⁷², out of which three-quarters live in rural areas¹⁷³. Its economy is mainly based on small-scale agriculture, fishing, offshore financial services and tourism; its economic development is hampered by high dependence on few commodity exports, vulnerability to natural disasters, and long distances from main markets¹⁷⁴.

The National Constitution of Vanuatu recognises the rights and freedoms of all individuals without discrimination on the grounds of sex, race, place of origin, religious or traditional beliefs, political opinions or language. It also has provisions for the 'special benefit, welfare, protection or advancement of females, children and young persons, members of under-privileged groups or inhabitants of less developed areas' (Chapter 2, Clause 5 (1) (k)).¹⁷⁵ The Government of Vanuatu recognises that gender equality and women's empowerment are critical to the achievement of the

^e Article 54 of the Constitution guarantees equal rights to women and men to own property and real estate holdings and) law number 13/2017 guarantees the rights of women and recognizes the equality of land ownership.

national development vision: “a Just, Educated, Healthy and Wealthy Vanuatu”. One of the key “society goals & policy objectives” identified in the National Sustainable Development Plan 2016 to 2030 is an “inclusive society which upholds human dignity and where the rights of all Ni-Vanuatu including women, youth, the elderly and vulnerable groups are supported, protected and promoted in our legislation and institutions”¹⁷⁶. In line with these commitments, Vanuatu introduced the National Gender Equality Policy (NGEP), to “promote equal rights, opportunities and responsibilities among men and women and to eliminate all forms of discrimination and violence against women and girls”¹⁷⁷. It was one of the first policies to explicitly address gender inequality in the country. Following endorsement by the Council of Ministers, the NGEP was launched in 2016¹⁷⁸. It lapsed in 2019, and was replaced by the National Gender Equality Policy 2020–2030.

Women in Vanuatu are more affected than men to climate change due to many interconnected factors, such as their low participation in the parliament and decision-making bodies, the low access to financial services and productive assets, and a high rate of gender-based violence. This differentiated vulnerability is recognised in the Vanuatu Climate Change and Disaster Risk Reduction Policy 2016-2030, where equity is one of six principles that direct the country’s climate change and disaster risk reduction efforts and gender and social inclusion are considered as one of the cross-cutting strategic priority issues¹⁷⁹.

While the country’s Constitution and key national plans and strategies articulate the importance of equality and rights, significant disparities continue to exist between men and women. Vanuatu Global Gender Gap Index is low, as the country scores 126 out of 153¹⁸⁰.

Health and gender-based violence. The 2011 National Survey on Women’s Lives and Family Relationships, the only comprehensive study on gender-based violence conducted in Vanuatu, highlighted that violence against women is pervasive and occurs across all socio-economic backgrounds, with peaks in rural areas¹⁸¹. Approximately 60% of women who have ever been married, lived with a man, or had an intimate sexual relationship with a partner, experienced physical and or sexual violence in their lives, of whom 21% were left with permanent injuries and 68% were subjected to psychological violence^{182,183,184}.

Women experience violence before, during and after disasters: it was reported that domestic violence increased by 300% after two tropical cyclones in 2011¹⁸⁵. UN Women found that, 98% of victims of violence do not access the formal justice system, for lack of economic resources or for the fear of being stigmatized if they leave the husband¹⁸⁶. The marginalization of women and lack of a gender inclusive frameworks prevent policy-makers from recognizing and tackling violence against women and girls.

Education. Literacy rates vary between urban and rural areas, and between ages. There aren’t strong gender-based differences in the adult population where the literacy rate for adult men stands at 87% and for adult women at 88%¹⁸⁷. However, only 87% of youth female (aged 15-24) are literate compared to 96% of youth male¹⁸⁸. Furthermore, 7 out of 10 women have either only completed primary education or have not had any schooling at all¹⁸⁹. In rural areas 80% of women reported being literate compared to 98% in urban areas¹⁹⁰. Often, older people can only communicate using the local vernacular language. Notably, the education curriculum includes life skills and gender-based violence awareness and prevention.

Political participation. Women are excluded from decision making and are absent from leadership positions: Vanuatu is one of the two countries in the world (together with Papua New Guinea) where there are no women in parliaments¹⁹¹. Since its independence in 1980, only 5 women have been elected into the national parliament^{192,193}.

The Constitution of Vanuatu recognizes gender equality as part of the fundamental rights, recognize the right to non-discrimination on the basis of sex and provides for affirmative actions for the advancement of women. However, women have a limited access to decision making process in formal governance structures; this is mainly due to women's care obligations and discriminatory custom norms. For example, women are excluded from direct roles in the village decision-making body, the *nakamal*^f, or the basic unit of *kastom*^g governance; in some cases women are invited to attend but decisions had already been formulated or made by the men¹⁹⁴. This exclusion continues at higher levels, where regional and national governance structures are normally male dominated.

Women's lack of direct participation in decision-making is reinforced by other barriers, such as low literacy, limited access to health and maternal mortality¹⁹⁵. In turn, Vanuatu women are marginalized from discussions and decisions on areas of social and economic development, environment, governance and human rights at all levels. It is common, during public discussion focused on these topics, to note the under-representation of women and the silencing of their inputs.

At the policy making level there is limited technical capacity for gender analysis and gender-responsive planning, and the country lacks accountability measures to support gender mainstreaming. This background is exacerbated by the lack in collection and analysis of sex disaggregated data¹⁹⁶.

Economic participation and labour. Women represent 61% of the labor force compared to 79% for men^{197,198} and 50% of urban women are employed versus 71% of men and in rural areas, 13% of women and 26% of men are employed¹⁹⁹. More than 40% of working-age women declared reported that family work is their main status in employment, compared to about 20% of working-age men (Government of Vanuatu, 2016). Of the total workforce, 63.6% of women engaged in unpaid reproductive work²⁰⁰. More women than men (49% and 41% respectively) are involved in the subsistence economy²⁰¹, which makes them more susceptible to poverty, climate change, disasters and other livelihood stresses. Levels of economic development are low, but rural and coastal communities are even more vulnerable to disasters-caused economic shocks, due to damages in property and food crops and reduced income from tourism²⁰².

Women are less financially included than men: only 32% of adult women report owning a bank account, compared to 41% of men. Furthermore, even if women are less likely to be formally employed and to have an account at a formal institution, women appear to deposit and withdraw more frequently than men do, with 65% of women reported saving in the past year, compared with 53% of men²⁰³.

The contribution of rural women to the agricultural sector is significant: women are represent about half of the workforce in agriculture (49.5%) and spend more time than men on agricultural production and income generating activities²⁰⁴. The division of labour in agriculture reflect

^f A space where all village decision-making is done.

^g Customary beliefs, values, governance and practices.

traditional gender roles: while both men and women practice daily farming, women are in charge of selling products, even among fishing communities²⁰⁵. For instance, women have a heavier workload than men due to socially contracted norm that assign to women, in addition to agricultural work, responsibilities for small-scale marketing, handicraft production, caregiving and housework²⁰⁶.

6. GENDER ISSUES IN THE CONTEXT OF THE CRPP

As outlined in the introduction, the CRPP comprises two separate but interlinked components; the CRPP Trust Fund (TF), which focuses on upstream support to strengthen the enabling environment required for financing and implementing local adaptation measures at scale; and the CRPP Investment Fund (IF) which focuses on the efficient roll-out of local adaptation measures through downstream investments. The TF will support implementation of outputs 1, 2 and 3 of the wider CRPP program. The IF, which will be financed by GCF and administered by ADB, will provide grant and loan resources to implement output 4, 5 and 6 of the wider CRPP program. Accordingly, this section is divided into two subsections. The first sub-section describes some of the key barriers to scaling-up investments in women-focused investments in climate resilience, and the types of interventions that will be implemented through the CRPP TF Gender Window. The second sub-section highlights some of the key issues that contribute to women's vulnerability to climate change in the context of the CRPP IF and its key areas of intervention, and the types of interventions that can contribute to building their adaptive capacity. The two sub-section provide the basis for the CRPP Gender Action Plan.

6.1 The CRPP Trust Fund

The CRPP Gender Window will support the identification, development, and implementation of investments that have the explicit objective of strengthening the resilience of women to climate change. As such, support will be provided through (i) knowledge support and technical assistance for identifying investment opportunities and creating an enabling environment for financing women-focused adaptation measures; (ii) transactional technical assistance for preparing investment projects; and (iii) grant financing, on a selective basis, for implementing investments identified and prepared under the technical assistance. This section identifies some of the specific activities that the CRPP can support under these modalities, and the rationale for including them in the CRPP.

6.1.1. Output 1: Knowledge and action research on climate risk-informed pro-poor community level solutions strengthened.

Substantial gaps remain in many countries to generate the necessary political will, and commitment from governments and communities to invest in women-focused adaptation measures that help tackle climate change. There is thus a need to generate a deeper understanding of women's vulnerability context in CRPP countries, and generate an evidence base for climate adaptation interventions that are effective in reducing these vulnerabilities and building the resilience of women. Furthermore, there is a need to demonstrate how women's contributions can strengthen the effectiveness of climate change measures in general and, where women are implementing adaptation initiatives that are having positive impacts, how they can be scaled up.

To this end, the CRPP will support governments and communities in **undertaking action research studies that address gaps in understanding** around women and resilience in CRPP countries and make the case for women-focused adaptation investments. In doing so, the CRPP

will contribute to a deeper and more contextualized understanding of the specific situation of women, their diverse conditions, and how they are impacted by climate change in both the short and long run, including identifying issues related to structural inequality between men and women in specific geographical areas that contribute to women's vulnerability to climate change. This activity area will help strengthen the evidence base for appropriate and effective women-centred adaptation, by collecting data and information to build evidence, and compiling lessons learned from related initiatives to demonstrate the potential for bankability of women-focused adaptation investments. This in turn should inspire political will and community support for investing in women-led, local-level adaptation at scale.

As part of the evidence generating and lesson learning process, the CRPP will also **support innovative adaptation initiatives and pilot activities at the community level** to identify and test women-focused adaptation measures that could be scaled up through larger investments in the future. These initiatives, which will focus on building skills, capacity, and knowledge rather than civil works, and they will be designed and led by women, providing a dedicated space for women to trial and test adaptation measures with them at the centre. This will provide an opportunity for women to demonstrate their ability to build resilience to climate change and present their achievements to local governments so that, in future, women can increasingly be recognized as partners for resilience building by local governments.

Increasing investment in local-level, women-focused adaptation goes against the dominant trends in many CRPP countries, and thus will require getting key stakeholders to transform current business as usual practice. Work is needed to build the capacity of ministries and sector agencies, as well as other key actors, such as the private sector, to understand the importance of promoting women-focused investments in climate resilience, and how this can be achieved. It is also imperative that women, especially at the grassroots level, have a voice in shaping adaptation programming, and that their practical needs, interests, vulnerabilities, and capacities are represented in policy, plans, and standards that inform women-centered adaptation investments. The CRPP will therefore support **policy dialogues and forums for exchange at the regional, national, and local levels** that bring together grassroots women's organizations, government ministries from various sectors, the private sector, academic institutions, and other civil society organizations, providing a platform for exchange and collaboration between actors that may not traditionally work closely together. By bringing together these key stakeholder groups for multiple iterations of dialogue, CRPP support through the Gender Window can contribute to a change in culture within governments and civil society that is more supportive of women-focused investments in adaptation and recognizes their value in shaping resilient development. The long-term objectives being to help create a movement for locally appropriate women-focused adaptation investments in CRPP countries and to promote women's leadership in adaptation.

6.1.2. Output 2: Institutional and community capacity to develop and deliver climate adaptation investments strengthened.

Creating policy incentives and procedures for the inclusion of women in adaptation decision-making at the local level and building their capacity to use participatory tools and procedures for climate risk informed development planning and implementation can empower local women to act on adaptation. The CRPP Gender Window will therefore support the development of climate risk assessment and decision support tools and training and capacity-building for designing and implementing adaptation investments at the community level that respond to the specific needs of women, and integrate design features that promote women's engagement in adaptation decision-making at the local level. Increasing women-focused investments will also require improved coordination and communication at all levels of government and also at the project level.

This will require greater vertical integration among municipal, provincial, and national governments on policies and practices that aim to build women's resilience to climate change²⁰⁷. In addition, ensuring the meaningful participation of women in adaptation decision-making beyond the program life-cycle will require strengthening the institutions and processes that provide the necessary space to support and deliver such an approach. The CRPP will thus support governments to create new and/or improve existing policy frameworks and communication and coordination mechanisms to stimulate coherence and convergence across gender, climate, and key sectors such as agriculture, land-use planning, and urban planning.

6.1.3. Output 3: Inclusive and pro-poor adaptation investment projects identified and prepared.

The CRPP will seek to increase the climate resilience of women through leveraging sizable investment in adaptation projects and replicating effective interventions. As such, the CRPP Gender Window will support the preparation of large-scale investments that explicitly strengthen the climate resilience of women, particularly poor women, and/or adaptation investments that are led by women. Women will be at the centre of the project design process and engaged at all levels of decision-making, using inclusive design processes. A number of key design features and approaches have been identified as critical to ensuring that women are at the centre of the design process. These include;

- The collection and analysis of **sex-disaggregated data** to understand the specific vulnerability characteristics of women so that they can be addressed through the project activities,
- Identifying **outputs and related activities that address structural inequalities between men and women**, challenge social and gender norms, increase women's empowerment and contribute to transformative change (e.g policy reforms, revision of regulations, information management, the provision of equal wages, ensuring participation of women in meetings where public works are designed and selected etc);
- Integrating approaches that aim to **create a structural, cultural, and financial base that can facilitate the desired change over time** such as improving organizational and individual capacity across the public and private sectors and civil society; developing leadership for women-focused adaptation at all levels; producing champions and supporting change makers in positions of influence that can drive change forward and advance the progress made under the interventions.
- Including **sex-disaggregated indicators in monitoring and evaluation frameworks** for all CRPP-supported projects and increase women's capacity to assess the efficiency and effectiveness of adaptation measures and institutionalize learning and information generation practices.

6.2. The CRPP Investment Fund

The CRPP IF will provide grant and loan financing to seven selected DMCs, namely, Cambodia, Indonesia, Lao PDR, Pakistan, PNG, Timor-Leste, and Vanuatu, to implement local adaptation measures as part of ADB financed sub-projects that are targeted at the poor and vulnerable population. The CRPP IF will focus on the efficient roll-out of local adaptation measures through downstream investments according to three climate related outputs. This subsection presents the assessment of gender concerns in the context of the key activity areas that will be the focus of each of the CRPP IF outputs. The gender assessment is structured according to the key activity areas, although many of the issues discussed are applicable to other activity areas.

It should be noted that although the Gender Window is funded by the CRPP TF, it can be leveraged to enhance gender equality outcomes of the Investment Fund in CRPP countries. Examples include; (i) generating the necessary political will, and commitment from governments and communities to invest in women-focused adaptation measures that help tackle climate change; (ii) creating policy incentives and procedures for the inclusion of women in adaptation decision-making at the local level and building their capacity to use participatory tools and procedures for climate risk informed development planning and implementation which can empower local women to act on adaptation; and (iii) place women at the centre of the design of large-scale investments that explicitly strengthen the climate resilience of women, particularly poor women, and/or adaptation investments that are led by women.

6.2.1. Output 4: Information and systems for delivering applied climate-risk informed investments at scale improved.

A. Women's access to climate information and early warning

Women typically have less access to all types of information including climate information and early warning^{208,209,210}. As detailed in the previous section, gender-based differences in literacy rates and access to education are present in most CRPP countries, with differences often more pronounced in rural than urban areas and in some countries more than others. In **Pakistan** for example, where less than half of women are literate (46%), compared with 71% of men, and in **Cambodia**, where the gender gaps in education are most prominently seen in rural areas. These lower levels of awareness and literacy exacerbate women's exposure and vulnerability to climate hazards and limit their adaptive capacity, because implementing effective strategies to adapt to climate change requires a robust understanding of current and future risk as the starting point. Understanding the nature of gender-based vulnerabilities in specific contexts is crucial to recognizing the different impacts that climate change has on women and men and thus informing the generation and delivery of climate information services. There is thus a need to generate a deeper understanding of women's vulnerability context in CRPP countries, and create an evidence base for climate adaptation interventions that are effective in reducing these vulnerabilities and building the resilience of women. Research also suggests that women have a deeper understanding of vulnerability than men, largely due to the role they play in households and the wider community. Because vulnerability links closely to livelihoods and well-being, tapping that knowledge is critical to identifying the most suitable interventions to address climate risk. The collection and analysis of gender-disaggregated data to understand the specific vulnerability characteristics of women so that they can be addressed through the project activities will be a key part of the design and implementation of investments. The CRPP will also take specific steps to improve women's access to climate information as part of its investments in improving climate information services.

B. Women's access to adaptive social protection

Social protection programmes can increase resilience to climate change through enhancing early adaptation actions and livelihoods improvement and facilitate impact mitigation during climate shocks, offering direct support to affected individuals, and preventing negative coping mechanisms. In the long run, these programmes can support livelihoods diversification, poverty and vulnerability reduction and increasing adaptive capacity²¹¹. This is especially true when linked with apprenticeship schemes and skills transfer programmes and social and financial inclusion initiatives²¹². Despite increasing awareness of the need to explicitly target women beneficiaries, women are often disadvantaged in social protection systems, experiencing lower coverage rates and substantially lower benefit levels²¹³. Experience from two CRPP countries with well-

established social protection programs, namely Indonesia and Pakistan, shows that significant gaps remain in reaching women. In **Indonesia**, insufficient use of up to date sex-disaggregated data for monitoring and targeting of beneficiaries, complicated information about protective measures and limited assistance available from official sources, and inadequate support to complete enrolment procedures have been identified as important issues²¹⁴. In **Pakistan**, large gender gaps in mobile phone ownership and national ID possession, has meant that women are at risk of being disproportionately excluded from the Ehsaas Emergency Cash Program program, which was a key part of the government's Covid-19 response²¹⁵. Experience from **Indonesia** also shows that women must feel safe to travel to collect social protection payments and this must be considered in the design of distribution networks, as security and safety of women in relation to GBV is known to increase in the aftermath of environmental shocks²¹⁶.

As part of its investments in climate risk informed adaptive social protection programs, the CRPP will seek to tackle the limitations of existing schemes in terms of reaching women, ensuring that social protection programs include specific identification of, and outreach to, women, using sex-disaggregated data for targeting beneficiaries. It will acknowledge and respond to differences in women's and men's roles and opportunities and ensure that women are linked to people that provide the various services they need. Investments will seek to contribute to transformative gender outcomes recognizing that adaptive social protection can be a powerful tool to redress women's socio-economic disadvantage, better responding to women's rights. Women's empowerment will be an explicit goal, and education and training specifically for women will be provided as part of the related activities. This can include, linking cash transfers to resilient livelihoods training and sensitizing beneficiary households, programme managers and service providers about harmful social norms and equal sharing of responsibilities. Women beneficiaries will be involved in the design, implementation and monitoring of investments in adaptive social protection so that they are explicitly designed with women's rights in mind, tackling the limitations of existing schemes. The CRPP will also seek to build partnerships between government and community-based women's groups to effectively deliver social protection programs empowering these local organizations and strengthening the transformative potential of the social protection program.

C. Women's participation in decision-making

The meaningful participation of women is crucial to the design and implementation of efficient resilience interventions at all levels. These interventions need to take into consideration the specific situation of women, their diverse conditions and how they are impacted by disasters and climate change in both the short and long run. However, as the review in the previous section shows, women in many CRPP countries are largely excluded from decision-making and are absent from leadership positions. This is particularly true at the local level. This is often due to discriminatory customary norms and cultural and structural barriers that prevent women from participating, such as in **Vanuatu** where women are excluded from direct roles in the village decision-making body, and in some cases women are invited to attend but decisions are already made by the men, or in **Timor-Leste** where 95% of the village chiefs are men, despite having a constitution that affirms the right of equal participation of women and men in political life. Even in countries such as **Indonesia**, **Lao PDR** and **Pakistan** where women participate actively as political party members and a proportion of elected officials in the national parliament are women, at the provincial and district level the participation of women is low. The CRPP recognizes that involving women in designing investment projects for resilience is critical, and so investments will be designed through inclusive processes where women are engaged at all levels of decision-making. Furthermore, the implementation of investments will need to include activities to strengthen women's self-reliance, leadership and confidence. Experience from **Indonesia** shows

that fostering collaboration between community-based women's groups and local governments in undertaking disaster risk assessments and resilience priority setting can empower women and increase women's leadership role, and crucially can result in disaster resilience elements being embedded into development plans, budgets and programs²¹⁷. Investments in climate responsive decentralization processes will particularly present opportunities for increasing women's participation in the design and prioritization of adaptation measures at the community level and ensuring that their knowledge and experiences are captured in local level risk assessments.

6.2.2 Output 5: Climate resilient pro-poor livelihoods investments implemented.

D. Women's access to agricultural extension services

The review presented in section 6 clearly highlights how women are overrepresented in agricultural livelihoods and that the majority of rural women in CRPP countries rely on subsistence farming for their livelihoods and food security. The agricultural sector is often the hardest hit by extreme weather events and this reliance on natural resource-based livelihoods makes women more vulnerable when these resources are depleted or become scarce, as a result of changing environmental conditions. Therefore, strengthening the resilience of agro-ecological systems to climate change is critical to protecting and improve the livelihoods of rural women. The role of agricultural extension and rural advisory services is crucial for supporting women to adopt improved climate resilient production practices. However, gender gaps in access to such services are prevalent in many CRPP countries, such as in Cambodia, where women receive only 10% of agricultural extensions services despite comprising 51% of the primary workforce in subsistence agriculture and 57% of the workforce in market-oriented agriculture²¹⁸. This is despite evidence that suggests women are more likely than men to use the information they receive²¹⁹. Unequal access to agricultural extension services compounds gender inequality in livelihood opportunities²²⁰. However, structural gender inequalities can be addressed over time if extension approaches are designed to consider women's views and priorities, be flexible with their time preferences and mobility constraints, and ensure their full engagement in disseminating information and adopting new technologies²²¹. CRPP investments in climate resilient livelihoods must therefore take specific steps to understand the differences in resources and technology needs of male and female farmers and include targeted interventions for women that improve their access to and use of knowledge, inputs, information and technologies.

E. Women's access to microfinance for disaster resilience

Financial inclusion is central to managing risk as it means that individuals have access to useful and affordable financial products and services such as savings, credit and insurance, to meet their financial needs in times of crisis. Access to microfinance products that can provide financial relief is thus an essential element of women's ability to face climate shocks and stresses. Targeted products such as gender-sensitive climate risk insurance (CRI) strengthen women's resilience against both the immediate impacts and the long-term consequences of disasters and climate change. Micro-insurance (such as crop-insurance and home-insurance) allows the implementation of protection measures enhancing livelihoods and economic opportunities, supports climate-smart agricultural practices and could be linked to social protection interventions. Lower access to financial services is one of the factors that result in women being more likely to be affected, and to be the least prepared for climate shocks and stresses. As highlighted in the review presented in section 6, social, cultural and legal barriers prevent women from accessing such financial services in many CRPP countries, such as **Pakistan, PNG, Timor-Leste, and Vanuatu**. Women have limited ownership of productive assets, such as agricultural land and property and the consequent lack of collateral represents a barrier for women's access to formal

financial services. Other barriers include; lack of collateral and ID documents that hinder the access to formal financial services; low digital literacy and mobile phone ownership, that deter women's outreach through digital financial services; mobility constraints that reduce their opportunities for economic activities outside their houses. Women may also have low levels of self-confidence as well as trust in commercial financial service providers such as banks. Similarly, women are frequently excluded from the insurance market: as inclusive insurance is normally attached to a loan, it is usually the head of household (more often a man than a woman) who is covered by a credit-linked insurance²²². Having fewer resources to invest in precautionary measures, and to support their post-disaster-recovery, mean that women have fewer opportunities to build resilience.

Financial inclusion interventions need to be designed to increase women resilience and need to carefully consider the barriers that women face in accessing financial services. Recognizing that financial inclusion is key to strengthening women's resilience to climate related shocks and stresses, the CRPP will seek to expand women's access to instruments such as indexed-based microinsurance, through targeted financial products and services as part of its investment sub-projects that can be paired with capacity building on savings and investments, supporting preparedness to shocks while empowering women financially, socially and politically. In countries where the levels of women's financial inclusion is low compared to men, the market maybe underserved and there would be significant potential to expand women's access through targeted financial products and services²²³.

D. Women's role in agriculture supply chains

Informal employment is the dominant form of employment in Asia, and women, especially from poor communities in CRPP countries, have a high dependence on the informal economy, that is characterised by low access to decent work with poor working conditions that can contribute to economic and welfare losses, damage to health and productivity, and forced labour migration²²⁴. There is a significant disparity between men and women in earnings, the quality of jobs, and voice and representation in leadership positions in many CRPP countries, and women's economic activities often involve low value-added activities. In Timor-Leste, for example, women are more likely to be employed in the informal economy (69%) compared to men (71%), and in Papua New Guinea there is a strong gender gap in employment with only 14% of women as waged workers compared to 33% of men, and 85% of women are in vulnerable employment compared with 65% of men²²⁵. In Lao PDR, 95.5% of women are informally employed and dominate small-scale retail and vending enterprises²²⁶. Under-employed workers are also more likely to be women as they juggle employment and family responsibilities. In Lao PDR, women spend significantly more time on unpaid domestic labour (4 hours per day on average) compared to men (30 minutes per day)²²⁷. Agriculture is a sector characterised by informality and decent work deficits and the vulnerabilities of workers in the agriculture sector are magnified due to climate impacts²²⁸. Women's role in agriculture and food security is not limited to planting, cultivation and harvesting. Women have significant involvement in storage, processing, logistics and sales, making them crucial stakeholders in the wider food and agricultural supply chain, the links of which can all be impacted by climate shocks and stresses. Heat extremes for example have been shown to cause exhaustion, reduce working times, and lower incomes of informal workers throughout the agricultural value chain. Flooding has been shown to significantly disrupt informal food vendors sales and lead to heavy losses as perishable produce end up being sold at lower prices or discarded²²⁹.

Sufficient and stable incomes and decent work are critical for climate resilience. Climate induced loss of livelihoods, reduction in incomes, or deterioration of working conditions in the agriculture

sector have particular implications for women. CRPP investments in increasing the resilience of agricultural supply chains provides an opportunity to address some of the specific structural inequalities that increase women's vulnerability to the impacts of climate change and strengthen women's economic empowerment. CRPP investments in this regard must seek to address gender-related factors that influence women's ability to participate in value chains, increase access to and control over income, promote women's engagement in agricultural entrepreneurship orientated activities, deepen women's chances to increase value of their agricultural activities, as well as diversify their income-earning opportunities, and provide women with leadership roles. Investments should also integrate initiatives that reduce women's time poverty and burden of care in ways that reduce and redistribute labor and time commitments to benefit them directly. Investments in labor-saving assets (e.g., running water, electricity) that allow women to take advantage of new income-earning opportunities are likely to promote better results for their overall well-being and for climate resilience. Many of the interventions in this case may require physical (structural) solutions such as providing child day rooms in agro-processing units, which largely employ women and which can be included as part of project outputs.

6.2.3 Output 6: Climate adaptation pro-poor infrastructure implemented.

E. Women's access and control over natural resources.

Through its investments in nature-based solutions for building resilience at the community level, the CRPP can promote women's access and control over natural resources. Women's reliance on natural resource-based livelihoods makes them more vulnerable when these resources are depleted or become scarce. Women's adaptive capacity is also limited by the fact that, due to socio-cultural norms and patriarchal power structures they often face barriers to accessing and control over natural resources. Women are underrepresented in environmental decision-making and natural resources management, such as forest user groups, water user associations, and environmental policy-making. Due to these disadvantages, women can be less likely to benefit from projects that promote nature-based solutions. Addressing restrictive social and cultural norms to make sure women and girls can access these resources is therefore essential to building resilience and must be integral parts of these types of investments²³⁰. Nature-based infrastructure can provide equitable benefits for women, and integrating gender considerations in initiatives related natural resource management can make them more effective and sustainable²³¹. If planned correctly, such investments can also have a co-benefit of reducing the daily workload of women associated with care and household work and reduce health and safety risks as well as productivity losses. For example, NBS that mimic natural processes to enhance water availability improving access to water for household use, making it easier and faster for women to collect water and minimizing their exposure to health and security risks on the way²³².

F. The participation of women in the identification, construction/installation and operation of small-scale adaptive infrastructure.

The identification, construction/installation and operation of small-scale adaptive infrastructure presents an opportunity to address gender inequalities and build the resilience of women beyond the primary purpose of reducing their physical exposure to climate related hazards. Thus, specific measures to increase gender equality and women's empowerment will be included in the design and implementation of such interventions. For example, gender disparities in employment remain extensive in many CRPP countries, primarily because of traditional attitudes about "appropriate" occupations for women and men. This is particularly true in the construction sector, where it is believed that women either cannot or do not wish to engage in manual labor. Ensuring participation of women in meetings where public works are designed and selected, including

capacity-building activities that specifically develop women's technical and leadership skills, and the provision of equal wages should be cornerstones of such investments.

G. Access to safe shelter

As the frequency and intensity of extreme climate events are expected to increase with global warming, responses to flood and cyclone hazards are likely to require increasing numbers of people to evacuate their homes in seek of safe shelter. Studies have shown that women are often reluctant to use emergency shelters, and there are frequent examples of women refusing to use emergency shelters due to either design concerns or because of their bad experiences from the past²³³. This is often because basic gender needs like separation toilet facilities, dedicated space for women, and private space for pregnant woman and nursing mothers are rarely factored into the design and management of emergency shelters. Evidence shows that women and girls are at greater risk of gender-based violence (GBV), including sexual harassment and domestic violence during emergency situations and in the aftermath of disasters in part because of poor emergency-shelter design and inadequate coordination by disaster responders^{234,235}. In countries such as Papua New Guinea, Pakistan and Vanuatu where GBV, including domestic violence is already pervasive even before emergencies, insecure physical environments resulting from environmental shocks adds further risks for women. In **Vanuatu**, UN Women reported that domestic violence increased by 300% after two tropical cyclones in 2011²³⁶. In **Pakistan**, Oxfam estimated that levels of threats, harassment, community-level disputes and domestic violence in Pakistan all increased during emergencies, and one of the major reasons for the increase in domestic violence during emergencies were said to be lack of proper shelter²³⁷.

CRPP supported investments in safe shelter must therefore ensure that shelter design includes features that respond to specific needs of women, and increases their access. Collecting sex-disaggregated data as part of initial assessments and ensuring that women of all ages, including those from marginalised groups, are properly engaged to understand their concerns and needs and that designs respond accordingly. As such women should be involved at all stages of the shelter programme, including settlement planning, beneficiary selection, design and implementation of all shelter activities. Women should also be equally represented in governance structures such as shelter or community committees, to support empowerment through participation and addressing gender norms in terms of decision-making. Disaster preparedness and evacuation planning must also take into account women's specific situation and ensure that appropriate measures are included.

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Annex 8.2

Gender Action Plan

31 May 2023

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1. ABOUT THE ANNEX

This annex presents the gender action plan for the Community Resilience Partnership Program (CRPP). The accompanying Gender Assessment describes critical gender issues related to the types of investments that are likely to be supported by the CRPP, and how such investments can address women's vulnerability and help women to build their resilience to climate change. As such, the gender assessment provides the backdrop for the gender action plan, presented in this annex, which identifies specific action, design features, and monitoring indicators and targets that will be implemented through the CRPP as part of its supported investments.

2. INTRODUCTION

The CRPP is a regional partnership program of the Asian Development Bank (ADB) which aims to help countries and communities in Asia and the Pacific region scale up investments in climate adaptation, especially investments at the community level, that explicitly target the **nexus between climate change, poverty, and gender**. Building women's resilience to climate change and empowering women through transformational investments in climate adaptation at the community level are therefore core pillars of the CRPP. The CRPP is operationalized through the Community Resilience Financing Partnership Facility (CRFPF) which was established by ADB in August 2021 and comprises two separate but interlinked components; the **CRPP Trust Fund (TF)** focusing on upstream support to strengthen the enabling environment required for implementing local adaptation measures at scale; and the **CRPP Investment Fund (IF)** focusing on the efficient roll-out of local adaptation measures through downstream investments.

The TF provides technical assistance and grant resources financed by development partners and administered by ADB, to selected developing member countries (DMC) of the ADB to implement the following three **outputs**: (1) Knowledge and action research on climate risk informed pro-poor community-level solutions strengthened; (2) Institutional and community capacity to develop and deliver climate adaptation investments at community-level strengthened; and (3) Inclusive and pro-poor adaptation investment projects identified and prepared. The TF includes a dedicated Gender Window with funds earmarked specifically for providing technical assistance and grants for identifying, developing, and implementing investments that explicitly strengthen the resilience of women, particularly poor women, and/or adaptation investments led by women that have a specific focus on supporting women to build their resilience.

The IF will provide grant and loan financing to seven selected DMCs, namely, Cambodia, Indonesia, Lao PDR, Pakistan, PNG, Timor-Leste, and Vanuatu, to implement local adaptation measures as part of ADB financed sub-projects that are targeted at the poor and vulnerable population. The IF will deliver three climate related **outputs** (4) information and systems for delivering applied climate-risk informed local investments at scale improved; (5) climate resilient pro-poor livelihoods investments implemented; and (6) pro-poor climate adaptation infrastructure investments implemented.

It should be noted that although the Gender Window is funded by the CRPP TF, it can be leveraged to enhance gender equality outcomes of the Investment Fund in CRPP countries. Examples include; (i) generating the necessary political will, and commitment from governments and communities to invest in women-focused adaptation measures that help tackle climate change; (ii) creating policy incentives and procedures for the inclusion of women in adaptation decision-making at the local level and building their capacity to use participatory tools and procedures for climate risk informed development planning and implementation which can empower local women to act on adaptation; and (iii) place women at the centre of

the design of large-scale investments that explicitly strengthen the climate resilience of women, particularly poor women, and/or adaptation investments that are led by women.

3. ACTION PLAN IMPLEMENTATION, MONITORING AND REPORTING

The CRPP Gender Action Plan (CRPP GAP) will come into effect on approval of the program by GCF. By including gender-related activities that are in line with the sectors of intervention of the program and related to program activities, the CRPP program level GAP presented in this annex will serve as a guiding framework for sub-projects, which will in turn be required to formulate their own GAP according to ADB and GCF gender policies. The sub-project level GAPs will be submitted to GCF in batches on an annual basis along with the approved sub-project documents.

ADB as the Accredited Entity and Executing Entity will implement, monitor, and report the CRPP GAP, consistent with the CRPP log-frame presented in the Funding Proposal. ADB program and project monitoring systems will be followed, in accordance with ADB's policies, procedures, and guidelines related to gender mainstreaming, to track implementation progress of the plan. Numeric targets have been included in the CRPP's GAP where possible to measure the GAP progress and the gender equality outcomes of the program. In some cases, current targets are very conservative and may not reflect the implementation of all activities in all CRPP countries. This is because detailed in country consultations and situation analysis in the context of specific activities is ongoing and a more detailed understanding of the country situation is required before it is known which activities are feasible in each country. It may not be feasible or practical to undertake some activities in every country. In the same regard, some indicators may be improved to be more precise once sub-project Gender Analyses and GAP preparation starts. For example, at this stage it is not possible to estimate at a program level how many women farmers may benefit from capacity building and agricultural extension services, but it will be possible to estimate the numbers of women benefiting from activities as detailed sub-project development advances. Current targets should thus be considered as indicative and will be updated based on the sub-project specific GAPs developed during their detailed design phase. The Annual Performance Reports which will be prepared by ADB to GCF, will capture the implementation progress of the GAP.

The CRPP investment eligibility criteria require all supported sub-projects to be classified as either effective gender mainstreaming (EGM) or gender equity theme (GEN). In line with ADB standard gender policy, this classification explicitly necessitates the development of a Gender Action Plan based on a detailed gender analysis, and therefore the necessary budget is allocated accordingly. The proposed gender category is determined by the resident mission's (country office) gender specialist in the focal country, at the concept stage when the project is in the pipeline. The proposed gender category is confirmed by ADB's gender equality thematic group, which has a member in the CRPP secretariate. At that point, during due diligence, the team proceeds with project preparation, which includes the gender analysis and the development of the GAP.

The gender analysis is based on desk research, gender diagnostics, consultations at the community level, household surveys and national level data. Based on the findings of the gender analysis, the gender action plan activities and targets are identified, and the gender analysis and data collected during project preparation serve as the baseline for establishing these targets. The GAP will also identify the relevant stakeholders responsible for implementation of different activities and verification sources for tracking implementation. A budget for GAP implementation is always allocated, but the exact budget will vary according to the size of the project, the coverage area, and the types of gender activities. The resident missions have close contact with their country counterparts during this process, promoting close collaboration in the design, implementation and monitoring of the GAPs

ADB resident missions which lead the project design have gender specialists and there is at least one gender focal point for each department, sometimes two gender focal points, and these work full time on gender, as well as the gender specialists in resident missions who work full time on gender. They are engaged in project processing, monitoring and project completion reporting. Quarterly reports are submitted per project, and these are tracked by the resident mission and the department. Gender consultants with experience in the sector (ie. gender and livelihoods specialist) are usually contracted to conduct the consultations with the community which include consultations with women and women's groups and support the development of the gender analysis and action plan. For EGM and GEN projects, it is a requirement that a gender specialist be hired. Budget is allocated from the project preparation resources.

Project level monitoring and reporting will combine ADB policies and protocols with GCF criteria to ensure complementarity to existing requirements, without adding unnecessary extra reporting burden. In line with standard ADB practice, a gender specialist is hired for monitoring of sub-project GAPs, and gender specialists participate in team monitoring missions. Mid-term reviews are conducted where the status of the GAP implementation and progress is reviewed. The GAP is included in the Project Administration Manual, as well as in the loan covenants. No changes to the GAP are allowed without engagement from the Gender Equality Thematic group and only under very specific circumstances.

CRPP GENDER ACTION PLAN

Impact Statement: Climate resilient sustainable development in Asia and the Pacific region is advanced.

Outcomes:

- (i) the enabling environment for knowledge, finance and implementation of gender-responsive and/or women led local adaptation measures is improved;
- (ii) finance for pro-poor, gender-responsive and/or women led adaptation is mobilized at scale;
- (iii) food security of poor and vulnerable women is increased; and
- (iv) exposure of poor and vulnerable women to climate shocks and stresses is reduced.

CRPP Outputs	Gender Activities	Targets/Indicators	Timeline	Responsibilities	Cost
Output 1: Knowledge and action research on climate risk-informed pro-poor community level solutions strengthened.					
Activity 1.1 Knowledge and evidence generation.	1.1.1 Carry out action research to generate knowledge and evidence to make the case for women-focused adaptation investments. 1.1.2 Support small-scale innovations and pilot projects focused on building women's resilience.	1.1.1 Research reports presenting evidence that makes the case for women-focused adaptation investments in at least 5 CRPP countries. 1.1.2 Small-scale pilot projects focused on building women's resilience implemented in at least 3 CRPP countries.	Year 1-5	CRPP Secretariat	\$ 1.375 million ¹
Activity 1.2 Evidence sharing and awareness raising	1.2.1 Host multi-stakeholder policy dialogues and knowledge-sharing forums to build a culture conducive to increased investment in women's climate resilience. 1.2.2 Implement skills development trainings for women to build their capacity to effectively participate in multi-stakeholder policy dialogues and knowledge-sharing forums	1.2.1 Stakeholders from at least 7 CRPP countries attend multi-stakeholder policy dialogues and knowledge-sharing forums to build a culture conducive to increased investment in women's climate resilience. 1.2.2 Women from 7 CRPP countries report improved skills and capacity to effectively participate in multi-stakeholder policy dialogues and knowledge-sharing forums.			

¹ The CRPP Gender Window has an allocation of approximately 15% of the total CRPP Trust Fund budget. This is broken down between the three Trust Fund outputs as follows; Output 1 12.5%, Output 2 12.5%, Output 3 75%

Output 2: Institutional and community capacity to develop and deliver climate adaptation investments strengthened.					
Activity 2.1 Development of climate risk assessment and decision support tools.	2.1.1 Integrate gender concerns into climate risk assessment and decision support tools and processes. 2.1.2 Implement skills development trainings for women to build their capacity to effectively participate in climate risk assessments.	2.1.1 100% of climate risk assessment and decision support tools and processes designed through the CRPP integrate gender responsive features. 2.1.2 Women report improved skills and capacity to effectively participate in climate risk assessments.	Year 1-5	CRPP Secretariat	\$ 1.375 million
Activity 2.2 Creation of new and/or improve existing institutional frameworks to promote coherence and convergence across gender, climate, and key sectors.	2.2.1 Support governments to create new and/or improve existing policy frameworks that promote coherence and convergence across gender, climate, and key sectors such as agriculture, land-use planning, and urban planning. 2.2.2 Support governments to create new and/or improve existing communication and coordination mechanisms between gender, climate, and key sectors such as agriculture, land-use planning, and urban planning.	2.2.1 Policy frameworks that promote coherence and convergence across gender, climate, and key sectors such as agriculture, land-use planning, and urban planning are prepared in at least 3 CRPP countries. 2.2.2 Communication and coordination mechanisms are created (new) or improved (existing) in at least 1 CRPP country.			
Output 3: Inclusive and pro-poor adaptation investment projects identified and prepared.					
Activity 3.1. Preparation of large-scale adaptation investment projects	3.1.1 Support the preparation of large-scale investments that explicitly strengthen the climate resilience of women, particularly poor women, and/or adaptation investments that are led by women. 3.1.2 Carry out gender assessments and develop Gender Action Plans for all CRPP IF sub-projects.	3.1.1 100% of CRPP investment projects have at least one outcome indicator explicitly related to building women's resilience. 3.1.2 100% of investments supported by the CRPP use sex-disaggregated climate risk information and data. 3.1.3 100% of investments prepared with support of the CRPP TF Gender Window include outputs that address structural inequalities between men and women.	Year 1-5	CRPP Secretariat	\$ 8.5 million

	3.1.3 Revise and update the CRPP Program Level Gender Assessment and Action Plan on a yearly basis.	3.1.4 100% of investments supported by the CRPP have specific sex-disaggregated indicators integrated into their monitoring and evaluation frameworks. 3.1.5 Gender assessments are carried out for 100% of CRPP IF sub-projects. 3.1.6 Gender Action Plans are developed for 100% of CRPP IF sub-projects. 3.1.7 The CRPP program level gender assessment and gender action plan is updated annually.			
Output 4: Information and systems for delivering applied climate-risk informed investments at-scale improved					
4.1 National frameworks for CIS are set-up or improved, and are gender responsive	4.1.1 Include sex-disaggregated data in new and improved national frameworks for CIS. 4.1.2 Develop dissemination channels for CIS and multi-hazard early warning systems that are designed to meet women's specific needs². 4.1.3 Engage with women's organizations as platforms for CIS and early warning dissemination and trainings for women to demonstrate its application to manage farm level climate risk.	4.1.1 100% of CIS frameworks that have been introduced or improved as part of CRPP investments include metrics that use sex-disaggregated data. 4.1.2 100% of CIS and multi-hazard early warning systems that have been introduced or improved as part of CRPP investments include information dissemination channels that are designed to meet women's specific needs. 4.1.3 70% of women participants in trainings report improved knowledge	Year 2-8	PMU ³	Covered under PIC ⁴

² Considering gender differentiated needs for CIS, differences in determinants of access to CIS, and use of technology, i.e. mobile phone ownership and use.

³ The Project Management Unit (PMU) of the related investment projects will be responsible for implementing and monitoring gender activities at the project level. The CRPP secretariat will be responsible for ensuring the CRPP Gender Action Plan at the program level.

⁴ Whenever implementation of the gender activities/actions are part of implementation of the regular project activities, they will be covered under the corresponding budget for these project activities, of which up to 5% can be used to make special arrangements (e.g. trainings; different venue; child supervision) and/or develop targeted materials where and when needed to ensure women can fully participate in and benefit from the project activities, and more particularly to ensure gender targets included in this project gender action plan are met.

		and/or skills related to the application of CIS for managing farm level risk.			
4.2 Social protection systems are climate risk informed.	4.2.1 Include sex-disaggregated data in the targeting and monitoring of beneficiaries. 4.2.2 Design and implement complimentary livelihood and financial literacy trainings specifically for women farmers. 4.2.3 Build partnerships between government and community-based women's groups to effectively deliver social protection programs.	4.2.1 100% of new or improved adaptive social protection systems use sex-disaggregated data. 4.2.2 Women farmers report improved knowledge, attitudes and skills related to livelihoods and financial literacy as a result of CRPP investments in at least 1 country. 4.2.3 Perception from women of improved collaboration between government and community-based women's groups for the delivery of social protection programs as a result of CRPP investments in at least 1 country.	Year 2-8	PMU	Covered under PIC
4.3 Climate risk informed local government development plans.	4.3.1 Prepare climate risk and vulnerability assessment tools that capture gender-based vulnerabilities. 4.3.2 Conduct trainings, workshops and consultations that build the capacity of women to use the results of climate risk and vulnerability assessments for local adaptation planning. 4.3.3 Identify barriers to women's participation in resilience related decision-making bodies in different CRPP countries and implement activities to strengthen public participation processes to engage local women in decision-making related to resilience building. 4.3.4 Implement skills and advocacy activities to increase understanding and	4.3.1 Gender-based vulnerabilities are captured using new/improved climate risk and vulnerability assessment tools developed through CRPP investments in at least one country. 4.3.2 As a result of CRPP investments in at least one country, women report improved knowledge, skills and attitudes for using the results of climate risk and vulnerability assessments for local adaptation planning. 4.3.3 As a result of CRPP investments in at least 1 country, women report improved participation in local decision-making related to resilience building. 4.3.4 Local planners and community leaders report increased, knowledge	Year 2-8	PMU	Covered under PIC

	capacity of gender mainstreaming of local planners and community leaders.	attitudes and skills related to gender mainstreaming as a result of CRPP investments in at least one country.			
Output 5: Climate resilient pro-poor livelihoods investments implemented.					
Activity 5.1 Climate resilient agroecological systems.	5.1.1 Deliver agriculture extension services to women farmers, for new climate resilient crop varieties and production systems 5.1.2 Households headed by women use small grants for implementing new diversified cultivation practices in selected CRPP countries. 5.1.3 Support national institutions to offer technical and vocational education and training (TVET) that responds to the needs of women employed in agriculture, to increase climate resilient livelihoods	5.1.1 100% of CRPP sub-projects that include components on delivering agriculture extension services for new climate resilient crop varieties and production systems, include related interventions designed specifically for women farmers. 5.1.2 At least 50% of households that use small grants for implementing new diversified cultivation practices in selected CRPP countries are headed by women. 5.1.3 100% of TVET training programs implemented under the CRPP service the specific needs of women.	Year 2-8	PMU	Covered under PIC
Activity 5.2 Information technology and risk management services for climate resilient livelihoods.	5.2.1 Invest in information technology systems to facilitate rapid information flows to small-holder women farmers. 5.2.2 Develop low cost risk transfer products that respond to the specific needs of women small-holder farmers. 5.2.3 Design and deliver capacity building programs on financial literacy for women insurance customers.	5.2.1 Small-holder women farmers use information technology systems to facilitate rapid information flows as a result of CRPP investments in at least 1 CRPP country. 5.2.2 Women small holder farmers use low-cost risk transfer products that respond to their specific needs as a result of CRPP investments in at least 1 CRPP country. 5.2.3 Women insurance customers have improved knowledge, attitudes and skills on financial literacy as a result of CRPP investments in at least 1 CRPP country.	Year 2-8	PMU	Covered under PIC

Activity 5.3 Climate resilient agriculture supply chains.	5.3.1 Integrate gender responsive design features into upgrading of product processing facilities. 5.3.2 Improve women's access to markets, ensuring that it is not interrupted by climate shocks and stresses. 5.3.3 Promote technological solutions within the agricultural value chain that can help to reduce women's time poverty and burden of care.	5.3.1 100% of new/improved facilities installed through CRPP sub-projects respond to the specific needs of women. 5.3.2 Women's perception of market accessibility is improved as a result of CRPP investments in at least 1 country. 5.3.3 Women report reduced time poverty and burden of care as a result of CRPP investments in climate resilient agriculture supply chains in at least 1 country.	Year 2-8	PMU	Covered under PIC
Output 6: Climate adaptation pro-poor infrastructure implemented.					
Activity 6.1: Ecosystem-based infrastructure	6.1.1 Consult with women as part of the process to identify locally appropriate green infrastructure. 6.1.2 Undertake trainings for local governments, communities and community-based organizations in planting, rehabilitation and maintaining green infrastructure that respond to needs of women. 6.1.3 Provide small grants to local women's groups for planting and rehabilitation of green infrastructure; 6.1.4 Provide leadership roles for women in natural resource management committees. 6.1.5. Implement skills development trainings for women to build their capacity to effectively carry-out their roles in planning and decision-making structures.	6.1.1. At least 50% of participants in consultations/workshops on locally appropriate green infrastructure undertaken through CRPP sub-project implementation are women. 6.1.2 Women show improved knowledge, attitudes and skills related to planting, rehabilitating and maintaining green infrastructure as a result of CRPP investments. 6.1.3 Small grants are provided to local women's groups for green infrastructure through CRPP sub-project activities in at least 1 country. 6.1.4 Women occupy decision-making positions in natural resource management committees as a result of CRPP sub-project activities in at least 1 country. 6.1.5 Women report improved skills and knowledge for participating in planning and decision-making structures as a result	Year 2-8	PMU	Covered under PIC

		of CRPP activities in at least 1 CRPP country.			
Activity 6.2 Flood and landslide protection infrastructure.	6.2.1 Women consultations for the identification and prioritization of small-scale protective infrastructure as part of local government planning processes. 6.2.2 Recruitment and training of women to undertake civil works for small scale flood protection and/or landslide protection measures. 6.2.3 Design and implement gender responsive trainings for local governments and communities, in maintenance of protective infrastructure.	6.2.1 At least 50% of participants in participatory planning meetings for CRPP sub-project activities are women. 6.2.2 At least 50% of small-scale protective infrastructure projects selected for financing were prioritised by women. 6.2.3 At least 50% of the paid workforce for undertaking civil works for small scale flood protection and/or landslide protection measures through CRPP sub-projects are women. 6.2.4 100% of women contracted through CRPP project activities receive equal pay to men. 6.2.5 At least 50% of participants in local governments and community trainings on operation and maintenance of protective infrastructure implemented through CRPP sub-projects are women.	Year 2-8		
Activity 6.3: Multi-purpose emergency shelters	6.3.1 Collect and analyse sex-disaggregated data and information as part of gender responsive needs assessment for multipurpose emergency shelters. 6.3.2. Implement skills development trainings for women to build their capacity to effectively carry-out their roles in planning and decision-making structures.	6.3.1 100% of needs assessments for multipurpose emergency shelters constructed under the CRPP collect and use sex-disaggregated data and information. 6.3.2 Women report improved skills and knowledge for participating in planning and decision-making structures related to emergency shelters as a result of CRPP activities in at least 1 CRPP country.	Year 2-8	PMU	Covered under PIC

	6.3.3 Set-up gender representative community committees as governance structures for emergency shelters. 6.3.4 Construct emergency shelters with gender inclusive and socially appropriate design features. 6.3.5 Integrate gender considerations into disaster preparedness and evacuation planning capacity building and training materials.	6.3.3 At least 50% of the members of emergency shelter committees established through the CRPP sub-projects are women. 6.3.4. 100% of multi-purpose emergency shelters constructed through CRPP sub-projects integrate gender inclusive and socially appropriate design features. 6.3.5 Women show improved knowledge, attitudes and skills related to disaster preparedness as a result of CRPP investments in at least 2 countries. 6.3.6 100% of community evacuation plans formulated through CRPP sub-projects respond to specific needs of women.			
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